



Transportation Management (TM)

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Forwarding Order Management

Use

You can use the **Forwarding Order Management** component in Transportation Management (TM) to create, edit, and confirm forwarding orders from your ordering parties. You can also already specify the route and have the transportation charges calculated here. The orders then form the basis for transportation planning.

In addition to creating the forwarding order, you can also enter the data as a forwarding quotation and send it to the ordering party. You can then create a forwarding order based on the forwarding quotation.

Note that the following function is not currently released:

- Integration of the forwarding order and the FWSD in TM with Credit Management in SAP S/4HANA

Integration

The TM component **Forwarding Order Management** is integrated with the following other TM components:

Component	Integration Type
Freight Order Management	You can create freight orders and freight bookings from forwarding orders.
Planning	Forwarding orders form the basis for planning and executing the transportation.
Forwarding Settlement	Based on the forwarding order, you can create a forwarding settlement to trigger the invoice creation process.
Master Data	As a prerequisite for creating forwarding orders, the necessary master data must have been created.

Forwarding Order Management is also integrated with the dangerous goods management and global trade processes. For more information, see [Considering Dangerous Goods](#) and [Global Trade](#).

Features

You can use the business documents from Forwarding Order Management to enter all the information required for planning and executing the transportation of goods, such as pick-up and delivery addresses, dates/times, and goods to be transported. You can enter the data manually or transfer it automatically via PI interfaces.

After you have edited the data, you can send a confirmation to your ordering party. Moreover, you can trigger transportation planning from the forwarding orders and generate the following objects, for example, to further plan and execute the transportation:

- [Freight Units](#)
- [Freight Orders](#)

More Information

Forwarding Order

Definition

An order from an ordering party to a carrier or logistics service provider with respect to the transportation of goods from a shipper to a consignee according to the agreed-upon conditions.

Use

You can use the forwarding order business document in Transportation Management (TM) to enter data for the transportation of goods that you have agreed upon with an ordering party. You can enter the data manually (for example, after a telephone call) or you can transfer it automatically (via PI interfaces, EDI). The forwarding order is the basis for further transportation planning. It contains the most important data for planning and executing the transportation, such as the pick-up and delivery address, dates/times, and the goods that are to be transported. From the forwarding order, you can determine and define the transportation route, and create freight units and a transportation plan. You can also have the system calculate the transportation charges and start the invoicing.

Structure

The design of the user interface for the forwarding order is determined by the following factors:

- Transportation mode (air, sea, or land)
- Forwarding order type that you define in Customizing (see [Settings for the Forwarding Order](#))
- Individual screen settings in the apps for forwarding orders from the launchpad

On the initial screen of the forwarding order, you can specify the forwarding order type and the transportation mode (if you have not already predefined this in the forwarding order type). The user interface for the forwarding order is automatically adjusted to the requirements of the transportation-mode-specific business processes. The forwarding order type also predefines which additional information or special tab pages are displayed for each business process (such as displaying dangerous goods information for the order items).

You can also configure your settings in the apps to define whether a tab page, assignment block, or specific fields in the forwarding order are displayed or hidden. The forwarding order generally has the following basic structure:

- Areas for general data, business partners, locations, and dates/times

In these areas of the forwarding order, you enter data (or the system displays data) that is valid for all items in the order. This data includes, for example, the sales organization, Incoterms, and service level. In addition, the totaled quantity data and weight data for the order items is displayed.

Furthermore, you can specify attributes for process control, such as the transportation mode and shipping type.

For more information, see [Creation of a Forwarding Order](#) and [Attributes for Process Control](#).

- Ordered route and actual route

In these assignment blocks or on the tab pages, you can enter the route requested by the ordering party and the actual route with the individual transportation stages. For more information, see [Determination of the Route](#).

- List of order items

In this table, you enter the item-specific data in the form of an item hierarchy. For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

- Document flow

The system displays here which business documents have been entered into the forwarding order and which business documents were created based on it. You can also display the follow-on documents from here, and call a freight unit, for example.

- Output management

Here the system displays which documents were already printed or sent (by e-mail or fax, for example) for this forwarding order. Moreover, you can print documents such as bills of lading from here. For more information, see [Output Management](#).

- Status

Here the system displays the different statuses of the document, such as the life cycle status and the blocking status. For more information, see [Status of a Forwarding Order](#).

- HBL or HAWB

You can display data for house bills of lading (HBLs) or house air waybills (HAWBs) in this assignment block or on the tab. In addition, you can make individual allocations of waybill numbers to freight units or remove the allocations again. For more information, see [Building and Printing of House Bills of Lading and House Air Waybills](#).

- Profitability

You can display a profitability analysis in this assignment block or on the tab page. From here, you can also navigate to the individual business documents. For more information, see [Profitability Analysis](#).

- Specialist, process-dependent tab pages

In addition to those described above, the system also displays additional tab pages or assignment blocks if you have activated the corresponding business process in the document type, such as:

- Charge calculation and internal charge calculation

The system displays the data from an (internal) transportation charge calculation that you have made for the forwarding order. Moreover, you can enter the transportation charges manually as well. For more information, see [Charge Calculation](#).

- Air cargo security

You can enter the data required if you want to perform an air cargo security check. For more information, see [Air Cargo Security](#).

- Instructions

If you have defined instructions for executing services in Customizing (such as an instruction for cleaning a container), the system displays these here. You can also change or delete instructions or create new instructions on this tab page. For more information, see [Management of Instructions](#).

- Change documents

If you have activated change tracking in Customizing, the system displays any changes to your documents here. For more information, see [Change Tracking](#).

When you create air forwarding orders, you can enter the nature of goods on the **Nature of Goods** tab page. You can enter the nature and quantity of goods, which is printed on air waybills, and the nature of goods for manifest, which is printed on the security manifest and cargo manifests. For more information, see [Nature of Goods](#).

In addition to the comprehensive view of the forwarding order user interface, you can also use the [Page](#) function to access a view for fast order entry. This view contains all input fields that are required to complete a forwarding order on a single screen (without tab pages). You can also enter values more quickly by using the keyboard (with the exception of the list of order items).

Integration

The business document for the forwarding order is integrated with further business documents in TM, such as the following:

- [Forwarding Quotation](#)

If you have sent a forwarding quotation to an ordering party and it was accepted by the ordering party, you can create a forwarding order based on this forwarding quotation.

- [Freight Unit](#)

You can create freight units for this forwarding order from the order.

- [Forwarding Settlement Document](#)

You can create a forwarding settlement document for this forwarding order from the order.

More Information

[Creation of a Forwarding Order](#)

[Editing of a Forwarding Order](#)

Status of a Forwarding Order

The following document describes important statuses of a forwarding order, which are mainly displayed on the [Status](#) tab page.

Life Cycle Status

Status	Description
New	A new forwarding order was created, and no freight units or transportation units have been created for this order yet.
In Planning	The forwarding order is in the planning process, and at least one freight unit or transportation unit has been created for the order.
Planned	The forwarding order is in the planning process. All freight units and transportation units assigned to it have been planned.
In Execution	The forwarding order is in the execution process. At least one follow-on document (not necessarily a freight unit or a transportation unit) is in execution. For more information about the execution of a forwarding order, see the execution status of the forwarding order or of the follow-on documents.
Executed	Execution of the forwarding order has been completed. Execution of all planning documents assigned to the forwarding order has been completed.
Partially Invoiced	The forwarding order is in the invoicing process. One or more forwarding settlement documents were created, but the invoicing has not been carried out completely yet.
Completed	Processing of the forwarding order has been completed. Execution has finished, the invoicing has taken place, and existing instructions have been completed (if this was specified in the Customizing settings for the forwarding order type). The status is also set by the system if the forwarding order was created only for

Status	Description
	settlement purposes (for example, as a service order) and the settlement documents were created in their entirety.
Determination Pending	This status is set temporarily by the system if the forwarding order was reopened after it had been assigned the status Complete . If you make changes to the forwarding order and save it again, the status also changes again (for example, back to In Execution , Partially Invoiced , or Completed depending on the type of change).
Canceled	The forwarding order was canceled.
Draft	The system sets this status if an import forwarding order is created or updated. You can change the sales organization but the forwarding order is locked for all other changes.

Planning Status

Status	Description
Planning Not Started	The planning process has not yet been started for the forwarding order.
In Planning	The forwarding order is in the planning process. At least one freight unit or transportation unit has been created for the forwarding order; however, a freight document has not yet been assigned.
Partially Planned	The forwarding order is in the planning process. At least one freight unit (FU) or transportation unit (TU) has been created for the forwarding order. Moreover, at least one freight document has been assigned to the forwarding order; however, not all FU or TU stages have been planned yet.
Planned	The forwarding order is in the planning process. All freight units and transportation units assigned to it have been planned.

Execution Status

Status	Description
Execution Not Started	Execution of the forwarding order has not yet been started.
In Execution	The forwarding order is currently being executed. At least one of the planning documents linked to the forwarding order is in the execution process.
Executed	Execution of the forwarding order has been completed. Execution of all planning documents assigned to the forwarding order has been completed.

Confirmation Status

Status	Description
Not Confirmed	No forwarding order confirmation has been created yet, or an existing confirmation was reset because data in the forwarding order has changed.
Confirmed	A forwarding order confirmation has been created.

Approval Status

This status is only displayed if you have activated the approval workflow.

Status	Description
Not Checked	The approval limits were not yet checked for this forwarding order.
No Approval Required	The approval check was conducted for this forwarding order. The defined limits were not exceeded.
Approval Required	The approval check was conducted for this forwarding order, and the defined limits were exceeded. The forwarding order was blocked for further processing.
Approved	The approval workflow was executed, and the forwarding order was approved by the person responsible. The block of the forwarding order for further processing was removed.
Not Approved	The approval workflow was executed, and the forwarding order was not approved by the person responsible. The forwarding order remains blocked for further processing.

Delivery Status

The status shows at header level of the forwarding order whether the freight units and transportation units assigned to the forwarding order were actually delivered. This data originates from the associated freight document. In addition to this status at header level, there is an additional status at order item level that specifies whether the freight units and transportation units for the order item were delivered (cargo receipt status).

Status	Description
Delivered	All freight units and transportation units that are assigned to the forwarding order have been delivered.
Partially Delivered	At least one but not all freight units and transportation units in the forwarding order requests were delivered.
Not Delivered	None of the freight units or transportation units in the forwarding order requests were delivered.

FWQ Assignment Status

This status indicates whether the forwarding order was created on the basis of a forwarding quotation or whether a forwarding quotation was assigned to it subsequently. For more information, see [Subsequent Assignment of Forwarding Quotations](#).

Organization Interaction Status

This shows the status of the interaction between organizational units, for example, between a sales organization and a planning and execution organization. It is available for a stage of the route and also as an aggregated status.

For more information, see [Interaction Between Organizational Units](#).

HBL or HAWB Status

This status is displayed on the [General Data](#) and [HBL or HAWB](#) tabs of a forwarding order. It supports, for example, communication between a customer service agent and a capacity manager at the gateway. The status shows you whether the house bills of lading (HBLs) or house air waybills (HAWBs) assigned to the forwarding order have been completed by the customer service agent ([Finalized](#)) and are available for subsequent processes.

You can set the status manually for the forwarding order using a menu option and reset it to [Not Finalized](#). If the HBL or HAWB number has not been assigned, this is done automatically by setting the status.

If you set the status for a forwarding order with the forwarding order type [Direct Shipment](#), it is set for both the HAWB and the master air waybill (MAWB).

Status	Description
Finalized	All house air waybills in the forwarding order have been completed.
Partially Finalized	At least one but not all house air waybills in the forwarding order requests were completed.
Not Finalized	No house air waybill in the forwarding order has been completed.

Status for Air Cargo Security and Compliance with Foreign Trade Regulations

This status is displayed on the **Status** tab if you have activated the corresponding processes in Customizing for the forwarding order type. For more information, see [Settings for the Forwarding Order](#), [Air Cargo Security](#), and [Trade Compliance Check](#).

Blocking the Forwarding Order

In addition to the forwarding order statuses described above, the system also displays the block status for this order on the **Status** tab page. You can also set blocks manually. For more information, see [Blocking Information of a Forwarding Order](#).

Blocking Information of a Forwarding Order

Use

You can set blocks manually or have the system set blocks automatically for the following functions and processes in the forwarding order:

- Planning of the forwarding order, in other words, creation of freight units and transportation units, and creation of freight orders
- Execution of the forwarding order and associated planning documents
- Confirmation of the forwarding order
- Creation of the forwarding settlement document

In Customizing you can specify the successor documents to which the blocks are copied. For more information, see [Blocking Information of Freight Units and Capacity Documents](#).

Prerequisites

You have defined block reasons in Customizing. For more information, see Customizing for **Transportation Management** under **Basic FunctionsGeneral SettingsDefine Block Reason Codes**.

Features

You can display planning, execution, confirmation, and settlement blocks on the **Status** tab page in a forwarding order. Furthermore, you can set or remove blocks manually:

- Manual setting of blocks

When you set a block manually, you can select a block reason from the values defined in Customizing. You can also set, for example, multiple planning or execution blocks and choose different block reasons. All the block reasons are displayed both in the aggregated view of the blocks next to the block status and in the successor documents.

- Automatic setting of blocks

Blocks are set by the system, for example, if the forwarding order does not yet contain all the required information and is therefore incomplete for planning. In the block details you can get information about which fields have still to be filled.

When you choose **Check** to check the completeness of the forwarding order, the system also checks the block status and displays the same detailed messages.

You can use a Business Add-In (BAI) to change the standard logic for checking the completeness of a forwarding order. This means that you can, for example, specify that certain fields do not have to be filled in contrast to the standard system (such as the movement type). Moreover, you can include additional fields in the check. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) [Forwarding Order Management](#) [Forwarding Order Management](#) [BAI: Enhancements of FWO Completeness Checks](#).

The system sets further automatic blocks when it executes the approval workflow or during the dangerous goods check, for example.

- Removal of blocks

When you remove a block, you can enter a comment. The system also logs and displays who removed the block and when.

Note that you require the authorization to remove blocks. For more information, see [Authorizations](#).

You cannot create a forwarding settlement document if the settlement block is set.

i Note

Note that changes to a business partner block are not always immediately visible in the forwarding order. If you have opened a forwarding order and have in the meantime made changes to the business partner's master data, you do not see these changes immediately. You first have to close the forwarding order and open it again. It is not sufficient to update the screen data.

Settings for the Forwarding Order

The following describes the general settings that you can or must make for a forwarding order. Specific settings for individual functions or processes can be found in the individual chapters.

Profiles

You have created the following profiles in Customizing. Creating these profiles is optional, but you do need them for important follow-up actions such as planning or printing:

- Planning profile

You use the planning profile to determine how transportation planning proceeds further, which you can trigger from the forwarding order, or on what basis the system determines the routes. For more information, see [Planning Profile](#).

- Output profile

You can use the output profile to determine which outputs are possible for the forwarding order (print documents and EDI messages). Standard profiles are available, but you can also define your own profile.

For more information, see Customizing for [Cross-Application Components](#) under [Processes and Tools for Enterprise Applications](#) [Reusable Objects and Functions for BOPF Environment](#) [PPF Adapter for Output Management](#) as well as [Output Management](#).

- Dangerous goods profile

If you transport dangerous goods with the forwarding order, you can use the dangerous goods profile to make settings for the processing of the dangerous goods items, such as for the output of error messages.

For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Dangerous Goods](#) > [Define Dangerous Goods Profile](#) as well as under [Considering Dangerous Goods](#).

- Partner Determination Profile

You can use this profile to indicate that the system is to assign business partners to your forwarding order automatically.

For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Business Partner](#) > [Define Partner Determination Profiles](#) and under [Business Partner Determination](#).

Other Customizing Activities

- You have defined the **forwarding order type**. You use this document type to determine a series of characteristics for the forwarding order. In this way, you can assign the profiles described above and the freight unit building rule or condition to the forwarding order type, and make certain default settings, such as the following:
 - Transportation mode (air, sea, land)
 - Define default shipping type, traffic direction and sales organization
 - Automatic confirmation of the forwarding order
 - Automatic freight unit building
 - Activate an approval workflow and a credit limit check
 - Activation and deactivation of charge calculation and internal charge calculation
 - Activation and deactivation of forwarding settlement and internal settlement
 - Automatic charge calculation and internal charge calculation
 - Define and activate process-specific parameters and functions, such as the air waybill type (AWB type) and air cargo security check
 - Activate change tracking

In addition, you use the forwarding order type to determine the appearance of the user interface. This means for instance that the input fields and menu options are tailored to the transportation mode. The system only displays dangerous goods information for the order items, for example, if you have assigned a dangerous goods profile to the forwarding order type.

For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#).

- If required, you have extended the standard checks that the system executes to decide whether a freight unit or transportation unit must be rebuilt during business document maintenance. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BADIs\) for Transportation Management](#) > [Forwarding Order Management](#) > [Freight Unit Building](#) > [BAdI: Extension of Checks for Triggering FU Building from TRQ](#).
- You have defined a **number range interval**, within which the system creates the identifiers (IDs) of the forwarding orders of a previously defined type. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Number Range Intervals for Forwarding Order Management](#).
- You have defined the **transportation mode**, such as sea, air, and road. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Transportation Network](#) > [Transportation Lane](#) > [Define Transportation Mode](#).

- If you want to use Forwarding Order Management to support an air or sea freight-specific process, you have defined **transportation mode-specific codes**, such as airline codes or standard carrier alpha codes (SCAC) in Customizing and assigned them to your business partners as required. For more information, see [Transportation-Mode-Specific Codes](#) and [Business Partner](#).
- You can specify that the **transportation service levels** available in a forwarding order are assigned to the same transportation mode as that for the forwarding order, or are not assigned to any specific transportation mode. If you do not make any setting here, the selection of service levels in the forwarding order is not restricted. For more information, see Customizing for [Transportation Management](#) under ► [Forwarding Order Management](#) ► [Define Transportation Service Level Codes](#) ► as well as [Transportation Service Levels](#) .
- If you want to use your own **shipping types**, you have defined shipping types in Customizing and assigned them transportation modes if required. You can then select these shipping types in Customizing for forwarding order types and forwarding quotation types using the input help on the user interface for the forwarding order and forwarding quotation. For more information, see Customizing for [Transportation Management](#) under ► [Forwarding Order Management](#) ► [Define Shipping Types](#) ►.
- If you want to define reasons for blocking a forwarding order, you have defined **block reasons** in Customizing. For more information, see Customizing for [Transportation Management](#) under ► [Basic Functions](#) ► [General Settings](#) ► [Define Block Reason Codes](#) ►.

Master Data

- You have defined the **business partners and locations** as well as data such as Incoterms or service level. For more information, see [Master Data](#).
- If you want to have an **approval workflow** for your forwarding order, you have activated this workflow in Customizing for the document type (see above) and have made the required settings for the workflow. For more information, see [Approval Check](#).
- If you want to support a process for which a house bill of lading (HBL) or house air waybill (HAWB) is required, you have assigned **number ranges for house bills of lading or house air waybills** to sales organizations and ordering parties in the master data for Transportation Management. On the user interface of the forwarding order, you can then assign these HBL or HAWB numbers to a forwarding order or assign a number from the corresponding number range.

If for particular scenarios you want to exclude being able to assign an HBL or HAWB number from a number range in the forwarding order document, you can deactivate this function in Customizing for the forwarding order type.

For more information, see [Waybill Stock Definition](#).

Other Settings

- You have optionally defined a **freight unit building rule** (FUB rule) and determined a condition for determining the FUB rule. For more information, see [Freight Unit Building Rule](#).
- You have made the settings for defining stages of a **route**. For more information, see [Determination of the Route](#).
- You have made the settings for creating **order items**. For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

Creation of a Forwarding Order

Use

In Transportation Management (TM), you can create forwarding orders (see [Forwarding Order](#)) both manually and automatically. You can create a forwarding order in the following ways:

- On the basis of a quotation

You first create a forwarding quotation and use the data entered there to create a forwarding order.

You can also subsequently assign a forwarding quotation to a forwarding order. However, this is subject to certain prerequisites such as corresponding data between the documents.

- On the basis of a template

A template can contain the most important default data, such as sales organization or Incoterms, and thus facilitates the creation of new orders.

You can create a new template or create a template based on an existing forwarding order (from the forwarding order or on the order management overview). The system copies the data in the forwarding order, such as document type and order data, into the template (with the exception of dates/times and planning and execution data).

- As a copy of an existing forwarding order

If you select **Copy** in the forwarding order, the new order already contains a lot of the data that has been copied, including data about the route. However, it does not contain any dates/times and no follow-on documents, such as freight units or freight orders, are assigned to it. You can also copy the forwarding order without transferring the data about the actual route. The life cycle status is **New**.

- From the order management overview

You create a new forwarding order without reference to another object.

Instead of creating the forwarding order fully, you can also first enter the order in a fast entry screen and save it there. This screen provides an overview of the most important data. You can call up the screen again later on and add further data.

- From a forwarding agreement

You can create a forwarding order from a forwarding agreement or with reference to a forwarding agreement. A forwarding agreement always consists of an ID and a version. If you create a forwarding order from a forwarding agreement item or with reference to a forwarding agreement item, service items in the forwarding agreement can be transferred to the forwarding order. In addition, further information from the forwarding agreement is automatically transferred to the forwarding order, such as transportation mode, shipping type, and sales organization.

- Using PI interfaces

You receive the data directly from the ordering party by means of an SAP NetWeaver Process Integration (SAP NetWeaver PI) interface.

Note that the enterprise services of forwarding order management are only released with restrictions. For more information, see [Released with Restrictions](#).

The following describes how you create a new forwarding order manually. For more information about creating forwarding quotations and their conversion to forwarding orders, see [Creation and Editing of a Forwarding Quotation](#).

Prerequisites

You have made the settings required for the forwarding order. For more information, see [Settings for the Forwarding Order](#).

Process

1. Entering the Data on the Initial Screen

On the initial screen of the forwarding order, you specify important data for the forwarding order, such as the forwarding order type and the transportation mode. You can also specify whether you want to create the forwarding order with reference to other documents, as a copy of other documents, or as a template. Note the following:

- If you do not enter a forwarding order type, the system uses the default type defined in Customizing for forwarding order types.
- You cannot change the forwarding order type subsequently in the document itself. However, you can still change some of the other parameters in the document, such as the sales organization. This depends on whether you have predefined a value in Customizing that can still be changed.
- If you have already defined the transportation mode in the forwarding order type, this is displayed and can no longer be changed. If you specify a transportation mode first, the input help for the forwarding order type only allows you to select order types that are valid for this transportation mode. If you have not defined the transportation mode in the forwarding type, you can enter it manually, or you can leave it empty and enter it later.
- If you copy an existing forwarding order, you can enter a new forwarding order type for the new forwarding order you are creating. This can, however, lead to inconsistencies with copied data, if, for example, the copied data has been determined from the Customizing settings for the forwarding order type (such as the transportation mode and the shipping type). If the data for the new forwarding order type is unique, the system updates the data automatically. Otherwise you have to change the inconsistent data manually.
- You can enter a reference to a specific version of a forwarding agreement. If you also specify the forwarding agreement item, the system automatically transfers service items that are flagged as flow services from the forwarding agreement item to the forwarding order. For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

You can also specify an additional forwarding agreement, version, and item that are to be used for the corresponding import or export business document. For example, if you create an export forwarding order, you can enter the additional forwarding agreement that is to be referenced by the corresponding import forwarding order.

- You can enter a default route on the initial screen. The system takes this route into account when it creates the requested route. In addition, the system checks whether the data for the default route matches the other data that you have entered on the initial screen, for example, the transportation mode or shipping type.

If you create a forwarding order based on an item from a forwarding agreement (FWA), you can select a default route that has been defined for the FWA item.

The default route that you enter or select here is displayed in the general data area of the forwarding order. You can still change it here. In this case, the system again checks whether the new default route corresponds to the data in the forwarding order or to an assigned FWA item.

The forwarding order that has the life cycle status **New** then appears.

2. Entering the General Forwarding Order Data

You first create the general data for the forwarding order such as data for the sales organization, process control attributes and Incoterm data. Note the following:

- If you have defined this data in Customizing for the forwarding order type, the system has already automatically determined the sales organization data.
- If a transportation mode has not already been specified, you can enter it here. If you leave the transportation mode empty, any fields that are dependent on the transportation mode (for example, shipping type, default route, and service level) are then also empty, but can be entered manually. However, this may restrict the available transportation modes. For example, if you enter the shipping type, you can then specify only the transportation mode that is assigned to the shipping type in Customizing.
- If you enter the transportation mode manually, you may be able to change it at a later stage of document processing. However, this depends on several factors. For example, you cannot change the transportation mode if

follow-on business documents exist for the freight units (FUs) or transportation units (TUs), and the execution of at least one main carriage stage of one of these FUs or TUs has already started.

If you change a manually entered transportation mode from one category to another (for example, from sea to air), the system clears the values in all fields that are not relevant for the new transportation mode and saves the document automatically. It then loads the UI that corresponds to the new transportation mode. If, however, you have defined a Web Dynpro application configuration in the FWO type, the system does not make any changes to the UI when the mode is changed. The ordered route, actual route and any dependent freight documents are not changed automatically. You must update them manually, if required.

- o In addition to the transportation mode, you can specify a series of attributes for process control, such as the shipping type, movement type, and air waybill type. Depending on the Customizing settings, these attributes are already automatically pre-defined. For more information, see [Attributes for Process Control](#).
- o The payment terms are considered during planning and invoicing. Depending on your Customizing settings for the forwarding order type, you can enter Incoterms or freight terms. If you do not enter any payment terms here, the system determines the corresponding forwarding agreement based on the ordering party and it creates only one forwarding settlement document for the ordering party.

Note that if you define party roles for payers in a forwarding order, the system then takes the payment terms of these payers into account when it calculates charges instead of the payment terms of the agreement parties. The system transfers the payment terms that were defined in forwarding orders to the subsequent processes. When it calculates charges, the system does not take into account any changes to payment terms that were made in settlement documents.

For more information, see [Incoterms in Forwarding Orders](#) and [Freight Terms in Forwarding Orders](#).

- o If the carrier to whom the main carriage of the forwarding order is assigned performs invoicing directly with the ordering party based on a special freight agreement, you must specify this in the forwarding order. For more information, see [Uncontrolled Transportation](#).
- o If you want to consolidate goods from different suppliers that are intended for one consignee before the main carriage and load them in one container, you can specify that this is to be **Buyer's Consolidation**. You can also consolidate goods that originate from a supplier and that are intended for several consignees (**Shipper's Consolidation**). For more information, see [Buyer's Consolidation \(BCO\) in Export/Import](#) and [Shipper's Consolidation \(SCO\) in Export/Import](#).
- o You can enter or change a reference to a forwarding agreement or additional forwarding agreement on the **General Data** tab page in the **Forwarding Agreement References** screen area. Here, a link is provided to the relevant version of the referenced forwarding agreement.

3. Entering the Business Partner, Locations and Dates/Times

Enter data for your business partner, such as the shipper, consignee and ordering party and define the location and dates for the entire forwarding order. Note the following:

- o If you have defined this data in Customizing, the system determines the business partner automatically using the partner determination profile specified in the forwarding order type. For more information, see [Business Partner Determination](#).
- o If you specify a location and have activated this in movement type Customizing, the system automatically determines the associated shipper or consignee from the master data. If you specify a business partner, the system determines the associated location data, as long as this data is unique (in other words, only one location exists for which the business partner has been specified in the master data).
- o You can manually overwrite the address of the business partner that was automatically determined. It is marked as a deviating address and copied to the follow-up documents. You can also already define a business partner in the master data as a one-time business partner without address data. The address entered is then also a one-time address. For more information, see [Deviating Business Partner Addresses](#).

- You can enter additional print-specific (such as L/C-compliant) address information for business partners. The system uses this address information automatically in print forms.
- If you have specified this in Customizing for the forwarding order type, you can make a credit limit check for a business partner (that is, for the ordering party or other paying business partners). For more information, see [Credit Limit Check](#).
- You can also define alternative business partners, locations and dates for individual items in your forwarding order (see process step 6).

For more information about business partner master data, see [Business Partner](#).

4. Entering and Changing the Business Process-Dependent Data

If you want to use the forwarding order in specialist business processes and have activated these in the Customizing settings for the forwarding order type, you can enter and change additional data, such as:

- Air cargo security

You can enter data for the air cargo security check. For more information, see [Air Cargo Security](#).

- Instructions

You can display instructions that have been defined in Customizing. You can also create, change and delete them. For more information, see [Management of Instructions](#).

- Export/Import processing

For export/import processing, you can predefine the traffic direction in Customizing of the forwarding order type or you can specify it in the forwarding order itself. You must also observe certain specifications when defining business partners and sales organizations in your forwarding order. For more information, see [Export/Import Processing](#).

- Handling codes

Handling codes allow you to specify certain properties of your transportations and goods to be transported (such as dangerous goods) that may also be relevant for planning. For more information, see [Handling Codes](#).

5. Determining the Route

If you have already received information about the ordered route from the ordering party, you can create the stages of the route. For more information, see [Determination of the Route](#).

6. Creating the Order Items

You create the items of a forwarding order in an item hierarchy consisting of container, packaging, and product, for example. Only the top level of this hierarchy is relevant to planning. The quantities and weights of these levels are totaled and copied to the header data.

For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

7. Saving the Forwarding Order

You save the forwarding order. The system now assigns the document an ID from the number range for this document type. In addition, the system checks the entered data for completeness and consistency. The system displays the results of these checks in the form of error messages and as a completeness status.

If you have defined this in Customizing, the system performs a series of automatic functions, such as the following:

- Automatic creation of freight units
- Automatic confirmation of the forwarding order, with sending of the confirmation to ordering party
- Automatic calculation of transportation charges

- Check whether the limits for triggering an approval workflow have been exceeded and, if necessary, trigger the workflow.
- Check of the credit limit of the business partners
- Check whether the foreign trade regulations have been followed in the forwarding order (see [Trade Compliance Check](#)).

More Information

[Editing of a Forwarding Order](#)

[Confirmation and Completion of a Forwarding Order](#)

[Creation of a Forwarding Order for Rail Transportation](#)

Attributes for Process Control

Use

In Transportation Management (TM), you can already specify several attributes when entering order data in the forwarding order or forwarding quotation. These attributes then control the further process of order creation and execution. They are entered in the header data of the business document in the **General Data** area and are valid for the whole business document.

The following section briefly describes these attributes. For more information, see the individual chapters in which the process is described.

Prerequisites

You have defined transportation modes and movement types in Customizing. For more information, see Customizing for **Transportation Management** under the following Customizing activities:

- [Master Data](#) > [Transportation Network](#) > [Transportation Lane](#) > [Define Transportation Mode](#) >
- [Forwarding Order Management](#) > [General Settings for Forwarding Order Management](#) > [Define Movement Types](#) >

If required, you have already defined the transportation mode, the traffic direction and the shipping type in Customizing for the forwarding order or forwarding quotation type. If you use this option, the attributes are automatically entered in the business document and cannot be changed there. For more information, see Customizing for **Transportation Management** under the following Customizing activities:

- [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#) >
- [Forwarding Order Management](#) > [Forwarding Quotation](#) > [Define Forwarding Quotation Types](#) >

If you want to use your own shipping types, you have defined shipping types in Customizing and assigned them transportation modes if required. You can then select these shipping types in Customizing for forwarding order types and forwarding quotation types using the input help on the user interface for the forwarding order and forwarding quotation. For more information, see Customizing for **Transportation Management** under [Forwarding Order Management](#) > [Define Shipping Types](#) >

i Note

If you want to create a forwarding order based on a forwarding quotation and predefine the shipping type and the transportation mode in Customizing for the forwarding quotation type, these settings must be the same as those in the

forwarding order type.

Features

Transportation Mode

This attribute determines how the freight is transported, that is, via sea, air, or road transportation, or by train. It is valid for all items and main carriage stages of the business document and is taken into account during manual planning as well as during the creation of freight bookings and freight orders.

Traffic Direction

This attribute determines whether the transport is an import or export from the point of view of the sales organization entered in the header data of the forwarding order or forwarding quotation. This information then determines, for example, the stage for which the sales organization and the processor of the business document is responsible (for example, up to the main carriage that starts in a different country or region).

Shipping Type

This attribute determines how the planning and execution of the transportation of the freight is performed for a transportation mode. For example, you can define the following information for the individual transportation modes:

- **Sea:** Transport of cargo as a full container load into which no other package or product may be loaded (**Full Container Load, FCL**), or as individual packages and products (**Less Than Container Load, LCL**).
- **Air:** Transport of cargo in unit load devices (**ULDs**) or as loose cargo (**Loose**)
- **Rail:** Transport of cargo as a full carload (**Carload**), individual cargo items (**Intermodal Rail**), or as a complete train (**Unit Train**)
- **Road:** Transport of cargo as a full truck load (**Full Truck Load, FTL**) or as individual packages and products (**Less Than Truck Load, LTL**)

The shipping type allows you to specify, for example, that when you enter items, only the container (or ULD) may be entered as a planning-relevant item (top level of the item hierarchy) or that no container items may be created at all.

Movement Type

This attribute allows you to specify for which stages the logistics service provider is responsible as well as which stage types are allowed for a forwarding order or a forwarding quotation. You can, for example, define the following movement types:

- **Door to Door:** The logistics service provider is responsible for picking up the goods at the shipper (source location) and delivering to the ship-to party (destination location).
- **Port to Port:** The logistics service provider is only responsible for the sea transportation of the freight. The transportation of the freight from the shipper (source location) to the port of loading and from the port of discharge to the ship-to party is arranged by the sold-to party.

Depending on your Customizing settings for the movement type, certain stages of the actual route are displayed automatically. Additionally, the stage types that you can enter may be restricted. For more information, see [Determination of the Route](#).

If you change the movement type, the system automatically adjusts the actual route to the new movement type. To transfer the existing route to the new route, the system applies standard logic. It determines which stages can remain in the new route, which have to be deleted, and which have to be created. You can use a Business Add-In (BAI) to change this logic. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#)

Air Waybill Type

You use this attribute to control the air freight-specific consolidation processes. You can also restrict the consolidation options and define how many forwarding orders you can assign to a freight booking or to a master air waybill. You cannot assign any additional forwarding orders to a freight booking or a master air waybill if a forwarding order is marked as **Back-to-Back**.

You can predefine the attribute in Customizing for the forwarding order type or forwarding quotation type; or you can define it directly in an air forwarding order or quotation. For more information, see [Air Freight-Specific Consolidation Processes](#).

Effects on the Follow-On Processes

The attributes mentioned are obligatory for follow-on processes such as transportation planning or the creation of freight orders and freight bookings. If you change the transportation mode, shipping type, or movement type, the assigned freight units (FUs) or transportation units (TUs) will be created again and the freight costs will be calculated again. The transportation mode and shipping type cannot be changed if follow-on business documents exist for the FUs or TUs and the execution of at least one main carriage stage of one of these FUs or TUs has already started.

The attributes (except for air waybill types) are not copied directly to the freight orders and freight bookings, but rather the system determines them again based on the business documents.

Uncontrolled Transportation

Use

You use this function to specify that the carrier assigned to the main carriage stage of a forwarding order bills the sold-to party directly based on a special freight agreement between the carrier and the sold-to party. In this case, you are still responsible for organizing the transportation of the goods and for planning all stages including the main carriage stage; however, you can calculate charges and create settlement documents only for the pre-carriage and on-carriage stages.

Activities

To specify that transportation is uncontrolled, deselect the **Controlled** checkbox on the **General Data** tab page of the forwarding order UI. You can then enter the ID of the external freight agreement that exists between the sold-to party and the carrier. Deselecting the checkbox and entering the freight agreement ID has the following effect on further processing:

- When you assign freight units that belong to the forwarding order to a freight booking, the system displays the **Controlled** checkbox and the **External Freight Agreement** field on the **Terms and Conditions** tab page of the freight booking. Both fields are read-only. They are visible in a freight booking only if the **Controlled** checkbox has been deselected in the corresponding forwarding order.

Note the following:

- You can assign freight units from other forwarding orders to this freight booking only if the forwarding orders contain the same external freight agreement ID.
- You cannot consolidate forwarding orders in the case of a less than container load (LCL) freight booking.
- The system determines a suitable forwarding agreement or forwarding agreement item where you deselected the **Controlled** checkbox. For the system to be able to calculate the charges for an uncontrolled transportation, you must specify a carrier for the stage or an external freight agreement in the forwarding order or freight agreement. Note that the system can determine the forwarding agreement or forwarding agreement item even if you have not specified the external

freight agreement for the forwarding agreement or forwarding agreement item. The system cannot calculate the charges for an uncontrolled transportation if the system cannot determine a suitable forwarding agreement or forwarding agreement item.

- When you calculate charges and carry out forwarding settlement or freight settlement, the system ignores the main carriage stage. You can calculate charges and create settlement documents only for the pre-carriage and on-carriage stages.

Incoterms in Forwarding Orders

Use

This function enables you to specify who pays for the individual stages of a transport.

In Transportation Management (TM), you create a forwarding settlement document (FWSD) when you need to send an invoice to a customer. The settlement is based on an associated forwarding order. To create a settlement, you must have a forwarding agreement between the business partner and sales organization in a forwarding order. This means you can have a forwarding order on the export side with a settlement to the shipper, and another forwarding order on the import side with a settlement to the ship-to party.

The Incoterms function addresses the following business requirements:

- Settlement with shipper or ship-to party

The settlement is independent of the organizational unit that organizes the parts of the transport.

❖ Example

In a full container load (FCL) transport with an Incoterm of FOB (Free On Board), the shipper pays for the pre-carriage to the export organization. The ship-to party pays the main carriage and on-carriage to the import organization. This applies even if the export organization organizes the main carriage.

- Settlement between the export and import organizations

The settlement depends on the organizational unit that organizes the parts of the transport.

❖ Example

In an FCL transport with FOB, the export organization that organized the main carriage actually pays for the main carriage. It then invoices the import organization to recover the main carriage costs from the ship-to party.

In this way, the Incoterm you specify in the forwarding order drives the overall settlement process.

Prerequisites

You have specified the agreement party role for a stage type and Incoterm combination. When you specify an Incoterm in a forwarding order, the system uses the agreement party role to identify the default agreement party for a stage.

For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Default Agreement Party Roles for Stages](#).

Features

Incoterms

You can use Incoterms to determine who pays for the stages of a transport. The following are the main party roles for Incoterms:

- Prepaid agreement party
- Collect agreement party
- Prepaid bill-to party
- Collect bill-to party
- Prepaid payer
- Collect payer

The prepaid party sends the goods; the collect party receives the goods. The party roles are not necessarily the shipper or ship-to party, and can be internal organizational units.

❖ Example

An internal organization can be the collect agreement party in a settlement between internal organizational units.

For an export forwarding order, the collect bill-to party and collect payer party can be the import organization.

You can specify the party roles and business partners for Incoterms on the **Business Partners** tab page. The system uses the parties you specify on this tab page to determine who pays for a stage when you create a settlement.

You can only specify agreement party roles in the **Stages** tab page. The system uses the information you specify on the **Stages** tab page to determine the agreement and therefore to calculate charges.

The following table explains how you assign party roles to business partners in the **Business Partners** tab page:

Business Partner	Party Role
BP1	Prepaid agreement party
BP2	Collect agreement party

The following table explains how the system determines the agreement party when you specify the agreement party role in the **Stages** tab page:

Stage Description	Stage Type	Agreement Party Role	Agreement Party
Stage 1	Pre-carriage	Prepaid agreement party	BP1
Stage 2	Main carriage	Collect agreement party	BP2

Stage Status

The system uses the following logic to determine the stage status:

- The planning and execution organization for a stage belongs to a company organization. A company organization is specified in the sales organization of the forwarding order. If these company organizations match, the stage status is relevant for planning and execution.

The system displays the appropriate status in the **Stage Status** field.

- If the company organizations do not match, the stage status is not relevant for planning and execution.

The system displays a status of **Not Relevant** in the **Stage Status** field.

When you manually create an FWSD, the system displays a dialog box in which you select the relevant business partner. You can create the FWSD for one of the new party roles. The party role that you select depends on the Incoterm specified in the forwarding order.

❖ Example

You have a forwarding order with an Incoterm of EXW (Ex Works) to which you have assigned a collect agreement party. You can create a collect FWSD for the collect agreement party.

You have a forwarding order with an Incoterm of DDP (Delivery Duty Paid) to which you have assigned a prepaid agreement party. You can create a collect FWSD for the prepaid agreement party.

You can create more than one settlement document for these forwarding orders.

More Information

For more information about forwarding orders, see [Forwarding Order Management](#).

For more information about forwarding settlement documents, see [Forwarding Settlement Document](#).

Freight Terms in Forwarding Orders

Use

In a transportation scenario, you can have the following business requirements:

- Agree transportation charges in advance

You agree the transportation charges in advance with your customer. The charges provide the basis for settling the overall transportation costs.

You can also agree on settlement terms for the individual parts of the freight cost in advance. You know in advance the individual parts of the transportation that are paid for by the prepaid payer or the collect payer.

- Capture the entire transportation in a single forwarding order

In the standard scenarios for logistics service providers (LSPs), you use two business documents, an export forwarding order and an import forwarding order, to deliver a forwarding order. For more information, see [Export/Import Processing](#). When you use a freight term, you work with just one forwarding order.

- Create forwarding settlement documents from the same forwarding order

You need a forwarding settlement document for the prepaid charges and another forwarding settlement document for the collect charges in the same forwarding order. For more information on the differences between a freight term process and an Incoterm process, see [Process Differences Between Freight Terms and Incoterms](#).

- Settle the prepaid charges with the prepaid payer

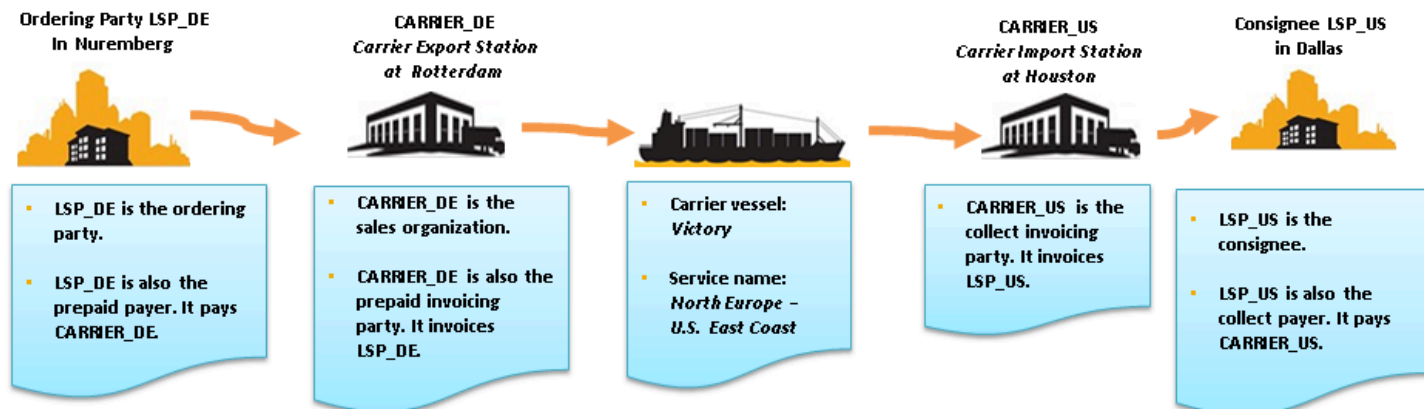
The prepaid payer is usually the organization of the sold-to party that is closest to the location of origin.

- Settle the collect charges with the collect payer

The collect payer is usually the organization of the ship-to party that is closest to the destination location.

Example

The following figure illustrates a simple carrier scenario with freight terms and is followed by an explanation:



An LSP, LSP_DE, orders a transportation service from you, a carrier. Your agency in Germany, CARRIER_DE, organizes and delivers the transportation service from Nuremberg to Houston. Your agency in the United States, CARRIER_US, organizes and delivers the transportation service from Houston to Dallas.

Forwarding Agreement Settings

You have one forwarding agreement, Forwarding_Agreement_1. The forwarding agreement contains the following details:

- Sales organization: CARRIER_DE
- sold-to party: LSP_DE

You have the following charge components in the agreement:

- Ocean freight charges
- Export handling charge at Rotterdam
- Import handling charge at Houston

You settle the charges as follows:

- Prepaid charges settled between the prepaid payer (LSP_DE) and the prepaid invoicing party (CARRIER_DE)
- Collect charges settled between the collect payer (LSP_US) and the collect invoicing party (CARRIER_US)

Calculation Sheet Settings

You have a calculation sheet item of EUR 1,500 for the charge type **Ocean Freight Charges**. The table explains the freight terms that you specify on the **Freight Terms** tab page of the calculation sheet item:

Freight Term	Payer
Prepaid/Prepaid (PP)	Prepaid payer
Collect/Collect (CC)	Collect payer

The table explains the freight terms for the charge type **Export Handling Charge** (EUR 150):

Freight Term	Payer
Prepaid/Prepaid (PP)	Prepaid payer
Collect/Collect (CC)	Collect payer

The table explains the freight terms for the charge type **Import Handling Charge** (USD 200):

Freight Term	Payer
Prepaid/Prepaid (PP)	Collect payer
Collect/Collect (CC)	Collect payer

You assign the calculation sheet to your forwarding agreement, Forwarding_Agreement_1.

Forwarding Order Settings

The table explains the settings you enter in the forwarding order:

Setting Name	Setting
Freight term	Prepaid/Prepaid (PP)
Sales organization	CARRIER_DE
sold-to party	LSP_DE
Prepaid bill-to party	LSP_DE
Prepaid payer	LSP_DE
Collect bill-to party	LSP_US
Collect payer	LSP_US
Prepaid invoicing party	CARRIER_DE
Collect invoicing party	CARRIER_US

Charge Calculation

The table explains how the system calculates the prepaid charges in the forwarding order (CARRIER_DE bills LSP_DE for these charges):

Charge Type	Amount
Ocean freight charges	EUR 1,500
Export handling charge	EUR 150

CARRIER_US bills LSP_US for USD 200, for the collect charge **Import handling charge**.

Forwarding Settlement Document

The system creates a settlement document for the prepaid charge types. The system includes the following details in the settlement:

- Invoicing party: CARRIER_DE
- Bill-to Party: LSP_DE
- Payer: LSP_DE
- Charge item: CARRIER_DE, Forwarding_Agreement_1, item_0001, against LSP_DE
- Ocean freight charges: EUR 1,500
- Export handling charge: EUR 150

The system also creates a forwarding settlement document for the collect charge types. The system includes the following details in the settlement:

- Invoicing party: CARRIER_US
- Bill-to party: LSP_US
- Payer: LSP_US
- Charge item: CARRIER_US, Forwarding_Agreement_1, item_0001, against LSP_US
- Import handling charge: USD 200

i Note

You can have different currencies in the prepaid and collect forwarding settlement documents. For more information, see [Freight Term Features in Forwarding Orders](#).

More Information

[Freight Term Prerequisites](#)

[Creation of a Forwarding Order](#)

Freight Term Prerequisites

Use

Standard Settings

You have specified the standard settings for charge calculation. For more information, see [Standard Charge Calculation](#).

Freight Term for Transportation Modes Settings

You have specified the appropriate freight term settings in Customizing for **Transportation Management** under **Basic Functions** **Charge Calculation** **Basic Settings** **Freight Term Settings for Transportation Modes** **Define Freight Terms for Transportation Modes**.

For example, the following table contains the freight term settings you can specify for a transportation mode category of **Sea**:

Freight Term	Description
CC	Collect/Collect

Freight Term	Description
CP	Collect/Prepaid
PC	Prepaid/Collect
PP	Prepaid/Prepaid

You can enter the freight terms you specify in this Customizing activity in the following business documents:

- On the **Freight Terms** tab page of a calculation sheet item (see [Calculation Sheet](#))
- In the **Freight Term** field on the **General Data > Settlement Terms** tab page of a forwarding order (see [Forwarding Order](#))

Charge Calculation

You have specified the following charge calculation settings in Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings**:

- **Define Charge Profiles**

You have specified a charge profile for sales. You have assigned the charge profile for sales to a sales organization in your company in Customizing for **Transportation Management** under **Master Data > Organizational Management > Organizational Model > Change Organizational Model** (transaction PPOME).

The system uses the charge profile when you calculate charges in a forwarding order, where the forwarding order contains the sales organization.

- **Define Charge Types**

You have specified the relevant settings for the charge types that you assign to calculation sheets and rate tables, including the freight term and payer for a charge type.

i Note

For a forwarding order that contains a freight term, the system automatically sets the calculation level to **Header Level Calculation**. You cannot change this setting. For more information, see [Calculation Profiles](#).

Party Roles

You have specified the appropriate business partners for the following party roles on the **Business Partner** tab page of a forwarding order:

- Prepaid payer
- Collect payer
- Prepaid invoicing party
- Collect invoicing party

Order and Quotation Types

- You have specified the settlement terms for the forwarding order type used when creating the forwarding order.

You specify the settlement terms in Customizing for **Transportation Management** under **Forwarding Order Management > Forwarding Order > Define Forwarding Order Types**.

- You have specified the settlement terms for the forwarding quotation type used when creating the forwarding quotation.

You specify the settlement terms in Customizing for [Transportation Management](#) under [Forwarding Order Management](#) [Forwarding Quotation](#) [Define Forwarding Quotation Types](#).

More Information

[Freight Terms in Forwarding Orders](#)

Freight Term Features in Forwarding Orders

Use

Examples of Freight Terms in a Forwarding Order

The following table contains an example of the freight terms you can assign in your calculation sheet. You assign the freight terms on the [Freight Terms](#) tab page of the calculation item.

Charge Type	Freight Term			
	Prepaid/Prepaid (PP)	Prepaid/Collect (PC)	Collect/Prepaid (CP)	Collect/Collect (CC)
Pick-up charges	Prepaid	Prepaid	Collect	Prepaid
Export clearance	Prepaid	Prepaid	Collect	Prepaid
Export handling charge at port of loading	Prepaid	Prepaid	Collect	Prepaid
Basic freight	Prepaid	Prepaid	Collect	Collect
Bunker adjustment	Prepaid	Collect	Prepaid	Collect
Currency adjustment	Prepaid	Collect	Prepaid	Collect
Import handling charge at port of unloading	Collect	Collect	Prepaid	Collect
Customs handling	Collect	Collect	Prepaid	Collect
Delivery charge	Collect	Collect	Prepaid	Collect

The following table explains the settings you enter in the forwarding order:

Setting Name	Setting
Freight term	Prepaid/Prepaid (PP)
Sales organization	CARRIER_DE
sold-to party	LSP_DE
Prepaid bill-to party	LSP_DE
Prepaid payer	LSP_DE

Setting Name	Setting
Collect bill-to party	LSP_US
Collect payer	LSP_US
Prepaid invoicing party	CARRIER_DE
Collect invoicing party	CARRIER_US

The following table contains an example of how the system determines who pays when you use the **Prepaid/Prepaid (PP)** freight term in a forwarding order:

Charge Type	Location of Charge	Payer	Invoicing Party
Pick-up charges	Nuremberg	LSP_DE	CARRIER_DE
Export clearance	Hamburg	LSP_DE	CARRIER_DE
Export handling charge at port of loading	Hamburg	LSP_DE	CARRIER_DE
Basic freight	Main stage	LSP_DE	CARRIER_DE
Bunker adjustment	Main stage	LSP_DE	CARRIER_DE
Currency adjustment	Main stage	LSP_DE	CARRIER_DE
Import handling charge at port of unloading	Houston	LSP_US	CARRIER_US
Customs handling	Houston	LSP_US	CARRIER_US
Delivery charge	Dallas	LSP_US	CARRIER_US

Forwarding Agreement

The system uses the sold-to party and the sales organization in the forwarding order to determine which forwarding agreement to use.

For a freight term, the calculation level is always header level. This means that the system uses just one forwarding agreement to calculate charges.

For more information on how the system determines which forwarding agreement to use, see [Agreement and Calculation Sheet Determination](#).

i Note

The document currency and term of payment in the agreement is copied into the settlement document. If you want to use different currencies and terms of payment in your scenarios for prepaid and collect business partners, do not enter the document currency and term of payment in the agreement.

Currency Determination

In a forwarding order that contains freight terms, the system uses the following approach to assign a currency to a collect charge item and a prepaid charge item:

1. The document currency specified in the agreement

2. The currency specified in the master data for the business partner under [Logistics > Transportation Management > Master Data > Maintain Business Partner](#)

For the business partner in focus (either prepaid or collect payer), the currency is maintained for the business partner on the [BP > Customer Org Data > Payment > Currency](#) tab page for the organization in focus (either prepaid or collect invoicing party).

❖ Example

You specify the following party role and business partner settings on the [Business Partner](#) tab page of a forwarding order:

Party Role	Business Partner
Sales Organization	CARRIER_DE
sold-to party	LSP_DE
Prepaid Bill-to Party	LSP_DE
Collect Bill-to Party	LSP_US
Prepaid Payer	LSP_DE
Collect Payer	LSP_US
Prepaid Invoicing Party	CARRIER_DE
Collect Invoicing Party	CARRIER_US

The following table contains the settings you enter in the [Maintain Business Partner](#) (BP) transaction:

Setting	Entry
Business Partner	LSP_DE
Display in BP Role	Sold-to Party
Customer Org. Data > Sales Org. Settings > Sales Organization	CARRIER_DE
Customer Org. Data > Payment > Currency	EUR

The system uses the currency that you have assigned to CARRIER_DE in the forwarding order.

3. The local currency that you specify in the charges profile that you assign to the sales organization in the forwarding order

For example, you specify a charges profile for sales organization CARRIER_DE (which is the prepaid invoicing party). Therefore the system uses the currency you specify for sales organization CARRIER_DE.

4. The currency that you specify for a company to which the sales organization in a forwarding order belongs

You can specify the currency in transaction PPOME under [Org. Data > Currency](#). For example, you do not specify a currency for sales organization CARRIER_DE (which is the prepaid invoicing party). Also, you do not assign a charges profile to CARRIER_DE. CARRIER_DE is a part of company COMPANY_DE. Therefore the system uses the currency you specify for company COMPANY_DE.

Credit Limit Check

The system performs a credit limit check on the business partners you specify in the party roles in a forwarding order. The system uses the following pairs:

- Sales organization or forwarding house represented by the prepaid invoicing party and the prepaid payer party
- Sales organization or forwarding house represented by the collect invoicing party and the collect payer party

For more information, see [Credit Limit Check](#).

Terms of Payment

In a forwarding order that contains freight terms, the system uses the following approach to assign terms of payment to a collect charge item and a prepaid charge item:

1. The terms of payment specified in the agreement
2. The terms of payment specified in the business partner master data of a sales organization

You specify the terms of payment in the business partner master data of a sales organization as follows:

- a. On the SAP menu, choose **►Logistics►Transportation Management►Master Data►Maintain Business Partner►**.
- b. In the **Display in BP Role** field, enter **Sold-to Party**.
- c. Choose the **Customer Org. Data** tab page.
- d. Choose **Show Detail Data** to display the details of the sales organization.
- e. In the **Payment** screen area, enter the terms of payment.

The sales organization is either the prepaid invoicing party or the collect invoicing party.

Note that you can also enter or edit a term of payment on the **Business Partner** tab page of a forwarding order for the party roles **Prepaid Payer** and **Collect Payer** to override the terms of payment determined by the system.

You define terms of payment in the **Define Terms of Payment** Customizing activity.

Forwarding Settlement Documents

In a forwarding order that contains freight terms, the system creates a forwarding settlement document for the prepaid payer and another forwarding settlement document for the collect payer. If you manually create a forwarding settlement document from the forwarding order, the system displays a dialog box with the following options:

- Prepaid party
- Collect party
- Both prepaid party and collect party
- Other party

When you create a forwarding settlement document against another party for unplanned charges, you can select either the prepaid invoicing party or the collect invoicing party as the sales organization that issues the forwarding settlement document.

More Information

[Freight Terms in Forwarding Orders](#)

Process Differences Between Freight Terms and Incoterms

Use

You can only enter either a freight term or an Incoterm in a forwarding order. You cannot enter both.

The following table describes the differences between entering a freight term or an Incoterm in a forwarding order:

Freight Term	Incoterm
You cannot use the export/import process.	You can use the export/import process (for more information, see Export/Import Processing).
The calculation level is always header level.	The calculation level is always stage level.
The forwarding settlement document for the prepaid party and the collect party are created from the same forwarding order. The forwarding order has a different prepaid sales organization and collect sales organization, which are derived from the prepaid invoicing party and collect invoicing party in the forwarding order.	The forwarding settlement document for the prepaid party and the collect party are created from the same forwarding order. The sales organization in the forwarding settlement document is the same as the sales organization in the forwarding order. If the export/import process is activated with a forwarding order with the traffic direction Export, on the import side the system creates a forwarding order with the traffic direction Import and the sales organization as the importing organization.
The system uses the sales organization and order party to determine the agreement used for the freight costs for both the prepaid payer and collect payer.	The system uses the sales organization and the prepaid agreement party to determine the agreement for the freight costs for the prepaid payer. The system uses the sales organization and the collect agreement party to determine the agreement for the freight costs for the collect payer.
The system uses the freight term settings that you define in the calculation sheet item to determine the payer of the charge.	The system uses the Incoterm settings that you define in the calculation sheet item to determine the payer of the charge.

If you do not specify either a freight term or an Incoterm in a forwarding order, the system uses the sold-to party to determine the appropriate forwarding agreement, and creates just one forwarding settlement document against the sold-to party.

More Information

[Freight Terms in Forwarding Orders](#)

[Freight Term Prerequisites](#)

[Freight Term Features](#)

Credit Limit Check

Use

You can use this function to determine the risk of losses on receivables from business partners and to make timely and efficient credit decisions.

Integration

Transportation Management (TM) integrates with the SAP Credit Management application to provide the credit limit check feature.

Features

Business Partner Party Role Checks

The system checks the business partners you specify in the party roles in a forwarding order during a credit limit check.

It checks in the following order of priority if you have specified an Incoterm.

1. Prepaid side
 - a. Prepaid payer
 - b. Prepaid agreement party
2. Collect side
 - a. Collect payer
 - b. Collect agreement party

The system checks in the following order of priority if you have not specified an Incoterm:

1. Payer
2. sold-to party

We deliver this party role checks as standard. You cannot change the settings for these checks in Customizing.

Credit Segment

When you perform a credit limit check on a forwarding order, the system checks the following conditions to identify the relevant credit segment:

- Has the **BAdI: Credit Segment Determination for Forwarding Orders Business Add-In** (BAdI) been defined?
- Has the sales organization in the forwarding order been assigned to a credit segment?
- Has the company organization to which the sales organization in the forwarding order belongs, been assigned to a credit segment? If one of these conditions is met, the system uses the relevant credit segment.

Check Enabling.

In Customizing, you can specify that the credit limit check feature can be enabled by forwarding order type. If you make this specification, the system automatically performs a credit limit check when you save a forwarding order. You can review the credit

limit check status for orders in the [Forwarding Orders](#) worklist, and perform a manual check in an individual forwarding order.

For more information about enabling the credit limit check for a forwarding order type, see Customizing for [Transportation Management](#) under. [▶ Forwarding Order Management ▶ Forwarding Order ▶ Define Forwarding Order Types ▶](#).

For more information about defining the credit segment IDs that the TM system uses to access the credit management functionality in the SAP Credit Management application, see Customizing for [Transportation Management](#) under. [▶ Basic Functions ▶ Credit Management Integration ▶ Integration with Credit Management ▶](#)

SAP Credit Management Integration

TM updates SAP Credit Management at the following points in the business transaction flow:

- When you create, change, or cancel a forwarding order with a credit limit check status of successful, or when you manually approve a credit limit check failure
- When you post a forwarding settlement document to Sales and Distribution (SD)
- When you post an SD billing document to Financial Accounting (FI)
- When you cancel a forwarding settlement document and the corresponding FI and SD documents are canceled

When you save, change, or cancel a forwarding order, the credit limit amount is updated to reflect the credit exposure corresponding to the forwarding order. The application updates the amount in the [Open Orders](#) exposure category from the forwarding order.

The TM system automatically checks the credit limit when you change the amounts or organizational entities in the forwarding order. For example, the system automatically checks the credit limit when you change the following settings:

- Sales organization
- Calculated amount or credit limit check amount
- Business partner
- Logistics data that can influence the calculated amount, such as stages, weights, and volumes

When you successfully create and post a forwarding settlement document to SD, the credit limit check amount is updated in SAP Credit Management. The application reduces the exposure value of the open order for the forwarding order by the amount of the forwarding settlement document. The value from the SD billing document corresponding to the forwarding settlement document is updated to the [Billing Document Value](#) exposure category. The system also performs these updates when you change and cancel a forwarding settlement document.

TM integrates directly with SAP Credit Management when you configure TM and SAP Credit Management on a single client. If you configure SAP Credit Management on a separate system or client, TM uses Web Services to interact with SAP Credit Management.

If you configure SAP Credit Management on a separate system or client, TM informs SAP Credit Management about the credit limit check amount approved for a customer in a specific forwarding order using the [Notify of Credit Commitment](#) (`CreditCommitmentNotification_Out`) service operation in the [Transportation Request Processing](#) process component.

If you configure SAP Credit Management on a separate system or client, TM informs SAP Credit Management about the credit limit check amount in billing documents that are related to an approved forwarding order using the [Notify of Credit Commitment](#) (`CreditCommitmentNotification_Out`) service operation in the [Customer Freight Invoice Request Processing](#) process component.

Check Failures.

If the credit limit check fails, the system uses the setting you specified in the **Act. If CLC Failed** (Action if Credit Limit Check Failed) field of the Customizing for **Define Forwarding Order Types**. You can choose to block the order or take no action.

You can review the results of a credit limit check failure, including details on the credit limit for the business partner, and the exposure amount. You can approve or reject an order that has failed. You must enter a reason for the approval or rejection.

When you approve the order, the system automatically removes the block, and you can process the order as normal. When you reject the order, the system cancels the order.

If you configure SAP Credit Management on a separate system or client, TM does the credit worthiness check using the **CreditWorthinessQuery_Out** service operations in the **Transportation Request Processing** process component.

More Information

For more information about SAP Credit Management, see SAP Library for SAP S/4HANA under. ► **Enterprise Business Applications** ► **Finance** ► **Financial Operations** ► **Receivables Management** ► **SAP Credit Management (FIN-FSCM-CR)** ►.

For more information about updating the credit exposure for a business partner in SAP Credit Management from TM, see [Updating of Credit Exposure](#).

Transportation Service Levels

Use

You use transportation service levels to describe the priority of transportation services. Examples are Express or Standard.

Features

A distinction must be made between the transportation service levels for the sales side and the purchasing side:

- Sales side

On the sales side, you specify transportation service levels for your customers. To do this, you define transportation service level codes in Customizing. For more information, see Customizing for **Transportation Management** under ► **Forwarding Order Management** ► **Define Transportation Service Level Codes** ►. If you then create one of the following documents the system offers you the transportation service level codes defined in Customizing for selection:

- Forwarding order
- Forwarding agreement (item level)
- Forwarding quotation
- Forwarding agreement quotation

If you are working with order- and delivery-based transportation requirements, you do not have to define any transportation service level codes. In this case, these are taken from the sales order (**Shipping Condition** field).

- Purchasing side

On the purchasing side, you define carrier service codes in the master data for your carriers. You do this on the **SAP Easy Access** screen under ► **Transportation Management** ► **Master Data** ► **Maintain Business Partner** ► (Carrier role, **Vendor Data** tab page). If you then create a freight document and enter one of these carriers the system offers you the corresponding values in the **Service Level** field for selection. This also applies to freight agreements (item level).

i Note

If you are a logistics service provider and can commit all the carriers who work for you to the same transportation service level, we recommend you make the following settings:

- Define transportation service level codes in Customizing. For more information, see Customizing for [Transportation Management](#) under ► [Forwarding Order Management](#) ► [Define Transportation Service Level Codes](#) ►.
- Enter one of these values as the standard service level when you define your freight order type. For more information, see Customizing for [Transportation Management](#) under ► [Freight Order Management](#) ► [Freight Order](#) ► [Define Freight Order Types](#) ►. The system then uses this value when you create a freight order.

Editing of a Forwarding Order

Use

After you have created and saved a forwarding order, you can use this business document to perform further actions, such as the following:

- Create freight units
- Determine route
- Start transportation planning
- Create freight documents
- Calculate transportation charges
- Create forwarding settlement document

Prerequisites

See [Creation of a Forwarding Order](#).

Process

The following describes an example of a process for editing the forwarding order and starting the transportation planning:

1. Creation of Freight Units

If the freight units are not created automatically after you save the forwarding order, you trigger the creation manually on the user interface for the forwarding order (**Follow-Up Actions** menu). In this case, the system considers either the freight unit building rule (FUB rule) or the condition for determining an FUB rule that you have specified in Customizing. A prerequisite for the creation of freight units (automatically or manually) is that all required data has been entered in the forwarding order.

You can have the system create the freight units per order item, for example, or define that the system merges as many order items as possible into one freight unit. For more information, see [Creation and Editing of Freight Units](#).

After the system has created the freight units, the life cycle status changes from **New** to **In Planning**.

2. Manual Transportation Planning

You call the transportation cockpit. You enter your planning profile and selection profile and start the planning. The freight units created by the system are displayed in the transportation cockpit.

You can now execute manual transportation planning by pulling the freight units to the resources via drag and drop. The system then creates a freight order, which is displayed in the document flow for the forwarding order.

For more information about transportation planning, see [Planning](#).

As an alternative to manual transportation planning, you can proceed as follows:

- o Determination of a route

You can have the system determine a route. In doing so, the system generates several alternative routes for each freight unit, one of which you can then accept. In Customizing for the forwarding order type, you can determine whether only the freight units (that is, only the route) are to be saved when you accept the route or whether the freight orders (that is, the planning) are to be saved as well. The system adjusts existing freight units.

If the freight orders are also saved, the life cycle status changes to **Planned** after you accept the proposed route. The route is displayed in the forwarding order in the form of stages. For more information, see [Determination of the Route](#) and [Generation of Transportation Proposals](#).

- o Automatic planning

You can use the VSR optimizer to perform automatic planning from the transportation cockpit. For more information, see [VSR Optimization](#).

- o Direct creation of freight documents

Directly from the forwarding order, you can already create freight documents, such as freight orders or freight bookings for the transportation stages defined in the document. You can also select available freight bookings. For more information, see [Direct Creation of Freight Documents and Selection of Schedules](#).

When all freight units of the forwarding order have been planned, the life cycle status changes to **planned**.

3. Calculating Transportation Charges

If you have activated transportation charge calculation in Customizing for the forwarding order type, and if the transportation charges are not calculated automatically when you save the forwarding order, you can allow the forwarding order to calculate these charges (internal and external charges). Moreover, you can enter the charges manually in the forwarding order. The system recalculates the transportation charges each time the forwarding order is saved again. However, transportation charges that you entered manually are fixed by the system and are kept during recalculation. For more information, see [Charge Calculation](#).

If you quickly want to estimate the transportation charges based on a minimum of data, you can use the **Estimate Forwarding Charges** function to do so. For more information, see [Charge Estimation](#).

You can also display a profitability analysis in the forwarding order (separate tab page). In the analysis, the system takes into account all costs and revenues associated with the execution of a forwarding order. For more information, see [Profitability Analysis](#).

4. Creating a Forwarding Settlement Document

If you have activated this function in Customizing for the forwarding order type, you can create a forwarding settlement document from the forwarding order based on the transportation charge calculation (**Costs/Settlement** menu). For more information, see [Forwarding Settlement](#) and [Internal Settlement Management](#).

5. Printing Documents

You can print the forwarding order directly or after previewing it. The default printout contains information such as item descriptions and weights. If you have entered your order items in an item hierarchy, information about the lower hierarchy levels are also displayed.

The print history is displayed in the forwarding order under **Output Management**. This history also contains information about all the documents you have faxed or sent by e-mail or EDI.

For more information, see [Output Management](#).

6. Editing or Canceling the Forwarding Order

You can still change most of the data in a forwarding order or quotation during the transportation process, such as dates, quantities, and locations, as well as add or delete items. This is also the case if freight units have already been built and planning has been completed or execution has started. The system updates the freight units when saving the data. Follow-on documents such as freight orders and freight bookings are also adjusted.

However, if data has already been transferred to the freight unit from the execution process and the status of the freight unit has been changed to **On Hand**, for example, the system no longer adjusts the follow-on documents automatically. In this case, the system prompts you to change the data manually.

If a forwarding settlement document or an internal settlement document already exists and you have activated the forwarding settlement change process, you can enter the changes directly in the forwarding order (for example, changes to quantities, to the shipper, or to the consignee). The change process triggers the necessary changes in the forwarding settlement process. If a forwarding settlement document or an internal settlement document already exists and you have not activated the forwarding settlement change process, you have to cancel the forwarding settlement document. You then enter the changes directly in the forwarding order and create a new forwarding settlement document that contains the changes. You can activate the forwarding settlement change process in the corresponding settlement profile by selecting **Reverse and Repost** or **Delta Only**. The system uses the organization or a combination of organization and business partner to assign the profile to the forwarding order. For more information, see [Automatic Change Management for Forwarding Settlements](#).

You can also cancel a forwarding order. Depending on your Customizing settings, you can enter a reason code that is then displayed in the **General Data** area of the forwarding order. For more information, see Customizing for **Transportation Management** under **► Forwarding Order Management ► Define Cancellation Reason Codes ►**.

After the cancellation, the planning is invalid and the system deletes the freight units. The life cycle status changes to **Canceled**.

i Note

The cancellation of a forwarding order cannot be changed. A forwarding order cannot be canceled if it has been assigned a freight document.

For more information, see [Processing of Changes and Deletion of Forwarding Orders](#).

7. Executing a Forwarding Order

You can use SAP Event Management to monitor the further execution of the forwarding order. To do this, you need to have set the corresponding indicator in Customizing for the document type. The forwarding order receives a tracking number (tracking ID) upon creation. You can provide this number to your ordering party, who can then use it to track the execution of the order on a special Web user interface. The system later creates event handlers for the follow-up documents freight unit and freight order. For more information, see [Integration with SAP Event Management](#).

Irrespective of SAP Event Management, the system displays the execution status in the forwarding order. It reflects the execution status of the freight order (or rather, the cumulated execution status if several freight orders exist for the forwarding order).

More Information

[Confirmation and Completion of a Forwarding Order](#)

[Forwarding Order Management Preparation](#)

Profitability Analysis

Use

You can view a profitability analysis in the forwarding order. The system takes into account all the costs and revenues associated with the execution of a forwarding order in the analysis. You cannot edit the analysis.

The system displays the revenues, costs, and profits for the planned profitability and estimated profitability. You can navigate to business document level in the analysis hierarchy, and you can open the individual business documents.

Sources of Costs and Revenue

The sources of costs are freight orders, freight bookings, and freight settlement documents. The sources of revenue are forwarding orders and the corresponding forwarding settlement documents. Internal settlement documents are costs for the company receiving the settlement and revenue for the company sending the settlement.

Calculation of Profitability

The system performs the following analysis:

- Planned profitability

The system calculates the revenue and cost charges from forwarding orders. It takes the planned revenue from the charge calculation done in the forwarding order.

It takes the planned cost from the internal charge calculation done in the forwarding order. This is the planned internal cost for the forwarding order. Note the system does not include the internal charges in the freight order or freight booking when it calculates profitability.

If you enable intercompany settlement, the system uses the cost from the cost distribution in the freight order for an intracompany stage and the cost from the internal charges for an intercompany stage. If you enable intracompany settlement, the system uses the cost from the internal charges for both an intracompany and an intercompany stage.

You can enable a sales organization for intracompany settlement in transaction **PPOME** [Org. Data \(Organization Data\)](#) [Detailed Settings](#) [Intracompany Settlement](#).

- Expected profitability

The system calculates the revenue and cost charges from settlement documents. It takes the expected revenue calculations from the charge calculation done in the forwarding settlement documents that are associated with the relevant forwarding order.

The system takes the expected cost from the cost distribution done on the relevant freight settlement documents. It also takes the expected cost from the relevant internal settlement documents.

Comparison of Profitability Charges

The system displays the results of the profitability analysis on the [Profitability Details](#) screen area of the [Profitability](#) tab page of a forwarding order. It groups the results in a hierarchy, by charge category. You can compare the profitability down to the level of charge types.

The following table contains an example of how the system displays the results:

Charge Category	Planned Cost (USD)	Planned Revenue (USD)	Planned Profit Amount (USD)
General Charges	100	150	50

Charge Category	Planned Cost (USD)	Planned Revenue (USD)	Planned Profit Amount (USD)
Sea Freight Charges	• 120	• 150	• 30
• CSI-BAF Bunker Adjustment Factor	• 350	• 225	• -75
• CSI-BASE Basic Freight Charge	• 100	• 75	• -25
• CSI-LOAD Loading/Unloading Charge			
Other Transportation Charges	1350	1500	150

In the forwarding order above, you incur a loss of USD 70 in sea freight charges. However, you make a profit of USD 50 in general charges and USD 150 in other transportation charges. This results in an overall profit of USD 130 ((50 + 150) -70).

Profitability Status

The system displays the profitability status on the **Profitability** tab page of a forwarding order. The following table explains what each profitability status means:

Status	Meaning
Not Yet Calculated	You have not yet calculated the profitability of the forwarding order.
Calculated	You have calculated the profitability and the calculation is up to date.
Recalculation Required	You have calculated the profitability, but there has been a change to one or more of the following elements since you performed the calculation: • Internal or external charges in the forwarding order or associated freight orders • Freight orders or freight bookings that contain the forwarding order • Freight settlement documents for the freight orders or freight bookings that contain the forwarding order • Forwarding settlement document for the forwarding order • Internal settlement documents for the forwarding orders You must calculate the profitability again.

You can calculate or recalculate the profitability by choosing **Charges/Settlement > Calculate Profitability** in a forwarding order. Alternatively, you can use the **Calculation of Profitability** (/SCMTMS/CALC_PROFITABILITY) background processing function to schedule a background function to calculate or recalculate the profitability. For more information, see [Calculation of Profitability](#).

Prerequisites

You have made the following settings in Customizing to enable a profitability analysis in a forwarding order:

- Enabled the relevant freight order or freight booking type for cost distribution. This is the freight order or freight booking that you use as a basis for determining the freight cost source for the expected profitability. For more information, see Customizing for [Transportation Management](#) under [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) and [Freight Order Management](#) > [Freight Booking](#) > [Define Freight Booking Types](#).
- If you use an internal agreement to calculate the planned costs, you have enabled the relevant forwarding order type for internal charge calculation. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#).
- Specified a cost distribution profile with distribution level of [Forwarding Order](#) for the purchasing organization. To specify a distribution profile, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Cost Distribution](#) > [Define Cost Distribution Profiles](#). You assign the cost distribution profile to the purchasing organization by assigning the distribution profile to the charges profile. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Charge Calculation](#) > [Basic Settings](#) > [Define Charges Profiles](#).
- Enabled the freight settlement document for cost distribution so the expected costs are available for the profitability analysis in the forwarding order. For more information, see Customizing for [Transportation Management](#) under [Settlement](#) > [Freight Settlement](#) > [Define Freight Settlement Document Types](#).

i Note

- The system only calculates the complete planned profitability after you have performed charge calculation in the following business documents:
 - Forwarding order
 - Freight order or freight booking in which you have enabled cost distribution at the level of [Forwarding Order](#)
- If you need to use an internal agreement to make the planned costs available to the profitability analysis, you must perform an internal charge calculation in the forwarding order.
- The system only calculates the complete expected profitability after you have completed the following tasks:
 - Generated all the necessary freight settlement documents for the freight order or freight booking that execute the relevant forwarding order
 - Enabled the freight settlement documents for cost distribution at forwarding order level or created internal settlement documents
- In an export-import situation, you create two forwarding orders; one for each of the export and import sides. The system restricts the profitability analysis to each of the forwarding orders individually. For example, it does not include the revenue and costs from the forwarding order on the import side in the profitability analysis of the forwarding order on the export side.

Example

You are sales organization Nuremberg. You receive a forwarding order, with an Incoterm of FOB (Free On Board), from a customer in Germany. The following table describes the planning, execution, and payment responsibilities.

Sales Organization (Company Code) (FWO Export/Import)	Stage	From/To	Planning and Execution Organization (Company Code)	Settlement Type	Billed By	Billed To	Reason
Nuremberg (0001) (Export)	Pre-carriage	Nuremberg to Hamburg	Nuremberg (0001)	Forwarding settlement	Nuremberg	Shipper	Nuremberg recovers the

Sales Organization (Company Code) (FWO Export/Import)	Stage	From/To	Planning and Execution Organization (Company Code)	Settlement Type	Billed By	Billed To	Reason
FWO)							cost of the pre-carriage from the shipper.
Nuremberg (0001) (Export FWO)	Main carriage	Hamburg to New York	Hamburg (0002)	Internal settlement	Hamburg	Nuremberg (0001)	Hamburg 0002 pays for the main carriage and recovers this cost from Nuremberg 0001.
Nuremberg (0001) (Export FWO)	Main carriage	Hamburg to New York	Nuremberg (0001)	Forwarding settlement	Nuremberg (0001)	New York (0003)	Nuremberg 0001 sends a forwarding settlement to New York 0003 to recover its costs.
New York (0003) (Import FWO)	Main and On-carriage	Hamburg to New York New York to ship-to party	New York (0003)	Forwarding settlement	New York	ship-to party	New York 0003 bills the ship-to party for the main carriage and on-carriage.

In the export forwarding order, the planned revenue for the Nuremberg sales organization is as follows:

- Charge in the forwarding order for the shipper for the execution of the pre-carriage. This is driven by the FOB Incoterms that the Nuremberg sales organization has for the shipper.
- Charge in the forwarding order for the destination organization, to recover the cost of the main carriage and any other charges at the origin or destination due back to the origin organization. The import organization recovers these charges from the ship-to party (customer) because of the FOB Incoterms.

The planned costs for the Nuremberg sales organization are as follows:

- Cost of pre-carriage
- Cost of main carriage

The system retrieves the planned cost as follows:

1. From the internal charge calculation for the pre-carriage and main carriage.
2. From the cost distribution of the charge calculation in the freight orders and freight bookings that are in place to execute the pre-carriage and main carriage.

For example, the system gets the cost of the main carriage from the purchasing organization in the freight booking. The purchasing organization in the freight booking belongs to the same company code as the sales organization in the forwarding order. This is an intracompany scenario, where the cost and the revenue are in the same organization in your company. For example, your company is located in Germany. Your sales organization is in Nuremberg and you also have a forwarding house in Hamburg. The forwarding house acts as a purchasing organization, and buys ocean carrier services.

The system determines the costs as follows:

- o If you do not enable intracompany settlement in Germany, the system uses the distributed cost in the freight booking at forwarding order level as the cost for the forwarding order.
- o If you enable intracompany settlement in Germany, the system uses the internal charges in the forwarding order that is in place between the purchasing organization in the freight booking (Hamburg) and the sales organization in the forwarding order (Nuremberg) as the cost for the main carriage.

Alternatively, the purchasing organization in the freight booking belongs to a different company code than the sales organization in the forwarding order. This is an intercompany scenario, where the cost and revenue are in different organizations in your company. For example, your company in Germany has a sales organization in Nuremberg. You have a forwarding house in Rotterdam, The Netherlands. The forwarding house acts as a purchasing organization, and buys ocean carrier services from the port of Rotterdam. The forwarding house in Rotterdam services shipments routed from Germany to Rotterdam. The forwarding house uses the internal settlement process to settle internally with the sales organization in Nuremberg. The system gets the cost of the main carriage from the internal charge calculation for the main carriage, in the internal settlement document.

The following table describes the profitability calculation.

Source	Planned Revenue (EUR)	Planned Cost (EUR)	Planned Profitability (EUR)
Charge for shipper in forwarding order	500	None	None
Charge for import organization in forwarding order	1500	None	None
Cost of pre-carriage (charge in freight order for pre-carriage)	None	400	None
Cost of main carriage (charge in freight booking for main carriage) This includes the distributed cost from the freight booking when you have not enabled intracompany settlement, as in this example. For more information about intracompany settlements, see Internal Settlement .	None	1200	None
Total	2000	1600	400

Some additional costs are incurred during execution, and these are included in the respective settlements. For example, a demurrage cost of EUR 50 is incurred at the pick-up location and is billed to the shipper.

The system retrieves the expected profitability from the settlement documents. The following table describes the expected profitability for the forwarding order.

Source	Expected Revenue (EUR)	Expected Cost (EUR)	Expected Profitability (EUR)
Settlement for shipper	500 + 50 = 550	None	None
Invoice for the main carriage for the import organization	1500	None	None
Freight settlement document for freight order	None	400 + 50 = 450	None
Freight settlement document for freight booking	None	1200	None
Total	2050	1650	400

Confirmation and Completion of a Forwarding Order

Use

After you have created and edited a forwarding order in Transportation Management (TM), you can confirm the data to the ordering party. You can execute the confirmation based on the following data:

- Data communicated by the ordering party
- Data entered manually or edited
- Data planned by the system

Prerequisites

See [Creation of a Forwarding Order](#).

In particular, you have defined the default confirmation type in Customizing for the document type, in other words, you have defined which data the confirmation to the customer is to be based upon. You have also defined whether the confirmation is to occur automatically upon saving. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#).

Process

The following section contains a description of an example process for forwarding order confirmation:

1. Manual Confirmation

If the confirmation was not already sent to the ordering party automatically upon saving the forwarding order as according to the Customizing settings, you confirm the order manually. You have the following options:

- If you have specified in Customizing that the confirmation is to occur based on the ordering party data or the planning data, choose **Confirm** and have the system enter the dates/times and quantities in the confirmation fields of the forwarding order according to the Customizing settings. You can also enter the dates manually and overwrite the Customizing settings.
- If you have specified in Customizing that the confirmation is to take place based on the manually entered data, enter the dates/times and quantities in the fields for confirmation of the forwarding order and choose **Confirm**.

i Note

If you have configured Customizing so that the confirmation occurs based on the planning data, a confirmation is only possible if complete planning data is available for the forwarding order. This means that the planning status of the forwarding order must be **Planned**.

2. Sending and Printing the Confirmation

The system automatically sends the confirmation to the ordering party. The sending type (for example, fax, e-mail) depends on your Customizing settings. You can also print out the forwarding order confirmation as a document. In the output history, you can display which confirmation documents were already printed or sent.

3. Changes After the Confirmation

You can still make changes to the forwarding order even if you have already sent a confirmation to the ordering party. The status of the order changes back to **Not Confirmed**, and you have to repeat the confirmation.

For more information about the effects of changes to the forwarding order on subsequent planning processes, see [Processing of Changes and Deletion of Forwarding Orders](#).

4. Completing the Forwarding Order

The system only assigns the status **Completed** to the forwarding order if the associated freight orders have been executed. If you have allowed the invoicing of the forwarding order in the Customizing settings for the forwarding order type, the invoicing must also have been completed. In this Customizing activity, you can also specify whether the instructions for the forwarding order must also have been completed before the entire order can be completed.

Once a forwarding order has been assigned the status **Completed**, you can reopen it if you have the appropriate authorization. This may be necessary if, for example, you want to add subsequent costs or attach documents. The life cycle status changes temporarily to **Determination Pending**. Depending on the changes you have made, once the order has been saved the status changes back to **In Execution**, **Partially Invoiced**, or **Completed**.

5. Archiving the Forwarding Order

Once the forwarding order has been closed or canceled, you can archive this business document. For more information, see [Archiving Business Documents for Forwarding Order Management \(TM-FWM\)](#).

More Information

[Confirmation of Forwarding Orders](#)

Creation of a Forwarding Order for Rail Transportation

Use

Transportation Management (TM) supports processes in which goods are transported by rail from a shipper to a consignee. The goods can be shipped from a source rail station, through intermediate rail stations, to a destination rail station (and can also be combined with pre-carriage and on-carriage stages by road). Different carriers can be responsible for execution and settlement on the different stages of the transportation process. The system therefore supports the transportation of full carloads and unit trains, as well as individual cargo items.

As a rail company or rail carrier transporting goods for your customers by rail, you can use forwarding order management in TM to create a forwarding order for rail transportation. In the process description below, which shows how to create and edit a rail forwarding order, rail companies and rail carriers are both referred to as **carriers**.

Prerequisites

You have configured the general settings for a forwarding order. For more information, see [Settings for the Forwarding Order](#).

You have also configured the following settings that are particular to the rail transportation process:

- You have defined the shipping type in Customizing. The following shipping types are available for rail transportation:
 - **Carload**: The shipper transfers an entire railcar to the carrier.
 - **Intermodal Rail**: The shipper transfers individual cargo items to the carrier, for example, containers, products, and trailers.
 - **Unit Train**: The shipper transfers an entire train to the carrier.

You assign the shipping type to the transportation mode **Rail**. You can also specify that a cargo item in the item hierarchy is always to be assigned to a railcar.

For more information, see Customizing for **Transportation Management** under **►►Forwarding Order Management ► Define Shipping Types ►**.

- You have defined an item type in Customizing. You assign the item category **Passive Vehicle Resource** and the resource class **Railcar** or **Trailer** to this item type.

For more information, see Customizing for **Transportation Management** under **►►Forwarding Order Management ► Define Item Types for Forwarding Order Management ►**.

- If you want to use vehicle groups and vehicle types, you have defined them in Customizing and assigned them to your item type. You can define your vehicle resources such as railcars and trailers in the master data as resources with the necessary details. Alternatively, you can also define a vehicle group and type for railcars and trailers that contains only general information about these vehicles.

For more information, see Customizing for **Transportation Management** under **►►Master Data ► Resources ► Define Equipment Groups and Equipment Types ►**.

- If you want to use a default route, you have defined it in the **Create Default Route** app on the launchpad under **Master Data**. For more information, see [Default Route](#).
- If you want the system to create railcar units automatically, you have defined a transportation unit type in Customizing and assigned it to your freight unit building rule. For more information, see Customizing for **Transportation Management** under **►►Planning ► Transportation Unit ► Define Transportation Unit Types ►** and under [Freight Unit Building Rule](#).

Process

The following section describes the special aspects of creating and editing a rail forwarding order. For more information about the standard process, see [Creation of a Forwarding Order](#) and [Editing of a Forwarding Order](#).

1. Entering the Data on the Initial Screen

On the initial screen of the forwarding order, you enter the basic data for the forwarding order. You enter **Rail** as the transportation mode and define a rail-specific means shipping type such as **Carload**. The user interface for the forwarding order dynamically adjusts to meet the requirements of a rail forwarding order based on the transportation mode.

You can also enter a default route at this stage that you have defined in the master data. If you want to create the forwarding order with a reference to the item of a forwarding agreement (FWA), you can select a default route that was defined for the FWA item. The system then takes into account the default route when it creates the transportation stages for the required route. The carrier data is also copied to the required route.

If you enter an FWA item as a reference, the system uses it to determine the agreed services and charges.

2. Entering General Forwarding Order Data, Locations, and Dates

You enter the data that you have received from the ordering party using the standard process. As the locations, you enter the source and destination rail stations.

3. Creating the Order Items

Depending on your Customizing settings, you can create a hierarchy consisting of railcar items and cargo items in the item table of the forwarding order. By combining the shipping type and the item types of the individual items in the forwarding order, you can specify that the railcar must be at the top level of the item hierarchy (shipping type **Carload**) or that the cargo items have to be on the top level (**Intermodal Transport**).

Your Customizing settings also determine the options that you have for entering information about your vehicles such as railcars:

- If you have defined a vehicle resource in the master data, the detail data for the resource is displayed in the item details of the forwarding order.
- If you have defined a vehicle group and type in Customizing and assigned them to the item type, this data is displayed in the details for the items created with the corresponding item type.
- You can enter the vehicle group and type directly in the item details of the forwarding order.

4. Determination of the Route

If you have entered a default route, the route is displayed on the **Ordered Route** tab page. Otherwise, you can enter the stages of the route with the source and destination rail stations. If the stages are to be executed or settled by different carriers, you can enter the executing carrier and the invoicing carrier in each case. You can also enter the external forwarding agreement for the invoicing carrier.

You can manually enter the train number for the relevant stages in the actual route. If you assign a schedule that contains a train number to a stage, the system automatically transfers the train number to the stage. In this case, the train number is read-only in the forwarding order. The train number is propagated to the freight unit, but not to the freight order. However, if you specify a train number when you create the freight order, the system displays the train number from the freight order in the forwarding order.

5. Creation of Railcar Units

If you have defined a transportation unit type in your freight unit building rule, the system creates railcar units instead of freight units. For more information, see [Creation and Editing of Freight Units](#) and [Railcar Unit](#).

6. Manual Transportation Planning

You plan the transportation as described in the standard process.

7. Exchange of Information Regarding Processing Progress

If other carriers are involved in the transportation process, you inform them of the transportation services that you are to provide. You can also inform each other about the processing progress.

8. Creating a Forwarding Settlement Document

According to the process that you have agreed with the ordering party and any other carriers involved, you have the following settlement options:

- All of the carriers involved in the transportation process invoice their transportation stages separately to the ordering party (for example, the rule 11 scenario in North America).
- You carry out the settlement process with the ordering party for the entire route. If necessary, the other carriers settle their stages with you. The system creates a forwarding settlement document for the entire forwarding order.
- Some stages of the route are invoiced to the ordering party by the executing carrier, while other stages are invoiced to the ordering party by one of the carriers on behalf of several executing carriers. This is a mixed settlement scenario.

More Information

For more information about a typical rail transportation process from the perspective of the shipper, see [Rail Transportation](#).

Direct Creation of Capacity Documents and Selection of Schedules

Use

When you create a forwarding order in Transportation Management (TM), you can already create capacity documents, such as freight orders, freight bookings, or transportation units, for the transportation stages defined in this document. You can also select existing capacity documents and schedules (departures).

You can only use these functions for stages of the actual route, that is, not for stages of the ordered route.

You can also create and select capacity documents directly from freight units and transportation units.

Prerequisites

- You have defined default capacity document types for stages in Customizing. This means that you have specified which type of capacity documents are to be created when you create capacity documents directly from the stage view of a forwarding order, a freight unit, or a transportation unit.

In this Customizing activity, you can also restrict which activities are to be possible from the user interface of the forwarding order (for example, selection of schedules).

For more information, see Customizing for Transportation Management under ► [Forwarding Order Management](#) ► [Forwarding Order](#) ► [Define Default Capacity Document Types for Stages](#) ►

- If you want default routes and schedules to be considered during the creation of capacity documents, you have specified this in advance. For more information, see [Schedule](#).

Features

Creation and Selection of Capacity Documents

On the [Actual Route](#) tab page of a forwarding order, you can create freight orders or freight bookings for the stages defined for the route, depending on the transportation mode. You can also select and assign a freight booking or freight order that already exists. Furthermore, you can create and select transportation units such as container units or package units.

If you create a capacity document from a forwarding order and have not defined a standard capacity document type in Customizing, a dialog box appears in which you can enter the required capacity document type. The system takes into account the planning profile that you specified in Customizing for the transportation unit type and freight unit type of the transportation units and freight units that are assigned to your forwarding order. If there is no planning profile or if the planning profile does not contain a standard capacity document type, the system takes the standard profile into account.

If a default route already exists for a stage, a capacity document cannot be created directly. In this case, you can select schedules for the stages of the default route. You can then create capacity documents for the stages of the actual route that were created based on the default route.

Once you have created the capacity document, the document ID is displayed as a link. You can then navigate to the document.

Creation of a Capacity Document for Several Stages

You can also create a capacity document for several consecutive stages or for stages of different freight or transportation units. This means that you can group several main carriage stages in a freight booking, for example.

The system first determines the capacity document type from your Customizing settings. The system then creates a freight order, a freight booking, or a transportation unit with this document type, depending on the transportation mode. The following prerequisites apply:

- A capacity document type must be defined for each stage selected.
- If you are creating freight orders, all selected stages must have the same freight order type.
- If you are creating freight bookings, all selected main carriages must have the same freight booking type and the same transportation mode.
- The following relevant data for the stage must be the same (or may not be defined):
 - Carrier
 - Voyage
 - Name of vessel
- The status of the stages must be **Unplanned**.

If you create a freight booking for several stages, you can change the freight document type that has been determined and, for example, choose document types with corresponding service levels at the port of loading and port of discharge.

Selection of a Schedule

You can select schedules that already exist for the stages of a forwarding order. If a default route exists, only the schedules that are available for the stages of this route are displayed. If fixed schedules are assigned to the default route, only the options for these schedules are displayed.

Once you have selected a schedule option, the data of the schedule is copied to your actual route. If the default route contains additional stages, they are also copied to the actual route.

If you have entered locations for transportation zones in the master data and use these zones in the default route instead of locations, the system takes them into account when it calculates the default route for the forwarding order.

Remove Assignment of a Capacity Document

You can also remove a capacity document that you have created for one or several stages or assigned to the stage. In this case you unassign it. The link to the capacity document is removed from the stage view.

If so defined in Customizing for the freight order type, the freight booking type, or the transportation unit type, the capacity document is canceled or deleted from the system once it has been unassigned.

Activities

- If you want to create or select capacity documents for a forwarding order, open the **Forwarding Orders - Worklist** or **Edit Forwarding Order** app on the launchpad. Open your forwarding order and select the **Actual Route** tab page.
- If you want to create or select capacity documents for a transportation unit, open the **Container Units - Worklist** or **Edit Container Unit** app on the launchpad. Open your transportation unit and select the **Stages** tab page.

- Depending on the function you require, you select one or several stages and choose **►Capacity Document ► Create One for Consecutive Stages ►** or **►Capacity Document ► Select ►**, for example.

The system creates a capacity document of the type that you specified in Customizing. For example, if you have defined that a freight booking is created for a transportation stage of the type **main carriage** with the transportation mode **sea**, the system takes this into account.

If you want to select a schedule or remove the assignment, select the stage and choose the corresponding pushbutton. You can also remove the assignment of several stages to a capacity document by selecting one or more of the assigned stages and choosing **►Capacity Document ► Remove ►** or **Remove for This Stage and Following Stages**. If you have only selected one of the stages, the system will ask if you want to remove the assignment of all stages. Only this variant is possible.

More Information

[Direct Creation of Freight Orders](#) (detailed information specific to freight orders)

[Determination of the Route](#)

For more information about the interaction of the default route and the determination of transportation distance and transportation duration, see [Determination of Transportation Distance and Transportation Duration](#).

Forwarding Quotation

Definition

Quotation from a carrier or logistics service provider to an ordering party with respect to the transportation of goods, which contains information about the price and further conditions regarding the transportation services.

Use

In Transportation Management (TM), the business document of the forwarding quotation serves to help the ordering party send the data of a potential forwarding order with the quotation price. When the ordering party places the order, you can then create a forwarding order in relation to the quotation.

Based on the forwarding quotation, you can have the system determine and define the route and calculate the transportation charges.

Structure

The forwarding quotation has a similar structure as the forwarding order, since it is the basis for the eventual forwarding order (see [Forwarding Order](#)). In addition, you can enter the following data:

- Validity Scope for the Quotation

When creating the forwarding order based on this quotation, the system checks whether the quotation is still valid.

- Quotation Price

Here you can enter a total price for the quotation, without specifying the transportation charges in detail. If you specify transportation charges, the system displays the total here in addition. You should note that the display of the quotation price is determined according to the calculation level that you have defined in the calculation profile. The quotation price is not displayed if you have defined a calculation at stage level in the calculation profile.

Status of the Forwarding Quotation

The statuses of the forwarding quotation represent for the most part those of the forwarding order (see [Status of a Forwarding Order](#)). Only the values of the life cycle status that are described below differ from those of the forwarding order. Some statuses for the forwarding quotation, which refer to planning and execution, have been removed.

Life Cycle Status

Status	Description
New	The forwarding quotation was newly created and not yet sent.
Not Transferred	The forwarding quotation that was already sent to the ordering party was changed. The changes were not yet sent.
Transferred	The forwarding quotation was sent to the ordering party.
Canceled	The forwarding quotation was canceled.

Locking the Forwarding Quotation

In addition to the forwarding order statuses described above, the system displays the locking status for this quotation. You can set the lock manually or have the system set it automatically. The system sets automatic locks when executing the approval workflow or during the dangerous goods check, for example. The reason for the lock is also displayed for the lock status. If you set the lock manually, you can use the value help to select the reason for the lock.

Integration

The business document for the forwarding quotation is integrated with the following business document in TM:

- [Forwarding Order](#)

You can convert a forwarding quotation that you had sent to an ordering party and which was accepted by this ordering party into a forwarding order, in other words, you create a forwarding order with reference to this quotation.

More Information

[Creation and Editing of a Forwarding Quotation](#)

Creation and Editing of a Forwarding Quotation

Use

You can use a forwarding quotation (see [Forwarding Quotation](#)) in Transportation Management (TM) to send an offer to your ordering party. You can have the system determine possible routes and calculate the transportation charges based on the selected route. You can send the quotation containing the transportation charge information to your ordering party via e-mail or fax, for example.

After the ordering party has accepted the quotation, you can create a forwarding order based on it. You can also subsequently assign a forwarding quotation to a forwarding order. However, this is subject to certain prerequisites such as corresponding data. For more information, see [Subsequent Assignment of Forwarding Quotations](#).

You can create the forwarding quotation manually. If you receive a forwarding request for quotation electronically (for example, within the tendering process), you can have the system create the forwarding quotation automatically based on the request for quotation. For more information about the tendering process, see [Freight Tendering](#).

Prerequisites

The same prerequisites apply for creating a forwarding quotation as for creating a forwarding order, especially if you later want to create a forwarding order based on the forwarding quotation. For more information, see [Settings for the Forwarding Order](#).

Alternatively or in addition to the forwarding order type, you must also define the **forwarding quotation type** in Customizing. You use this document type to determine a series of characteristics for the forwarding quotation and can make certain default settings. For example, you can predefine the sales organization or determine how many forwarding orders are allowed to be created based on a forwarding quotation of this type. Moreover, you can specify whether the forwarding quotation of this type is based on a forwarding request for quotation (quotation mode)

For more information, see Customizing for [Transportation Management](#) under [▶▶Forwarding Order Management ▶ Forwarding Quotation ▶ Define Forwarding Quotation Types ▶](#).

i Note

If you later want to create a forwarding order based on a forwarding quotation and predefine the shipping type and the transportation mode in Customizing for the forwarding quotation type, these settings must be the same as those in the forwarding order type.

If you want to specify reasons for the rejection or cancellation of a forwarding quotation, you have defined them in Customizing. For more information, see Customizing for [Transportation Management](#) under the following Customizing activities:

- [▶▶Basic Functions ▶ General Settings ▶ Define Rejection Reason Codes ▶](#)
- [Forwarding Order ManagementDefine Cancellation Reason Codes](#)

Process

1. Entering the Data on the Initial Screen, the General Data, and the Quotation Items

Since the forwarding quotation can be the basis for a forwarding order, the data structure and user interface for creating the quotation data is similar to that of the forwarding order. In addition, you can determine the validity scope of the quotation.

For more information, see [Creation of a Forwarding Order](#) and [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

If you have received a forwarding request for quotation from your ordering party, the forwarding quotation is automatically created based on this request and the data it contains. It is displayed in your personal object worklist (POWL), from where you can access and edit it.

2. Saving the Forwarding Quotation

If the forwarding quotation was not created automatically, save it. The system now assigns the document an ID from the number range for this document type. In addition, the system checks the entered data for completeness and consistency. The system displays the results of these checks in the form of error messages and as a completeness status.

If so defined in Customizing for the forwarding quotation type, the system automatically determines the transportation charges after saving.

3. Determining the Route

If you have already received information about the ordered route from the ordering party, you can create the stages of the route. For more information, see [Determination of the Route](#).

4. Calculating Transportation Charges

If this has not occurred automatically after saving, you can have the system calculate the transportation charges or you can enter them manually. The sum of transportation charges is displayed as quotation price. Instead of entering the transportation charges, you can also specify only the sum of the charges as the quotation price. For more information, see [Charge Calculation](#).

If you quickly want to estimate the transportation charges based on a minimum of data, you can use the **Estimate Forwarding Charges** function to do so. For more information, see [Charge Estimation](#).

5. Sending the Quotation to the Ordering Party and Printing the Document

You can send the forwarding quotation to your ordering party and print out the document immediately or after previewing it. The sending type (for example, fax, e-mail) depends on your Customizing settings. If you have created the quotation based on a request from your ordering party, you can dispense with this step.

The history of sending and printing quotation documents is displayed in the forwarding quotation on the **Output Management** tab. For more information, see [Output Management](#).

6. Accepting or Rejecting a Quotation

If your ordering party has already informed you whether he agrees with the quotation, or if the quotation is based on a forwarding request for quotation from your ordering party, you can accept or reject the forwarding quotation. If you reject it, you can also specify a rejection reason code. The quotation is then automatically cancelled.

The corresponding response code and if necessary reason rejection code is displayed in the **General Data** area of the forwarding quotation.

7. Creating the Forwarding Order Based on the Quotation, or Canceling the Quotation

Depending on whether your ordering party has accepted or rejected the quotation, or whether you accept the conditions of your ordering party contained in the forwarding request for quotation, you can proceed as follows:

- o You create a forwarding order based on the forwarding quotation, in other words, you create a forwarding order based on the data in the quotation. When you do so, the system also copies the route data and the price calculation from the quotation. The system also checks whether the quotation is still valid. For more information, see [Creation of a Forwarding Order](#).

You can also receive a forwarding order based on a forwarding quotation electronically (EDI).

- o You cancel the quotation. The life cycle status changes to **Canceled**. Depending on your Customizing settings, you can enter a cancellation reason that is displayed in the **General Data** area of the forwarding quotation.

Note that the cancellation is irreversible.

If you have specified in Customizing for the document type how many forwarding orders are allowed to be created from this forwarding quotation, the system checks this when you create a forwarding order.

8. Archiving the Forwarding Quotation

After the forwarding order has been canceled or if it is no longer valid, you can archive the business document. For more information, see [Archiving Business Documents for Forwarding Order Management \(TM-FWM\)](#).

Subsequent Assignment of Forwarding Quotations

Use

If you follow the standard process in Transportation Management (TM), you create a forwarding order with reference to an existing forwarding quotation (FWQ). However, you can also assign forwarding quotations to a forwarding order later. For example, you can create a forwarding order even if the corresponding forwarding quotation does not exist at that time.

The transportation charges entered in the forwarding quotation can be copied automatically to the forwarding order. Transportation charges that already exist in the forwarding order can be overwritten by the system.

Prerequisites

If you want to assign a forwarding quotation to a forwarding order subsequently, certain prerequisites must be fulfilled, which are checked by the system. When doing this, the system differentiates between hard and soft constraints. If these constraints are not fulfilled, the system proceeds as follows:

- **Hard Constraints**

The forwarding quotation cannot be assigned. These constraints are implemented as standard in TM. They cannot be changed.

- **Soft Constraints**

The forwarding quotation can be assigned, but the system does not copy the transportation charges to the forwarding order.

You can use a Business Add-In (BAI) to enhance these constraints. You can also replace the soft constraints with other constraints. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Forwarding Order Management](#) > [Assignment of Forwarding Quotation to Forwarding Order](#).

Features

Hard Constraints

The system checks the following constraints:

- **Forwarding order:**

- No forwarding settlement document has been assigned to the forwarding order as yet.

- **Forwarding quotation:**

- The forwarding quotation is valid and has been submitted (in other words it has the status **Submitted**).
- The forwarding quotation is not subject to a quotation block.
- The forwarding quotation can be assigned to another forwarding order.

This constraint relates to a Customizing setting with which you can restrict the number of forwarding orders that can be created based on one forwarding quotation. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Forwarding Quotation](#) > [Define Forwarding Quotation Types](#).

- **Matching data:**

The following data in the forwarding quotation and the forwarding order must match:

- Forwarding agreement
- Traffic direction
- Air waybill type (AWB type)

- Shipper and consignee
- Source location and destination location

Soft Constraints

The system checks whether the following data matches in the forwarding quotation and the forwarding order:

- Sales organization and ordering party
- Additional forwarding agreement (import or export forwarding agreement)
- Incoterm data (Incoterm and free text for Incoterm location)
- Service level, transportation mode, shipping type, and movement type
- Amounts and quantities in the header data, for example **high-value cargo**
- Pick-up and delivery date in the header data
- Dangerous goods indicator

In addition, certain items or stages in the forwarding quotation for which transportation charges have already been calculated must exist in the forwarding order. Otherwise the system cannot copy the transportation charges from the forwarding quotation to the forwarding order.

Messages, Status Updates, and Options

The system displays the check results as system messages. In addition, it displays the status of the assignment on the **Status** tab of the forwarding order as follows:

Status	Description
No FWQ Assigned	No forwarding quotations have been assigned to the forwarding order; this is the initial status.
FWQ Subsequently Assigned	A forwarding quotation was assigned to the forwarding order subsequently, in other words after the forwarding order was created; the hard checks were successful; existing transportation charges from the forwarding quotation have been copied to the forwarding order.
FWQ Subsequently Assigned; Soft Check Failed	A forwarding quotation was assigned to the forwarding order subsequently; the soft checks were unsuccessful; any existing transportation charges were not copied to the forwarding order.
FOW Created from FWQ	The forwarding order was created based on a forwarding quotation, either manually or using EDI.

If the soft check fails, you have the following options:

- You can change the data of the forwarding quotation or forwarding order in such a way that the prerequisites are fulfilled. Then you can run another check. This does not change the status.
- You can copy the transportation charges from the forwarding quotation to the forwarding order. The status then changes to **FWQ Subsequently Assigned**.

Activities

You can subsequently assign a forwarding quotation to a forwarding order as follows:

- From the user interface of the forwarding order, by choosing ► **Forwarding Quotation** ► **Assign FWQ** ►
- From your personal worklist (POWL) (same navigation path)

In the POWL you can also find forwarding orders with assigned forwarding quotation based on their status.

If you want to recheck the prerequisites or subsequently copy the transportation charges, choose ► **Forwarding Quotation** ► **Recheck Constraints** ►, or ► **Forwarding Quotation** ► **Copy Transportation Charges** ►.

If you want to replace an assigned forwarding quotation by another one, you can assign the new forwarding quotation as described above, by choosing ► **Forwarding Quotation** ► **Assign FWQ** ►.

i Note

- If transportation charges already exist in the forwarding order, the system asks if you want to overwrite them. The forwarding quotation is only assigned to the forwarding order if you agree to this.
- The transportation charges copied to the forwarding order are fixed.

More Information

[Creation and Editing of a Forwarding Quotation](#)

[Creation of a Forwarding Order](#)

Creation of Forwarding Order Items or Forwarding Quotation Items

Use

When you create a forwarding order or a forwarding quotation in Transportation Management (TM), you can enter the items of this business document in an item hierarchy. The hierarchy can contain the following levels:

- Railcar
- Trailer
- Container
- Package
- Product
- Service

The planning always occurs on the highest level, for example, on the level of the container or package. Information about the packages in a container or the products contained in a package is additional information for the planner, and does not play an essential role in the planning by the system. This information does, however, have an impact on a potential transportation charge calculation. This is also valid for service items.

Prerequisites

- If you want to provide information about railcars, trailers, or containers when creating the order items, you have defined **equipment groups and types** as well as ventilation settings in Customizing. You can define transportation mode-specific equipment groups and equipment types, for example, for sea freight containers and unit load devices (ULDs). You can also enter information about packages by defining package types.

This data serves as additional information for the planner and is not relevant for the planning by the system. For more information, see Customizing for [Transportation Management](#) under the following Customizing activities:

- [Master Data](#) > [Resources](#) > [Define Equipment Groups and Equipment Types](#) >
- [Forwarding Order Management](#) > [Define Package Type Codes](#) >
- [General Settings for Order Management](#) > [Define Ventilation Settings](#) >
- In Customizing, you have specified **item types** and assigned them to document types. You can use the item type to predefine various parameters for an item, such as the item category, equipment group and equipment type, as well as default units of measure. Additionally, you can already assign a dangerous goods UI profile, with which you can make settings for the display of dangerous goods information for items.

Assigning item types to document types allows you to specify which item types are allowed for a document type. You can also specify a default item type for items with a certain item category that are created for a document with a certain document type.

The default item type is used when forwarding orders and forwarding quotations are received electronically. If you create items manually and have only defined one default item type for a combination of the item category and document type, the system enters the item type automatically.

You can also define that the system automatically creates initial items using specified item types when you manually create a forwarding order or forwarding quotation.

For more information, see Customizing for [Transportation Management](#) under the following Customizing activities:

- [Forwarding Order Management](#) > [Define Item Types](#) >
- [Forwarding Order Management](#) > [Forwarding Order](#) > [Assign Item Types to Forwarding Order Types](#) >
- [Forwarding Order Management](#) > [Forwarding Quotation](#) > [Assign Item Types to Forwarding Quotation Types](#) >

If you want to use railcars or trailers in your item hierarchy, you have assigned the item category **Passive Vehicle Resource** to your item type along with resource class **Railcar** or **Trailer**.

If you have defined the item type for the **Service** item category, and you want to assign service types to the service items, you have defined service types in Customizing. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [General Settings](#) > [Define Service Types](#) >.

- You have defined **shipping types** in Customizing. For the shipping type, you have specified whether cargo items such as containers, packages, and products can or must be assigned to equipment items such as railcars. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Shipping Types](#) >.
- If you want to use **commodity codes** to classify goods, you have made settings in Customizing for [Transportation Management](#) under [Master Data](#) > [Classification of Goods](#) >. For more information, see [Classification of Goods](#).
- If you want to define that you can specify **locations and business partners** for an item that differ from the header data in the forwarding order, you have defined this accordingly in Customizing for the forwarding order type or the forwarding quotation type. The default setting is that you cannot specify locations and business partners that differ from the header data. For more information, see the Customizing activities mentioned above.

Features

You can create items in the forwarding order and forwarding quotation business documents as follows:

- **Item Hierarchy**

You can create the items individually or as a hierarchy consisting of the levels railcar, trailer, container, package, product, and service. The sequence of the levels is predefined, in other words, you cannot define a product as highest level with a lower-level package.

Depending on the Customizing settings for your shipping type that you enter in the forwarding order or forwarding quotation, you can or must assign cargo items such as containers, packages, and products to equipment items such as railcars. In this way you can define, for example, that if the shipping type is **Carload**, the railcar must be at the highest level in the item hierarchy.

- **Item Type**

You must enter an item type for each item. If you leave this field empty, a unique default item type for the combination of the item type (for example, product) and the document type must be defined in Customizing (see above under *Prerequisites*). In this case, the system determines the default value from the Customizing settings.

Once you have specified an item type for an item, it cannot be changed.

- **Service Items**

You can define the services in a service product catalog and include the service products in the forwarding agreements. You can also include services that are not part of a service product catalog. The services are always assigned to a forwarding agreement item.

When you specify a reference to a forwarding agreement item in your business document (that is, forwarding quotation or forwarding order), the system automatically transfers the service items from the forwarding agreement item to the business document. In the **Items** screen area on the **Service Details** tab page, the system sets the corresponding reference type, that is, **Reference to FWA Item** or **Reference to Additional FWA Item**. If required, you can change this to **No Reference**. If you delete the reference to the forwarding agreement item, the system sets **No Reference** for all service items.

Note the following:

- The system transfers services only if they have been flagged as flow services in the forwarding agreement. If a service is also flagged as mandatory in the forwarding agreement, you cannot subsequently delete this service item in the business document.
- Before the system transfers the services, it checks if the business document already contains non-referenced service items. If a non-referenced service item in the business document has the same service type as a service item in the forwarding agreement item, the system reuses the non-referenced service item by setting the relevant reference type in the service details (for example, **Reference to FWA Item**). It does not create a new service item in this case.

If you have defined an item type for the **Service** item category in Customizing, you can create service items in your forwarding order or forwarding quotation. You can specify the following features:

- Service type, for example, cleaning and fumigating a container. For example, the service type can be assigned to an instruction set that contains the step-by-step instructions for fumigating a container.
- Reference (for example, **Reference to FWA Item**). You can specify a reference only if there is a service item with the same service type in the forwarding agreement item.

Service items are used mainly for charge calculation and settlement. However, they are not carried over to follow-up documents or included in scheduling. On the **Charges** tab page, you can enter the quantity for a service item below the main item. The system uses the logistics data of the service item to calculate the charges. If you do not enter the quantity for the service item, the system calculates the charges based on the quantity that you entered for the higher-level logistics item.

You can also create a forwarding order that contains only service items (for settlement purposes, for example). This means that the forwarding order does not contain cargo items at the same time.

For more information, see [Service Product Catalogs](#) and [Management of Instructions](#).

• Equipment Type

If you have made the corresponding settings in Customizing, you can assign equipment types (for example, freezer containers and 20-foot containers) as well as package types (for example, pallets) to the items. The following also applies:

- If you have assigned an equipment group to the item type in Customizing, it is entered automatically for an item with this item type and equipment types of this group are displayed for selection. You can, however, overwrite the equipment group.
- If you have specified a default package type and a default equipment type in Customizing for the item type, this is displayed automatically.

For containers, you can enter data in the detail data, for example, about temperature control. If you have defined temperature data for this container type in Customizing for the equipment type, the system checks your entries based on the Customizing data. You can also select ventilation settings that you have predefined in Customizing. This data is information for the planner.

Furthermore, you can already enter data about seals for the container. This data is then automatically copied to the successor freight booking or the freight order. For more information, see [Use of Seals](#).

• Commodity Codes

If you have defined these in Customizing, you can assign commodity codes to the items of an air forwarding order or forwarding quotation. You use these codes to classify and evaluate your goods to be transported (commodities) according to characteristics of your choice. When classifying goods, you can take into account transportation-relevant product characteristics, such as weight, density, loading properties, and handling (of dangerous goods, for example). You can use this information, which is also transferred to follow-on documents such as freight orders or freight bookings, as an additional description or a basis for decisions about packaging and so on.

For more information, see [Classification of Goods](#).

• Quantities, Weights, and Volumes

You can enter quantity, weight and volume data for the items you entered (except for service items for which you can enter only quantity data). These are partly aggregated across the different levels of an item hierarchy automatically. In addition, any discrepancies between the required quantities and actual quantities are displayed. For more information, see [Creation of Quantity, Weight and Volume Data of an Item](#).

• Automatic Checks

In Customizing you can specify that you want the system to automatically check whether a specific product exists and that you want the system to automatically calculate the weight and volume of the item based on the quantity and unit of measure entered in the product master data. You can also specify that the system is to check whether a package exceeds certain dimensions. If necessary, you can check the dimensions of a package manually. Additionally, a container number that has been entered is checked to see if it meets international standards.

• Copy and Move

You can copy existing items and move items via drag and drop to another position in the item hierarchy.

If you create a forwarding order based on a forwarding quotation, the item types that have been defined are copied to the forwarding order.

• Splitting Items

You can also split items into several items. The number of the split items depends on the number of pieces contained in the item. A new item is created for each piece. The quantities are also split.

An item can only be split if it meets the following prerequisites:

- It has not yet been planned.
- It contains more than one piece.
- It does not have any subitems.
- It is not a service item.

- **Locations and Business Partners**

If you have not specified otherwise in Customizing, you can define your own source and target locations, pick-up and delivery dates as well as business partners in the detail data of items or subitems in a forwarding order or forwarding quotation. The following business scenarios can provide a reason for doing so:

- You want to pick up the goods for a forwarding order from different locations or different shippers and transport them together in a main carriage.
- You want to deliver the goods from one forwarding order that you have transport in a main carriage to different locations or consignees.

A combination of these two scenarios is also possible. Moreover, for business partners, you can enter one-time locations and one-time addresses that differ from those in the header data in a forwarding order or quotation.

i Note

- If you have specified different shippers or consignees for multiple forwarding order items, an automatic settlement is performed only for those shippers or consignees that you have specified in the header data of the forwarding order (or for the corresponding prepaid agreement parties and collect agreement parties from the header data).
- You can perform the settlement for the different business partners by creating forwarding settlement documents manually.

More Information

[Example for the Creation of Items](#)

Creation of Quantity, Weight and Volume Data of an Item

Use

When you create a forwarding order or forwarding quotation in Transportation Management (TM), you can create quantity, weight and volume data for the order items, and have the system calculate the data automatically. The following is a description of the special features that apply to this.

For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

Prerequisites

- If required, you have assigned a **default unit of measure** for weight, volume, and quantity as well as for length, width, and height in Customizing for the business document type that you created for the forwarding order or forwarding quotation. The system uses this unit of measure if the following applies:
 - You have not entered a unit of measure for the item on the user interface
 - You have not defined a unit of measure in the item type
 - The system calculates the total quantity of the planning-relevant order items

For more information, see Customizing for [Transportation Management](#) under the following Customizing activities:

- [Forwarding Order Management](#) > [Forwarding Order](#) > [Define Forwarding Order Types](#) >
- [Forwarding Order Management](#) > [Forwarding Quotation](#) > [Define Forwarding Quotation Types](#) >
- If you want the system to automatically total the **gross volume** of an item from lower item levels, you can define this in Customizing. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Item Types for Forwarding Order Management](#) >.
- If an item has the category [Product](#) and this product exists in the product master data, you can specify that you want the system to automatically calculate the weight and volume of the item based on the quantity and unit of measure entered in the product master data. To do this, set the [Product Check](#) checkbox in Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Item Types for Forwarding Order Management](#) >.

Features

Aggregation of the Data and Automatic Calculations

- The quantities, weights and volumes of the planning-relevant items (in other words, the items on the highest level of an item hierarchy) are totaled by the system and displayed in the base unit of measure in the header data of the document. However, the totaling is only possible if the units of the items can be converted to the required unit. If this is not the case, the total stays empty. All other items are additional information and can be included for instance on the print documents as well.
- The quantities and weights of the lower-level items in an item hierarchy are aggregated by the system to the next-higher level of the hierarchy. In a three-level hierarchy consisting of container, package, and product levels, the system aggregates the data as follows:
 1. If you specify the gross weight on the product level, the system aggregates the data to the net weight of the next-higher levels, for example of the package level. The net weight is no longer changeable.
 2. If you specify on package level the weight of the packaging (tare weight) in addition to the net weight, the system adds the data to the gross weight of the package.
 3. The gross weight of the package is aggregated by the system to the net weight (loading capacity) of the container and is then no longer changeable. If you additionally specify the tare weight of the container, the system totals the net weight (loading capacity) and the tare weight to the gross weight (total weight) of the container. The total weight is likewise no longer changeable. It is also displayed in the document header.
- If you have activated this in Customizing of the forwarding order type (see below), the gross volume data of the lower-level items in an item hierarchy is aggregated by the system to the next-higher level of the hierarchy. Once the value of an item from lower item levels has been totaled, it can no longer be changed. You can only process the value at higher items levels if the item does not have any subitems with volume data.

The value in the **gross volume** field of an item represents either the external volume or the volume of the contents, depending on the item type. It can be interpreted as follows:

- Container or ULD: Gross volume of the contents of a container or ULD

- o Package: Gross volume of the package that, for example, is defined by the external dimensions of the package (length, width and height)
 - o Product: Gross volume of the product
- After you have entered the equipment group, the equipment type and the number of containers, the system calculates the tare weight and the capacity of the container in TEU based on the Customizing settings.
- If you do not specify a unit of measure, the system determines the default unit of measure from the item type or, if not available, from the document type (if defined in Customizing).
- If you enter the length, height, and width of a package item, the system calculates the gross volume automatically. However, this is only possible if no volume data from lower item levels has been totaled. In addition, the **gross volume** field must either be empty before the calculation, or the value entered in the field must correspond to the product regarding length, height and width (meaning it has probably already been calculated).
- If you enter only the gross weight, the system calculates the net weight and the empty weight. It does so as follows:
 - o Containers and railcars: The system assumes that the gross weight is the same as the empty weight and that the net weight is 0.
 - o Packages and products: The system assumes that the gross weight is the same as the net weight and that the empty weight is 0.

Actual Quantities and Discrepancies

- If the actual received quantities for an item have already been entered in the freight document, the cargo receipt status of the freight document (for example, **Delivered**) is displayed in the item table. The detailed data for the item then displays the quantities requested by the ordering party as well as the confirmed actual quantities of the freight document.
- If there are discrepancies between the requested and actual quantities, or other discrepancies, this is displayed with a status in the item table as well as on a separate tab in detail. You can then clarify the discrepancies and set them to solved for subsequent planning. For more information, see [Handling Discrepancies](#).

Example for the Creation of Items

Use

The following describes an example of how you can create a hierarchy of items of a forwarding order or forwarding quotation in Transportation Management (TM). For more information, see [Creation of Forwarding Order Items or Forwarding Quotation Items](#).

Example

Item Level 1	Item Level 2	Item Level 3	Item Description	Item Type	Package Type	Container	Gross Weight	UoM	Net Weight	UoM	Tare Weight
Container				CN		HJCU4572305	5,900.00	KG	3,900.00	KG	2,000.00
	Package			PKG	Europallet		3,900.00	KG	3,650.00	KG	250.00
		Product	Spare Part	PRD			3,650.00	KG			
Container				CN		MWCU6059784	9,800.00	KG	7,800.00	KG	2,000.00

Item Level 1	Item Level 2	Item Level 3	Item Description	Item Type	Package Type	Container	Gross Weight	UoM	Net Weight	UoM	Tare Weight
	Package			PKG	Europallet		3,900.00	KG	3,650.00	KG	250.00
		Product	Spare Part	PRD			3,650.00	KG			
	Package			PKG	Europallet		3,900.00	KG	3,650.00	KG	250.00
		Product	Spare Part	PRD			3,650.00	KG			

As the table shows, you create two items for containers on the highest hierarchy level. The first container HJCU4572305 has a subitem on the second hierarchy level with a package of package type **Europallet** and a subitem on the third hierarchy level with a product of the description **Spare Parts**. The second container MWCU6059784 has two subitems on the second level, each with a package of package type **Europallet**. Each package has one additional subitem with a product of the description **Spare Parts**.

For the container MWCU6059784, the system aggregates the weight data as follows:

- The container has a gross weight of 9,800 KG. Each package has a gross weight of 3,900 KG. The products each have a gross weight of 3,650 KG.
- The gross weight of the package of 3,900 KG is the sum of the net weight of 3,650 KG aggregated from the product and the weight of the packaging (tare weight), which is 250 KG. The gross weight of the container of 9,800 KG is the sum of the net weight of 7,800 KG that was aggregated from the packages and the tare weight of 2,000 KG of the container.

Only the two container items that were defined on the first hierarchy level are relevant for planning. Therefore, 15,700 KG is displayed in the header data of the forwarding order as the total gross weight of the order (sum of the gross weight of 9,800 KG and 5,900 KG of both containers). The subitems only serve as additional information, however, they can have an effect on the calculation of transportation charges.

Handling Discrepancies

Use

In a forwarding order, you enter the quantities and attributes requested by the ordering party for the items to be transported. When the actual quantities are reported, there can be discrepancies between them and the requested quantities. You can enter these quantity discrepancies in the freight document, just as you enter other discrepancies, such as packaging damage or missing documents. These discrepancies are then also displayed in the forwarding order.

Using the displayed data, you can check discrepancies and, for example, clarify after discussion with the ordering party whether the discrepancies are acceptable. Then, for further planning, you can set discrepancies to resolved, or block stages of the forwarding order that were not yet executed.

In addition, you can search for forwarding orders that contain discrepancies. You can also display discrepancies in your personal object worklist for discrepancies.

Process

The following example illustrates the process for handling quantity discrepancies and other discrepancies.

1. The employee, who receives the goods, discovers a quantity discrepancy or other discrepancy for an item. This person enters the discrepancy in the freight order or in the freight booking. For more information see [Discrepancies and Events](#).

2. In the forwarding order, the discrepancy is shown by a status in the table of order items (**Discrepancy Exists**).
3. You display the detail data for the item. Here you can check the actual quantities against the ordered quantities. The system also displays here whether certain transportation-relevant attributes, such as **Overdimensional Cargo** or **Dangerous Goods**, deviate from the requested data and whether a specific discrepancy needs to be resolved.
4. On the **Discrepancies** tab page of the detail data, you check the details of the various discrepancies. It is possible that the discrepancy type, the discrepancy subcategory, and other additional information was entered in the freight document.
5. If your ordering party accepts the discrepancies in quantities and attributes, you can choose **Copy Actual Data** > **Copy Actual Quantities** to copy the actual quantities to the ordered quantities in the item table. You can also copy the actual attributes in the same way.

If you want to copy to the order data only single items that deviate from the actual data, you can also change the order data manually.

6. You can set the discrepancy to "Resolved" on the **Discrepancies** tab page. You can also define an execution block for stages of the forwarding order that have not been executed yet. If you set the discrepancy to resolved, you indicate to the subsequent planning stages that no further action is required (for example, if the discrepancy was accepted by the ordering party). This does not mean that the discrepancy was corrected. For more information, see [Resolution of Discrepancies](#).

i Note

If you want to resolve the discrepancy **Missing Cargo** and copy the actual data to your order data, you have to set the discrepancy to "Resolved" first; you can then copy the actual quantities and attributes to your order data. There are other restrictions that apply to the copying and manual changing of attributes. For more information, see [Handling Discrepancies in Transit Warehousing Process](#).

After you have set the status **Discrepancy Resolved**, you can also enter a note for it.

7. The status **Discrepancy Resolved** that is set in the forwarding order and any additional information you may enter are displayed in the freight document. The reverse is also true - the forwarding order shows when a discrepancy was already set to resolved in the freight document.

i Note

You can still change data in the forwarding order after the cargo receipt status has been set to **On Hand**. However, these changes are not copied to follow-on documents. If a settlement document has already been created, first you have to cancel this document, then you can make the changes, and finally you create a new settlement document.

Search for Forwarding Orders with Discrepancies

If you want to directly process discrepancies in forwarding orders, you can search for forwarding orders with discrepancies in your personal worklist as an additional process step. That is, you define a query for forwarding orders that have the status **Discrepancy Exists**. A forwarding order receives the overall status **Discrepancy Exists**, if there is a discrepancy for at least one item of the order. A forwarding order does not receive the status **Discrepancy Resolved** until all discrepancies that existed are resolved.

You can display the discrepancies together with all the relevant data in your personal object worklist for discrepancies. Furthermore, you can solve discrepancies directly and access the related documents such as forwarding orders and freight orders.

More Information

[Discrepancies and Events](#) (freight order)

[Quantity Discrepancies](#) (freight order)

[Other Discrepancies](#) (freight order)

Determination of the Route

Use

When you create a forwarding order or a forwarding quotation in Transportation Management (TM), you can already specify the route requested by the ordering party. You can also define the route in more detail manually, as well as edit the requested route.

To do this, you can enter the individual stages of the route in the forwarding order or the forwarding quotation. The system then considers the route during further transportation planning and settlement. If necessary, you can also have the system determine the route and then copy the result to your order or quotation. The result of transportation planning done at a later time is also displayed in the forwarding order.

Prerequisites

Along with the general settings for a forwarding order or a forwarding quotation, you have made the following settings in Customizing:

- You have defined **stage types**. In doing so, you define whether a stage is a pre-carriage, main carriage, or on-carriage. This categorization is required for the processes that follow, such as charge calculation, settlement, and the creation of freight documents.

You can also define an instruction set in the stage type that contains work instructions for a stage (see [Management of Instructions](#)).

- You may have additionally specified which **transportation modes are permitted for certain stage types**. You can also define a default transportation mode for a stage type.
- You have defined **movement types**. In doing so, you specify the stages for which the logistics service provider is responsible as well as which stage types are permitted for a forwarding order or a forwarding quotation.
- If required, you have defined how the system determines **stages in forwarding orders or forwarding quotations**. You have the following options:

- Determination of stages based on a stage profile

In a stage profile, you define which stage types are permitted and which are required in a forwarding order or a forwarding quotation, which sequence is to apply to which stage types, and the properties of the stages. This allows you to define a rule to determine the stages that are relevant for planning, for example.

You can assign the stage profile directly to the forwarding order or forwarding quotation type in Customizing. Alternatively, you can first create a condition and then assign the stage profile to the condition (condition type [Determine Stage Profile](#), /SCMTMS/TRQ_STG_PRF). With the condition, you can configure the settings for determining the stages to allow a flexible combination of movement type, transportation mode, and traffic direction.

- Determination of stages based on the movement type

You define which stage types are permitted and which are required for a specific movement type. As in the stage profile, you also define the sequence for the stage types and specific properties of the stages. However, in contrast to the stage profile, the settings only apply to a specific movement type.

i Note

These Customizing settings are not relevant for stages for empty provisioning items and empty return items. In this case, there are no restrictions as regards the stage type and stage sequence and the system does not automatically enter any stages.

For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) and the following Customizing activities:

- Define Stage Types
- Define Allowed Transportation Mode for Stage Types
- Define Movement Types
- Define Stage Type Sequence for Movement Types
- Define Stage Profiles

In Customizing you have also defined **default agreement party types** for a combination of the stage type and Incoterm. The system uses this information to automatically determine agreement parties for stages. For more information, see Customizing for [Transportation Management](#) under [Forwarding Order Management](#) > [Define Default Agreement Party Types for Stages](#).

If you want to work with **Standard Carrier Alpha Codes (SCACs)** or **Airline Codes** for carriers and airlines, you have defined this in the business partner master data (see [Definition of Business Partners](#)). You can also specify SCACs and airline codes in Customizing for the display in the input help. For more information, see Customizing for [Transportation Management](#) under [Master Data](#) > [Business Partners](#) > [Define Standard Carrier Alpha Codes](#) and [Define IATA Airline Codes](#).

If required, you have defined a **default route** in the master data. For more information, see [Default Route](#).

Features

Ordered Route

In the forwarding order or forwarding quotation, you can enter the stages of the route requested by the ordering party on the tab page or in the assignment block entitled **Ordered Route**. If you have already entered the source and destination locations on the [Locations and Dates/Times](#) tab, this is displayed as a **Complete Carriage**. If you enter a default route on the initial screen of the forwarding order or forwarding quotation, or if you select a default route based on a forwarding agreement item, the system displays the stages from this route.

Otherwise you enter the source and destination locations on the **Complete Carriage** line of your route. The system then automatically transfers this data to the general data on the [Locations and Dates/Times](#) tab. To create additional transportation stages for the ordered route, you have the following options:

- As intermediate stages of the complete carriage

If the route only consists up to now of a source location and a destination location and you would like to add an intermediate location, select the **Complete Carriage** line and choose **Split**. The system inserts two additional lines with stages in the table. The system copies the data for the source location and the dates of the complete carriage for the first transportation stage, and it leaves the destination location data empty. For the second transportation stage, the system copies the data of the destination location and leaves the source location data empty.

You can enter the new source location data and the destination location data in the empty fields. If the complete carriage previously went from location A to location C, you now have, for example, one transportation stage from A to B and another from B to C.

- As additional stages for existing stages

You select the line of an existing stage and choose **Split**. The system divides this line into two lines and proceeds as described above.

Actual Route

Manual Entry of the Actual Route

You can also enter the actual route separately from or in addition to the route requested by the ordering party on the tab page or in the assignment block entitled **Actual Route**. You can also have the system determine the actual route using **Route**  **Transportation Proposal**  or select a default route using **Route**  **Apply Default Route**  and copy it to the table of the stages. If you copy only the route proposed by the system, you can still make changes later. For you to create the actual route, it is not necessary for the items and freight units to already exist in your forwarding order or forwarding quotation.

Depending on your Customizing settings as well as the entries made in the header data of the forwarding order or forwarding quotation, specific data for your stages is already displayed as follows:

- Movement type, traffic direction, and stage profile

You enter the movement type and traffic direction in the header data for the whole business document. You may also have defined a stage profile with the document type. Depending on your settings for the movement type or in the stage profile, certain stages of the actual route are determined automatically. Additionally, the stage types that you can enter may be restricted. Along with the traffic direction (export or import), the movement type also determines the transportation stages for which you, as the logistics service provider or document processor, are responsible. This is displayed with the **Stage Status**.

- Transportation mode

You enter the transportation mode in the header data for the whole business document. However, it only refers to the main carriage (or the main carriage stages) of the route. If you enter the transportation mode manually for a stage or if you have defined it for the stage type in Customizing, the system considers the entries in the following sequence:

1. Manual entry for the stage
2. Customizing settings for the stage type
3. Transportation mode in the header data

- Agreement party and agreement type

The system uses the stage type and the Incoterms entered in the header data to automatically determine the agreement party and the agreement party type, determining who is responsible for settlement.

- SCAC or airline code

If you enter an SCAC and no carrier, the carrier is determined in the subsequent planning processes. If you enter a carrier directly, the system automatically determines the related SCAC and enters the value in the corresponding field of the stage. For more information, see [Carrier Categorization](#).

In Customizing and in the master data, you can define airline codes for air cargo carriers, and you can specify whether the carrier is a member of the International Air Transport Association (IATA). When you create an air forwarding order or quotation, you can enter these codes. The system then automatically determines the related prefix for the master air waybill (MAWB).

If the transportation stage is checked by the planning and execution organization, the system determines the organization interaction status (for example, from the default route). For more information, see [Interaction Between Organizational Units](#).

You can use the **Determine Distance and Duration** pushbutton to have the system determine the transportation distance and duration of a stage. You can also enter the distance and duration manually or change the data determined by the system. For more information, see [Determination of Transportation Distance and Transportation Duration](#).

You can also determine the detailed route data by selecting a schedule and copying it to your actual route data. For more information, see [Direct Creation of Freight Documents and Selection of Schedules](#).

Actual Route Resulting from Transportation Planning

If you have performed transportation planning based on the forwarding order (see [Editing of a Forwarding Order](#)), the actual route is displayed and can no longer be changed.

As an alternative to transportation planning, you can create freight orders and freight bookings directly from the forwarding order for the stages of the route. The freight document ID as well as other related data, such as information about ship connections, are displayed for each stage. For more information, see [Direct Creation of Freight Documents and Selection of Schedules](#).

Copying the Actual Route to the Ordered Route

The ordered route is not updated automatically by the data of the actual route. If you would like this to happen because you want print up-to-date documents, you can choose **Copy from Actual Route** on the **Ordered Route** tab page to copy the data to the route ordered by the ordering party. However, note that the ordered route is no longer visible.

Detail Data

You can enter and display detail data, such as delivery dates and address data, for a specific transportation stage. If the system has already determined the address data from the master data of the location, you can overwrite this data for the complete carriage of the ordered route.

The system displays the detail data of the actual route on three different tab pages:

- **Ordered Data:** Data that is provided by the customer and that was entered in the freight unit
- **Planning/Execution Data:** Data that originates from successor documents, such as freight orders, and cannot be changed
- **Additional TU Dates/Times:** Dates for transportation units

The **Planning/Execution Data** and **Additional TU Dates/Times** tab pages are displayed only if the corresponding data is available (for example, if freight documents have already been created).

The following dates for the transportation stages, among other things, are displayed in the detail data or can be entered there:

- Requested pick-up date and delivery date

The pick-up date and delivery date requested by the ordering party and specified in the ordered route.

- Planned departure and arrival date

The departure and arrival date entered manually in the actual route or by the system after transportation planning. If planning has not yet been performed, the planned departure and arrival date corresponds to the requested date.

If a freight document has not been created and transportation planning has not been performed, you can still change the dates.

- Actual departure and arrival date

The actual departure and arrival date (including the time) entered manually in the freight document or reported by SAP Event Management.

- Confirmed pick-up date and delivery date (valid for complete carriage of the ordered route)

The pick-up date and delivery date confirmed to the ordering party.

- Cut-off dates

Cut-off dates are read from the freight document and indicate the latest time by which goods or specific documents must be available. The document cut-off date, for example, indicates the date by which all freight-relevant documents must be made available to ensure that transportation can be prepared and goods can be loaded on time.

If you have entered a default route that is used to determine the route, the system calculates the departure and arrival dates based on the specified transportation duration. The transportation duration is the total length of time needed to complete a transportation run by a single transportation resource, such as a truck. It includes the length of stay at each location in the run. For more information about the interaction of the default route and the determination of transportation distance and transportation duration, see [Determination of Transportation Distance and Transportation Duration](#).

Stage View and Item View

For the stage table display, you can switch between the stage view and the item view. These views have the following characteristics and advantages:

- Stage view

In this view, the stages are displayed at the top level of a hierarchy and the items belonging to a stage are displayed below. This has the advantage that you can, for example, easily perform the following for a stage:

- Create a new stage for several items
- Edit the data of a stage for several items
- Create or select freight documents for a stage, as well as select schedules

If not all of the freight units contain the same data, the sequence of the stages is determined based on the following criteria:

1. The sequence specified for the movement type in Customizing
2. Item number
3. ID of the freight unit

- Item view

In this view, the items are displayed at the top level of the hierarchy and the related stages are displayed below. If your forwarding order or forwarding quotation contains several freight units, you can see a list of the items with the corresponding freight units. The list contains an entry for each item that is assigned to a freight unit. You can also make changes to an item that apply for all related stages.

Both stage table views are divided into the following areas (only displayed if the corresponding stages exist):

- **Route**: Contains the general stages of the route.
- **BCO**: Contains the stages of the route that relate to the buyer's consolidation process.
- **SCO**: Contains the stages of the route that relate to the shipper's consolidation process.
- **Empty Provisioning and Empty Return**: Contains the stages of the route that relate to the corresponding process.

Determination of Locations and Business Partners at Item Level

For **individual items**, you can also define source and destination locations as well as business partners that differ from the locations in the header data and the complete carriage. For example, you can define that the items that are transported together in the main carriage are picked up from or delivered to different locations.

In certain cases, you can also define different source and destination locations as well as business partners for **subitems**. This is valid for the buyer's consolidation (BCO) and shipper's consolidation (SCO) processes. The following options are available:

- If you enter the value SCO in the **Buyer's or Shipper's Consolidation** field for a forwarding order and have defined Export as the traffic direction, you can define an alternate destination location for a subitem in a container. The system automatically creates the corresponding stage for the subitem in the ordered route. You can then still change the stage manually in the stage table.

When you create freight units for a forwarding order, the system copies the stages for the subitems to the actual route. If you then change the stages in the ordered route, the changes are not copied to the actual route. However, you can change the stages separately in the actual route.

- In the BCO process, an import forwarding order is created automatically from an export forwarding order. For this import forwarding order, the system defines the value BCO in the **Buyer's or Shipper's Consolidation** field and Import as the traffic direction. If the source location of the subitem in the forwarding order differs from the higher-level container item, then the stage for the subitem is automatically created in the ordered route and the actual route. You can no longer change the stage manually in the stage table.

As described above under *Prerequisites*, the system determines the stage types of the automatically created pre-carriage and on-carriage stages from the movement type or stage profile that was assigned to the forwarding order type.

For more information, see [Buyer's Consolidation \(BCO\) in Export/Import](#) and [Shipper's Consolidation \(SCO\) in Export/Import](#).

Copying the Route Data

If you copy a forwarding order or use a forwarding quotation as a template for creating a forwarding order, the route data (both the requested route and the actual route) is also copied. The following data is also copied:

- Source location and destination location
- Stage types
- Transportation mode

All other route data, for example, dates and information about ship connections, are not copied, otherwise they would mean a restriction for later transportation planning. Therefore, you are still able to replan.

Air Freight-Specific Consolidation Processes

Use

You can use Forwarding Order Management in Transportation Management (TM) to consolidate goods from several forwarding orders or house air waybills (HAWBs) in a single air freight booking or a master air waybill (MAWB). A capacity manager at the gateway can for example combine several forwarding orders for an air freight booking (gateway consolidation). You can also create multiple air freight booking for a forwarding order.

However, it can also be necessary to restrict the consolidation options, because for example consolidation is not admissible in a country. Therefore, TM also supports the following air freight-specific processes in which goods cannot be consolidated in the main carriage:

- Back-to-Back Process
- Direct Shipment
- Internal Co-Load

Prerequisites

In Customizing for the forwarding order type or forwarding quotation type, you have optionally defined the air waybill type (AWB type). For more information, see Customizing for [Transportation Management](#) under the following Customizing activities:

- [▶ Forwarding Order Management ▶ Forwarding Order ▶ Define Forwarding Order Type ▶](#)
- [▶ Forwarding Order Management ▶ Forwarding Quotation ▶ Define Forwarding Quotation Type ▶](#)

If you want to define which AWB types are allowed for particular countries, you have assigned AWB types to individual countries in Customizing. For more information, see Customizing for [Transportation Management](#) under [▶ Forwarding Order Management ▶ Define Allowed AWB Types for Country ▶](#).

Features

You control air freight-specific consolidation processes by assigning the AWB type attribute to the forwarding order or forwarding quotation. You can predefine this attribute in Customizing for the forwarding order type or forwarding quotation type; or you can define it directly in the forwarding order or quotation.

This attribute for example controls how many forwarding orders or house bills of lading can be assigned to an air freight booking or master air waybill. This means you can only assign a forwarding order that is defined as **Back-to-Back** or a house bill of lading to an air freight booking or a master air waybill.

The attribute also has the following features:

- You define the air waybill type in a forwarding order or a forwarding quotation. The system transfers this to the air freight booking where it is read-only. This means you cannot modify the AWB type in a booking.
- If you have predefined a value in Customizing, the system displays this value in the business document but you cannot change it there. Otherwise, the field in the business document is empty and you can enter a value.
- If the AWB type for a forwarding order or forwarding quotation is not predefined in Customizing, you can also change this subsequently yourself, directly in the document. This changes the consolidation options accordingly. If for example you change a forwarding order that is defined as **Back-to-Back** to a forwarding order with **Standard Consolidation**, you can assign any number of additional forwarding orders to the air freight booking associated with this forwarding order.
- If you define or change an AWB type in a forwarding order, or enter or change the destination location of a main carriage, the system checks whether the AWB type is allowed for the country in which the destination location is found.
- You can search for (air) forwarding orders, forwarding quotations and air freight bookings with this attribute in your personal worklist (POWL).

More Information

[Back-to-Back Process](#)

[Direct Shipment](#)

[Internal Co-Load](#)

Back-to-Back Process

Use

You use a back-to-back process to negotiate with the ordering party that consolidation cannot be performed for the main carriage of the route. This can have the following reasons:

- Legal reasons (consolidation is not admissible in some countries)
- Special features of the goods involved (such as for dangerous goods)
- Customer request
- Customs declaration

In this case, you can only assign a back-to-back house air waybill (HAWB) to a master air waybill (MAWB). In Transportation Management (TM), you can specify accordingly that an air freight booking can only be assigned to a forwarding order that is defined as a **back-to-back** order.

Prerequisites

You have assigned number ranges for house air waybills to your sales organization and ordering party involved in the TM master data. For more information, see [Waybill Stock Definition](#).

Process

To define and execute a back-to-back process, proceed as follows: This process description only covers the special features of the back-to-back process. For more information about creating forwarding orders and quotations, see [Creation of a Forwarding Order](#) and [Creation and Editing of a Forwarding Quotation](#).

1. You create an (air) forwarding order and define the **Back-to-Back** air waybill type (AWB type). If you have predefined the air waybill type in Customizing for the forwarding order type, the system displays this as the default value.

If you first want to create a quotation for your ordering party, you can also create a forwarding quotation with the AWB type **Back-to-Back**.

2. You assign a house air waybill number to the forwarding order or assign a number from the corresponding number range.
3. Define the stages of the route and create an air freight booking for the main carriage of your route. Alternatively, you can assign an existing air freight booking to the main carriage.
4. The system transfers the data in the forwarding order to the booking. This includes the AWB type **Back-to-Back**. The system also determines business partners in the booking automatically on the basis of the business partner in the forwarding order.
5. You can then add to and change the data in the booking. However, the AWB type is read-only in the booking and can no longer be changed. If you want to subsequently change the AWB type, you should change this in the forwarding order.

Result

- You cannot assign an additional forwarding order to a booking to which a back-to-back forwarding order has already been assigned.
- You can print the house air waybill from the **Output Management** tab. The transportation charges are also included. For more information, see [Printing](#).

Direct Shipment

Use

In a direct shipment process, the transportation process is performed by the shipper and the carrier. Goods can for example be transported to the airport directly by the shipper. This means that a house air waybill (HAWB) is not required and transportation activities are performed on the basis of the master air waybill (MAWB) for the airline. Goods are not consolidated in the main carriage.

In Transportation Management (TM), you can mark a forwarding order for the direct shipment process using the air waybill type. When you create an air freight booking for this forwarding order, the data in the order is transferred automatically to the booking and you cannot assign any other forwarding orders to the booking. The system does not create a house air waybill, but instead creates an air waybill (AWB) that is based on the master air waybill.

There is also a special user interface for the direct shipment process. When you create a forwarding order with the AWB type **Direct Shipment**, a corresponding air freight booking is created automatically for the forwarding order. You can enter the data on a consolidated screen for the forwarding order and freight booking, and start follow-on processes such as calculation of costs for both documents. The following process description applies when you create a forwarding order with the AWB type **Direct Shipment** for the direct shipment process.

Prerequisites

- In Customizing, you have defined a forwarding order type with the following settings:
 - **Same Locations and BPs** checkbox is selected
 - **Automatic Freight Unit Building** checkbox is selected
 - Freight unit building rule (FUB rule) that does not permit freight units to be split

The system supports just one freight unit per direct shipment.
 - Transportation mode **Air** (optional)
 - AWB type **Direct Shipment** (optional)

For more information, see Customizing for **Transportation Management** under ► **Forwarding Order Management** ► **Forwarding Order** ► **Define Forwarding Order Types** ►.

- You have specified in the stage profile or for the movement type that the forwarding order must contain a main carriage stage. For more information, see Customizing for **Transportation Management** under ► **Forwarding Order Management** ► **Define Stage Profiles** ► or under ► **Forwarding Order Management** ► **Define Stage Type Sequence for Movement Types** ►. You have then specified in the Customizing settings for the forwarding order type how the stages are to be determined, that is, using the stage profile or the movement type.
- You have defined a freight booking type in Customizing and indicated in the **Service Definition** area that consolidation is not to take place for the source and destination location of the main carriage. For more information, see Customizing for **Transportation Management** under ► **Freight Order Management** ► **Freight Booking** ► **Define Freight Booking Types** ►.
- In Customizing, you have defined the freight booking type that you created as the default freight document type. You have defined this default freight document type for the forwarding order type that you defined for the direct shipment. You have also selected **Main Carriage** at the stage type. For more information, see Customizing for **Transportation Management** under ► **Forwarding Order Management** ► **Forwarding Order** ► **Define Default Capacity Document Types for Stages** ►.

Process

To define and execute a direct shipment process, proceed as follows: This process description only covers the special features of the direct shipment process. For more information about creating forwarding orders and quotations, see [Creation of a Forwarding Order](#):

1. On the initial screen for the forwarding order, use the input help to select a forwarding order type that is suitable for the direct shipment process. If this has not already been predefined in Customizing, choose the transportation mode [Air](#) and the AWB type [Direct Shipment](#).
2. When you have specified in the Customizing settings for the forwarding order type that the stages are to be determined based on the movement type, you select the relevant movement type using the input help.
3. The system creates an air forwarding order and the corresponding air freight booking. The system determines the freight booking type based on your Customizing settings for the default freight document type.

The forwarding order and freight booking data that you predefined in Customizing are displayed on the user interface for the forwarding order with the AWB type [Direct Shipment](#). The freight booking is also displayed in the stage view of the forwarding order for the main carriage stage.

4. You enter the airport data for the freight booking. You can then save the forwarding order.
5. You specify additional data for the forwarding order and freight booking depending on your requirements. Note the following:
 - The general data as well as the business partner, stage, and item data relate to the forwarding order.
 - The location data for the transportation stages in the section view are determined by the system based on the booking data. You can change this data only on the tab page or in the assignment block [Booking Data](#).
 - The handling codes also relate to the forwarding order and are not copied to the booking.
 - If you copy an existing forwarding order with AWB type [Direct Shipment](#), only the forwarding order data is copied.

You can also copy a forwarding order with the AWB type [Direct Shipment](#) to a forwarding order with AWB type [Normal Consolidation](#) or vice versa. However, freight bookings assigned are not also copied.
6. You can draw an MAWB number from the waybill stock for the creation of an air waybill. If there is a carrier change, you can return the number to stock and specify a new carrier.
7. You can calculate the costs for the forwarding order and freight booking separately or together for both documents. Both the forwarding charges and the freight costs are displayed.
8. If you want to use additional input and editing options, you can open the forwarding order or the air freight booking separately from the user interface for the forwarding order with the AWB type [Direct Shipment](#).

Result

- You cannot assign any other forwarding orders to the air freight booking that is indicated as [Direct Shipment](#).
- If you cancel a forwarding order that is indicated as a [Direct Shipment](#), the system also automatically cancels the associated booking. The MAWB number is returned to the waybill stock automatically.
- You can search for forwarding orders with the AWB type [Direct Shipment](#) in your personal object worklist.

Internal Co-Load

Use

In this process, you can make cargo capacity that you have already booked available to another organizational unit in your company. This business unit can also be a legally independent unit.

As a consolidating business unit (consolidator), you are responsible for the freight booking and the master air waybill (MAWB). You can calculate the internal freight costs and perform an internal settlement with the other business unit (coloader). As a coloader,

you conduct the negotiations with the customer and are responsible for executing the forwarding order to the customer and for the house air waybill (HAWB). You also perform the settlement with the customer.

Prerequisites

- You have created a suitable organizational structure in Customizing. You must assign your organizational unit to a company organization so that the system can, for example, determine whether your sales and purchasing organizations belong to the same company code.

For more information, see Customizing for [Transportation Management](#) under [►►Master Data► Organizational Management► Organizational Model► Create Organizational Model](#).

- You have configured the necessary settings for the external and internal charge calculation and settlement. In particular, you have activated the internal charge calculation in Customizing for the forwarding order type and the charge calculation in Customizing for the freight booking type. For more information, see Customizing for [Transportation Management](#) under [►►Forwarding Order Management► Forwarding Order► Define Forwarding Order Types](#) and under [►►Freight Order Management► Freight Booking► Define Freight Booking Types](#). For further information, see also [Charge Management and Service Product Catalogs](#).
- There is an internal agreement between the two organizational units between which you want to perform the co-load process. There is also an external agreement with the customer of the coloader. For more information, see [Agreement](#).

Process

- As a **consolidator**, you create an air freight booking and therefore reserve capacity for a particular flight. You define the consolidating business unit as a purchasing organization. For more information, see [Freight Booking](#).
- As a **coloader**, you create a forwarding order. You define the co-loading business unit as a sales organization.
- You define the stages of the route in the forwarding order. For the main carriage, you specify the source and destination locations as well as the pick-up dates/times and delivery dates/times according to the freight booking already created whose capacity you want to use. You also define the consolidating business unit as a planning and execution organization.
- You assign the freight booking whose capacity you want to use to the main carriage of the forwarding order.
- From the forwarding order, you calculate the charges that the customer must pay to the co-loading business unit based on the agreement reached between the two parties. For more information, see [Charge Calculation](#).
- You calculate the internal charges that the co-loading business unit must pay to the consolidating business unit.
- You create a settlement document for the settlement between the coloader and their customer from the forwarding order. For more information see [Forwarding Settlement](#).
- As a **consolidator**, you create an internal settlement document for the settlement with the coloader from the overview of the settlement documents. For more information, see [Internal Settlement Management](#).

Interaction Between Organizational Units

Use

You can use this function to model how organizational units interact with each other, for example, how a sales organization interacts with a planning and execution organization. This interaction is especially important if you have to transport certain goods in a specific transportation stage, for example, dangerous goods.

Features

The sales organization is responsible for entering the order, for example, but is not allowed to create freight bookings for specific transportation stages or send them to a carrier. It can therefore propose how to transport the goods (by specifying a route and a schedule, and assigning a freight order or a freight booking) and sets the organization interaction status to transfer the affected stages to the planning and execution organization to be checked. The planning and execution organization checks the proposal and the transportation stage details in the transportation cockpit. It can then confirm the data exactly as proposed in the forwarding order, change data such as the departure and then confirm the proposal, or reject the proposal outright. The status for the stage in the forwarding order then changes to **Confirmed**, **Confirmed with Deviations**, or **Rejected** respectively.

i Note

The functions for confirming or rejecting proposals are not displayed as standard, and so the planning and execution organization must first select an appropriate layout set in the transportation cockpit. For more information, see [Use of Profile and Layout Sets](#).

The planning and execution organization can also create a freight booking or a freight order for the transportation stage in the transportation cockpit, or assign the transportation stage to a document that already exists. In this case, the system sets the organization interaction status automatically from **To Be Checked** to **Confirmed**. The sales organization can then confirm the transportation stages with the sold-to party.

Furthermore, the planning and execution organization can select an existing freight booking in addition to the route and schedule and confirm the details for the carrier.

Even if a stage has already been confirmed, the sales organization can still make changes to the order. In this case, the system sets the status of the stage to **Changed After Confirmation**.

If the planning and execution organization rejects a stage, the sales organization can submit a new proposal by changing the status of the stage back to **To Be Checked**.

i Note

If the planning and execution organization rejects unconfirmed stages that are assigned to a freight booking via an auto-confirmation profile these unconfirmed stages are automatically unplanned.

If the sales organization wants to cancel a stage, this must also be approved by the planning and execution organization. The sales organization therefore sets the status of the relevant stage in the forwarding order to **Cancellation Requested**. In the transportation cockpit, the planning and execution organization chooses **Confirm** for the stage and the status in the forwarding order subsequently changes to **Cancellation Approved**.

The sales organization can also remove the default route from a stage in the forwarding order by selecting the stage and choosing **►Schedule ► Unassign ►**.

If the sales organization wants to delete the stages of the default route from the actual route, it can do so by selecting one of the relevant stages and choosing **Delete Stage**. The system then replaces the stages containing the default route by the original stages.

Automatically Setting the Organization Interaction Status

The sales organization can set the organization interaction status manually from **Confirmed** or **Rejected** to **To Be Checked**, **Not Yet Requested**, or **Cancellation Requested**.

You define criteria for automatic confirmation in a profile, which the system determines based on the freight booking type. These criteria can include, for example, the security status, volume, or weight. If a transportation stage meets the criteria defined in the profile, the stage can be confirmed automatically (for example, if the volume and weight defined in the profile are less than the volume and weight defined in the transportation stage).

You can use the Business Add-In **BAdI: Change Organization Interaction Status** (/SCMTMS/TOR_ORG_INT_STATUS) to control the following cases:

- A transportation stage can be confirmed automatically (that is, the organization interaction status is set automatically from **To Be Checked** to **Confirmed**), such as when the forwarding order does not contain any dangerous goods.
- A confirmed transportation stage must be rechecked (that is, the organization interaction status is set automatically from **Confirmed** or **Rejected** to **To Be Checked**), for example, when quantities are changed.

Aggregated Organization Interaction Status

The **Statuses** tab page of the forwarding order shows the aggregated organization interaction status of the forwarding order. This is an overall status that is based on the statuses of the individual stages.

If all of the stages have the same status, this status is shown as the aggregated organization interaction status (unless all stages have the status **Confirmed with Deviations**, in which case the aggregated organization interaction status is set to **Confirmed**). If all of the stages have a status that is either **To Be Checked** or **Changed After Confirmation**, the aggregated organization interaction status is set to **Changed After Confirmation**.

In cases where the status of one stage differs from all of the others, the system sets the aggregated organization interaction status based on the following rules, which it applies in the sequence specified:

Sequence	Status of at Least One Stage	Aggregated Status
1	Cancellation Approved	Cancellation Partially Approved
2	Cancellation Requested	Cancellation Partially Requested
3	Rejected	Partially Rejected
4	Confirmed	Partially Confirmed
5	Confirmed with Deviations	Partially Confirmed
6	Changed After Confirmation	Changed After Confirmation
7	To Be Checked	To Be Checked
8	Not Yet Requested	Not Yet Requested

More Information

[Auto-Confirmation Profile](#)

[Coordination Between Customer Service Agent and Gateway Agent](#)

Auto-Confirmation Profile

Definition

A user-specific grouping of parameters that are taken into account in the interaction between organizational units (in particular, between customer service agents and gateway). The profile is only relevant for freight units for which interaction between organizational units is mandatory. You define whether interaction between organizational units is mandatory in the default route or in Customizing under **► Transportation Management ► Freight Order Management ► Define Freight Booking Types ►**.

The settings that you configure in the auto-confirmation profile are checked by the system when the freight booking is assigned to the freight unit stage. If all criteria are fulfilled, the assignment of the freight booking is confirmed automatically.

To define auto-confirmation profiles, open the [Create Auto-Confirmation Profile](#) app on the launchpad under **Profiles and Settings**.

Structure

In the auto-confirmation profile, you define the following, for example:

- General data

Here, you specify a technical name and description for the profile that you want to create. In addition, you can configure whether automatic confirmation for this profile is to be activated and can choose a transportation mode. If automatic confirmation is activated, the criteria for capacity and utilization limits and excluded security statuses must be met so that the freight unit stages are confirmed automatically.

- Capacity limits

Here you define hard capacity limits that the system takes into account for automatic confirmation. You can define a maximum gross weight and a maximum gross volume for the freight unit. You can also define a maximum weight capacity and a maximum volume capacity for the freight booking as percentage values. The utilization of the freight booking after the freight units have been assigned must not exceed the limits defined in the profile.

- Excluded security statuses

Freight units for which at least one of the excluded security statuses is specified are not confirmed automatically. The security status is only relevant for the **Air** transportation mode.

If all parameters are observed, the assignment of the freight unit stage to the freight booking is confirmed automatically.

Administrative data is also displayed.

Coordination Between Customer Service and Gateway Agent

Use

If you are a customer service agent and receive a forwarding order, you can assign it to a freight booking. A gateway agent can then check this assignment (in other words, confirm or reject it).

Prerequisites

- In the [Create Auto-Confirmation Profile](#) app from the launchpad, you have defined auto-confirmation profiles.
- You have defined rejection reason codes. For more information, see Customizing for [Transportation Management](#) under **Basic Functions** > **General Settings** > **Define Rejection Reason Codes**.
- You have defined freight booking types. For more information, see Customizing for [Transportation Management](#) under **Freight Order Management** > **Freight Booking** > **Define Freight Booking Types**.

Process

1. Creating the forwarding order

As a customer service agent, you create a forwarding order and enter information about the business partners and locations involved and the items to be transported.

2. Creating the transportation stages (optional)

You create the transportation stages and select a predefined schedule.

3. Defining the planning and execution organization

You define the planning and execution organizations that are responsible for the transportation stages on the actual route. You also have the option of assigning a transportation stage to a freight booking.

4. Starting the organization interaction process automatically

- If you have executed step 2, you proceed as follows: If the organization interaction is activated for some stages on the actual route and you have finished working on the stages, you change the organization interaction status from **Not Yet Requested** to **To Be Checked**.
- If you have **not** executed step 2, you proceed as follows: If you have assigned a stage to a booking for which the organization interaction is activated, the system automatically sets the status to **To Be Checked**.

5. Starting the organization interaction process manually

You can also start the organization interaction by manually setting the status to **To Be Checked**.

6. Changing the organization interaction status to **Confirmed**

If a booking is assigned to a stage for which an auto-confirmation profile exists, the system checks the rules that are defined in this profile. If the rules defined in the profile have been fulfilled, the system automatically sets the organization interaction status to **Confirmed**.

7. Gateway Agent Check

If the freight unit stages are not confirmed automatically, the gateway agent of the planning and execution organization must do this. As a gateway agent, if you accept the proposals provided by the customer service agent without changes, the system sets the status to **Confirmed**. If you change the data, for example, dates/times or quantities, the system sets the status to **Confirmed with Deviations**.

If the customer service agent has not made an assignment to a freight booking, you, as the gateway agent, must do this.

If you reject the proposals provided by the customer service agent, the system sets the status to **Rejected**. You can enter a reason for the rejection.

8. Monitoring of the organization interaction status by the customer service agent

As a customer service agent, you can check whether the gateway agent has accepted or rejected your proposals and can clarify any necessary changes with your customer.

More Information

[Auto-Confirmation Profile](#)

[Interaction Between Organizational Units](#)

Building and Printing of House Bills of Lading and House Air Waybills

Use

You can use this function to create and print house bills of lading (HBLs) and house air waybills (HAWBs) for the following business documents:

Document	Supported Business Documents
House bill of lading	Freight orders Ocean freight bookings Freight units
House air waybill	Air forwarding orders Air freight bookings Freight units

Prerequisites

Building of House Bills of Lading and House Air Waybills

- In Customizing, you have specified a **process controller strategy** for your forwarding order type, freight order type or freight booking type for the building of house bills of lading and house air waybills. For more information, see Customizing for [Transportation Management](#) under:
 - [Forwarding Order Management](#) ► [Forwarding Order](#) ► [Define Forwarding Order Types](#) ►
[Freight Order Management](#) [Freight Order](#) [Define Freight Order Types](#)
 - [Freight Order Management](#) [Freight Booking](#) [Define Freight Booking Types](#)

Which strategy is used depends on the business document from which you build the house bill of lading or house air waybill. If you do not define a strategy in Customizing, the system uses the relevant standard strategy for forwarding orders or freight documents (see below).

- You have defined a **waybill stock**. For more information, see [Waybill Stock Definition](#).
- You have activated the **use of waybill stocks** in forwarding order type Customizing (see below). If you deactivate this function, you can only enter waybill numbers in the forwarding order manually, and the system does not check waybill numbers entered against the waybill stock.

Printing of House Bills of Lading and House Air Waybills

- You have made the **general settings** for printing documents. For more information, see [Output Management](#) and [Printing](#).
- In Customizing for the freight unit type, which you use for freight unit building, you have defined the **output profile** /SCMTMS/TOR_PRINT_AIR_FU. If you do not specify a freight unit building rule in Customizing for your forwarding order type, and consequently do not define a special freight unit type, the system uses the standard freight unit type. For more information, see Customizing for [Transportation Management](#) under ► [Planning](#) ► [Freight Unit](#) ► [Define Freight Unit Types](#) ►.
- You have made the basic settings for the **charge calculation** that you need for your business process. For more information, see [Charge Management and Service Product Catalogs](#). You have made the following particular settings for printing the house bills of lading and house air waybills:
 - In Customizing you have defined and appropriately classified **charge types** for weight-dependent charges, for other charges and for valuation charges. This classification is necessary as the charge types appearing in the printing of a house bill of lading or house air waybill are detailed separately. Note that the charge types classified as **other**

charges must contain the two digit code for IATA other charges and also entries for the charge due (carrier or agent).

For more information, see Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Charge Types**, and also under [Charge Calculation](#).

- o You have created a **calculation profile** in Customizing. You should note in this connection that the system for printing a house bill of lading or a house air waybill supports both the calculation of charges at stage level and the calculation at item level. If you use an Incoterm in the forwarding order, the system switches to a calculation of charges at stage level irrespective of the setting in the calculation profile.

For more information, see Customizing for **Transportation Management** under **Basic Functions > Charge Calculation > Basic Settings > Define Calculation Profiles**, and also under [Calculation Profiles](#).

- o You have created a **calculation sheet**. To do this, open the **Create Calculation Sheet** app on the launchpad under **Charge Management**. If you want to print house bills of lading or house air waybills, you must have entered one of the following values as the calculation resolution base for weight-dependent charge items:
 - **PRODUCT** (product)
 - **MAIN_ITEM** (main item of the document)
 - **PACKAGE** (package)
 - **CONTAINER** (container)

The weight charges for products, packages or containers are listed depending on the calculation resolution basis that you have defined. The calculation resolution basis should be the same for all charge items, as otherwise you may experience a mixture of package and product items, and the totals formation is rendered incorrect.

For more information, see [Calculation Sheet](#).

Features

Building of House Bills of Lading and House Air Waybills

The system automatically groups all items that do not already belong to a fixed house bill of lading or house air waybill in accordance with the process controller strategy you chose.

SAP provides the following strategies as part of the standard delivery.

- **Build House Bill of Lading or House Air Waybill by Shipper/Consignee** (HBL_SHPCNS)

With this strategy, the system groups together all items that have the same shipper and consignee.

This is the standard strategy that the system uses for freight documents if you do not specify a strategy in Customizing of the freight order type or the freight booking type. The system also uses this strategy if you build house bills of lading or house air waybills for freight units or items.

- **Build House Bill of Lading or House Air Waybill by Forwarding Order** (HBL_TRQID)

With this strategy, the system groups together all items that belong to the same forwarding order.

This is the standard strategy that the system uses for forwarding orders if you do not specify a strategy in forwarding order type Customizing.

- **Build House Bill of Lading or House Air Waybill by Shipper/Consignee/Container** (HBL_SCCONT)

With this strategy, the system groups together all items that have the same shipper and consignee, and are to be transported in the same container.

- [Build House Bill of Lading or House Air Waybill by Freight Documents \(HBL_FDOC\)](#)

With this strategy, the system groups together all items that have the same freight documents in the main carriage.

- [Build House Bill of Lading or House Air Waybill by Shipper/Consignee with Address \(HBL_SHPCNA\)](#)

With this strategy, the system groups together all items in which the shipper's and consignee's addresses are identical.

Printing of House Bills of Lading and House Air Waybills

You can print or dispatch the house bills of lading or house air waybills you have built from the [Output Management](#) tab. For more information, see [Output Management](#).

The printout also contains information about the transport charges. You can define in an air forwarding order on the [Charges](#) tab that the transportation charges as agreed or the other charges and their totals are to be printed. If the transportation charges **as agreed** are printed, then all weight-dependent charges, charges, valuation charges and rates as agreed between the business partners are printed (meaning not their currency amounts).

Activities

You create house bills of lading or house air waybills as follows:

- Forwarding orders

You can create house air waybills by entering the waybill number manually or with [HAWB > Draw HAWB Number](#) from the waybill stock. You can only draw a house air waybill (HAWB) number from the waybill stock if this involves an internal number range and you have activated waybill stocks in Customizing (see under *Prerequisites*).

You can display data for bills of lading on the [HBL or HAWB](#) tab, and make individual allocations of waybill numbers to freight units or items or remove the allocations again.

- Freight orders

You create house bills of lading by choosing [Follow-up Actions > Build House Bill of Lading](#). You display house bills of lading on the [Items](#) tab page by choosing [Bill of Lading](#) as the hierarchy type.

i Note

The [Items](#) tab page replaces the [Cargo](#) tab page. You can personalize your user interface in such a way that the system displays the old [Cargo](#) tab page.

- Ocean freight bookings

You create house bills of lading by choosing [Follow-up Actions > Build House Bill of Lading](#). You display house bills of lading on the [Items](#) tab page by choosing [Bill of Lading](#) as the hierarchy type.

i Note

The [Items](#) tab page replaces the [Cargo Management](#) tab page. You can personalize your user interface in such a way that the system displays the old [Cargo Management](#) tab page.

- Air freight bookings

You create house air waybills by choosing [Follow-up Actions > Build House Bill of Lading](#). You display house air waybills on the [Capacity and Cargo](#) tab page by choosing [Air Waybill](#) as the hierarchy type.

- Freight units

You create house bills of lading or house air waybills by choosing ► **Follow-up Actions** ► **Build HBL or HAWB** ►. The information for the house bill of lading or house air waybill is displayed at header level. A house bill of lading or house air waybill can have one or more freight units.

For more information about hierarchy types, see [Use of Hierarchical Views in FOM](#).

More Information

[Process Controller Configuration](#)

Nature of Goods

Use

The nature of goods is a short and meaningful description of the cargo to be transported. Although you can enter the nature of goods only in business documents that are relevant for air transportation, the information is used along the entire transportation chain. The business documents (for example, air forwarding orders and air bookings) and the corresponding print documents (that is, air waybills and manifests) can contain the same nature of goods information or different information, according to your requirements.

You enter the nature of goods as free text within the allowed space on the UI, which reflects the requirements for the number of lines and characters. You can also define standard texts for nature of goods information that you use on a regular basis.

In Transportation Management (TM), there are two types of nature of goods:

- Nature and quantity of goods

This information can be entered in air forwarding orders and air bookings. You enter it on header level on the **Nature of Goods** tab page in the air forwarding order and on the **Operations** tab page in the air booking. You can enter 12 lines with a maximum of 20 characters per line.

The nature and quantity of goods specified on the **Nature of Goods** tab page in the air forwarding order or on the **Operations** tab page in the air booking are synchronized with the **Air Waybill View** on the **Charges** tab page. When you modify the nature and quantity of goods information on one of the UIs, the system automatically updates the other UI (for example, if you update the nature and quantity of goods on the **Air Waybill View** in the freight booking, the system automatically updates the nature and quantity of goods on the **Operations** tab page in the freight booking). The system not only updates the UIs, but also centrally updates the nature of goods data in the backend system for the relevant business document.

The nature and quantity of goods is printed in the house air waybill (HAWB), master air waybill (MAWB), and air waybill for a direct shipment. It is used to provide additional information about the weight-based charge items in air waybills.

- Nature of goods for manifest

This information can be entered in air forwarding orders, freight units, air bookings, and freight orders that cover pre-carriage or on-carriage for the air booking. You can enter 9 lines with a maximum of 65 characters per line. In the air forwarding order, you enter this information on header level on the **Nature of Goods** tab page. In the air booking, you enter it on item level on the **Capacity and Cargo** tab page, and in the freight order, on item level on the **Items** tab page (**Generic Item View** hierarchy type).

The nature of goods for manifest is printed in the security manifest and cargo manifests, which you print from the air booking.

Note that you can also enter the nature of goods on the **Notes** tab page of the air forwarding order or air booking. However, we recommend that you use the dedicated screen areas for nature of goods (for example, the **Nature of Goods** tab page in the air forwarding order).

Prerequisites

To use standard texts, you have done the following:

- Assigned the text schema FWOAIRHDR to the relevant forwarding order types (**Text Schema** field)
- Assigned the text schema CBAIR to the relevant booking types (**Text Schema** field)
- Defined standard texts in transaction S010 using text type ST

For information about defining your own text types for the nature of goods, see SAP Note [2006789](#). This SAP Note also contains information about enhancing the pushbuttons for copying the nature of goods in air forwarding orders and air bookings.

Process

1. You enter the nature and quantity of goods as well as the nature of goods for manifest on the **Nature of Goods** tab page in your air forwarding order.

You can copy the nature of goods for manifest from the nature and quantity of goods. When you do this, the system concatenates the text, but you can then adjust it as required.

The nature and quantity of goods is printed in the house air waybill (HAWB), which you print from the air forwarding order. After you have calculated the charges, you can use the **Air Waybill View** on the **Charges** tab page to align the nature and quantity of goods text so that it appears next to the correct weight-based charge items in the HAWB printout.

The nature of goods for manifest is not used to print documents from the air forwarding order, but is transferred automatically to the follow-on documents.

2. You create freight units (FUs).

The nature of goods for manifest from the air forwarding order is displayed in the corresponding freight units. If required, you can edit the text to create FU-specific information.

3. You create an air booking and you create a freight order for pre-carriage and on-carriage.

The nature of goods information is created as follows:

- Nature and quantity of goods (air booking only)

You enter this information on the **Operations** tab page. If required, you can copy it from the air forwarding order to the air booking (**Copy from Requirement Documents**). If more than one air forwarding order is assigned to the air booking, the system copies the first 2 lines of text from the first 6 air forwarding orders. The text on the **Operations** tab page is then also displayed on the **Charges** tab page.

- Nature of goods for manifest (air booking and freight order)

This information is copied automatically to the air booking and freight order from the freight unit. It is displayed in the details area of a freight unit that you have selected on the **Capacity and Cargo** tab page in the air booking or on the **Items** tab page in the freight order (**Generic Item View** hierarchy type).

For both types of nature of goods, you can change the copied text. For example, you can replace the text with something that is specific to a certain stage, country or region, or you can write the text in another language. You can also insert a standard text that you have defined beforehand.

4. You print the air waybill (master air waybill or air waybill for a direct shipment) or manifest (security manifest or cargo manifest) from your air booking.

Similar to the air forwarding order, you can use the [Air Waybill View](#) on the [Charges](#) tab page to align the nature and quantity of goods text so that it appears next to the correct weight-based charge items in the air waybill printout.

Note that when you print an air waybill or manifest from the air booking, the system uses the following logic to derive the nature of goods:

- a. It checks the air booking.
- b. If the air booking does not contain any nature of goods information, it checks the freight units.
- c. If the freight units do not contain any nature of goods information, it checks the air forwarding order.

The HAWB always contains the nature and quantity of goods from the air forwarding order.

i Note

- You can specify the nature and quantity of goods only for weight-based charge lines under charge items with the transportation mode [Air](#).
- If there are two stages with the transportation mode [Air](#), then there can be two charge lines in the air waybill view for the same logistical item. In order to avoid printing the duplicate charge line in the master air waybill or house air waybill, you must mark one of these charge lines as inactive in the [Grouped View](#) or [Ungrouped View](#) and adjust the amount in the other charge line manually, so that the overall charges remain the same.
- The order of charge lines in the [Grouped View](#) or [Ungrouped View](#) and the [Air Waybill View](#) are not the same. In the [Air Waybill View](#), the charge lines are sorted by the position numbers and the nature of goods lines remain as defined in the data of the air forwarding order or air booking. In the [Grouped View](#) or [Ungrouped View](#), charge lines are sorted based on the charge line numbers and the nature of goods lines align with the air waybill view.

More Information

[Printing](#)

Empty Provisioning and Empty Return

Use

As a logistics service provider, you may be requested by your ordering party to provide the requisite empty containers or railcars (empties) in addition to transporting goods. This means that you may be requested to provide more or fewer empties than you actually needed to transport the goods. The ordering party can also request that you provide empties independent of a transportation.

Transportation Management (TM) supports these different requests. The system also supports the return of empties as well as different variants of these business processes. For example, you can specify that the ordering party organizes the pick-up and delivery of the empties themselves.

The following describes how you can map your requests for the provisioning and return of empties in forwarding order management.

i Note

Empty provisioning and return is supported only for container units and railcar units. It is not supported for trailer units and package units.

Process

Provisioning and Return of Empties with Reference to a Goods Transportation

If your ordering party has requested that you also provide the **required number** of empties in addition to transporting goods, you can map this as follows in forwarding order management from TM:

1. Create a forwarding order with the corresponding container items or railcar items for the transportation of goods. For more information about creating forwarding orders, see [Creation of a Forwarding Order](#).

You can also request empty provisioning and return in forwarding quotations. If you create a forwarding order with reference to a quotation, the system copies the entries for empty provisioning and return to the forwarding order. For more information about creating forwarding quotations, see [Creation of a Forwarding Quotation](#).

2. In the **Empty Provisioning** field, select the option **Requested (X)** for your cargo item. This indicates that your ordering party requests the provisioning of the same number of empties as you have defined as the item quantity.
3. If your ordering party also requests the return of the provided empties, select **Requested (X)** in the **Empty Return** field. Otherwise, select the default value **Not Requested (empty)**.
4. The system displays information about the provisioning and return of empties in the item details on the **Details of Empty Provisioning** and **Details of Empty Return** tab pages.

The system enters automatically the source location of the item as the destination location of the provisioning information and the destination location of the item as the source location of the return information. You can also enter a source location for the provisioning of empties and a destination location for their return. You can even leave this information open in the forwarding order and add it at a later date. Furthermore, you can change the pick-up and delivery dates for the provisioning and return of empties to meet your individual requirements.

Information about empty provisioning and empty return is also displayed in the stage tables of the ordered route and the actual route.

5. The system creates separate freight units and transportation units (container units and railcar units) for the transportation of goods and the provisioning or return of empties.

Note that you can search for forwarding orders and forwarding quotations in your personal object worklist (POWL) using empty provisioning and empty return information.

Provisioning and Return of Empties Without Reference to a Goods Transportation

If your ordering party has requested that you provide or return **empties only** without transporting goods, you can map this as follows. You can also use this to map scenarios in which your ordering party requests that you provide or return more or fewer empties than are actually needed for the transportation of goods:

1. Create a main item for the provisioning or return of empties. Create a container item or railcar item for this main item and enter the required quantity.
2. Select the option **Provisioning Only** or **Return Only** in the **Empty Provisioning** or **Empty Return** field of this item.
3. The system displays information about the provisioning and return of empties in the item details on the **Details of Empty Provisioning** and **Details of Empty Return** tab pages. Here you can enter the source and destination locations for the provisioning or return of empties as well as the corresponding pick-up and delivery dates.

Information is also displayed in the stage tables of the ordered route and the actual route.

4. The system creates transportation units for the provisioning and return of empties.

Pick-Up and Delivery of Empties by Ordering Party

Your ordering party may want to organize the pick-up and delivery of empties themselves. This means that as the logistics service provider you are only responsible for providing the empties. You are not responsible for their pick-up or delivery. You specify this as follows:

1. Specify the provisioning or return of empties for cargo items or separately as described above.
2. In the detailed data about the provisioning or return of empties select the **Organized by Ordering Party** checkbox.
3. The system marks the corresponding stages of the pick-up and delivery route for empties as not relevant for planning (planning block).

More Information

[Empty Provisioning and Return in Export/Import Processing](#)

[Triangulation](#)

[Calculation of Charges for Empty Provisioning and Empty Return](#)

Empty Provisioning and Return in Export/Import Processing

Use

Empty provisioning and return can be planned and executed as part of export/import processing. In this case, the following applies:

- The export organization is responsible for planning and executing empty provisioning (because the empty containers or railcars are required before transportation starts), but the import organization may be responsible for the transportation charges.
- The import organization is responsible for planning and executing empty returns (because the empty containers or railcars are returned after transportation has taken place), but the export organization may be responsible for the transportation charges.

Empty provisioning and return can be requested in export forwarding orders and quotations, and import forwarding orders and quotations. The following are example processes for export/import processing with empty provisioning and return.

Process

Empty provisioning and return **with reference to the transportation of goods** can be handled as follows in export/import processing:

1. The export organization creates the export forwarding order and requests empty provisioning and empty returns for the cargo items.
2. The system does the following:
 - Creates container units or railcar units for the empty provisioning items so that the export organization can plan and execute empty provisioning

- Copies information about the requested empty returns to the item details of the main cargo item

If required, the export organization can then calculate the charges for the empty return items.

3. The system creates the import forwarding order. It does the following for empty provisioning and empty returns:

- Copies information about the requested empty provisioning to the item details of the main cargo item

If required, the import organization can then calculate the charges for the empty provisioning items.

- Creates container units or railcar units for the empty return items so that the import organization can plan and execute empty returns

For empty provisioning and return **without reference to the transportation of goods**, the following applies:

- Since the export organization is responsible for planning and executing empty provisioning, you can request empty provisioning without reference to a main cargo item only in the export forwarding order.
- Since the import organization is responsible for planning and executing empty returns, you can request empty returns without reference to a main cargo item only in the import forwarding order.

More Information

[Export/Import Processing](#)

Triangulation

Use

As a carrier, you can combine a request for empty provisioning and a request for empty return to save transportation charges.

❖ Example

You have to organize the pickup of an empty container that is no longer required from a consignee and the subsequent transportation of this empty container to the container yard. At the same time, you have to organize the pickup of a second empty container of the same type from the container yard and the provisioning of this empty container to the customer. Instead of having the system execute these two orders separately, you can combine them into one order. This means that you have the first empty container directly transported to the customer.

i Note

Triangulation is supported for container units and railcar units only. For the sake of simplicity, the term transportation units is used below even though this term includes trailer units. Triangulation is not supported for trailer units.

Prerequisites

- You have created at least two forwarding orders. For one of these forwarding orders you have selected the option **Requested** in the **Empty Provisioning** field for the corresponding cargo item. For the other forwarding order you have selected the option **Requested** in the **Empty Return** field for the corresponding cargo item.
- There are at least two transportation units: one for the empty provisioning and one for the empty return (movement category **Empty Return** and **Empty Provisioning**). The two transportation units must have the same equipment type. The transportation unit for the empty provisioning must be unplanned.

- The requested return of the empty transportation unit must be before the provisioning of the empty transportation unit.

Features

You can combine a transportation unit for the empty provisioning (empty provisioning TU) and a transportation unit for the empty return (empty return TU) in a single transportation unit (triangulation transportation unit). From a technical point of view, the system keeps the empty return TU and deletes the empty provisioning TU. The new triangulation transportation unit (triangulation TU) now refers to two forwarding orders and is assigned the movement category **Triangulation**. As with any other transportation unit you can plan this transportation unit on a freight order. When you revoke triangulation, the system dissolves the triangulation TU into the two original transportation units.

Triangulation is supported for unplanned transportation units only. If the empty return TU was already planned before triangulation, the triangulation TU is also planned. The system updates the locations only.

If you change one of the assigned forwarding orders, the system tries to keep the triangulation TU. However, if you make critical changes (by changing the equipment type or canceling the provisioning, for example) the system may have to dissolve the triangulation TU into the two original transportation units.

To enable you to recognize whether an empty provisioning or empty return item belongs to a triangulated transportation unit, the text (**Triangulation**) is added to the item text that is displayed in the actual route on the **Stages** tab page of the forwarding order.

Activities

To perform triangulation, select a container unit or a railcar unit in the personal object worklist for container units or railcar units. To do this, open the **Railcar Units** or **Container Units** app on the launchpad under **Planning**.

Then choose ► **Triangulation** ► **Create Triangulation** ►. If you have selected the option for a container unit or railcar unit for the provisioning, the system automatically proposes only container units or railcar units for the return, and vice versa.

Alternatively, you can directly select two container units or railcar units (one for the provisioning and one for the return) and then choose ► **Triangulation** ► **Create Triangulation** ►.

To revoke triangulation, choose **Revoke Triangulation**.

The options to create and revoke a triangulation are available on the user interfaces where you create and change container units and railcar units as well as in the transportation cockpit.

You can search for triangulation TUs in your personal object worklist using the movement category. In the selection criteria to start the transportation cockpit, you can also search for triangulation TUs using the movement category. Note that TUs are blocked for planning if you have not entered a source location or destination location. In this case, you must select the value **Include in Selection** in the selection criteria under **Documents Blocked for Planning** to ensure that these triangulation TUs are found.

More Information

[Empty Provisioning and Empty Return](#)

Processing of Changes and Deletion of Forwarding Orders

Use

The forwarding order data received from your ordering party can change during the process. The change controller in Transportation Management (TM) checks whether forwarding orders or forwarding quotations have changed and propagates the changes to the freight unit. You can specify how the change controller is to carry out the changes. The following options are available:

- Synchronous
- Asynchronous
- Synchronous with fallback to asynchronous

Prerequisites

- You have defined the way the system propagates the changes to successor documents in Customizing for **Transportation Management** under:
 - ▮ **Forwarding Order Management** ▸ **Forwarding Order** ▸ **Define Forwarding Order Types** ▮
 - ▮ **Forwarding Order Management** ▸ **Forwarding Quotation** ▸ **Define Forwarding Quotation Types** ▮
- You have implemented the batch report **Background Job to Process Triggers Periodically** (/SCMTMS/PROCESS_TRIGGER_BGD), in case you process the changes asynchronous.
- You have created the necessary change controller strategies and change controller conditions in Customizing for **Transportation Management** under ▮ **Basic Functions** ▸ **Process Controller** ▸ **Define Strategy** ▮.
- You have specified whether freight units can be deleted or canceled in Customizing for **Transportation Management** under ▮ **Planning** ▸ **Freight Unit** ▸ **Define Freight Unit Types** ▮.

Features

Changed Forwarding Order or Forwarding Quotation Data

You can still change most of the data in a forwarding order or quotation during the transportation process, such as dates, quantities, and locations, as well as add or delete items. This is also the case if freight units have already been built and planning has been completed or execution has started. The system updates the freight units when saving the data. Follow-on documents such as freight orders and freight bookings are also adjusted.

However, if data has already been transferred to the freight unit from the execution process and the status of the freight unit has been changed to **On Hand**, for example, the system no longer adjusts the follow-on documents automatically. In this case, the system prompts you to change the data manually.

If a forwarding settlement document or an internal settlement document already exists and you have activated the forwarding settlement change process, you can enter the changes directly in the forwarding order (for example, changes to quantities, to the shipper, or to the consignee). The change process triggers the necessary changes in the forwarding settlement process. If a forwarding settlement document or an internal settlement document already exists and you have not activated the forwarding settlement change process, you have to cancel the forwarding settlement document. You then enter the changes directly in the forwarding order and create a new forwarding settlement document that contains the changes. You can activate the forwarding settlement change process in the corresponding settlement profile by selecting **Reverse and Repost** or **Delta Only**. The system uses the organization or a combination of organization and business partner to assign the profile to the forwarding order. For more information, see [Automatic Change Management for Forwarding Settlements](#).

The system implements the changes synchronously or asynchronously as follows depending on your Customizing settings:

- Asynchronous changes

The system can change the freight unit even if it is blocked by another process. It updates the freight unit as soon as the freight unit is unlocked.

The system processes the changes in the background using the report [Background Job to Process Triggers Periodically \(/SCMTMS/PROCESS_TRIGGER_BGD\)](#). You can check in the application log if the system correctly propagated asynchronous processed changes. To do so from the SAP Easy Access menu, choose [►►Logistics►Transportation Management►Administration►Display Application Log Entries►](#).

- Synchronous changes

The system informs you immediately if a freight unit is blocked. You cannot propagate the changes.

In case the forwarding order or forwarding quotation has the approval status **Approval Needed** and with your change the approval becomes redundant, the system deletes the workitem from the workflow.

Special Features When Changing Data

The following special features apply to the changing of forwarding order data and forwarding quotation data:

- Deleting items

When you delete items the system deletes the corresponding freight units provided they are still in planning. The system does not delete freight units that are already in execution.

- Changing an item hierarchy

When you change an item hierarchy the system does not delete freight units that are in planning or are already in execution.

- Changing the shipping type, movement type, and transportation mode

You cannot change these parameters if the following applies:

- Shipping type: The main carriage stage of the route is already in execution or there is a valid forwarding settlement document (in other words, this has not been canceled).
- Movement type: A follow-on document has already been assigned to a stage of the route or there is a valid forwarding settlement document.
- Transportation mode: The main carriage stage of the route is already in execution, there is a valid forwarding settlement document, or invoicing has already been completed.

Canceled Forwarding Orders or Forwarding Quotations

You can cancel forwarding orders and forwarding quotations. The system either deletes or cancels the freight unit. If you have specified, that the system can only cancel freight units, the freight unit remains with the status **Canceled**. Otherwise, the system deletes the freight unit. If the execution has already started for an assigned freight order or freight booking, you cannot cancel the forwarding order.

If the forwarding order or the forwarding quotation has the approval status **Approval Needed**, the system deletes the work item from the workflow.

More Information

For more information about changes to freight document data, see [Change Controller](#).

For more information about changes within the integrated ERP process, see [Integration of Changed and Deleted Orders and Deliveries](#).

Logistics Integration

You can integrate Transportation Management (TM) with components in the same system or an external system. Alternatively, you can integrate a decentralized EWM and TM system with an external system.

For more information see:

- [Internal TM Component Integration](#)
- [External TM System Integration](#)
- [Advanced Shipping and Receiving](#)
- [Advanced SR for Decentralized EWM and TM](#)

Internal TM Component Integration

Use

You use integration of orders, deliveries, and JIT calls to facilitate transportation planning and execution in Transportation Management (TM) based on orders created in Sales and Distribution (SD), Materials Management (MM), deliveries created in Logistics Execution (LE), and JIT calls created in Next Generation Just-In-Time (nJIT).

With the Advanced Shipping and Receiving process, you can have enhanced integration between Transportation Management (TM) and Extended Warehouse Management (EWM). A new or updated sales order or sales scheduling agreement triggers the creation of LE deliveries. The integration uses the warehouse request and its handling units in EWM and freight units in TM to allow TM to update predecessor documents and also trigger actions such as goods movements or reversal of goods movements.

Integration with nJIT enables you to conduct the entire process on the basis of a JIT call. This includes the execution of sequenced and forwarded JIT calls. A JIT call can also trigger the creation of inbound deliveries, and integration between components ensures that changes status updates are communicated between nJIT, LE, and EWM.

For more information, see [Advanced Shipping and Receiving](#).

You can also integrate a decentralized EWM and TM system with an external ERP system and perform transportation planning based on outbound delivery orders.

The TM component creates or updates freight units directly for orders or deliveries which you have defined in Customizing as relevant for TM integration. If such orders or deliveries are created or changed, the system creates or updates freight units in the TM component once you save the order or delivery. The integration allows you to continuously react to changes made to orders and deliveries in SD, MM and LE. In particular, integration with these components enables you to:

- Collectively plan transportation for all orders and scheduling agreements
- Perform transportation planning in TM based on sales or purchase orders, prior to delivery creation, thus enabling you to get an earlier forecast of future transportation and resource demands
- Perform tendering, subcontracting, and execution in TM based on sales and purchase orders
- Perform sales order scheduling
- Propose and create deliveries based on dates and quantities according to transportation planning results, taking into account transportation constraints such as resource availability and transportation durations.
- Perform transportation planning and execution of JIT calls.

Features

Order Integration

Orders that are created in SD or MM can be integrated into TM. The following types of order can be integrated into TM:

- Sales order
- Customer returns
- Purchase order
- Stock transport order
- Returns purchase order
- Returns stock transport order

Scheduling Agreement Integration

Scheduling Agreements that are created in SD or MM can be integrated into TM. The following types of order can be integrated into TM:

- Sales Scheduling Agreement
- Purchase Scheduling Agreement
- Stock Transport Scheduling Agreement

JIT Call Integration

JIT calls created in nJIT can be integrated into TM. The following types of calls can be integrated into TM:

- Sequenced JIT Call
- Forwarded JIT Call

Delivery Proposals

TM can generate a proposal for a delivery to be created in LE based on the order-based transportation requirement. For more information, see [Creation of Deliveries from TM](#).

Delivery Integration

Outbound deliveries and inbound deliveries that are created in LE can be integrated into TM, regardless of whether the delivery creation was triggered by TM or not. Deliveries can also be integrated into TM, even if an order that relates to the delivery has not been sent to TM.

You can also define the transportation relevance of your deliveries.

Transportation Relevance

You can define the transportation relevance of sales, purchase, and delivery documents.

Integration Time Range

You can define an integration time range for purchase scheduling agreements, which includes an integration start date, to control the creation of freight units (FU) in TM.

Dangerous Goods

If a product contained in an order or delivery is classified as dangerous goods, TM considers this information to ensure a safe and compliant transportation of the product. For more information, see [Dangerous Goods Checks](#).

Configuring Integration of Orders and Deliveries

Use

This section describes the additional settings that you need to configure to integrate orders, scheduling agreements, deliveries, and JIT calls.

Prerequisites

You have run the location creation report /SAPAPO/CREATE_LOCATION.

Activities

You must complete the activities in Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration** as follows:

- **Define Control Keys for Document Integration**

You can define control keys that determine which documents in a document chain are to be integrated into TM. You can define as many control keys as required, with different combinations of relevant documents. You need to specify the integration mode: you can select the internal TM component or choose to work with an external TM system.

- **Define Logistics Integration Profile**

You can use the [Logistics Integration Profile for SD, MM, and LE](#) to control the integration of SD, MM, and LE documents into TM. You can create multiple profiles and assign these to TM relevant business document types.

For more information, see [Logistics Integration Profile for SD, MM, and LE](#).

You can use the [Logistics Integration Profile for nJIT](#) to control the integration of JIT calls into TM.

For more information, see [Logistics Integration Profile for nJIT](#).

- **Activate Integration of Sales Documents**

You activate the integration of sales documents into TM by assigning a control key to a combination of the required sales document types, sales area (sales organization, distribution channel, division), and shipping condition. The sales documents can be sales orders, or customer returns. The control key determines whether a sales document and the corresponding outbound delivery are to be integrated into TM.

When you create a sales document for which the integration has been activated, the system assigns the control key to the sales document (parameter TM_CONTROL_KEY) and to all relevant documents in the document chain, for example, the outbound delivery.

- **Activate Integration of Purchasing Documents**

You activate the integration of purchasing documents into TM by assigning a control key to a combination of the required document types, purchasing order, purchasing organization, and purchasing group. The purchasing documents can be purchasing orders, stock transport orders, returns, or scheduling agreements. The control key determines whether a purchasing document and the corresponding inbound delivery or outbound delivery are to be integrated into TM.

When you create a purchasing document for which the integration has been activated, the system assigns the control key to the document (parameter `TM_CONTROL_KEY`) and to all relevant documents in the document chain, for example, the inbound delivery.

- **Activate Integration of Delivery Documents**

You activate the integration of delivery documents into TM by assigning a control key to a combination of the relevant shipping point, delivery types and shipping condition.

Note that you can also activate the integration of the delivery using the control key assigned to the preceding document in the document chain, for example, a sales order or a purchase order. In this case, the delivery inherits the control key from the preceding document and the system ignores any settings that you make in this Customizing activity. This is needed only in cases where a delivery is to be integrated into TM but the corresponding order has not been integrated into TM.

- **Determine Transport-Driven Process Relevance for Deliveries**

You can define which outbound deliveries are relevant for the transport-driven process. In a transport-driven process, the delivery is blocked for warehouse execution while transportation planning takes places. When you release the delivery for warehouse processing, it is then unblocked, and warehouse execution can start.

For more information about the transport-driven process, see [Transport-Driven Execution](#).

- **Define Rule Groups to Determine Integration Time Range of Purchase Scheduling Agreements**

You can create rule groups and assign rules to rule groups. Your rule groups determine the span of the integration time range of your purchase scheduling agreement. Integration time ranges, on the other hand, define the time span within which freight units for purchase scheduling agreements are created.

For more information, see [Integration Time Range for Purchase Scheduling Agreements](#).

- **Assign SD/MM Text Types to TM Text Types for SO/PO**

You can define which types of text entered in a Sales and Distribution (SD) or Materials Management (MM) document are displayed on the Notes tab page of the corresponding order in Transportation Management (TM).

- **Assign Logistics Execution Text Types to TM Text Types for Deliveries**

You can define which types of text entered in a Logistics Execution (LE) document are displayed on the Notes tab page of the corresponding delivery-based transportation requirement (DTR) in Transportation Management.

- **Define Usage of Packaging Information**

By configuring shipping planning and delivery blocks, you define how delivery blocks are handled for specific document types in TM, specifying whether a delivery block is a planning block, an execution block or both. Blocks prevent further planning and/or execution in TM.

Delivery Profile

Definition

A group of user-specific settings that the system takes into account during delivery creation.

Structure

A delivery profile enables you to make the following settings:

- You can specify that during delivery creation the system fixes the planning results for freight units, freight orders, or freight bookings that have already been planned.

- You can define how the system creates deliveries when a user triggers delivery creation in Logistics Execution (LE) for freight units that have not been planned. You can choose to create deliveries according to:
 - freight unit (default)
 - order
 - order item
- You can specify incompatibility settings. In the incompatibility settings, you can define attributes that prevent freight units from being combined into one delivery, for example. For more information about incompatibilities, see [Incompatibilities](#).
- You can include or exclude freight units and freight orders that are blocked for planning or blocked for execution in the delivery creation process.
- For freight unit items belonging to the same order item, you can amalgamate into the same delivery item or create separate delivery items.
- You can indicate when packaging information is to be transferred from a freight order or consignment order to an LE delivery:
 - When you create or update an LE delivery or when an LE delivery is created automatically
 - When you set the **Warehouse Processing Status** to **Ready for Warehouse Processing** for a freight order or consignment order

Note that this field is only relevant in the Advanced Shipping and Receiving inbound process. For more information, see [Advanced Shipping and Receiving](#).

For more information about these settings, see the corresponding field help.

i Note

You define delivery profiles in the [Create Delivery Profile](#) app, which you can open from the launchpad.

Source and Destination Location Determination

Source and destination determination refers to the technical process in logistics integration by which software logic identifies and transmits business entities and their locations from ERP documents, such as purchase orders, to the corresponding freight units.

Specifically, these can be business partners, plants, shipping points or storage locations in ERP documents from SD, MM, LE, or nJIT – the latter only in Advance Shipping and Receiving (Adv. SR) scenarios. In some cases, you can specify a document-specific address in the business transaction document, which is also considered in the source and destination determination.

With the details extracted from ERP documents, the determination identifies and can, if necessary, updates the corresponding locations or creates new locations.

Prerequisites

- Optionally for stock transport orders (STO) and returns STOs, you have specified an address for your storage location in Customizing for **Enterprise Structure** under [Definition](#) [Materials Management](#) [Maintain Storage Location](#).
- Optionally, in an internal setup, you have activated Improved Location Determination in Customizing for **Transportation Management** under [Integration](#) [Logistics Integration](#) [Internal TM Component Integration](#) [Define Logistics Integration Profile for SD, MM, and LE](#).

Improved Location Determination / Advanced Shipping and Receiving

The customizing setting provides improvements concerning the determination of the receiving location in non-Adv. SR scenarios. In Advanced Shipping and Receiving (Adv. SR) scenarios, these improvements are active by default.

Stock Transport Orders, returns Stock Transport Orders, Stock Transport Scheduling Agreements

By default, the determination will select business partner addresses as destination location. But you have the option of activating Improved Location Determination to use shipping points as destination location.

Purchase Orders, Stock Transport Orders, returns Stock Transport Orders, Purchase and Stock Transport Scheduling Agreements

By default, when an address for the storage location is available, a one-time location with the respective address is used as destination location. But you have the option of activating Improved Location Determination to use shipping points as destination location. In such cases, the item's delivery address tab in the purchase document displays the address of the storage location, while the freight unit displays the address of the receiving point, which can differ.

One Time Locations

You can enter new addresses in your business transaction documents. As a result, the system creates a one-time location, which is used as source or destination in the freight unit.

Related Information

[One-Time Location](#)

[Logistics Integration Profile for SD, MM, and LE](#)

Logistics Integration Profile for SD, MM, and LE

Definition

A group of user-specific settings that control the integration of Sales and Distribution (SD), Materials Management (MM), and Logistics Execution (LE) documents into Transportation Management (TM).

Structure

You define logistics integration profiles in Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration**.

When working with the internal TM component you can control the integration of documents from LE, SD, or MM using the logistics integration profile. In the profile you can configure order-related settings and common settings, which are both order-related and delivery-related:

Common Settings

- **Freight Unit Building Rule**

Specifies the freight unit building rule that is used when freight units are automatically created for business documents.

- **Condition for Freight Unit Building Rule Determination**

Specifies settings that are used when freight units are created for business documents. The corresponding condition determines a freight unit building rule (FUB rule) based on the business document details. For more information, see [Freight Unit Building Rule](#).

- **Automatic Freight Unit Building**

Specifies whether freight unit building is performed automatically after the system saves the relevant business document (order or delivery), or if it needs to be triggered manually in a separate step.

Stage Building Variant

Specifies whether you use the **Static** or **Advanced** stage building variant.

- Specifies whether you use the **Static** or **Advanced** stage building variant.

If you select the **Static** stage building variant, you can specify a stage profile and stage profile condition to determine how the system creates stages for freight units. You can define which stage(s) will be built if the Incoterm location is used for stage building.

If you select the **Advanced** stage building variant, the system considers Incoterm locations 1 and 2 when it builds the stages for freight units. You can also choose whether to use only planning-relevant stages in stage building, or static stages as well.

Order-Related Settings

- **Plan on Requested or Confirmed Quantities**

Specifies whether freight units are built based on requested or confirmed quantities.

- **Freight Unit Building for Partly Complete Sales Documents**

Enables the creation of freight units for complete items in a partly complete sales document.

- **Rule Group to Determine Time Range of Purchase Scheduling Agreements**

Enables the use of time ranges that limit the creation of freight units for scheduling agreements.

Integration Based on Actual Quantity

Allows delivery integration based on actual quantities, which are determined during the execution phase.

Improved Location Determination

Designates the destination location based on the shipping point.

Advanced Shipping and Receiving

- **Advanced Shipping and Receiving(Advanced SR)**

Enables compliance checks for business documents in the Advanced Shipping and Receiving process.

Logistics Integration Profile for nJIT

Definition

A group of user-specific settings that control the integration of nJIT with TM.

Structure

You define logistics integration profiles in Customizing for [Transportation Management](#) under [Integration](#) [Logistics Integration](#) [Internal TM Component Integration](#).

When working with the internal TM component you can control the integration of S2P calls from nJIT using the logistics integration profile. In the profile you can configure common settings and stage building:

Common Settings

- [Freight Unit Building Rule](#)

Specifies the freight unit building rule that is used when freight units are automatically created for business documents.

- [Condition for Freight Unit Building Rule Determination](#)

Specifies settings that are used when freight units are created for business documents. The corresponding condition determines a freight unit building rule (FUB rule) based on the business document details. For more information, see [Freight Unit Building Rule](#).

Stage Building Variant

The system considers Incoterm locations 1 and 2 when it builds stages for freight units. You can choose whether to only use planning-relevant stages in stage building or both planning-relevant and static stages as well.

Logistics Integration Profile for Decentralized EWM

Definition

A group of user-specific settings that control the integration of decentralized EWM with TM.

Structure

You define logistics integration profiles in Customizing for [Transportation Management](#) under [Integration](#) [Decentralized EWM-TM Integration](#).

When working with the internal TM component you can control the integration of outbound delivery orders from decentralized EWM using the logistics integration profile. In the profile you can configure common settings, stage building, and compliance checks:

Common Settings

- [Freight Unit Building Rule](#)

Specifies the freight unit building rule that is used when freight units are automatically created for business documents.

- [Condition for Freight Unit Building Rule Determination](#)

Specifies settings that are used when freight units are created for business documents. The corresponding condition determines a freight unit building rule (FUB rule) based on the business document details. For more information, see [Freight Unit Building Rule](#).

Stage Building Variant

The system considers Incoterm locations 1 and 2 when it builds stages for freight units. You can choose whether to only use planning-relevant stages in stage building or both planning-relevant and static stages as well.

Compliance Check

Enables compliance checks for business documents in the Advanced Shipping and Receiving process.

Related Information

[Advanced SR for Decentralized EWM and TM](#)

Reports for Logistics Integration

This overview lists reports available in the logistics integration area of Transportation Management (TM). These reports assist with integration and post-integration processes between TM and Sales and Distribution (SD), Materials Management (MM), Logistics Execution (LE), and nJIT (Just-In-Time).

Along side the report names, the list contains technical names and short descriptions. For more information, see the individual report documentation in the system.

You can access the reports through your SAP Easy Access screen under ► **Logistics** ► **Transportation Management** ► **Administration** ► **Logistics Integration** ►.

Reports

Activation of Purchase Scheduling Agreements Without Delivery Integration

- **Technical Name:** /SCMTMS/ACTIVATE_MMSAG_WO_DLV
- **Description:** Run to activate the integration of purchase scheduling agreements without integrating the corresponding inbound deliveries.

Activation of Documents for Transportation Management

- **Technical Name:** /SCMTMS/LI_ACTIVATE
- **Description:** Run to activate the integration of business transaction documents into TM in an internal integration setup.

Adjustment of Integration of Purchase Scheduling Agreements Without Delivery Integration

- **Technical Name:** /SCMTMS/ADAPT_START_MM_WO_DLV
- **Description:** Run to adjust the integration start date and reduction quantity for a purchase scheduling agreement item without delivery integration.

Create Deliveries in ERP

- **Technical Name:** /SCMTMS/DLV_BATCH
- **Description:** Run to create deliveries based on existing freight units.

Deactivation of Purchase Scheduling Agreements

- **Technical Name:** /SCMTMS/DEACTIVATE_MMSAG_W_DLV
- **Description:** Run to deactivate your purchase scheduling agreements.

Prepone End Dates for Items in Purchase Scheduling Agreement

- **Technical Name:** /SCMTMS/UPDATE_INT_TIMERANGE
- **Description:** Run to verify end dates and reschedule them for an earlier date.

Reactivation of Purchase Scheduling Agreements

- **Technical Name:** /SCMTMS/REACTIVATE_MMSAG_W_DLV
- **Description:** Run to reactivate the integration of purchase scheduling agreements into TM.

Setting and Fixing of Integration Time Range End Date for Purchase Scheduling Agreements

- **Technical Name:** /SCMTMS/FIX_INT_TIMERANGE
- **Description:** Run to set a fixed end date for the end of the integration of purchase scheduling agreements.

Subsequent Freight Unit Building in Internal TM Component Integration

- **Technical Name:** /SCMTMS/SUBSEQUENT_FUB_LOGINT
- **Description:** Run to trigger the creation of freight unit in an internal TM component integration setup.

Update of Purchase Scheduling Agreements Based on TM Planning Results

- **Technical Name:** /SCMTMS/TM_SAG_PLANNING_SYNC
- **Description:** Run to synchronize the transportation planning date of freight unit items with the delivery date of the corresponding schedule line in the purchase scheduling agreement.

Business Add-Ins for Logistics Integration

The following Business Add-Ins (BAI) are available in the logistics integration area. Expand each element for detailed information.

The BAIs and detailed documentation are available in Customizing for **Transportation Management (TM)** under [Integration](#)  [Logistics Integration](#) .

BAI List

BAI: Adaptation of Error Handling in Logistics Integration

- **Technical Name:** /SCMTMS/BADI_LI_ERROR_HNDL
- **Description:** Modifies error handling behavior.

BAI: Adaptation of Incoming Business Transaction Document Data in TM

- **Technical Name:** /SCMTMS/BADI_LI_BT_D_CHG
- **Description:** Enhances the integration of logistics documents into TM by adapting the input data of business transaction documents.

BAI: Change Transient TRQ Data

- **Technical Name:** /SCMTMS/BADI_LOGINT_CHG_TTRQ
- **Description:** Allows you to adapt the data in the transient transportation request after the data from the business transaction document has already been mapped to the transient transportation request in an internal integration setup.

Deliveries updates employ one of two update paths. Depending on the warehouse processing status, either the Freight Unit Builder or the Logistics Integration engine is called. For this reason, if you implement this BAdI, you must also implement [BAdI: Change TRQ Data \(LE and EWM\)](#).

BAdI: Changing Source/Destination of Business Transaction Documents

- **Technical Name:** /SCMTMS/LI_SRC_D_CHG
- **Description:** Changes the source and destination data of a freight unit after determination.

BAdI: Change TRQ Data (LE and EWM)

- **Technical Name:** /SCMTMS/BADI_LI_UH_MAP
- **Description:** Allows you to adapt the data in the transportation request update after the data from the business transaction document has already been mapped to the transportation request in an internal integration setup.

Deliveries updates employ one of two update paths. Depending on the warehouse processing status, either the Freight Unit Builder or the Logistics Integration engine will be called. For this reason, if you implement this BAdI, you must also implement [BAdI: Change Transient TRQ Data](#).

BAdI: Extend TM-LE Interface

- **Technical Name:** /SCMTMS/BADI_LI_TM_LE
- **Description:** Supports the extension of delivery functions such as images, updates, and creation in the integration between Transportation Management (TM) and Delivery Management (LE-SHP).

BAdI: Invoice Type Determination

- **Technical Name:** /SCMTMS/BADI_LI_TM_SD_BIL
- **Description:** Influences the determination of invoice type, date, and sales organization during invoice creation.

BAdI: Override Charge Recalculation Logic at Save

- **Technical Name:** /SCMTMS/TRQ_CALC_CHARGE_SAVE
- **Description:** Controls the charge recalculation behavior of transportation and internal charges when you save forwarding orders, forwarding quotations, as well as order-based and delivery-based transportation requirements.

Integration of Orders and Deliveries in Transportation Planning

Use

You can integrate orders and deliveries from Sales and Distribution (SD), Material Management (MM), and Logistics Execution (LE) into Transportation Management (TM).

You can view status and block information, and document-flow information for TM-relevant SD/MM orders and TM-relevant deliveries as follows:

- Consolidated status information on header and item level displaying
 - life cycle status
 - execution status
 - confirmation status
 - tendering status
 - subcontracting status
 - invoicing status
 - execution block information
 - planning block information
- Document flow which includes
 - SD/MM orders
 - deliveries
 - follow-on TM documents

You can navigate into any document that you display.

Process

The process for integrating SD/MM orders and LE deliveries with transportation planning in TM is as follows:

1. Create order (SD/MM)

An order is created in SD or MM containing items and schedule lines. When the order has been created, it is sent to the system, which creates freight units in TM. For more information, see [Integration of Orders](#).

2. Perform transportation planning (TM)

TM performs transportation planning based on the freight units. The planning results include pickup dates and delivery dates, resource assignments, and routes. For more information about transportation planning see [Planning](#).

3. Create delivery proposal (TM)

TM creates delivery proposals based on the transportation planning results, triggering the creation of deliveries in LE to create a delivery. It is also possible to create delivery proposals based on freight units, if transportation planning has not been performed. For more information, see [Creation of Delivery Proposals](#).

4. Create deliveries (LE)

When LE receives the request to create a delivery, LE creates the delivery. This can be an inbound delivery or an outbound delivery. For more information about the process of creating deliveries in Logistics based on delivery proposals from TM, see [Creation of Deliveries from TM](#). It is also possible to create deliveries in LE without having sent a proposal from TM. For more information, see [Integration of Deliveries](#).

5. Reassign freight units (TM)

TM determines the order that corresponds to the delivery order and reassigns the freight units from that order to the delivery.

6. Execute transportation

Transportation is executed based on the planning results.

i Note

The TM document flow for a specific order or delivery can be displayed in SD, MM or LE. TM provides information about the documents created for an order or delivery, for example, freight units, freight orders, and freight settlement documents, and the status of the documents, for example, life cycle status and execution status. This gives the user an overview of the entire process for an individual document and enables the user to track the status of document processing in TM from SD, MM or LE.

Integration of Orders

Use

In this process, a customer bases their transportation processes on an order created in Sales and Distribution (SD) or Materials Management (MM).

Prerequisites

For information about setting up Transportation Management (TM) for order integration, see [Configuring Integration of Orders and Deliveries](#).

Process

The process for transportation planning based on an SD/MM order is as follows:

1. Create order (SD/MM)

An order is created in SD/MM containing items and schedule lines. The item contains information regarding the material and the requested/confirmed quantity, and the desired delivery date. The quantity of an item can be subdivided into multiple schedule lines, which, for example, can have different delivery dates. When the order has been created, it is integrated into TM.

2. Create freight units (TM)

Once the order is saved in LE, TM creates freight units containing items that correspond to the schedule line items of the SD/MM order. Depending on your settings in Customizing for the logistics integration profile, TM can build freight units automatically when the order is created and saved; can use either the requested quantities or the confirmed quantities in an order when creating freight units; and can create freight units for complete items in a partly complete sales document.

TM only creates freight units for transportation-relevant orders.

3. Perform transportation planning (TM)

TM can consolidate FUs from multiple orders, even if the base documents in SD/MM have different order types. For example, the transportation demands of purchase orders and sales orders can be consolidated during transportation planning.

Result

After transportation planning, TM can either execute transportation or trigger the creation of deliveries in LE. For more information, see [Creation of ERP Deliveries from TM](#).

Related Information

[Subcontracting of Purchase Order](#)

Integration of Stock Transport Orders

In Transportation Management (TM), a stock transport order (STO) enables the transfer of goods and materials from one plant to another with the same company code.

In a two-step process, outbound and inbound deliveries are created, and the freight units (FU) are reassigned from the outbound to the inbound delivery. This process affords you visibility once goods issue has been posted and goods are in transit.

Prerequisites

General Prerequisites

- Delivery Management (LE-SHP), Extended Warehouse Management (EWM), and TM must run on a single system.

Other Prerequisites

- You have activated the improved location determination used for STOs. To do this, select the **Improved Location Determination** checkbox under the **Order-Related Settings** section of the logistics integration profile. For more information, see Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Logistics Integration Profile for SD, MM, and LE►](#).
- You have defined the control keys that determine system behavior after integration. For more information, see Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Control Keys for Document Integration►](#).
- You have determined purchase document relevance for TM by assigning control keys to the required document types. Optionally, you can assign a logistics integration profile. For more information, see Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Transportation-Relevance of Purchasing Documents►](#).
- You have defined a profile for the direct integration with Extended Warehouse Management (EWM) or an external warehouse management system. For more information, see Customizing for **Transportation Management** under [►Integration►Integration with Extended Warehouse Management►Define EWM Integration Profile►](#).
- You have defined the shipping point in Logistics Execution. For more information, see Customizing for **Transportation Management** under [►Logistics Execution►Shipping►Basic Shipping Functions►Shipping Point and Goods Receiving Point Determination►Assign Goods Receiving Points for Inbound Deliveries►](#).

Use

You can execute an STO as order- or delivery-based processes. After you create an STO, the system can either create FUs automatically or based on the outbound delivery depending on your Customizing settings. Additionally, you can choose between warehouse- and transport-driven executions.

STOs in the Advanced Shipping and Receiving Scenario

The posting of goods issue triggers the creation of an inbound delivery and the automatic reassignment of the FUs from the outbound to the inbound delivery.

As soon as the FU is assigned the inbound delivery as its predecessor document, the warehouse processing status is set to **In Transit**. For more information, see **Warehouse Processing Status** in [Statuses in Advanced Shipping and Receiving](#).

i Note

You must deselect the **One Step** checkbox for the process to occur. For more information, see Customizing for **Materials Management** under **Purchasing > Purchase Order > Set Up Stock Transport Order > Assign Document Type, One-Step Procedure, Underdelivery Tolerance**.

At both ends of the process, outbound and inbound, you can choose between EWM- or LE-managed warehouses or storage locations, or a combination of both. FUs grouped in a delivery, however, must be assigned to the same warehouse or storage location — that is, either EWM- or LE-managed.

For more information about the STO process in warehouse integration based on EWM transportation units (TU), see [Stock Transport Order Process](#).

STOs in Other Scenarios

Here, there is no communication between the inbound delivery and the freight order. An inbound delivery is created but not shown under the **Document Flow** tab in TM. Therefore, FUs are not reassigned to the inbound delivery.

Handling of Discrepancies

In cases where goods issue has been posted and a correction is required, you will need to delete the inbound delivery before reprocessing.

Corrections can be made as soon as discrepancies are identified before the truck has departed the warehouse, and at the latest, when in **Ready for Warehouse Processing**, but before any changes to quantity or packaging have been made on the inbound side. If set, you would need to reverse **Ready for Warehouse Processing**, delete the inbound delivery, and cancel goods issue.

To delete the inbound delivery, open your **Freight Units – Worklist**. For more information about deletions, see [Delivery Deletion](#).

Alternatively, you can make the adjustments on the inbound side by adjusting the quantities directly in the inbound delivery.

i Note

Once such adjustments are made, reversing previously executed steps or making subsequent changes on the outbound side is no longer possible.

Creation of Deliveries from TM

Use

You use this process if you want to create a delivery proposal in Transportation Management (TM) and send the proposal to Logistics Execution (LE) where the delivery is created. The delivery proposal is based on the results of transportation planning such as consolidation and dates, taking into account transportation-planning-relevant information such as resource availability and transportation durations. TM can also trigger the creation of deliveries prior to transportation planning.

Process

The process for creating delivery proposals in TM and the subsequent creation of deliveries in LE is as follows:

1. Select input (TM)

The user can start delivery creation by using a background report or interactively. For more information, see [Creation of Delivery Proposals](#).

TM considers data for delivery creation from the available freight units, freight orders, and freight bookings..

2. Generate delivery proposals (TM)

Based on the chosen input, TM creates delivery proposals. A proposal consists of groups of freight units that can be delivered together. Depending on the input method used, TM either sends the proposals automatically to LE to create deliveries, or the user can check the proposals and then send them to LE.

If the user specified a delivery profile that requires the planning results to be fixed for planned freight units, freight orders, or freight bookings, the system fixes the planning results during delivery proposal creation. Otherwise, the system does not fix the planning results.

3. Create deliveries (LE)

When LE receives the request to create deliveries, LE creates the deliveries. These can be inbound or outbound deliveries. LE might split the proposed deliveries further, for example, based on additional split criteria. However, LE does not change the dates that were proposed by TM nor consolidate proposed deliveries.

Creation of Delivery Proposals

Use

Transportation Management (TM) can propose inbound deliveries and outbound deliveries and, based on such proposals, trigger the creation of deliveries in Logistics Execution (LE).

Delivery creation is possible:

- through a dialogue process based on the selection of freight units that are assigned to Sales and Distribution (SD) or Materials Management (MM) orders
- as a batch process based on the selection of SD/MM orders
- out of the transportation cockpit based on the selection of freight orders
- out of the freight order UI

Prerequisites

- Freight units exist for SD/MM orders.
- The order is not blocked for delivery.
- A delivery is expected for the order in LE. In Customizing for TM you have specified a control key for the order that has been configured to make deliveries TM-relevant.
- If you want to use a selection profile to select the relevant input for the delivery proposal, you have defined a selection profile (see [Selection Profile](#)).
- If you want to use a delivery profile during delivery creation, you have defined a delivery profile (see [Delivery Profile](#)).
- If you want to send delivery proposals to LE from the freight order UI, you have assigned a delivery profile to the relevant freight order types. You do this in Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types**.
- If you want to handle actual amounts that differ from the planned amount, you have to activate this functionality in the relevant logistics integration profile. For more information see [Logistics Integration Profile for SD, MM, and LE](#)

When using the background report, a delivery proposal must not have already been made for the freight units. This includes freight units for which a delivery failed to be created.

Features

Background Report

The user can create delivery proposals for order-based transportation requirements by using a background function (program /SCMTMS/DLV_BATCH). The user can enter selection criteria or use selection profiles. In both cases, the user can also enter a delivery profile. The system selects the documents, creates the delivery proposals, and creates the deliveries.

For more information about the background function, see [Creation of Deliveries in SAP ERP](#).

Interactive Delivery Creation

The user can create deliveries interactively as follows:

- From the **Delivery Proposal Builder** app, which you can open from the launchpad..

The user specifies one or more profiles according to which the system selects and displays a list of business documents. The user then selects the relevant documents in the list and triggers delivery proposal creation. The user can review the delivery proposals before creating the deliveries.

- From the **Freight Orders - Worklist** app, which you open from the launchpad.

The user selects the relevant business documents in the overview and chooses the **Create Deliveries** pushbutton. The system displays a list of freight units related to the selected business documents. The user selects the relevant freight units and triggers delivery proposal creation. The user can review the delivery proposals before creating deliveries.

- From the **Transportation Cockpit**

The user triggers delivery proposal creation for specific business documents after carrying out planning in the **Transportation Cockpit**. The user selects the business documents in the **Transportation Cockpit** and chooses the **Create Delivery Proposals** pushbutton. The system automatically creates and displays the delivery proposals. The user can review the delivery proposals before creating deliveries.

Source of Data

TM collects data from the relevant documents as follows:

- Dates

TM uses dates from the freight order or freight booking. If no freight order or freight booking exists, then TM uses dates from the freight units.

- Quantities

TM uses quantities from the freight units.

Consolidation

If freight orders or freight bookings exist for the transportation requirements, TM uses only one freight order or one freight booking for each delivery proposal. That is, only freight units that belong to the same freight order or freight booking can be consolidated into a single delivery proposal. In this way, freight units from various orders can be consolidated into a single delivery proposal if they belong to the same freight order or freight booking. If, however, freight units from a single order were distributed across multiple freight orders or freight bookings, then multiple deliveries are proposed for the freight units of this one order.

If order combination is not allowed for an order, TM does not group freight units from this order together with freight units from other orders. If freight units for such orders have been consolidated into the same freight order or freight booking during planning, TM does not include them in the same delivery proposal, but instead creates multiple delivery proposals.

In addition, freight units can only be consolidated into the same delivery proposal if the following data is the same:

- Source location
- Destination location
- Incoterms
- Shipping conditions
- Document type, such as purchase order or sales order
- Start stop (if planning has already been done)
- Destination stop (if planning has already been done)

If an order has not been planned in TM, and no freight order or freight booking exists, TM can also create delivery proposals for the corresponding freight units. In this case, TM creates delivery proposals according to the settings in the delivery profile. For more information about the delivery profile, see [Delivery Profile](#).

Bill of Materials (BOMs)

Either a BOM header or BOM items can be goods-movement-relevant. If a BOM header is goods-movement relevant, TM can, if necessary, distribute the quantity across multiple delivery proposals. If BOM items are delivery relevant, TM expects a common delivery group for all items and does not distribute the items or their quantities across multiple delivery proposals.

Consumption of Schedule Lines in LE

The delivery proposal in TM takes into account that LE consumes order schedule lines in chronological order. Therefore, TM does not propose a delivery for a schedule line before another schedule line of the same order item that has an earlier delivery date has been used for a delivery proposal. If freight unit selection results in this situation, TM excludes the affected freight units from the selection.

If an error occurs in LE when creating a delivery, then LE does not create deliveries that correspond to later schedule lines of the same order item, or the delivery is created but excludes the later schedule lines of that item.

Execution and Planning Blocks in LE

You can use the delivery profile to determine how, during delivery creation, the TM system deals with objects that are blocked for planning or execution. You can choose either or both of the following options:

- Exclude freight units or freight orders that are blocked for planning
- Exclude freight units or freight orders that are blocked for execution

The system considers freight units and freight orders to check whether they are blocked for planning or blocked for execution. If a freight order or its assigned freight units are blocked, the system – depending on the settings you have chosen in the delivery profile and depending on how you start delivery creation – prevents the user from creating a delivery. The system runs the check based only on the object (freight orders, bookings, transportation units, freight units) that you have selected as the starting point for creation of delivery proposals.

❖ Example

A freight unit is assigned to a pre-carriage and a main-carriage freight order. The main-carriage freight order is blocked for execution, but the freight unit has no blocks. The delivery profile has been configured to exclude freight units or freight orders that are blocked for execution.

If you start creation of delivery proposals with the pre-carriage freight order, the system allows creation of deliveries.

If you start creation of delivery proposals with the main-carriage freight order, which is blocked for execution, the system prevents you from creating a delivery.

The system checks for blocks in freight orders only when you start delivery creation in a freight order context, for example, from the freight order POWL or the freight order UI or by using a freight order selection profile in the report /SCMTMS/DLV_BATCH. Otherwise only the blocks in freight units are considered.

Delivery Proposals with Freight Document Hierarchies

With transportation units, TM provides a three-tier hierarchy of freight documents. This can comprise a freight order on the highest level, transportation units on the middle level, and freight units on the lowest level. For two-tier hierarchies, the highest level can include freight orders, freight bookings and transportation units. On the lowest level, you can have freight units and automatically generated transportation units. In addition, automatically generated transportation units and automatically generated freight orders can represent a one level hierarchy.

You can create delivery proposals for each of these hierarchy types. If not explicitly stated, the term **freight unit** can also mean **automatically generated freight order** or **automatically generated transportation unit**, in order to represent objects on the lowest hierarchy level. In the same way the term **freight order** can also mean **freight booking** or **transportation unit** in order to represent objects on the highest hierarchy level.

Delivery Proposals for Bulk Material

Bulk materials are fluids and materials like sand or coal, that are given in base units of measure of volume or weight. You can use TM to deal with the differences in actual and planned quantity that occur naturally during the loading of liquid or dry bulk material.

In the Freight Order UI you can enter the volume and/or the weight of the actual quantities that were measured during or after loading. If these quantities differ from the planned quantities that were received from the order, the system creates a delivery proposal with the actual quantities. Subsequently, LE creates a delivery accordingly. Differing quantities do not lead to split Freight Units.

Integration of Deliveries

Use

You use this process if a customer creates a delivery in Logistics Execution (LE) and integrates it into Transportation Management (TM).

Process

Integration of LE Deliveries Without Integrating Order into TM

The process for a delivery that is created in LE for an order that is not relevant for TM runs as follows:

1. Create order (SD/MM)

An order is created in SD/MM, but is not relevant for TM due to settings in Customizing.

2. Create deliveries (LE)

Deliveries for the order are created in LE.

3. Create freight units (TM)

Since the order is not TM-relevant, TM creates new freight units for the delivery according to the freight unit building rule specified.

4. Perform transportation planning (TM)

5. Execute transportation (TM)

Integration of Changed and Deleted Orders and Deliveries

Use

Logistics integration supports changes to and deletion of orders in Sales and Distribution (SD) and Materials Management (MM), and changes to, deletion, and splitting of deliveries in Logistics Execution (LE). If an order or delivery is changed the freight units are updated accordingly in Transportation Management (TM). If an order is deleted, the corresponding freight units in TM are deleted. If a delivery is deleted, the corresponding freight units are either reassigned to the order if the order is TM-relevant, or if there is no preceding order, they are deleted.

Prerequisites

You have specified whether TM deletes or cancels empty freight units. You make this setting in Customizing for **Transportation Management** under **►►Planning►Freight Unit►Define Freight Unit Types►**.

More Information

For more information see the following sections:

- [Processing Changed or Deleted Orders](#)
- [Processing Changed, Deleted, or Split Deliveries](#)

Processing Changed or Deleted Orders

Use

If a user changes the data in an order in Logistics Execution (LE), (for example, quantities, dates and locations can be changed), Transportation Management (TM) creates the required freight units.

If an order is deleted in LE, TM deletes the corresponding freight units.

Process

Update of an Order

The process when an order is changed is as follow:

1. Change order (Sales and Distribution (SD) / Materials Management (MM)

Changes are made to an order in SD/MM.

2. Adapt freight units (TM)

TM updates the freight units according to the assigned freight unit building rule. If the order quantity has been increased and a delivery has already been created, TM creates new freight units to cover the resulting open quantity of the order.

Deletion of an Order

The process when an order is deleted runs as follows:

1. Delete order (Sales and Distribution (SD) / Materials Management (MM))

SD/MM triggers the deletion of the freight units in TM. Note that if a delivery has already been created in Logistics Execution (LE), the order cannot be deleted in SD/MM.

2. Adapt freight units (TM)

If the freight unit was only relevant for the canceled order the freight unit becomes empty. If the freight unit was assigned to multiple orders, the quantity in the freight unit is reduced according to the canceled order.

3. Delete or cancel freight units (TM)

If the freight unit is empty, TM either deletes or cancels the freight unit according to your settings for the freight unit type.

Processing Changed, Deleted, or Split Deliveries

Use

If a delivery is changed, split, or deleted in Logistics Execution (LE), Transportation Management (TM) adjusts the corresponding freight units accordingly.

Note that changes or splits can originate in LE or in Extended Warehouse Management (EWM). Changes or splits in EWM are always integrated into TM via LE.

The following sections describe how TM handles these changes.

Process

Update of a Delivery

The process for updating a delivery in LE after delivery creation is as follows:

1. Update delivery (LE)

2. Adapt freight units (TM)

TM adjusts the freight units according to the changes in the delivery.

If the quantity of the delivery has changed **and** the delivery is assigned to freight units that were consumed from an order, TM processes the freight units as follows:

- If the quantity of the delivery has **decreased**, the open quantity of the corresponding order is increased accordingly. It is necessary to reflect this in TM, so the freight units of the delivery are adapted and because the open quantity of the order has changed, the freight units of the order are also adapted.

- If the quantity of the delivery has **increased**, and there are still open quantities in the corresponding order, the system reassigns the freight units from the order to cover this increased delivery quantity. The system then reduces the open quantity of the freight units that are assigned to the order.

Deletion of a Delivery

The process for deleting a delivery in LE after delivery creation is as follows:

1. Delete delivery (LE)
2. Reassign freight units to order (TM)

If corresponding order-related freight units exist, the freight units are reassigned to this order.

i Note

If the process runs without order integration, the freight units are adapted and, if necessary, are canceled or deleted depending on your settings for the freight unit type.

Split of a Delivery

For more information, see the following:

- [Delivery Splits Triggered by SAP ERP](#)

Delivery Splits Triggered by Logistics Execution (LE)

Use

You can split an outbound delivery in Logistics Execution (LE) or in Extended Warehouse Management (EWM) and trigger the required changes in Transportation Management (TM). If EWM triggers the split, the delivery split is processed in LE.

Prerequisites

- You have created freight units in TM based on an outbound delivery from LE.

Process

1. Split delivery (LE)

LE splits a delivery for which TM has already created freight units. The existing delivery is changed and a new delivery is created in LE.

2. Adapt freight units (TM)

TM deletes the freight units which represent the items / quantities which are to be split. TM then creates new freight units and assigns these to the new delivery.

Delivery Splits and Updates in Logistics Execution Triggered by TM

You can split an outbound delivery in Logistics Execution (LE) based on planning changes in Transportation Management (TM). You can also update inbound deliveries and outbound deliveries from TM.

When LE receives a request for a split or update, it makes the necessary changes in outbound delivery processing and sends the delivery data back to TM. TM can then make the required changes to the freight units.

TM can trigger the split or update of outbound deliveries only. It cannot trigger the split or update of inbound deliveries.

Prerequisites

- You have created freight units in TM based on an LE outbound delivery.
- For outbound delivery splits, the distribution status of the outbound delivery in LE has not been set to **Distributed**.

Process

Outbound Delivery Split

1. Perform transportation planning (TM)

You perform planning or change the existing planning of freight units that are based on an outbound delivery. When you save the planning results, the system determines the delivery split relevance. If planning assigns freight units of a delivery to different transports for the first or last stage those freight units are not transported together on these stages so a split is necessary.

TM determines whether the LE delivery can still be changed by checking the delivery status. If the delivery can be changed, TM sends the required delivery split to LE.

2. Split outbound delivery (LE)

Based on a split request received from TM, LE creates a new outbound delivery and changes the original outbound delivery (that is, it removes the relevant items from the original delivery and creates one or more new deliveries for these items). LE splits the delivery according to the settings in the split profile.

3. Create new delivery and update original delivery (TM)

TM updates the freight units of the original delivery and reassigns the appropriate freight units to the new delivery.

Outbound Delivery Update

1. Perform transportation planning (TM)

You perform planning or change the existing planning of freight units that belong to an outbound delivery. Dates are changed during planning. Alternatively, you change the dates of the relevant freight order.

When you save the planning results or the changed freight order, TM determines that the dates in the freight order no longer match the dates in the freight units and the original LE delivery.

TM determines whether the LE delivery can still be changed by checking the delivery status. If the delivery can be changed, TM sends the changed dates to LE and requests the update of the outbound delivery

2. Update outbound delivery (LE)

LE updates the relevant deliveries.

3. Update Freight Units (TM)

TM updates the freight units.

Delivery Creation and Split for Package Units with Multiple Assignments

You can configure the system to allow a freight unit to be assigned to several package units when you create package units from a freight order.

This means that the items of a freight unit may be spread across multiple package units and multiple freight orders. If load planning is not finalized, this can lead to:

- unnecessary delivery splits
- unnecessary freight unit splits
- creation of unnecessary deliveries

To avoid unnecessary splits, or creation of unnecessary deliveries, the system does not consider a freight order for delivery creation until load planning for the freight units of that freight order is finalized. The system treats such freight units as unplanned freight units. When load planning is finalized for all freight units of a freight order, the system considers that freight order and can create, split, or update deliveries.

Prerequisites

You have specified that a freight unit can be assigned to multiple packages. For more information see Customizing for [Transportation Management](#) under [Freight Order Management](#) > [Transportation Unit](#) > [Define Transportation Unit Types](#) > [\(Assgnmt of Pred.Docs](#) (Assignment of Predecessor Documents) field)

Features

When load planning for a freight order is finalized, the system identifies all required delivery splits, and then splits and updates deliveries accordingly.

Related Information

[Creating a Package Unit](#)

[Package Unit](#)

[Load Planning](#)

Delivery Updates in TM Triggered by Logistics Execution

In Advanced Shipping and Receiving (Advanced SR) processes, you can change specific fields in outbound deliveries in Logistics Execution (LE-SHP) after freight units (FU) have been created in Transportation Management (TM) and deliveries have been distributed to Extended Warehouse Management (EWM).

In embedded and decentralized scenarios, two aspects of the process influence whether the specific fields can be edited. First, you must consider whether you are in a warehouse- or transport-driven process, and second, what the execution status is at that point in time. If any fields can be edited, both the outbound delivery document and the FUs are locked to ensure no data is lost. Any changes you make to the outbound delivery in LE are communicated to TM via Extended Warehouse Management (EWM).

You can find a comprehensive list of the changeable fields in [Delivery Updates](#).

Prerequisite

- You have created outbound deliveries in LE.
- You have defined text types in TM in Customizing for [Cross-Application Components under Processes and Tools for Enterprise Applications](#) under [► Reusable Objects and Functions for BOPF Environment ► Dependent Object Text Collection ► Maintain Text Schema ►](#).
- You have assigned EWM text types to TM text types in Customizing for [Transportation Management](#) under [► Integration ► Logistics Integration ► Internal TM Component Integration ► Assign EWM Text Types to TM Text Types for Deliveries ►](#).

Process

The process varies depending on whether you are in a transportation- or warehouse-driven scenario.

1. Update Outbound Deliveries

To make changes to an outbound delivery the system first checks whether TM allows these changes.

If, for example, you want to update textual information, you can do so at any point in time regardless of the execution status or scenario. If, however, you would like to change the shipping type, the system will allow it as long as the execution status is not yet set to [Ready for Warehouse Execution](#) in a transport-driven scenario. In a warehouse-driven scenario, the same is true if the status is [In Warehouse Execution](#) and no warehouse tasks exists.

2. Update Outbound Delivery Order

EWM updates the outbound delivery orders and communicates the changes to TM.

3. Update Freight Units

TM updates the freight units.

Goods Movement in Logistics Execution (LE) Triggered by TM

The goods issue of an outbound delivery can be triggered as soon as the execution of a freight order starts. For inbound deliveries, goods issue can be triggered once the execution of a freight order ends.

You can use transportation-related events to control and trigger the execution of goods movements automatically.

You can control goods issue of a freight order using transportation-related events such as:

- Loading begin
- Loading end
- Departure

Goods issue for an outbound delivery can only be associated with the execution of a freight order covering the first stage of the transport.

You can control goods receipt of a freight order using transportation related events such as:

- Arrival at destination
- Unloading start
- Unloading end

Goods receipt for an inbound delivery can only be associated with the execution of a freight order covering the last stage of the transport.

Prerequisites

You have defined the transportation-related events in the freight order type. For more information see Customizing under ► [Transportation Management](#) ► [Freight Order Management](#) ► [Freight Order](#) ► [Define Freight Order Types](#) ►.

Integration of Scheduling Agreements

Integration between Materials Management (MM), Sales and Distribution (SD), and Transportation Management(TM) enables the integration of purchase, sales, and stock transfer scheduling agreements.

Features

Integration enables the planning and execution of scheduling agreement schedule lines, which contain the required quantities and delivery dates. You can perform the following operations:

- Forecasts
- Just in Time (JIT)
- Just in Sequence (JIS)
- Planning Delivery

In addition to new scheduling agreements, you can also integrate existing scheduling agreements with TM. Deliveries are created for requirements in a schedule line that lie outside the scheduling horizons. The system automatically calculates and updates the open quantity.

Related Information

[Integration of Purchase Scheduling Agreements](#)

[Integration of Sales Scheduling Agreements](#)

[Integration of Stock Transport Scheduling Agreements](#)

Integration of Purchase Scheduling Agreements

A scheduling agreement is an outline agreement containing delivery quantities and dates within a specified time period. These are then entered as schedule lines in a delivery schedule.

You can integrate purchase scheduling agreements from Materials Management (MM) with Transportation Management (TM). The integration is based on delivery schedules.

The integration can be triggered by the creation or change of schedule lines.

With the Advanced Shipping and Receiving process, you can have enhanced integration between Transportation Management (TM) and Extended Warehouse Management (EWM). For more information see [Advanced Shipping and Receiving](#).

Prerequisites

You have completed Customizing for integration of purchase scheduling agreements. For more information, see Customizing for [Transportation Management](#) under [Integration](#) > [Logistics Integration](#) > [Internal TM Component Integration](#) > [Define Transportation-Relevance of Purchasing Documents](#).

Features

Create or Update Schedule Lines

Use an MRP run or manual planning to create, change, or delete schedule lines of an purchase Scheduling Agreement. Once changes to the purchase scheduling agreement are saved, TM creates, changes or deletes freight units according to the schedule lines of the scheduling agreement items.

Process Planning

Plan freight units on, for example, existing or newly created freight orders. The planning can be done manually or by the optimizer. After transportation planning has finished, the planning result can, optionally, be synchronized with the materials planning using a batch run. MM creates, updates, or deletes schedule lines based on the TM planning results. You can use the planning time fence to restrict the updates from TM to schedule lines. Updates are only allowed within the horizon fixed by an MRP run. Update of schedule lines based on TM planning results is possible only for freight units that are not assigned to a delivery. Once deliveries are created for freight units, no further update to schedule lines is possible.

Communication to Suppliers

MM creates releases, based on schedule lines then sends these releases to a supplier. This means that the information the supplier receives is based on TM planning results, rather than on material planning results.

Create Deliveries

MM creates or updates inbound deliveries based on an ASN received from the supplier. When a new delivery is saved, TM reassigns freight units, which are assigned to the related scheduling agreement, from the scheduling agreement to the inbound delivery.

Integration with JIT and nJIT

In a standard process, TM creates FUs for items in schedule lines of a scheduling agreement that is relevant for TM. However, if a purchase scheduling agreement has Kanban indicator Z and has been assigned to a control cycle with TM relevance, TM creates FUs based on the S2P JIT call regardless of the scheduling agreement's TM relevance.

For a control cycle to have TM relevance, its [Replenishment Strategy](#) field must have a TM control key. A Z in the Kanban field in the scheduling agreement indicates that the agreement is relevant for nJIT processing.

Communication between supplier and customer happens via ASNs. In the standard process, the supplier sends the buyer JIT call number (BTD type code 3018) as internal JIT Call number in the ASN. When the internal JIT Call number is not available, the outbound messages sends the external JIT Call number in the internal JIT Call number field.

In an nJIT-based scenario, however, only the internal, or buyer, JIT call number is used.

Restrictions

1. Acknowledgments of internal SA releases have no influence on which data is transferred to the TM system.
2. Replenishment with summarized JIT calls is out of scope.
3. Replenishment based on outbound sequenced JIT calls is out of scope.

4. Two-level goods receipts process is supported only in the Advanced Shipping and Receiving Process for agreements that are relevant for TM. For more information, see [Advanced Shipping and Receiving](#).

Integration Time Range for Purchase Scheduling Agreements

You can control how many schedule lines are included in the transportation process. This allows you to improve system performance when there are many schedule lines in an agreement. Once included in the process, Transportation Management (TM) can create and edit freight units (FUs) for the schedule lines. SAP refers to this as integrating the schedule line into TM.

To determine which schedule lines it must integrate, TM counts the number of schedule lines in a specified time range. Based on this count, it determines the time range for integration starting from today or in the recent past, and ending at a defined time in the future. It then integrates all schedule lines that are due for delivery in the time range for integration. You configure the start and end dates of both time frames dynamically in Customizing using rules and rule groups.

Prerequisites

- You have defined your rules and rule groups in Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Rule Groups to Determine Integration Time Range of SAGs►](#).
- You have indicated your preferred rule group for purchase scheduling agreements in Customizing for **Transportation Management** under [►Integration►Logistics Integration►Internal TM Component Integration►Define Logistics Integration Profile for SD, MM, and LE►](#).

Manually Changing the Integration End Date

If the settings in Customizing lead to an incorrect end date being determined for the integration time range, you can bring the end date forward using report /SCMTMS/UPDATE_INT_TIMERANGE. SAP recommends that you adjust the end date in Customizing before running the report. Running the report brings forward the end date of the integration time range.

Determining Time Ranges

When your rules and rule groups are configured, the system uses them to determine the start and end date of the following time ranges:

- Time range for schedule line count
- Time range for integration

Time Range for Schedule Line Count

The start date for the time range for schedule line count is always today's date. The end date is calculated by adding the number of days defined in the rule group to the start date.

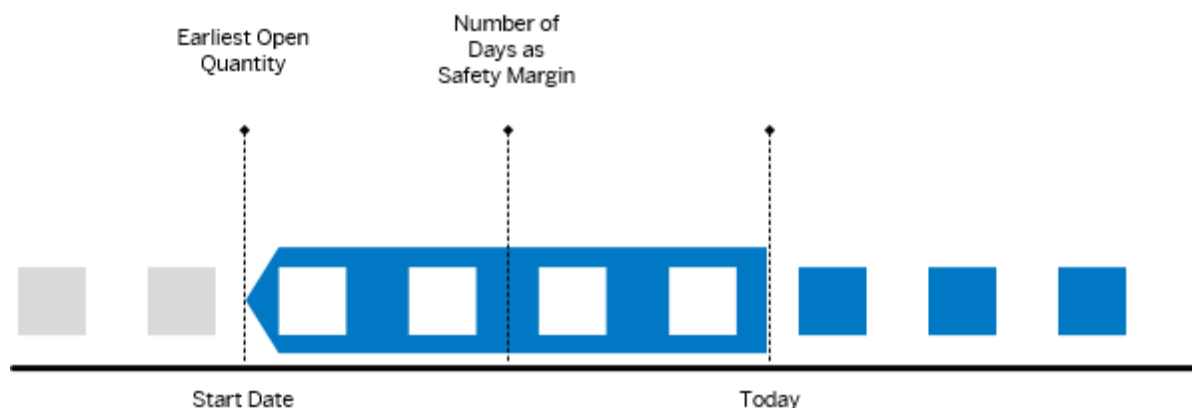
Time Range for Integration

Customizing enables you to set an integration time range for schedule lines. Integration time ranges are implemented through the determination of integration start and end dates, which in turn, determine or limit the number of freight units created within this time interval.

For every purchase agreement, the start date determination happens once when the purchase scheduling agreement is integrated and processed for the first time.

To determine the start date, the system takes into account open quantities at the schedule line level. The system locates the earliest schedule line with an open quantity and uses its date as the upper limit. If you have set a safety margin, the system chooses the earliest of the two dates.

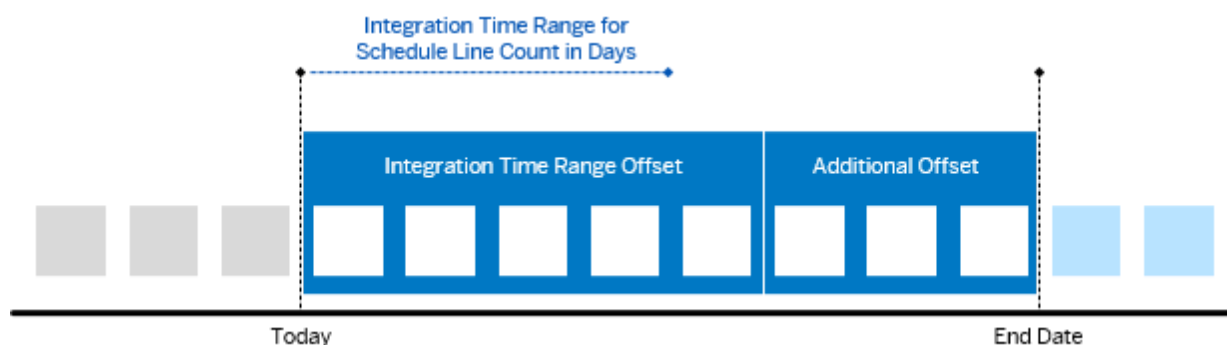
The start date is usually set before the current date and never in the future.



How the Start Date is Determined

To determine the end date, the system considers the number of schedule lines within the **Integration Time Range for Schedule Line Count**. Based on the number of schedule lines counted, the system identifies the corresponding rule and then sums the number of days comprised by the integration time range offset and the additional offset.

End date determination runs daily.



How the End Date is Determined

To establish the integration time range, first, you must customize your system so that it can determine start and end dates.

Setting an Integration Time Range: Rule Groups and Rules

Setting up your rule groups will enable the system to determine the start and end dates. To define your rule groups you will need the following:

- An integration time range offset in calendar days
- An additional offset type

The additional offset is an number of days added to the integration time range offset as a safety measure.

You can select between **Planning Time Fence** and **Planned Delivery Time in Days**. Alternatively, you can choose not to set an additional offset.

- A safety margin

This margin is set as the start date whenever there are no schedule lines with open quantity that predate the date determined by the safety margin, in which case, the start date is set back to the earliest of the two dates. The safety margin

always precedes the current date.

Rule groups consist of a set of rules. And to set up your rules, you will need the following:

- A minimum number of schedule lines
- The integration time range offset in days

Configuration

To configure the integration time range, you define rule groups and rules.

Rule Groups

In rule groups, you enter the following information:

Rule Group Configuration

Information	Description
Time range for schedule line count	TM adds the number of days entered here to the current date to determine the end date of the time range for schedule line count.
Additional offset for integration time range	This is applied on top of any offset defined in an individual rule. You can choose one of the following additional offset types: • No Additional Offset: No additional days are required. • Planning Time Fence: Material Master attribute visible on the MRP 1 tab which specifies, for a plant material combination, a period within which no automatic changes are to be made to the master plan. The planning time fence is defined in workdays. • Planned Delivery Time in Days: Material Master attribute visible on the MRP 2 tab which specifies, for a plant material combination, the number of calendar days needed to obtain the material or service if it's procured externally.
Safety Margin	This margin is set as the start date whenever there are no schedule lines with open quantity that predate the date determined by the safety margin, in which case, the start date is set back to the earliest of the two dates. The safety margin always precedes the current date.

→ Remember

Each rule group must have at least one rule with a minimum number of schedule lines equal to 0.

Rules

In each rule, you enter the following information:

Rule Configuration

Information	Description
-------------	-------------

Information	Description
Minimum number of schedule lines	When the system counts the number of schedule lines in the time range for schedule line count, it compares the count to this column. If the count is higher than a number in this column, it applies the relevant offset in the next column.
Offset for the integration time range	This is the number of days into the future that must be considered when integrating schedule lines. TM adds this number of days to the start date for integration to calculate the end date for integration.

Example

The following is an example of start and end date determination:

Start Date

The current date is 17 May. Your chose a safety margin of seven days so it falls on 10 May. The earliest schedule line with open quantity is 8 May. Because the earliest schedule line with open quantity precedes the safety margin, the system chooses 8 May as the start date.

End Date

Let's say you have selected **Planned Delivery Time in Days** as **Additional Offset Type**. The planned delivery time for this combination of plant and material is five days. And you have set the following rules:

Rule Group 1

Time Range for Schedule Line Count in Days	30
Additional Offset Type	Planned Delivery Time in Days
Number of Days as Safety Margin	7

Rule One

Minimum Number of Schedule Lines	0
Integration Time Range in Days	40

Rule Two

Minimum Number of Schedule Lines	6
Integration Time Range in Days	20

Rule Three

Minimum Number of Schedule Lines	21
Integration Time Range in Days	15

The determination would proceed as follows:

1. With a time rage for schedule line count of 30 days, the schedule lines within this 30-day time range are counted. Let's imagine this time range hold 10 schedule lines.

2. Next, the relevant rule is determined. Because there are more than six schedule lines, the system selects rule 2.
3. The integration time range offset in days for rule two is 20 days, and the additional offset, because of planned delivery time, is five days.
4. Therefore, the system determines the end date is 25 days in the future from the current day.

Integration of Purchase Scheduling Agreements without Inbound Delivery Integration

In an order-only integration scenario, freight units are created for items in a purchase scheduling agreement without inbound delivery integration. Since the freight units (FU) that are created for items in such purchase scheduling agreements are never transferred to a delivery, transportation execution must rely on the order-based FUs and the purchase scheduling agreement.

Unlike in delivery-based integration, order-based FUs are never transferred to the corresponding deliveries, even when they have been created.

To integrate existing purchase scheduling agreements based only on the order, you must consider how you wish to deal with order quantities that have already been delivered. You can do this by deciding on a start date and setting a reduction quantity.

Prerequisites

- You have defined transportation relevance of sales documents. For more information see Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal Component Integration > Define Transportation Relevance of Purchase Documents**.
- You have defined your rules and rule groups in Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration > Define Rule Groups to Determine Integration Time Range of Purchase SAGs**.
- You have indicated your preferred rule group for purchase scheduling agreements in Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration > Define Logistics Integration Profile for SD, MM, and LE**.

Key Features

Report **Activation of Purchase SAGs Without Delivery Integration** (/SCMTMS/ACTIVATE_MMSAG_WO_DLV) enables you to activate purchase scheduling agreements without integrating existing or future deliveries. It also enables you to indicate the integration start date and reduction quantity.

In an order-only integration scenario, the system does not reduce open quantity based on goods receipt or existing deliveries. However, if you decide to activate purchase scheduling agreements with existing deliveries into TM, you probably don't want to create freight units for order quantities that have been delivered and completely processed before the activation is done. For the remaining deliveries, you should decide whether order-based FUs should be built for the corresponding quantities in the agreement. This is done in the following way:

- While running the activation report, a start date is automatically determined. When the agreement is integrated and FUs are built, the report doesn't build FUs for order quantities from dates before the start date. The start date usually corresponds to the next date with order quantity for which goods receipt isn't yet booked.
- It might be that there are delivered quantities on or after the start date, either with or without goods receipt booked. As the system builds, by default, order-based FUs for all quantities after the start date you need to decide whether the system should ignore some of the already delivered quantities. In order to do this, you can choose a reduction quantity strategy when you integrate the purchase scheduling agreement. The reduction quantity dictates which quantities the system takes

into account when calculating open quantity for schedule lines. The reduction quantity influences the creation of FUs upon every integration. You can use one of the following strategies when running the report:

- Don't consider any deliveries
- Consider only goods receipt quantities at or after the start date as reduction quantity
- Consider all delivered quantities (goods receipt and deliveries) at or after the integration start date as reduction quantity
- Every time FUs are created for your purchase scheduling agreement the system checks whether a reduction quantity was set during activation. From the start date on, in that order, the system subtracts the reduction quantity before it builds FUs for the remaining quantity.
- Functionality regarding end date determination remains unchanged. In case your customizing allows the application of an integration rule and its corresponding rule groups, an end date is also set during every integration which prevents the creation of FUs for quantities beyond the end date.

Report [Adjustment of Integration of Purchase SAG Without Delivery Integration \(/SCMTMS/ADAPT_START_MM_WO_DLV\)](#) enables you to adjust the integration start date and reduction quantity for a purchase scheduling agreement without delivery integration. In the activation report, you can't change the start date and only have limited options for determining reduction quantity. However, in this report, you can freely adjust both the start date and the reduction quantity.

Related Information

[Integration of Purchase Scheduling Agreements](#)

[Integration Time Range for Purchase Scheduling Agreements](#)

Deactivation and Reactivation of Purchase Scheduling Agreement Integration

You can deactivate and reactivate the integration of purchase scheduling agreements (PSA) with integrated deliveries into Transportation Management (TM).

Prerequisites

- PSAs must have integrated deliveries.

Use

There are various scenarios in which you might want to deactivate or activate the transportation relevance of your document. For instance, when you deactivate a PSA because your supplier delivers the goods by the truckload without a separate transportation plan in TM. Or you want to transition from a PSA-based integration to an nJIT-based integration.

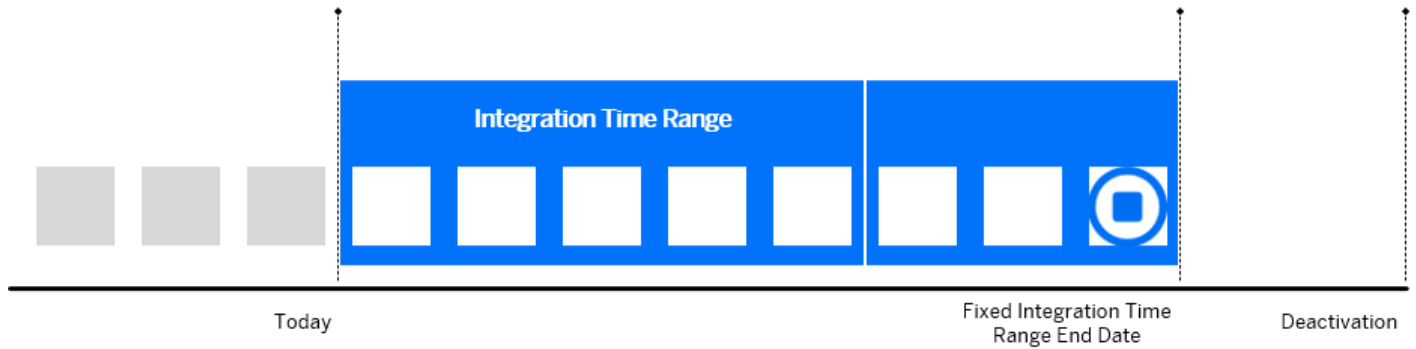
Note that the complete business document must be integrated. Item-specific integration is not supported.

Since existing PSAs are likely in execution and could have existing FUs, which could have already been planned, we recommend taking the following steps to avoid having duplicated FUs:

Establishing an Integration Time Range End Date

Before you deactivate the integration of PSAs, we recommend setting a fixed integration time range end date. The fixed end date ensures that all freight units (FU) are created up to but not beyond the fixed time point. That includes all FUs that had not yet been

created within the integration time range, but fall within the fixed end date. It is important to perform this step in advance to avoid planning or delivering schedule lines beyond the fixed end date.



To set an end date, use report [Setting and Fixing of Integration Time Range End Date for PSA](#) (/SCMTMS/FIX_INT_TIMERANGE).

Deactivating the Integration of Purchase Scheduling Agreements

Use report [Deactivation of Purchase Scheduling Agreements](#) (/SCMTMS/DEACTIVATE_MMSAG_W_DLV) to deactivate the integration of PSAs into TM.

If there are order-related freight units that have already been planned, the deactivation process will not be executed. Instead, the report will generate a list of the PSAs that have planned order-based freight units which are preventing the deactivation. You can proceed in one of the following ways:

- Create deliveries for the planned FUs
- Remove the freight units from your transportation plan
- Adjust the quantities on the relevant schedule lines, which leads to the deletion of the corresponding planned FUs
- In the report, choose to delete all planned order-based FUs without delivery assignment when deactivating the PSA

Additionally, the report can also be used to generate a list of PSAs that have not yet been deactivated and still have active control cycles.



Having ended the integration and deactivated your PSAs, you can proceed with the activation of nJIT integration. For more information about nJIT integration, see [Integration of Next Generation Just-In-Time](#).

Reactivating the Integration of Purchase Scheduling Agreements

Since previously deactivated PSAs are blocked for activation, they are not automatically reactivated upon integration of business documents. First, use report [Reactivation of Purchase Scheduling Agreements](#) (/SCMTMS/REACTIVATE_MMSAG_W_DLV) to remove the block set on deactivated PSAs. You can reactivate your PSAs as long as they are not assigned to an nJIT control cycle.



You must deactivate all PSAs used in the bill of materials (BOM) for the component group material linked to a control cycle, before you activate the control cycle. Likewise, you must deactivate all nJIT control cycles whose BOMs include materials in the PSAs you want to activate.

Activities

You can call the reports from the SAP Easy Access screen under ► **Logistics** ► **Transportation Management** ► **Administration** ► **Logistics Integration** ►.

- **Setting and Fixing of Integration Time Range End Date for PSA** (/SCMTMS/PSA_FIX_EDT)
- **Deactivation of Purchase Scheduling Agreements** (/SCMTMS/PSA_DEACTIVT)
- **Reactivation of Purchase Scheduling Agreements** (/SCMTMS/PSA_REACTIVT)

Related Information

[Integration of Purchase Scheduling Agreements](#)

[Integration Time Range for Purchase Scheduling Agreements](#)

[Integration of Next Generation Just-In-Time](#)

Integration of Sales Scheduling Agreements

You can integrate sales scheduling agreements from Sales and Distribution (SD) with Transportation Management (TM).

Delivery quantities and dates in a scheduling agreement are entered as schedule lines in a delivery schedule. The integration is based on delivery schedules and can be triggered by the creation or change of schedule lines.

With the Advanced Shipping and Receiving process, you can have enhanced integration between TM and Extended Warehouse Management (EWM). For more information, see [Advanced Shipping and Receiving](#).

Prerequisites

You have completed Customizing for integration of sales scheduling agreements. For more information, see Customizing for [Transportation Management](#) under [Integration](#) > [Logistics Integration](#) > [Internal TM Component Integration](#) > [Define Transportation-Relevance of Sales Documents](#).

Features

Create or Update Schedule Lines

The customer uses an MRP run or manual planning to create, change, or delete schedule lines of a Materials Management (MM) scheduling agreement.

Once, the customer transfers changes to the supplier, the sales scheduling agreement and schedule lines are updated. TM then creates, changes, or deletes freight units (FU) accordingly.

Freight Unit Building

TM builds freight units (FU) based on the relevant schedule line requirements and customizing. There are multiple ways in which you can arrange the creation of FUs. For example, when only JIT delivery schedules are designated relevant, TM only creates FUs for scheduled deliveries within the JIT horizon.

You set the freight unit building rules in your logistics integration profile. For more information, see [Logistics Integration Profile for SD, MM, and LE](#).

Create Deliveries

You create outbound deliveries directly from the sales or freight order document. When you save a new delivery, TM reassigns freight units from the scheduling agreement to the outbound delivery. Subsequently, data is transferred to an inbound delivery on the customer side.

Integration of Orders

You can also integrate existing sales scheduling agreements into TM. Integration into TM requires the creation and assignment of a logistics integration profile with order type, sales organization, distribution channel, and division. You can always reverse the integration of a scheduling agreement while no deliveries have been created.

For more information, see the corresponding customizing activity under [Prerequisites](#) above.

SACO

Transaction SACO allows you to copy schedule lines together with header and item data from an existing sales scheduling agreement to a new sales scheduling agreement. You can use the transaction to improve performance by reducing the number of documents referenced in the document flow.

Related Information

[Integration of Purchase Scheduling Agreements](#)

Integration Time Range for Sales Scheduling Agreements

You can control how many schedule lines are included in the transportation process. This allows you to improve system performance when there are many schedule lines in an agreement. Once included in the process, Transportation Management (TM) can create and edit freight units (FUs) for the schedule lines. SAP refers to this as integrating the schedule line into TM.

TM counts the number of schedule lines in a defined time range. Based on this count, it determines a second time range starting from today or in the recent past, and ending at a certain time in the future. It then integrates all schedule lines that are due for delivery in this time range. You configure the rules that determine when each period starts and ends in Customizing.

Prerequisites

- You've defined your rules and rule groups in Customizing for Transportation Management under [Integration > Logistics Integration > Internal TM Component Integration > Define Rule Groups to Determine Integration Time Range of SAGs](#).
- You've indicated your preferred rule group for sales scheduling agreements in Customizing for Transportation Management under [Integration > Logistics Integration > Internal TM Component Integration > Define Logistics Integration Profile for SD, MM, and LE](#).

Manually Changing the Integration End Date

If the settings in Customizing lead to an incorrect end date being determined for the integration time range, you can bring the end date forward using report [Prepone End Dates for Items in Scheduling Agreements](#) (/SCMTMS/UPDATE_INT_TIMERANGE). SAP recommends that you try to adjust the end date determination in Customizing before running the report.

Determining Time Ranges

When your rules and rule groups are configured, the system uses them to determine the start and end date of the following time ranges:

- Time range for schedule line count
- Time range for integration

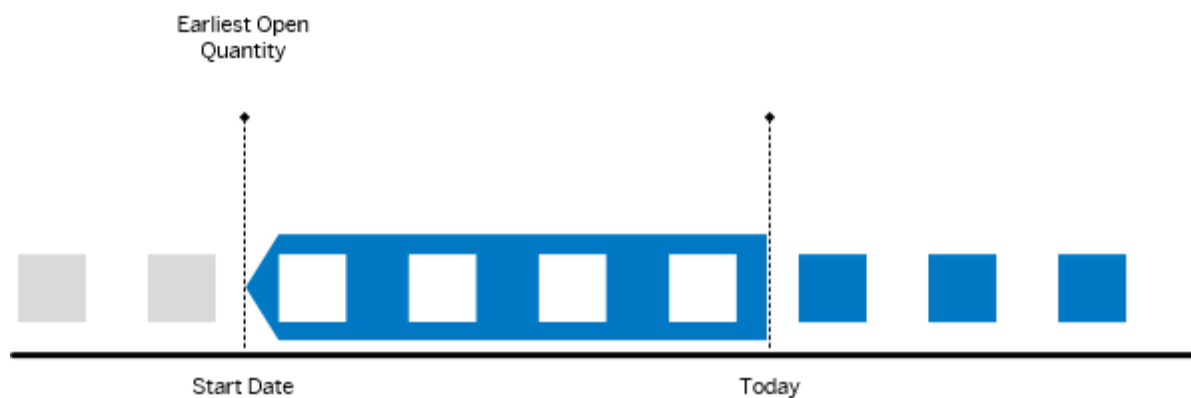
Time Range for Schedule Line Count

The start date for the time range for schedule line count is always today's date. The end date is calculated by adding the number of days defined in the rule group to the start date.

Time Range for Integration

To determine the start date, the system considers open quantities at the schedule line level. The system locates the earliest schedule line with an open quantity and uses its date as the upper limit. If there's no open schedule line quantity before today's date, it uses today's date instead.

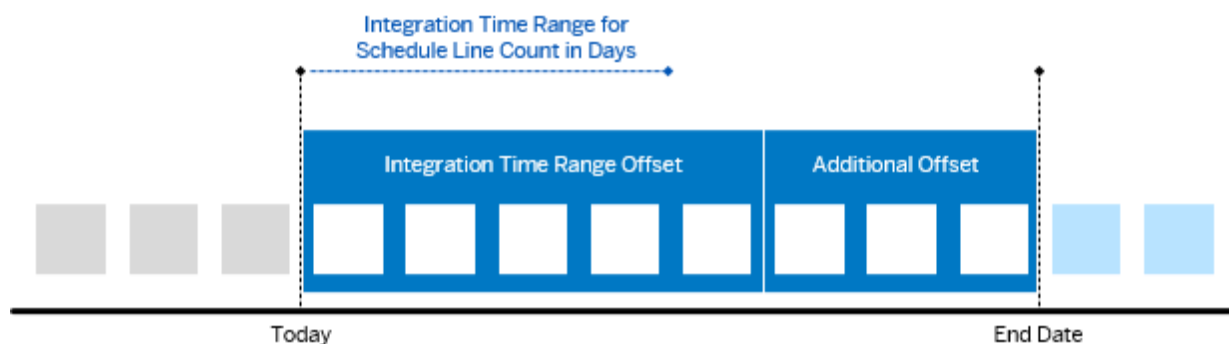
For every scheduling agreement, the start date determination happens once when the scheduling agreement is integrated and processed for the first time.



How the Start Date is Determined

The system calculates the start date by counting the number of schedule lines within a specified time range. Based on the number of schedule lines counted, TM identifies the corresponding rule. It adds the integration time offset in the rule to the additional offset in the rule group. It adds this sum to the current date to calculate the end date.

End date determination runs once per day, when you first update a scheduling agreement. If you change the scheduling agreement again on the same day, TM doesn't recalculate the end date.



How the End Date is Determined

Configuration

To configure the integration time range, you define rule groups and rules.

Rule Groups

In rule groups, you enter the following information:

Rule Group Configuration

Information	Description
Time range for schedule line count	TM adds the number of days entered here to the current date to determine the end date of the time range for schedule line count.
Additional offset for integration time range	This is applied on top of any offset defined in an individual rule. You can choose one of the following additional offset types: • No Additional Offset: No additional days are required. • Planning Time Fence: Material Master attribute visible on the MRP 1 tab which specifies, for a plant material combination, a period within which no automatic changes are to be made to the master plan. The planning time fence is defined in workdays. • Planned Delivery Time in Days: Material Master attribute visible on the MRP 2 tab which specifies, for a plant material combination, the number of calendar days needed to obtain the material or service if it's procured externally.
Safety Margin	i Note The safety margin isn't relevant for sales scheduling agreements. For more information, see the Rule Group Configuration table in Integration Time Range for Purchase Scheduling Agreements.

Rules

In each rule, you enter the following information:

→ Remember

Each rule group must have at least one rule with a minimum number of schedule lines equal to 0.

Rule Configuration

Information	Description
Minimum number of schedule lines	When the system counts the number of schedule lines in the time range for schedule line count, it compares the count to this column. If the count is higher than a number in this column, it applies the relevant offset in the next column.
Offset for the integration time range	This is the number of days into the future that must be considered when integrating schedule lines. TM adds this number of days to the start date for integration to calculate the end date for integration.

Example Implementation

The following is an example of start and end date determination:

Rule Group One

Time Range for Schedule Line Count in Days	30
--	----

Additional Offset Type	Planned Delivery Time in Days
------------------------	-------------------------------

Rule One

Minimum Number of Schedule Lines	0
Integration Time Range in Days	40

Rule Two

Minimum Number of Schedule Lines	6
Integration Time Range in Days	20

Rule Three

Minimum Number of Schedule Lines	21
Integration Time Range in Days	15

Start Date

The current date is 17 May. The earliest schedule line with open quantity is 8 May. Because the earliest schedule line with open quantity precedes the current date, the system chooses 8 May as the start date.

End Date

You've selected **Planned Delivery Time in Days** as the **Additional Offset Type**. The planned delivery time for this combination of plant and material is five days. The system adds five days to any rule defined in the rule group.

Determination

The determination proceeds as follows:

1. With a time range for schedule line count of 30 days, the schedule lines within this 30-day time range are counted. In this example, this time range holds 10 schedule lines.
2. TM determines the relevant rule. Because there are more than six schedule lines but fewer than 21, the system selects rule two.
3. The integration time range offset in days for rule two is 20 days, and the additional offset, because of planned delivery time, is 5 days.
4. Therefore, the system determines the end date is 25 days in the future from the current day to calculate 11 June.

Integration of Stock Transport Scheduling Agreements

A stock transport scheduling agreement is an internal (or intracompany) outline agreement containing delivery quantities and dates within a specified period. These are then entered as schedule lines in a delivery schedule.

Stock transport scheduling agreements cover internal procurement processes, such as when a production plant receives components from another plant within the same company. The requirements of such an agreement are defined by a material requirement planning (MRP) run.

You can use stock transport scheduling agreements in an Advanced Shipping and Receiving (Advanced SR) process. For more information see [Advanced Shipping and Receiving](#).

i Note

- Freight units (FU) can be assigned to either an Advanced SR or other warehouse or storage location. Combining FUs that are relevant and not relevant for Advanced SR in a single outbound order is not supported.
- In a stock transport scheduling agreement scenario, TM calculates open quantities based on schedule lines. The open quantity calculation is identical to that in external procurement.
- As with stock transport orders, TM considers receiving plants and their Advanced SR relevance.

Prerequisites

- Delivery Management and TM must run on the same system.
- Optionally, you have activated the improved location determination. To do this, select the **Improved Location Determination** checkbox under the **Order-Related Settings** section.

For more information, see Customizing for **Transportation Management** under [Integration > Logistics Integration > Internal TM Component Integration > Define Logistics Integration Profile for SD, MM, and LE](#).

- You have defined the control keys that determine system behavior after integration.

For more information, see Customizing for **Transportation Management** under [Integration > Logistics Integration > Internal TM Component Integration > Define Control Keys for Document Integration](#).

- You have assigned control keys to the required document types to determine whether purchase documents are relevant for TM. Optionally, you can assign a logistics integration profile.

For more information, see Customizing for **Transportation Management** under [Integration > Logistics Integration > Internal TM Component Integration > Define Transportation-Relevance of Purchasing Documents](#).

- You have defined a profile for direct integration with Extended Warehouse Management (EWM) or an external warehouse management system.

For more information, see Customizing for **Transportation Management** under [Integration > Integration with Extended Warehouse Management > Define EWM Integration Profile](#).

Sample Process

The following are the possible steps in a standard process:

1. Creating a stock transport scheduling agreement

In Material Management (MM), run transaction ME37 to create a stock transport scheduling agreement.

After, run transaction ME38 and fill in the scheduling agreement's delivery schedule. Enter schedule lines with quantity and delivery dates.

i Note

You must have set the confirmation control key at item level for the inbound delivery to receive a confirmation of the delivery date when the advanced shipping notification is sent. For more information, see the [Prerequisites](#) section above.

2. Creating freight units

Step 1 triggers the automatic creation of FUs in TM. You can open your [Freight Units Worklist](#) app to check that the FUs for the delivery schedules of the stock transport scheduling agreement have been created.

Additionally, go to the [Stages](#) tab in the FU and check the location determination for source and destination location is correct.

3. Planning transportation

You can assign the FU to a freight order (FO) in the [Transportation Cockpit](#) or the [Create Road Freight Order](#) apps.

4. Performing carrier subcontracting

Optionally, either select a carrier or tender the FO. For more information about tendering, see [Freight Tendering](#).

5. Changing the freight order status

In your FO, set the execution status to [Ready for Transportation Execution](#).

6. Editing the outbound delivery

Optionally, proceed with packaging the goods.

Insert the storage location first.

Pick the outbound delivery.

7. Changing the freight order status

In your FO, report the truck's arrival, check-in, check-out and departure at and from the supplying plant accordingly.

8. Posting goods issue

In Logistics Integration (LE), post goods issue.

The system creates the corresponding inbound delivery and reassigns the FUs automatically.

9. Managing the freight order

In the FO, check the truck in at the destination location.

If you are in the Advanced SR process, select [Set Ready for Warehouse Execution](#) to report the truck's departure from the destination location.

10. Making changes to the schedule lines

Adjust the schedule lines by changing the delivery date or quantity or by deleting or adding schedule lines.

Check whether the FU, FO, and outbound delivery have been updated or created accordingly.

Integration of Sales Document Scheduling

You can use **Transportation Requirement Routing and Scheduling (TRS)** when calculating the delivery dates of sales orders and scheduling agreements in Sales and Distribution. The scheduling of sales documents uses the determination capabilities of BPS, which rely on Transportation Management (TM) for duration determination and scheduling logic.

Prerequisites

- You've defined either a transportation duration determination profile or a requirement routing profile based on the scenario you want to use:

- Sales document scheduling based on TM Durations

You've defined a transportation duration determination profile.

In this profile, you can indicate one or more routing methods and the sequence in which the system performs them. You can also indicate a planning profile if you want to use the **Fixed Loading** durations and durations for **Prepare** and **Finalize** actions that you've defined there. To create a profile, open the **Create Transportation Duration Determination Profile** app from the launchpad.

i Note

- If the system determines durations using one routing method, these durations aren't changed by any subsequent routing method that you indicated.
- The system uses the default means of transport for determining the loading duration.

- Sales document scheduling based on TM Scheduling

You've defined a requirement routing profile.

In this profile, you can indicate one or more routing methods and the sequence in which the system is to perform them. Further, you can define additional parameters that influence routing and scheduling. You can also indicate a planning profile if you want to use the **Fixed Loading** durations and durations for **Prepare** and **Finalize** actions that you've defined there. To create a profile, open the **Create Requirement Routing Profile** app from the launchpad.

- You've defined a control key and indicated your TRS profile, that is, either a transportation duration determination profile or a requirement routing profile.

For more information, see Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal Component Integration > Define Control Key for Document Integration**.

- (Optional) You've defined the transportation relevance of sales documents.

For more information, see Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal Component Integration > Define Transportation-Relevance of Sales Documents**.

i Note

Sales Order Scheduling

Sales order scheduling only supports sales orders of the following sales document categories:

- Order
- Order Without Charge

You can use the scheduling capabilities of TM without having integrated your sales document.

For more information, see Customizing for **Sales and Distribution** under **Sales > Sales Documents > Sales Document Header > Define Sales Document Types**.

Use

When a customer places an order, the system first tries to deliver on the requested date. It does this through backward scheduling, working backward from that date to determine when the goods need to be shipped to arrive on time. However, if that delivery date can't be met, for example, because there isn't enough stock and part of the order is on backorder, the system switches to forward scheduling. In this case, it starts from today and calculates the earliest possible delivery date based on when the products are available.

Process

1. Create a sales order using [Create Sales Order](#) (transaction VA01) or a scheduling agreements using [Create Scheduling Agreement](#) (VA31).
2. Perform scheduling.

i Note

Sales Order Scheduling based on TM Durations

Press **ENTER** after entering the material and the related details to run the scheduling and receive a delivery date.

Sales Order Scheduling based on TM Scheduling

Choose [Item Availability Check](#) after selecting an item to register the entry so that the scheduling happens when you save the document. Alternatively, choose either [Document Check](#) or [Group Check](#) to receive a delivery date. In either case, the system performs a group check when you save the document.

3. After scheduling, the system displays the [Review Availability Check Result \(RACR\)](#) screen, where you can see the durations of the different stages as well as the determined means of transport and vehicle types.

For more information about the [RACR](#) screen, see [Review Availability Check Result](#).

4. Save the sales document.

Saving triggers the creation of freight units, and the delivery date is passed on.

i Note

If you change a sales document, SAP S/4HANA recalculates the delivery date.

Integration of Purchasing Document Scheduling

You can use **Transportation Requirement Routing and Scheduling (TRS)** when calculating the delivery dates of stock transport orders (STO) and scheduling agreements in Materials Management (MM). The scheduling of these documents uses the determination capabilities of BPS, which rely on Transportation Management (TM) for duration determination and scheduling logic.

Prerequisites

Prerequisites in Transportation Management

- In Customizing activity [Define Control Key for Document Integration](#), you've defined a control key and selected [transportation duration determination](#) as your TRS profile.

For more information, see Customizing for [Transportation Management](#) under [Integration > Logistics Integration > Internal TM Component Integration > Define Control Key for Document Integration](#).

- You've defined a transportation duration determination profile.

In this profile, you can specify one or more routing methods and the sequence in which the system processes them. You can also assign a planning profile if you want the system to use the [Fixed Loading](#) durations and durations for the activities [Prepare](#) and [Finalize](#) defined there.

To create a profile, open the [Create Transportation Duration Determination Profile](#) app from the launchpad.

i Note

If the system determines durations using one routing method, it doesn't overwrite them with results from any later methods defined in the sequence. The system determines the loading duration based on the default means of transport.

Prerequisites in ATP

- You've activated advanced Available-To-Promise (aATP).

For more information about activating aATP, see Customizing for [Cross-Application Components](#) under [Advanced Available-to-Promise \(aATP\) > Define Availability Checking Groups > Define Availability Checking Group and Behavior](#).

Prerequisites in Business Process Scheduling (BPS)

- You've activated BPS for STO and scheduling agreement scheduling.

For more information, see Customizing for [Cross-Application Components](#) under [Advanced Available-to-Promise \(aATP\) > Business Process Scheduling \(BPS\) > Activate BPS for Stock Transport Documents](#).

- In the Customizing for [Transportation Management](#) under [Basic Functions > General Settings > Business Process Scheduling \(BPS\) > Maintain BPS Schema](#), you have created a valid BPS schema.

SAP provides template ST0CK_TRANSP_TM_MD_SECS (Stock Transport Scheduling based on TM Durations) that you can use to create a valid BPS schema.

For more information about activity attributes, see [Activity Attributes for Business Process Scheduling \(BPS\)](#).

→ Remember

- TM and MM must operate in a single system.
- The function provides an estimate for relevant documents only. You can establish the relevance in Customizing.
- The TRS settings must match the settings of the corresponding TM control key.
- The choice of scheduling can impact system performance. Using BPS scheduling can result in longer runtimes.

Use

When you initiate a stock transfer to another storage location, the system attempts to organize the delivery for the requested delivery date. If it can't, for example, due to missing inventory, it schedules the delivery for the next available date.

Business Process Scheduling (BPS) performs the actual scheduling. It uses TM and ATP data to determine when the stock transport order can be delivered and sets the date in the order. If you change the stock transport order, the system recalculates the delivery date.

How Scheduling Works

Scheduling takes both the requested delivery date and material availability into account when determining the delivery schedule.

TM provides BPS with information about the durations of loading, transport, and unloading. To calculate the loading, transport, and unloading durations, it uses the routing methods that you selected in the TRS profile. For more information, see [Determination of Transportation Durations for BPS-Based Scheduling](#).

ATP provides information about the availability of materials. For more information about the scheduling process with ATP, see [Use BPS with Available-to-Promise, Sales, Stock Transport, and Outbound Delivery](#).

To calculate whether it can meet the requested date, the system uses backward scheduling. During backward scheduling, the system works backwards from the requested delivery date to determine the latest possible shipping time. It does this by considering how long each activity in the process takes and whether materials are available at the calculated times.

If the requested delivery date can't be met, the system uses forward scheduling. During forward scheduling, the system starts from the current date and calculates the earliest possible delivery date based on the material availability and transportation durations.

Customizing Input Data

You can implement BAdI /SCMTMS/IF_LI_TRS_BTDC_CHG (Input Data Changes for TRS Functionality) to adapt input data of business transaction documents arriving in TM before the scheduling and routing starts.

Business Process Scheduling

Business Process Scheduling (BPS) is a flexible, configurable scheduling solution used in various business documents. BPS relies on its BPS schema to estimate delivery dates. These schemas model the process from planning through to unloading.

For more information about BPS, see [Business Process Scheduling \(BPS\)](#).

For more information about BPS schemas, see [Defining Business Processes \(Schemas\)](#).

For more information about BPS determining attributes, see [Determining Attributes](#).

Process

1. Create a stock transport order using [Create Purchase Order \(ME21N\)](#) or scheduling agreement using [Create Scheduling Agreement \(ME37\)](#).

! Restriction

Scheduling doesn't work if you create the documents in a different app or transaction.

2. Save your purchasing document.

The system sets the calculated delivery date in the order.

Integration of Intercompany Business Processes

Transportation Management (TM) supports intercompany business processes in Sales and Distribution.

TM supports the following processes:

[Transportation Management in Advanced Intercompany Business Processes](#)

Transportation Management (TM) supports the business processes advanced intercompany sales and advanced intercompany stock transfer.

[Transportation Management in Multistage Intercompany Business Processes](#)

Transportation Management (TM) supports multistage intercompany sales and multistage intercompany stock transfer by enabling integrated stage building, harmonized Incoterm handling, and automated transfer of control management across entities in the logistics chain.

Transportation Management in Advanced Intercompany Business Processes

Transportation Management (TM) supports the business processes advanced intercompany sales and advanced intercompany stock transfer.

TM supports these advanced intercompany business processes in the following ways:

- One freight unit is created across the various business documents.
- Incoterm handling is harmonized between Sales, Sourcing and Procurement, Logistics Execution and TM.
- You use the Incoterm locations to build freight unit stages in TM.
- TM sets the transfer of control dates and times in the Logistics Execution delivery (LE delivery), based on transportation planning and actual events.
- Freight cost allocation posting is enabled for the advanced intercompany sales process and advanced intercompany stock transfer process. For more information, see [Posting of Allocated Freight Costs in Advanced and Multistage Intercompany Business Processes](#).

Prerequisites

- To use TM as part of the advanced intercompany sales process and advanced intercompany stock transfer process, you first need to set these processes up. For further information, see [Prerequisites for Advanced Intercompany Sales](#) and [Prerequisites for Advanced Intercompany Stock Transfer](#).
- Set up your TM processes, such as making settings in the Customizing activity [Define Logistics Integration Profile for SD, MM, and LE](#).

Advanced Intercompany Sales

The process for advanced intercompany sales provides functions that allow two affiliated companies to participate in a single intercompany process.

A sales organization assigned to the selling company creates a standard sales order, ordering goods from a plant assigned to another company code, which acts as the delivering company. To keep financial and material ledger postings in sync, a purchase order is created in the selling company that triggers the creation of an intercompany sales order in the delivering company. The delivering company delivers the goods to the customer. The delivering company invoices the selling company, and the selling company invoices the customer.

A standard sales order can initiate the entire process, which is integrated into value chain monitoring. For the transfer of control of goods between the affiliated companies and to the customer, the system uses valuated stock in transit.

Advanced Intercompany Stock Transfer

A purchasing organization assigned to the receiving company code creates an intercompany purchase order, ordering goods from a plant assigned to another company code.

The plant in the delivering company code delivers the goods to the plant for which the purchasing organization ordered the goods.

Because the two companies codes balance their accounts independently and have to fulfill legal requirements, the delivering company must bill the receiving company for the goods. This internal billing transaction is carried out by means of an

intercompany billing document, that is, an intercompany customer invoice. Based on the intercompany customer invoice, the system automatically posts an intercompany supplier invoice.

Incoterms and Incoterm Locations

International commercial terms (Incoterms) are a set of international rules for the interpretation of the commonly used terms in international trade. An Incoterm defines the terms of sale and the passing of risks for import and export of goods. The International Chamber of Commerce publishes Incoterms and specifies which Incoterms are allowed. You define Incoterms in business partners and business partner info records.

Incoterms describe the point at which the following responsibilities are passed from one party to another:

- Planning

Planning responsibility means that the party organizes the transport.

- Cost

Cost responsibility means that the party has the obligation to pay the costs of the transport.

- Risk

Risk responsibility means that the party bears the risk for the respective transport section.

You define Incoterm locations at which these responsibilities are transferred from one party to another. Incoterms can have two locations. Planning and cost responsibility are transferred for all Incoterms at Incoterm location 1. For risk responsibility, the system first tries to determine the transfer of control at Incoterm location 2. If Incoterm location 2 is not maintained, the system transfers risk at Incoterm location 1.

As a result, for non-C-clauses, risk responsibility is transferred at Incoterm location 1 because these clauses typically do not maintain Incoterm location 2. For C-clauses such as CIF, CIP, CFR, and CPT, risk responsibility is transferred at Incoterm location 2, provided that this location is maintained.

Some Incoterms are valid without a location 1 or location 2. The following table shows these Incoterms and the default Incoterm location 1, which is used if no locations are entered:

Default Incoterm Location 1

Incoterm	Default Incoterm Location 1
Delivered At Place (DAP)	Destination
Delivered Duty Paid (DDP)	Destination
Delivered at Place Unloaded (DPU)	Destination
Ex Works (EXW)	Start
Free Carrier (FCA)	Start

→ Recommendation

SAP recommends using Incoterms in accordance with their intended commercial meaning, as system behavior partially takes Incoterm semantics into account. For example, for Incoterm CIF, Incoterm location 1 is related to the port of discharge, while Incoterm location 2 is related to the port of loading. In such cases, both Incoterm locations should be maintained and should contain a different location. Otherwise, a different Incoterm may be more appropriate for the intended purpose.

Transfer of Control

Transfer of control dates refer to the date at which the risk responsibility is transferred between parties. For internal and external transfer of control, there's planned and actual transfer of control. The transfer of control dates and times in the LE delivery are set by TM, based on transportation planning and actual events. You can find the planned and actual internal and external transfer of control dates and times in the **Shipment** tab in the LE delivery. For more information, see [Transfer of Control of Goods \(Sales\)](#) and [Transfer of Control of Goods \(Stock Transfer\)](#).

TM registers the actual transfer of control date when you confirm that a certain transportation activity has happened, for example, Loading End or Arrived. The required activity depends on the Incoterm. The following table shows the Incoterms and the transportation activity required to save the actual transfer of control date:

Transportation Activities for Transfer of Control

Incoterm	Status
Cost, Insurance and Freight (CIF)	Loading End
Carriage and Insurance Paid To (CIP)	Arrived
Cost and Freight (CFR)	Loading End
Carriage Paid To (CPT)	Arrived
Delivered At Place (DAP)	Arrived
Delivered At Terminal (DAT)	Arrived
Delivered Duty Paid (DDP)	Arrived
Delivered at Place Unloaded (DPU)	Unloading End
Ex Works (EXW)	Loading Begin
Free Alongside Ship (FAS)	Arrived
Free Carrier (FCA)	Loading End
Free On Board (FOB)	Loading End

Configuring Transfer of Control

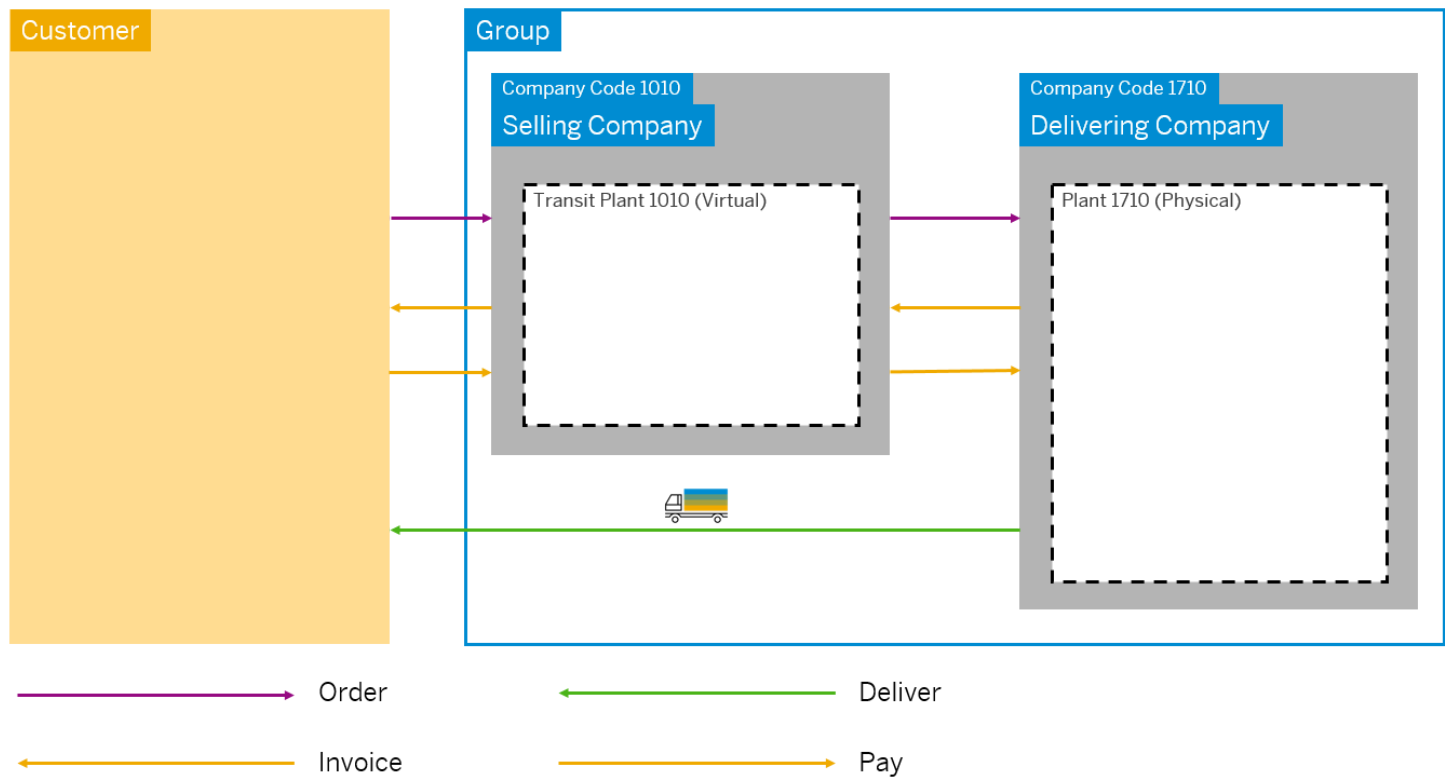
The values in the [Transportation Activities for Transfer of Control](#) table are the default values. When you perform a transportation activity, you must select the corresponding activity in the freight document. If you do not, the actual transfer of control date will not be saved. You can change the required transportation activity in table /SCMTMS/V_INCVST. Access the table using display mode in [Table View Maintenance](#) (transaction SM30) and make a new entry for the Incoterm.

Stage Building

Transportation stages show the stops (start/end locations) of a freight unit or a freight order. Stages indicate that certain locations must be visited in a certain sequence, or that certain transportation activities such as loading or unloading must be performed.

TM considers Incoterms during transportation planning for the purposes of stage building. TM uses Incoterm locations to build freight unit stages.

Example of an Advanced Intercompany Sales Process



A local sales office in Germany (as part of selling company 1010) receives a purchase order from a customer and creates a sales order. The sales order contains all relevant information for selling company 1010 to organize the transport and to deliver the sold goods to the customer in Walldorf, Germany.

This information comprises the following:

- Plant/shipping point (source location of the transport)
- Customer
- Ship-to (final recipient/final destination location of the transport)
- Incoterm and Incoterm location(s)
- Material (goods sold)
- Material availability date and requested delivery date

In this example, the sales order contains Incoterm DDP (Delivery Duty Paid). This means that the group is responsible for organizing the whole transportation route. In the sales order, Incoterm location 1 (in this case, the customer's address) is relevant for stage building in TM. The group carries the planning and cost responsibility for transporting the goods to the customer's address in Walldorf.

During sourcing, it is determined that the material shall be shipped from a plant in Palo Alto, California, which belongs to a company code different from the company code of the selling company in Germany.

An additional intercompany purchase order and intercompany sales order are created. The intercompany purchase order and intercompany sales order carry all information relevant for selling company 1010 and delivering company 1710 to (jointly) organize the transport and to deliver the sold goods to the customer.

This information comprises the following:

- Supplier (determines the delivering company)
- Plant (virtual plant which is only used for stock in transit postings)

- Incoterm and Incoterm location(s)
- Material (goods procured from delivering company)
- Loading date and delivery date

In this example, the Incoterm on the intercompany purchase order is CIF (Cost, Insurance and Freight), which means that delivering company 1710 is responsible for organizing the transport section from the plant to the port of discharge in Hamburg, Germany, while selling company 1010 is responsible for organizing the transport section from the port of discharge to the customer in Walldorf.

In the intercompany sales order, Incoterm location 1 (here the port of discharge) is relevant for stage building in TM.

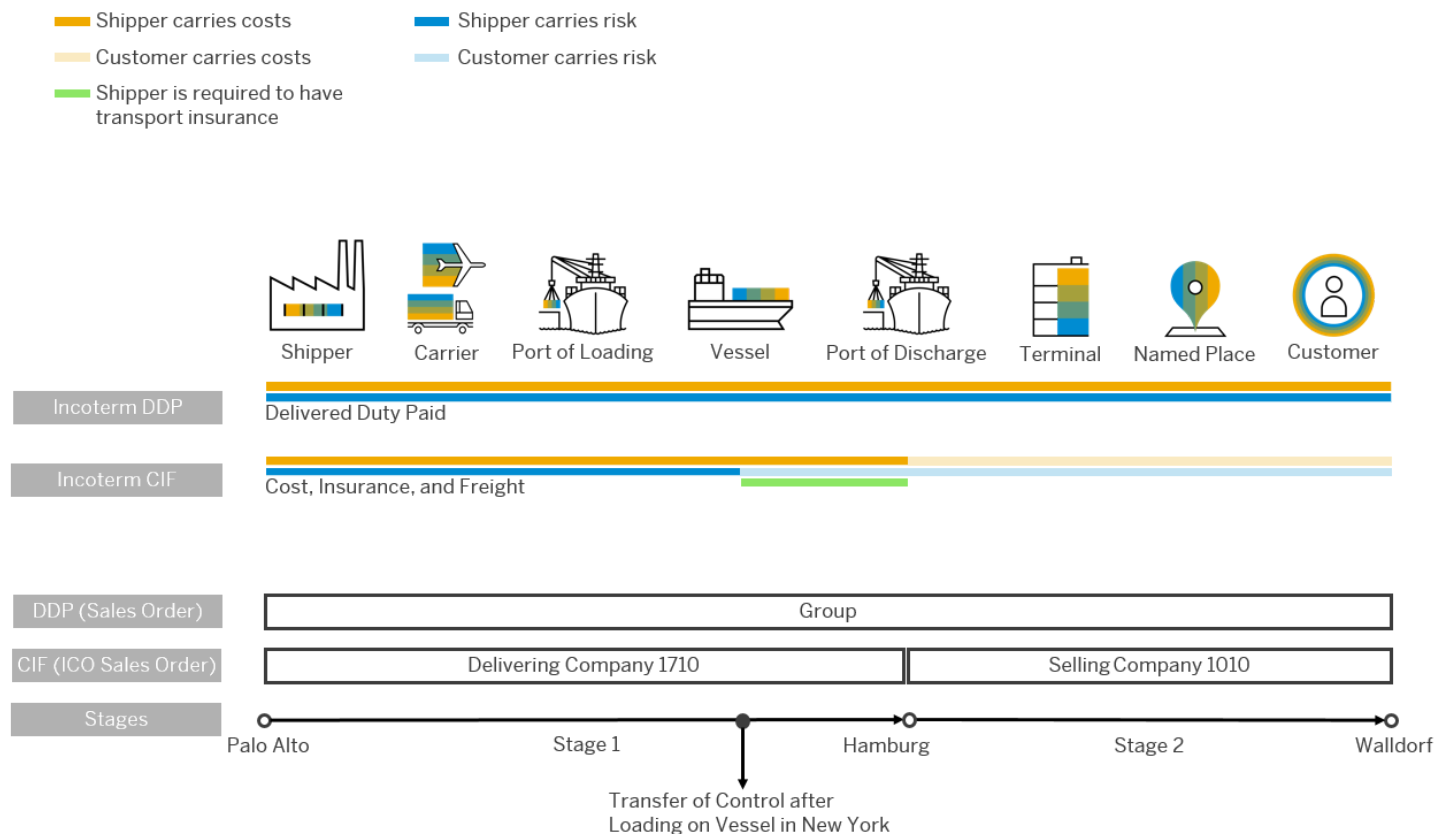
Consequently, two freight unit stages are created:

1. The first stage from the plant in Palo Alto to the port of discharge in Hamburg (Incoterm location 1 in the intercompany sales order), performed by delivering company 1710
2. The second stage from the port of discharge in Hamburg to the customer in Walldorf (Incoterm location 1 in the sales order), performed by selling company 1010

As the Incoterm in the intercompany sales order is CIF in this case, there is an internal transfer of control at Incoterm location 2. This is when the goods are loaded on to the vessel in New York. At this point, the risk responsibility is transferred from delivering company 1710 to selling company 1010. While the goods are on the vessel and until they reach the port of discharge in Hamburg, delivering company 1710 is still responsible for planning and cost, but no longer for risk.

The port of discharge is Incoterm location 1 for the intercompany sales order, so the remaining responsibilities of delivering company 1710 end there, and selling company 1010 takes over responsibility for planning and cost.

There is an external transfer of control when the goods are delivered to the customer at Incoterm location 1 for the sales order, which in this case is the customer's address in Walldorf. The responsibilities of selling company 1010 end there.



Related Information

[Posting of Allocated Freight Costs in Advanced and Multistage Intercompany Business Processes](#)

[Advanced Intercompany Sales](#)

[Advanced Intercompany Stock Transfer](#)

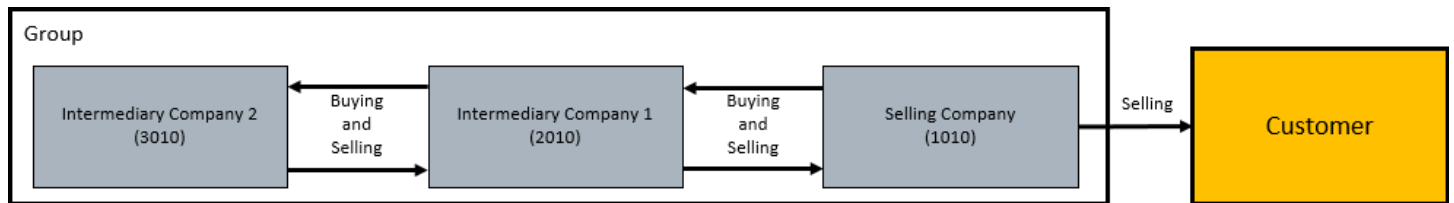
Transportation Management in Multistage Intercompany Business Processes

Transportation Management (TM) supports multistage intercompany sales and multistage intercompany stock transfer by enabling integrated stage building, harmonized Incoterm handling, and automated transfer of control management across entities in the logistics chain.

Multistage intercompany business processes allow you to manage and monitor complex financial chains and chains of sales documents. The following processes are available:

- Multistage intercompany sales allows you to sell products to customers that you don't sell or that you don't have in stock but that are available in other plants in your group.
- Multistage intercompany stock transfer allows you to transfer products from one plant to another in a different company code.

Multistage intercompany business processes allow for a 1-N-1 scenario, where you sell or transfer a product to a company but the product is sent along the financial chain through any number of organizations on the way to the recipient. For example, a 1-1-1 scenario has 1 sender, 1 recipient, and 1 intermediary company. The following image shows a simplified 1-2-1 multistage intercompany sales process as an example.



TM creates the resulting logistics chain. The logistics chain is decoupled from the financial chain and the goods are delivered from the final intermediary company to the receiving party. For more information about multistage intercompany business processes, see [Multistage Intercompany Sales](#) and [Multistage Intercompany Stock Transfer](#).

Prerequisites

1. To use TM as part of multistage intercompany business processes, you first need to set up these processes. For further information, see [Prerequisites for Advanced Intercompany Sales](#) and [Prerequisites for Advanced Intercompany Stock Transfer](#).
2. Ensure that TM is running in an embedded scenario.
3. Define TM integration profiles and freight unit building rules for order-based or delivery-based integration.
4. Maintain Incoterms and Incoterm locations in business partner master data and in the business partner info records.

Functions

TM supports multistage intercompany business processes in the following ways:

- Builds freight units with all transport-relevant stages based on Incoterms and Incoterm locations.
- Derives and manages planned and actual transfer of control (ToC) dates for each handover location and writes them back to outbound deliveries.
- Enables TM-driven freight cost allocation per responsible entity for each stage. For more information about settlement, see [Posting of Allocated Freight Costs in Advanced and Multistage Intercompany Business Processes](#)
- Allows you to plan and execute transport as normal using TM.

Multistage Intercompany Sales

In an intercompany sales scenario, you sell a product to a customer that isn't part of your own group. This section contains information specific to the multistage intercompany sales process.

TM integrates the customer sales order and then uses it and the related intercompany sales orders (ICSO) and intercompany purchase orders (ICPO) to derive transport-relevant stages from Incoterms and locations. Each stage corresponds to a handover point between entities in the value chain. In the multistage intercompany sales process, internal transfer of control occurs when goods are transferred from one company within the same group to another company within the group. External transfer of control occurs when the goods are delivered from the group to the customer.

TM uses the Incoterm in the customer sales order to calculate the external transfer of control. It derives internal transfer of control using the Incoterms in the ICPOs. Planned and actual transfer of control dates are managed per relevant Incoterm location and written back to outbound deliveries.

TM builds freight units with the correct stages based on the Incoterms in the business documents. You can plan, transport, and assign freight documents to carriers as normal.

Multistage Intercompany Stock Transfer

In an intercompany stock transfer scenario, you move a product to another company within your group. This section contains information specific to the multistage intercompany stock transfer process.

TM integrates the main stock transport order and related ICPOs and ICSOs and derives transport-relevant stages from Incoterms and locations. Each stage corresponds to a handover point between entities in the value chain. In the multistage intercompany stock transfer process, internal transfer of control occurs when goods are transferred from one company within the same group to another company within the group.

The Incoterms used to derive the internal transfer of control are the ones in the ICPOs. Planned and actual transfer of control dates are managed per relevant Incoterm location and written back to outbound deliveries.

TM builds freight units with the correct stages based on the Incoterms in the business documents. You can plan, transport, and assign freight documents to carriers as normal.

Incoterms and Incoterm Locations

International commercial terms (Incoterms) are a set of international rules for the interpretation of the commonly used terms in international trade. An Incoterm defines the terms of sale and the passing of risks for import and export of goods. The International Chamber of Commerce publishes Incoterms and specifies which Incoterms are allowed. You define Incoterms in business partners and business partner info records.

Incoterms describe the point at which the following responsibilities are passed from one party to another:

- Planning

Planning responsibility means that the party organizes the transport.

- Cost

Cost responsibility means that the party has the obligation to pay the costs of the transport.

- Risk

Risk responsibility means that the party bears the risk for the respective transport section.

You define Incoterm locations at which these responsibilities are transferred from one party to another. Incoterms can have two locations. Planning and cost responsibility are transferred for all Incoterms at Incoterm location 1. For risk responsibility, the system first tries to determine the transfer of control at Incoterm location 2. If Incoterm location 2 is not maintained, the system transfers risk at Incoterm location 1.

As a result, for non-C-clauses, risk responsibility is transferred at Incoterm location 1 because these clauses typically do not maintain Incoterm location 2. For C-clauses such as CIF, CIP, CFR, and CPT, risk responsibility is transferred at Incoterm location 2, provided that this location is maintained.

Some Incoterms are valid without a location 1 or location 2. The following table shows these Incoterms and the default Incoterm location 1, which is used if no locations are entered:

Default Incoterm Location 1	
Incoterm	Default Incoterm Location 1
Delivered At Place (DAP)	Destination
Delivered Duty Paid (DDP)	Destination
Delivered at Place Unloaded (DPU)	Destination
Ex Works (EXW)	Start
Free Carrier (FCA)	Start

→ Recommendation

SAP recommends using Incoterms in accordance with their intended commercial meaning, as system behavior partially takes Incoterm semantics into account. For example, for Incoterm CIF, Incoterm location 1 is related to the port of discharge, while Incoterm location 2 is related to the port of loading. In such cases, both Incoterm locations should be maintained and should contain a different location. Otherwise, a different Incoterm may be more appropriate for the intended purpose.

Transfer of Control

Transfer of control dates refer to the date at which the risk responsibility is transferred between parties. For internal and external transfer of control, there's planned and actual transfer of control. The transfer of control dates and times in the LE delivery are set by TM, based on transportation planning and actual events. You can find the planned and actual internal and external transfer of control dates and times in the **Shipment** tab in the LE delivery. For more information, see [Transfer of Control of Goods \(Sales\)](#) and [Transfer of Control of Goods \(Stock Transfer\)](#).

TM registers the actual transfer of control date when you confirm that a certain transportation activity has happened, for example, Loading End or Arrived. The required activity depends on the Incoterm. The following table shows the Incoterms and the transportation activity required to save the actual transfer of control date:

Transportation Activities for Transfer of Control

Incoterm	Status
Cost, Insurance and Freight (CIF)	Loading End
Carriage and Insurance Paid To (CIP)	Arrived
Cost and Freight (CFR)	Loading End
Carriage Paid To (CPT)	Arrived
Delivered At Place (DAP)	Arrived
Delivered At Terminal (DAT)	Arrived
Delivered Duty Paid (DDP)	Arrived
Delivered at Place Unloaded (DPU)	Unloading End
Ex Works (EXW)	Loading Begin
Free Alongside Ship (FAS)	Arrived
Free Carrier (FCA)	Loading End
Free On Board (FOB)	Loading End

Configuring Transfer of Control

The values in the [Transportation Activities for Transfer of Control](#) table are the default values. When you perform a transportation activity, you must select the corresponding activity in the freight document. If you do not, the actual transfer of control date will not be saved. You can change the required transportation activity in table /SCMTMS/V_INCVST. Access the table using display mode in [Table View Maintenance](#) (transaction SM30) and make a new entry for the Incoterm.

Stage Building

Transportation stages show the stops (start/end locations) of a freight unit. Stages indicate that certain locations must be visited in a certain sequence, or that certain transportation activities such as loading or unloading must be performed.

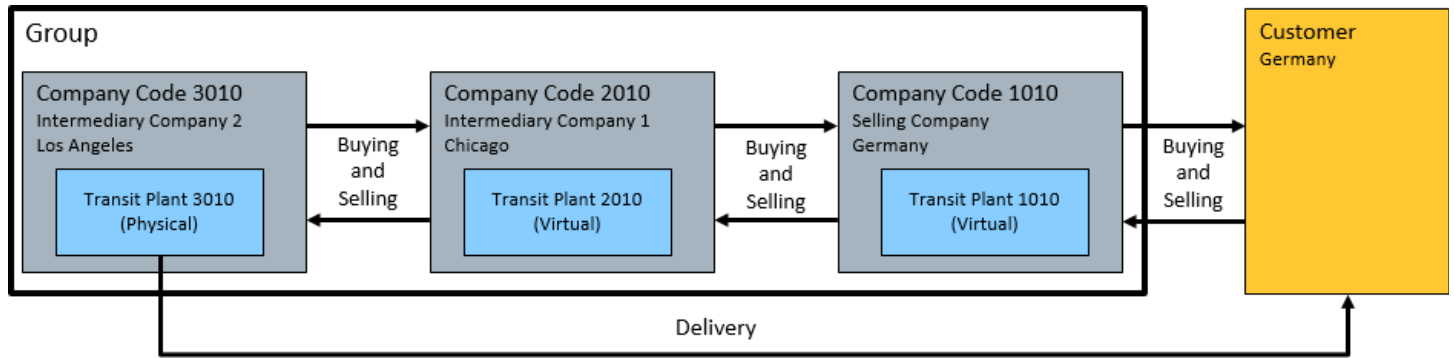
TM considers Incoterms during transportation planning for the purposes of stage building. TM uses Incoterm locations to build freight unit stages.

Example: Multistage Intercompany Sales

A local sales office in Germany (selling company 1010) sells to a customer. The sales order contains all relevant information for the selling company (1010) to organize the transport and to deliver the sold goods to the customer who is also in Germany. This includes the following information:

- Plant/shipping point
- Customer
- Ship-to party
- Incoterm and Incoterm locations
- Material
- Material availability date and requested delivery date

Building the Financial Chain

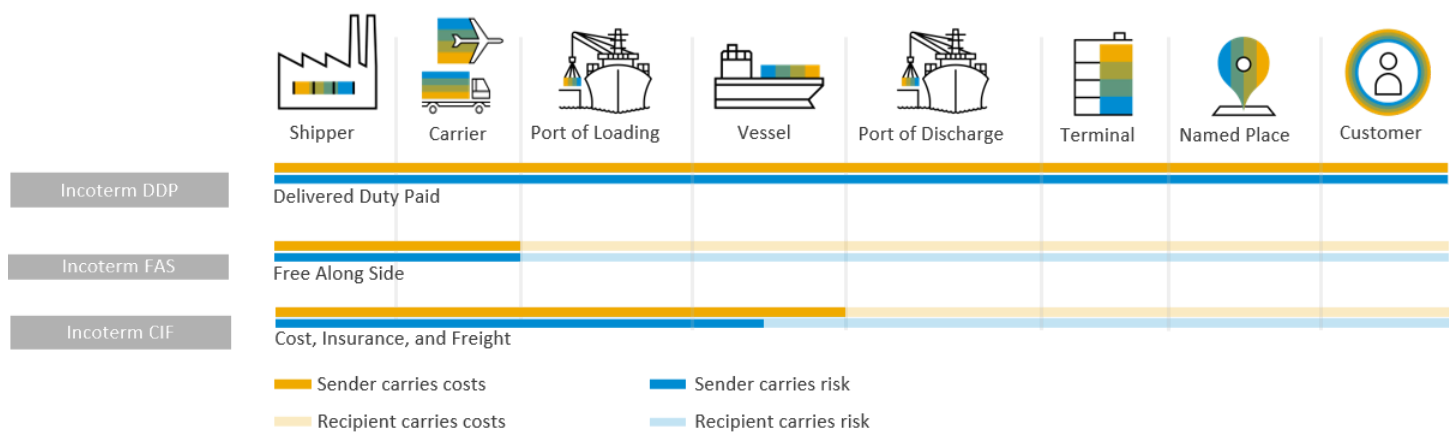


During sourcing, it's determined that the material must be shipped from a plant in Los Angeles, California, which belongs to a company code different from the company code of the selling company in Germany. Unfortunately, the selling company isn't allowed to purchase directly from intermediary company 2 (3010) and must purchase from the U.S. head office in Chicago (2010).

Additional ICPOs and ICSOs are created between the selling company and intermediary company 1 and between intermediary company 1 and intermediary company 2. The intercompany purchase orders and intercompany sales orders contain all information relevant for the selling company (1010) and intermediary company 2 (3010) to organize the transport and to deliver the sold goods to the customer. The sales orders contain the following information:

- Supplier (determines the delivering company)
- Plant (virtual plant that is only used for stock in transit postings)
- Incoterm and Incoterm locations
- Material (goods procured from delivering company)
- Loading date and delivery date

Incoterms



The sales order contains Incoterm Delivery Duty Paid (DDP). This means that the group is responsible for organizing the whole transportation route. In the sales order, Incoterm location 1 (in this case, the customer's address) is relevant for stage building. The group carries the planning and cost responsibility for transporting the goods to the customer's address in Germany.

From the ICPOs, TM derives the following Incoterms:

- Free Alongside Ship (FAS) for the handover from intermediary company 2 to intermediary company 1.

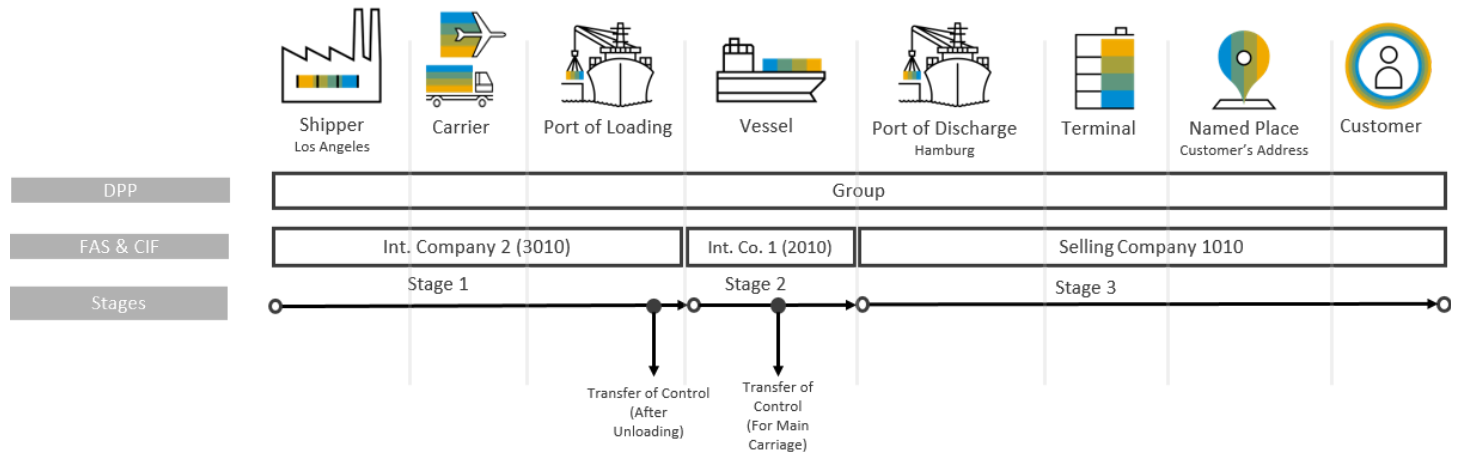
Incoterm location 1 (in this case the harbor in New York) is relevant for stage building. Intermediary company 2 carries the planning and cost for transporting the goods and unloading them alongside the ship.

- Cost, Insurance, and Freight (CIF) for the handover between intermediary company 1 and the selling company.

Incoterm location 1 (in this case the harbor in Hamburg) is relevant for stage building. Intermediary company 2 carries the planning and cost until the goods are unloaded from the vessel in Hamburg.

From the harbor in Hamburg, the selling company takes the remaining responsibility for planning and cost for the group.

Freight Unit and Stage Building



Based on the Incoterms in the documents, TM creates a single freight unit with the following stages:

1. Stage 1

From the plant in Los Angeles to the port of loading in New York (Incoterm location 1 in the intercompany sales order); organized by intermediary company 3010

2. Stage 2

From the port of loading in New York to the Port of Discharge in Hamburg (Incoterm location 1 in the intercompany sales order); organized by intermediary company 2010

3. Stage 3

From the port of discharge in Hamburg to the customer in Germany (Incoterm location 1 in the sales order); organized by selling company 1010

Related Information

[Posting of Allocated Freight Costs in Advanced and Multistage Intercompany Business Processes](#)

[Multistage Intercompany Sales](#)

[Multistage Intercompany Stock Transfer](#)

Integration of Next Generation Just-In-Time

Integration between Next Generation Just-In-Time (nJIT) and Transportation Management (TM) enables the creation of freight units (FU) based on sequenced JIT calls.

Integration with nJIT supports the execution of the inbound process in Advance Shipping and Receiving. As part of the Advanced Shipping and Receiving process, a key difference lies in the creation of FUs, which is based on the component group materials in a sequenced JIT call. The outbound process is conducted based on outbound deliveries from Delivery Management (LE-SHP) and doesn't require direct integration with JIT.

i Note

- All solutions must operate on a single system.
- SET materials from a summarized process are supported. The next generation JIT integration scenario centers around the integration of sequenced JIT calls. SET scenarios represent the only summarized calls supported in the integration between JIT and TM.

For more information about nJIT processes and setup, see [Next Generation Just-In-Time \(JIT\) Supply to Production](#).

Prerequisites

For more information, see [nJIT: Prerequisites](#).

Features**General**

FU created out of a sequenced JIT call are built out of the call's component group material and not the individual components. They contain JIT call reference number, receiving point, and unloading point. Updates and deletion of FUs based on changes to the JIT call are also enabled.

Supply to Production (S2P)

Supply to Production (S2P), or inbound process, represents the procurement process for acquiring manufacturing materials. In an S2P nJIT scenario, there is no initial package building available. Instead, packaging information and slot number are transferred together with the advanced shipping notification (ASN).

However, it is possible to do an initial freight unit building based on component group materials. Upon receipt of the ASN in the customer system, the system updates FUs built for S2P calls. You can make changes to or delete these FUs.

Supply to Customer (S2C)

Supply-to-customer (S2C), or the outbound process, represents the logistical process for delivering manufacturing materials to a customer.

In an S2C scenario, the system automatically creates FUs out of the outbound delivery in Delivery Management (LE-SHP).

i Note

The outbound process is a delivery-based integration only. As such, freight unit building can't be based on the S2C call.

Deliveries

You can trigger the creation of inbound deliveries from your consignment and freight orders.

Inbound deliveries display the package information based on the item structure. In the case of sequenced JIT, after you set the status to **Ready for Warehouse Execution**, Delivery Management (LE-SHP) converts components to delivery items and transfers them to the inbound delivery. That includes component group material and components. Additionally, deliveries contain JIT call internal and external reference IDs, slot number, and Incoterms.

Outbound deliveries are based on the S2C call, and the FUs built on its basis are created for component group materials. The FUs are updated whenever you make changes to the outbound delivery and include the S2C call reference number, slot number,

unloading point and delivery address from the ship-to-party business partner.

i Note

In the S2C scenario, creating deliveries via report /SCMTM/DLV_BATCH is not available.

Forwarded JIS Call

A forwarded JIS call represents the outsourcing of component manufacturing from one supplier to another. The process runs in the same manner as an S2P, but the communication takes place between two suppliers.

Advanced Shipping Notification

An advanced shipping notification (ASN) contains the component group material and JIT call number. An ASN can also be mixed. A single ASN can contain materials at component group level and others at component level. However, ASN data must have a one-to-one correspondence to the FU.

The receiving system relies on the internal or external JIT call number, supplier number, and unloading point to locate the preliminary FUs and update them. The update includes packaging information and package slot numbers.

In a forwarded JIS call, call numbers diverge. The second supplier has their own internal JIT call and uses the first supplier's forwarded JIS call number as the external number. Also, the second supplier has the original JIT call number from the customer.

Business Add-Ins (BAdI)

You can use the following BAdIs to further customize your system's behavior:

- **BAdI: Change Transient TRQ Data**
- **BAdI: Adaptation of Incoming Business Transaction Document Data in TM**
- **BAdI: Changing Source / Destination of Business Transaction Documents**

For more information, see Customizing for **Transportation Management** under **Business Add-Ins for Transportation Management** > **Integration** > **Logistics Integration**.

Related Information

[nJIT: Inbound Process](#)

[nJIT: Forwarded Inbound Process](#)

[nJIT: Creating Deliveries](#)

[nJIT: Web Services](#)

nJIT: Prerequisites

To work with Next Generation Just-In-Time (nJIT) you must have configured your system settings.

Logistics Integration

- You have set your Transportation Management (TM) Control Key to Internal TM Component and selected the relevant control keys. For more information, see Customizing for **Transportation Management** under **Integration** > **Logistics Integration** > **Internal TM Component Integration** > **Define Control Keys for Document Integration**.

→ **Tip**

- To activate supply to production (S2P) nJIT integration, you must have selected the **S2P nJIT to TM** and **Inbound Delivery to TM** checkboxes. You can use standard control key **0065** or create your own control key.

If a purchase scheduling agreement has been assigned to a control cycle that is TM-relevant, freight unit (FU) creation is suppressed for the purchase scheduling agreement. Instead, TM creates FUs based on the S2P JIT call.

If the scheduling agreement has not been assigned to a TM-relevant control cycle but has a TM control key, the system builds freight units based on the purchase scheduling agreement.

- To activate TM integration for an agreement with FUs already built for items in its schedule lines, the integration time range must be adapted to end before integration with TM starts. Otherwise, FUs are built twice. Shrinking the integration time range triggers the deletion of existing FUs that lie beyond the range. For more information about the integration time range, see [Integration Time Range for Purchase Scheduling Agreements](#).

i Note

For transportation-relevant nJIT calls, Materials Management (MM) integration via scheduling agreement must be deactivated. Otherwise, this could lead to the creation of duplicate freight units.

- You have configured your logistics integration profile for nJIT. For more information, see Customizing for **Transportation Management** under **Integration > Logistics Integration > Internal TM Component Integration > Define Logistics Integration Profile for nJIT**.
- You have defined the required actions. For more information, see Customizing for **Logistics Execution** under **Just-In-Time Processing (Next Generation) > Action Control > Define JIT Actions**.

i Note

Actions are provided, among others, to create, update, cancel, or to set a certain processing status for an nJIT call.

Actions are assigned to the respective action controls. The action controls represent the different nJIT scenarios for S2P and S2C.

nJIT: Supply-to-Production

- You have activated TM integration via the control cycle in nJIT. Fill out the **Replenishment Strategy for External Procurement** and **Integration Relevance** fields in the respective control cycle. Assigning a TM control key will trigger TM integration for component group materials assigned to the control cycle. TM builds freight units accordingly.

To create and modify control cycles, open the **Manage JIT Control Cycles** app from the launchpad.

→ **Tip**

- When creating a sequenced JIT call, select the control cycle category **Sequenced** and set the **Action Control** to **S2P_ACTRL**. For a forwarded JIS call, use action control **S2P_ACTFWD** instead.
- The system conducts an initial check to assure that the TM control key and logistics integration profile have been configured correctly.
- A Z in the Kanban field at item level in the purchase scheduling agreement indicates that the agreement is assigned to a control cycle. The plant and storage location in the purchase scheduling agreement must match that in the control cycle.

- Attempting to remove the TM control key and logistics integration profile will raise a warning message as it would lead to disruptions in execution. Also, we recommend creating a new replenishment strategy instead of making changes to the Customizing activities as any changes made after integration could lead to errors.
 - To switch integration type from a purchase scheduling agreement-based to an nJIT-based integration with TM, see [Deactivation and Reactivation of Purchase Scheduling Agreement Integration](#)
- You have defined the required communication groups using the **Manage Communication Group** app. Communication groups provide supplier and, when available, carrier information to the control cycle.

For more information about S2P configuration, see [Configuration](#).

nJIT: Supply-to-Customer

- You have defined your supply control using the **Manage JIT Customer Supply Controls** app from the launchpad.
- You have created packing group specifications. To create packing group specifications, open the **Manage Packing Group Specification for Sequenced JIT Calls** app from the launchpad.
- You have created delivery sequence groups. To do so, open the **Manage Packing Group Specification for Sequenced JIT Calls** app from the launchpad. To create a delivery sequence group, you must have first defined your supply control and created packing group specifications.
- You have assigned sequenced JIT call and packing teams. To do so, open the **Manage Teams and Responsibilities** app from the launchpad.

For more information about S2C, see [Configuration](#).

i Note

Stock transport orders and intercompany processes are not available within the scope of nJIT integration.

Related Information

[Integration of Next Generation Just-In-Time](#)

[nJIT: Inbound Process](#)

[nJIT: Forwarded Inbound Process](#)

[nJIT: Creating Deliveries](#)

[nJIT: Web Services](#)

nJIT: Inbound Process

The inbound process, also known as Supply to Production (S2P), represents the procurement process for acquiring manufacturing materials.

Sample Process

The following is an example of how an inbound process might develop:

1. Creating a Purchase Scheduling Agreement

Create a scheduling agreement with its corresponding confirmation control key number. Go to **Additional Details** and select the item. Then go to **Additional Data** and insert a Z under **Kanban Indicator**.

i Note

Whether a purchase scheduling agreement is TM relevant or not, TM builds freight units (FU) for agreements that are nJIT-relevant and have been assigned to a TM-relevant control cycle.

2. Creating Schedule Lines

Enter your schedule lines to be delivered within the integration time frame. In an nJIT scenario, the system doesn't create FUs for items in schedule lines. Instead, the system creates FUs based on the component group materials in an S2P call. This step is described below.

3. Creating an nJIT Control Cycle

Component group materials must be assigned to a control cycle before you execute the S2P process.

If you haven't yet, open your [Manage JIT Control Cycles](#) app to create a control cycle. For a sequenced JIT call, select the control cycle category **Sequenced** and set the **Action Control** to S2P_ACTRL. For a forwarded JIS call, use action control S2P_ACTFWD instead.

You must fill the [Replenishment Strategies](#) and [Logistics Integration Relevance](#) fields to activate TM relevance.

4. Creating a Sequenced JIT Call

A sequenced JIT call is created based on the planned order or automatically, as soon as a planned order is generated. Open your [Manage JIS Calls and Reorder](#) app from the launchpad to find the call.

For more information, see [Next Generation Just-In-Time \(JIT\) Supply to Production](#).

5. Finding Freight Units

The creation of an S2P JIT call triggers the creation of FUs. To confirm, open your [Freight Units Worklist](#) app from the launchpad, and enter the JIT call number into one of three search fields: **JIT Call**, **External JIT Call**, or **Predecessor Document**.

Your FUs contain component group materials for a sequenced JIT call.

6. Planning Transportation

Create a freight order (FO) and insert your FUs.

Create a consignment order (CO) and insert your FO. As do FUs, COs contain the component group materials.

Alternatively, you can customize your system so that consignment orders are created and FUs updated with the receipt of an advanced shipping notification (ASN).

7. Receiving an Advanced Shipping Notification

The supplier creates and sends an ASN, which contains such details as the corresponding JIT call number, package slot number, and component group materials.

If you have customized your system, when the supplier sends an ASN, the system creates a consignment order. Additionally, integration between TM and nJIT facilitates the exchange of ASN status information so when TM creates a consignment, the status is reflected in nJIT. The internal processing status under **Component** changes from **JIT/JIS Call Created** to **Advanced Shipping Notification Received**. If you delete the consignment, the system sends an update, and nJIT reverts the status.

8. Creating Inbound Deliveries

In your FO or CO, choose [Follow-Up](#) [Create Delivery](#) to create an inbound delivery. Once created, you can find the inbound delivery in the [Document Flow](#).

9. Advanced Shipping and Receiving

The remaining process steps correspond to the standard Advanced Shipping and Receiving process steps. For more information, see [Advanced Shipping and Receiving](#).

Related Information

[Integration of Next Generation Just-In-Time](#)

[nJIT: Prerequisites](#)

[nJIT: Forwarded Inbound Process](#)

[nJIT: Creating Deliveries](#)

[nJIT: Web Services](#)

nJIT: Outbound Process

The outbound process, also known as supply-to-customer (S2C), represents the logistical process for delivering manufacturing materials.

Sample Process

The following is an example of how an outbound process might develop:

1. Creating Outbound Deliveries

In the outbound process, the outbound delivery is created based on the S2C call directly from the **Manage Customer JIT Calls by Component Groups** app.

You can start the creation by selecting all component group materials to be delivered and choosing **Create Outbound Delivery – Sequenced**.

For more information about the outbound delivery, see [Integration of Next Generation Just-In-Time](#).

2. Creating Freight Units

The system automatically creates freight units (FU) for component group materials based on the outbound delivery. The FUs contain the S2C call number, slot number, unloading point and the ship-to-party's delivery address.

3. Planning Transportation

With FUs created, you can proceed with transportation planning. The execution can take place either as warehouse-driven or transport-driven.

For a description of the following stages of the process, see [End-to-End Process \(E2E\) for Advanced Shipping and Receiving](#) and [Transport-Driven Execution](#).

Related Information

[Integration of Next Generation Just-In-Time](#)

[nJIT: Prerequisites](#)

[nJIT: Inbound Process](#)

[nJIT: Forwarded Inbound Process](#)

[nJIT: Creating Deliveries](#)

[nJIT: Web Services](#)

nJIT: Forwarded Inbound Process

A forwarded JIS call is a type of inbound call based on a supply-to-customer (S2P) JIT call, where the primary supplier procures materials from a secondary supplier.

Sample Process

The following is an example of how a forwarded inbound process might develop:

1. Creating a Forwarded JIS Call

To create a forwarded JIS call, you must have first created a supply-to-customer (S2C) JIT call. The S2C call must have transmitted external status Z and call type **Sequenced JIT Call**. The system creates the forwarded JIS call out of the S2C JIT call automatically.

2. Finding a Forwarded JIS Call

Open your **Manage Forwarded JIS Calls** app from the launchpad. The status of the call should be set to **JIT/JIS Call Created**.

3. Finding Freight Units

Once the forwarded JIS call is created, TM creates freight units (FU), which you can find using your **Freight Units – Worklist** app.

4. Planning Transportation

Create a freight order (FO) and insert your FUs.

Create a consignment order (CO) and insert your FO. As do FUs, COs contain the component group materials.

Alternatively, you can customize your system so that consignment orders are created and FUs updated with the receipt of an advanced shipping notification (ASN).

5. Receiving an Advanced Shipping Notification

The supplier creates and sends an ASN, which contains such details as the corresponding JIT call number, package slot number, and component group materials.

If you have customized your system, when the supplier sends an ASN, the system creates a CO. Additionally, integration between TM and nJIT enables the communication of the ASN status so that when TM creates a CO, it is reflected in nJIT. Under **Component Group Material**, the status changes from **JIT/JIS Call Created** to **Advanced Shipping Notification Received**. If you delete the CO, the system sends an update, and nJIT reverts the status of the component group material.

By selecting the status, you can navigate to a detailed view of the component group material where you can also find a detailed view of the internal processing statuses.

From here on, the process resembles the standard Advanced Shipping and Receiving process.

For more information, see [Advanced Shipping and Receiving](#).

Related Information

[Integration of Next Generation Just-In-Time](#)

[nJIT: Prerequisites](#)

[nJIT: Inbound Process](#)

[nJIT: Creating Deliveries](#)

[nJIT: Web Services](#)

nJIT: Creating Deliveries

nJIT integration enables you to create inbound deliveries from Transportation Management (TM) with a JIT call as predecessor document. Inbound deliveries contain package slot number and Incoterms. It also enables you to create outbound deliveries out of S2C calls, which leads to the automatic creation of freight units (FU) in TM.

Prerequisites

For information about prerequisites in Delivery Management (LE-SHP), see [Integration of Next Generation Just-In-Time](#).

Features

Inbound Deliveries

You create inbound deliveries from your freight or consignment orders with component groups. Deliveries with component groups as delivery items have JIT calls as predecessor documents, while deliveries with components as delivery items must have scheduling agreements as predecessor document. When customized, your system creates inbound deliveries automatically after TM has created a consignment order based on the supplier's advanced shipping notification (ASN).

Since the deliveries reference the corresponding JIT call, LE notifies nJIT when deliveries are created for component groups and sends goods receipt notification for component groups to nJIT. LE also replicates inbound deliveries into Extended Warehouse Management (EWM).

You have the option to make changes to the consignment orders before the status is set to **Ready for Warehouse Execution**. The system automatically updates the deliveries. Once the status is set to **Ready for Warehouse Execution**, this information is communicated to both LE and EWM. The change in status also prompts the transfer of component materials as delivery items to the inbound delivery.

Finally, when you post goods receipt in EWM, EWM sends an update to LE. For partial deliveries, LE identifies the respective components and registers the delivered items.

You must process the movement of goods in either Inventory Management for components or EWM for component groups.

This feature is part of the Advanced Shipping and Receiving (Advanced SR) process. For more information about related features, see [Advanced Shipping and Receiving](#).

Outbound Deliveries

In the supply-to-customer (S2C) process, you create your outbound deliveries out of S2C calls, and the system automatically creates FUs based on component group materials. The FUs are updated automatically when you make changes to the deliveries.

FUs contain the S2C call reference number, slot number, unloading point, and delivery address from the ship-to-party business partner.

More Information

For more information about integration with Delivery Management (LE-SHP), see [Integration of Next Generation Just-In-Time](#).

For more information about integration with Extended Warehouse Management (EWM), see [Inbound Process for Supply-to-Production](#).

Related Information

nJIT: Web Services

You can send or receive advanced shipping notifications (ASN) as a supplier or customer.

Using an ASN, a supplier notifies the customer regarding the pending arrival of goods that have been shipped or that are about to be shipped.

In an nJIT scenario, the ASN is based on the supplier's consignment order (CO) and uses the *BusinessTransactionDocumentReference* (BTDRef) node with business transaction document ID, type code, and item ID. The standard type code for communication is 3018, which represents the internal, or customer reference, JIT call ID.

```
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>JITCALL</ID>
    <TypeCode>3018</TypeCode>
    <ItemID>10</ItemID>
  </BusinessTransactionDocumentReference>
</BusinessTransactionDocumentReference>
```

The inbound message enables the customer system to create a CO based on the business transaction document reference and to locate and assign the corresponding freight units (FU). To locate the FUs, the system relies on the internal JIT reference number and JIT reference item number.

```
<Item>
  <ID>10</ID>
  <CategoryCode>PKG</CategoryCode>
  <Description>T11_CFT0_1</Description>
```

The service also includes nJIT data such as the call type (S2C or S2P) and the delivered quantity with quantity unit.

```
<PiecesTransportationQuantity unitCode="EA">1000</PiecesTransportationQuantity>
```

Furthermore, it contains packaging information such as packaging material and slot number.

```
<Package>
  <TypeCode>PT00</TypeCode>
  <TrayID>SLOTNUMBER</TrayID>
  <RecipientPackagingMaterialInternalID>PACKAGINGMATERIALID</RecipientPackagingMaterialInternalID>
</Package>
```

Extensibility

In addition to the standard implementation, which uses BTDR type code 3018, BTDR type code 3020 for external JIT call numbers can be implemented by the supplier in the ASN outbound message.

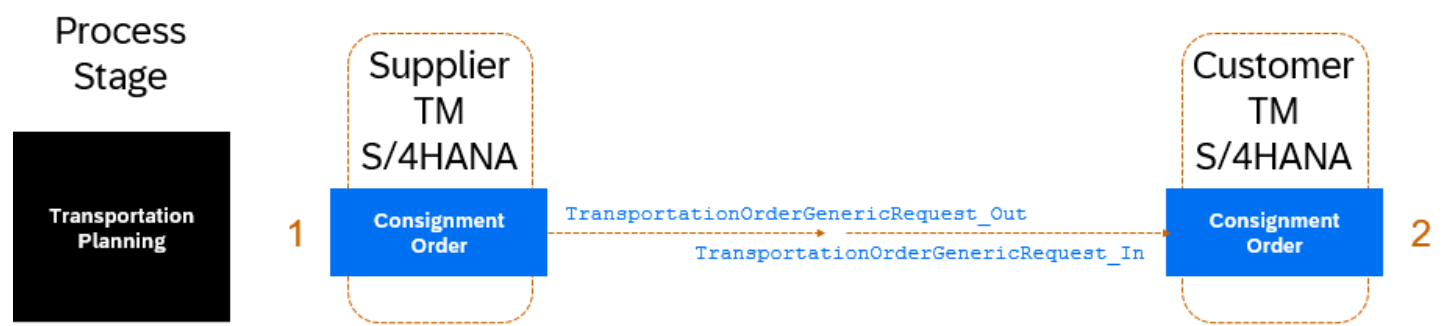
In such a case, the external JIT call number is used in the ASN inbound message when the internal number is missing. Type code 3020 doesn't require an item ID. Instead, it can resort to the external JIT reference item number, the supplier number, component group material, and unloading point.

```
<Product>
  <InternalID>PRODUCTID</InternalID>
  <ProductRecipientID>CUSTOMERID</ProductRecipientID>
</Product>

<UnloadingPointLocation>
  <InternalID>UNLOADINGPOINTID</InternalID>
</UnloadingPointLocation>
```

Process

The following graphic gives you an overview of the communication and data exchanged between a manufacturer's Transportation Management (TM) system and a supplier's TM system. The data are exchanged by using web services based on the SOA protocol.



Description of Web Services

Phase	Process Step	Details	Web Service
Transportation Planning	Step 1: Send ASN based on CO to customer	The CO contains the JIT data. That includes the component call type, internal number, delivery quantity and unit, packaging information, package slot number, and group materials.	TransportationOrderGenericRequest-Out
Transportation Planning	Step 2: Receive CO from supplier	Based on the BTDRef node, TM creates or updates the CO, and the system finds the corresponding FUs and assigns them to the CO.	TransportationOrderGenericRequest-In

Sample XML Payload

☰ Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2020-05-15T18:12:13.383Z</CreationDateTime>
    <SenderBusinessSystemID>ABC_123</SenderBusinessSystemID>
    <BusinessScope>
      <TypeCode>3</TypeCode>
```

```

    <ID>ASN</ID>
  </BusinessScope>
  <TestDataIndicator>>false</TestDataIndicator>
</MessageHeader>
<TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator:
  <ID>CONSIGNMENTORDERID</ID>
  <TypeCode>42C0</TypeCode>
  <CategoryCode>TU</CategoryCode>
  <ShipperParty>
    <InternalID>T42_SP42_1</InternalID>
  </ShipperParty>
  <AdditionalParty>
    <Party>
      <InternalID>T42_SP42_1</InternalID>
    </Party>
    <RoleCode>TM033</RoleCode>
  </AdditionalParty>
  <TransportationTerms>
    <TransportModeCode>01</TransportModeCode>
  </TransportationTerms>
  <BaseBusinessTransactionDocumentReference>
    <ID>CONSIGNMENTORDERID</ID>
    <TypeCode>3002</TypeCode>
    <VersionID>2</VersionID>
  </BaseBusinessTransactionDocumentReference>
  <Item>
    <ID>10000000</ID>
    <CategoryCode>SHP</CategoryCode>
  </Item>
  <Item>
    <ID>10</ID>
    <CategoryCode>PKG</CategoryCode>
    <Description>T42_CST0_1</Description>
    <GrossWeightTransportationQuantity unitCode="KGM">80</GrossWeightTransportationQuantity>
    <GrossVolumeTransportationQuantity unitCode="DMQ">0.010</GrossVolumeTransportationQuantity>
    <NetWeightTransportationQuantity unitCode="KGM">0.010</NetWeightTransportationQuantity>
    <PiecesTransportationQuantity unitCode="EA">1</PiecesTransportationQuantity>
    <HierarchyRelationship>
      <ParentItemID>10000000</ParentItemID>
    </HierarchyRelationship>
    <Package>
      <TypeCode>PT01</TypeCode>
      <TrayID>AB01</TrayID>
      <RecipientPackagingMaterialInternalID>T42_CST0_1</RecipientPackagingMaterialInternalID>
    </Package>
  </Item>
  <Item>
    <ID>20</ID>
    <CategoryCode>PRD</CategoryCode>
    <Description>T42_JIT_GROUP_A</Description>
    <GrossWeightTransportationQuantity unitCode="KGM">80</GrossWeightTransportationQuantity>
    <GrossVolumeTransportationQuantity unitCode="DMQ">0.010</GrossVolumeTransportationQuantity>
    <NetWeightTransportationQuantity unitCode="KGM">0.010</NetWeightTransportationQuantity>
    <PiecesTransportationQuantity unitCode="EA">1</PiecesTransportationQuantity>
    <DeliveryQuantity unitCode="EA">1</DeliveryQuantity>
    <HierarchyRelationship>
      <UnloadingPointLocation>
        <InternalID>T42_LP_ROT_L43</InternalID>
      </UnloadingPointLocation>
      <ParentItemID>10</ParentItemID>
    </HierarchyRelationship>
    <Product>
      <InternalID>T42_JIT_GROUP_A</InternalID>
      <ProductRecipientID>T42_JIT_GROUP_A</ProductRecipientID>
    </Product>
    <DeliveryTerms>
      <DeliveryPriorityCode>1</DeliveryPriorityCode>
    </DeliveryTerms>
    <Incoterms>
      <ClassificationCode>ddp</ClassificationCode>
      <TransferLocationName>berlin</TransferLocationName>
    </Incoterms>
    <TransportationTerms>
      <TransportServiceLevelCode>01</TransportServiceLevelCode>
    </TransportationTerms>

```

```

    <BusinessTransactionDocumentReference>
      <BusinessTransactionDocumentReference>
        <ID>DELIVERYID</ID>
        <TypeCode>1137</TypeCode>
        <ItemID>100002</ItemID>
      </BusinessTransactionDocumentReference>
    </BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
      <BusinessTransactionDocumentReference>
        <ID>JITCALL</ID>
        <TypeCode>3018</TypeCode>
        <ItemID>10</ItemID>
      </BusinessTransactionDocumentReference>
    </BusinessTransactionDocumentReference>
  </Item>
</TransportationOrder>
</n0:TransportationOrderGenericRequest>

```

Related Information

[Integration of Next Generation Just-In-Time](#)

[nJIT: Prerequisites](#)

[nJIT: Inbound Process](#)

[nJIT: Forwarded Inbound Process](#)

External TM System Integration

You can integrate orders and deliveries from external source systems such as SAP ERP or an external SAP S/4HANA system. The information that follows on external system integration functionality refers to SAP ERP but applies also to integration with an external SAP S/4HANA system.

Use

You use integration of SAP ERP orders and deliveries to facilitate transportation planning and execution in Transportation Management (TM) based on orders and deliveries that are created in an SAP ERP system. The orders and deliveries are represented by transportation requirements in TM. The integration allows you to continuously react to changes to orders and deliveries that occur in the SAP ERP system. In particular, integration with SAP ERP enables you to:

- Collectively plan transportation for all orders, across order types.
- Perform transportation planning in TM based on SAP ERP orders, prior to delivery creation, thus enabling you to get an earlier forecast of future transportation and resource demands.
- Perform tendering, subcontracting, and execution in TM based on SAP ERP orders prior to delivery creation.
- Propose and create deliveries based on dates and quantities according to transportation planning results, taking into account transportation constraints such as resource availability and transportation durations.

Integration

The TM component [ERP Logistics Integration](#) is integrated with the following TM components:

Component	Type of Integration
Planning	Transportation requirements can form the basis for transportation planning and execution.

Component	Type of Integration
Master Data	As a prerequisite for creating transportation requirements, the necessary master data must have been created.

Features

Order Integration

Orders that are created in SAP ERP can be transferred to TM, where they are stored as order-based transportation requirements. The following types of order in SAP ERP are supported by integration with TM:

- Sales order
- Customer returns
- Purchase order
- Stock transport order
- Returns purchase order
- Returns stock transport order
- MM Scheduling agreements

Delivery Proposals

TM can generate a proposal for a delivery to be created in SAP ERP based on the order-based transportation requirement. For more information, see [External TM System Integration](#).

Delivery Integration

Outbound deliveries and inbound deliveries that are created in SAP ERP can be transferred to TM, where they are stored as delivery-based transportation requirements. Deliveries can be transferred to TM regardless of whether the delivery creation was triggered by TM or not. Deliveries can also be transferred to SAP TM even if an order that relates to the delivery has not been sent to TM.

Dangerous Goods

If a product contained in an order or delivery is classified as dangerous goods, TM considers this information to ensure a safe and compliant transportation of the product. For more information, see [Dangerous Goods Checks](#).

ERP-TM Communication

Order and delivery data can be transferred between TM and SAP ERP using enterprise services.

Configuring Integration of Orders and Deliveries

Use

This section describes the additional settings that you need to make in Transportation Management (TM) and in SAP ERP to integrate orders and deliveries.

Prerequisites

- You have set up connections between SAP ERP and TM and configured SAP ERP for order creation and delivery creation.
- You have created master data in SAP ERP and TM.

For more information about master data in TM, see

- [Master Data](#)
- [Master Data Replication](#)

Activities

Integration of ERP Orders and Deliveries

Settings in TM

To set up integration of orders and deliveries in TM, you must complete the following activities:

- In Customizing for **Transportation Management**, choose **Integration > Logistics Integration > External TM System Integration > Order-Based Transportation Requirement > Define Order-Based Transportation Requirement Types**

In this activity the following fields are of particular relevance:

- **Plan on Req/Cnf Qty**
- **Automatic Freight Unit Building**
- **Freight Unit Building Rule**
- **FU Building Rule Condition**
- In Customizing for **Transportation Management**, choose **Integration > Logistics Integration > External TM System Integration > Delivery-Based Transportation Requirement > Define Delivery-Based Transportation Requirement Types**
- If you have specified a freight unit building rule in Customizing for an order-based transportation requirement type, you must have defined this freight unit building rule in the **Freight Unit Building Rule** app which you open from the launchpad..
- Define conditions for determining the corresponding order-based transportation requirement type to be used for orders. You do this by creating a condition of type /SCMTMS/OTR_TYPE in the **Create Condition** app which you open from the launchpad.
- Define conditions for determining the corresponding delivery-based transportation requirement type to be used for deliveries. You do this by creating a condition of type /SCMTMS/DTR_TYPE in the **Create Condition** app which you open from the launchpad.

For more information about conditions, see [Condition](#).

- Define a delivery profile for creating deliveries in SAP ERP. You do this in the **Create Delivery Profile** app which you open from the launchpad.

For more information, see [Delivery Profile](#).

Settings in SAP ERP

In SAP ERP, you can display available control keys for document transfer, which you need to use for the following Customizing activities in SAP ERP. For more information about the control keys delivered by SAP, see Customizing for **Integration with Other SAP Components**, under **Transportation Management > Logistics Integration > Define Control Keys for Document Transfer**

You must complete the activities in Customizing for [Integration with Other SAP Components](#) under [Transportation Management](#) [Logistics Integration](#) as follows:

1. To integrate sales orders, customer returns, and their corresponding deliveries, in the [Activate Transfer of Sales Documents](#) activity, assign a control key to your sales order document types. The control key that you assign should have the **SO to TM** checkbox and the **Outbd Del.** checkbox selected.
2. To integrate purchase orders, stock transport orders, return purchase orders, return stock transport orders, and their corresponding deliveries, in the [Activate Transfer of Purchasing Documents](#) activity, assign a control key to your purchasing document types. The control key that you assign should have the **PO to TM** checkbox and the **Inbd Del.** checkbox selected.

For more information about the necessary message control settings in SAP ERP, see SAP Help Portal at <http://help.sap.com/erp> under SAP Central Component. Select an SAP enhancement package for SAP ERP 6.0. In SAP Library, choose [SAP ERP Central Component](#) [Logistics](#) [Logistics - General \(LO\)](#) [Integration of SAP ERP with SAP Transportation Management](#) [Integration: Order and Delivery Processing - SAP TM](#) [Prerequisites for Order and Delivery Integration](#).

Integration of ERP Orders Without Sending Deliveries to TM

To configure this variant of integration, in your SAP ERP system, you make settings for orders such that corresponding deliveries are not sent to TM. You make these settings as follows in the SAP ERP system:

In Customizing for [Integration with Other SAP Components](#), under [Transportation Management](#) [Logistics Integration](#) [Activate Transfer of Sales Documents](#) and [Activate Transfer of Purchase Orders](#), assign a control key to the sales document type and the purchasing document type respectively. The control keys that you assign should **not** have the delivery checkboxes selected and should have integration mode **External TM System**.

Integration of ERP Deliveries Without Sending Orders to TM

To configure this variant of integration, in your SAP ERP system, you make settings for deliveries such that the preceding order documents are not sent to TM. You make these settings as follows in the SAP ERP system:

In Customizing for [Integration with Other SAP Components](#), under [Transportation Management](#) [Logistics Integration](#) [Activate Transfer of Delivery Documents](#), assign your control key to the delivery types. The control keys that you assign should **not** have the **SO to TM** checkbox or the **PO to TM** checkbox selected.

Delivery Profile

Definition

A group of user-specific settings that the system takes into account during delivery creation.

Structure

A delivery profile enables you to make the following settings:

- You can define that during delivery creation the system fixes the planning results for freight units, freight orders, or freight bookings that have already been planned.
- You can define how the system creates deliveries when a user triggers delivery creation in ERP for freight units that have not been planned (for example, one delivery per item in an order-based transportation requirement).
- You can specify incompatibility settings. In the incompatibility settings, you can define attributes that prevent freight units from being combined into one delivery, for example. For more information about incompatibilities, see [Incompatibilities](#).

- You can configure the delivery proposal to aggregate order items in a delivery. The system compares items in a freight unit to see if they have the same sales order or purchase order numbers and the same item type ID.
- You can exclude freight units that have been blocked for planning.
- You can exclude freight units that have been blocked for execution.

For more information about these settings, see the corresponding field help.

i Note

You define delivery profiles in the [Create Delivery Profile](#) app, which you can open from the launchpad.

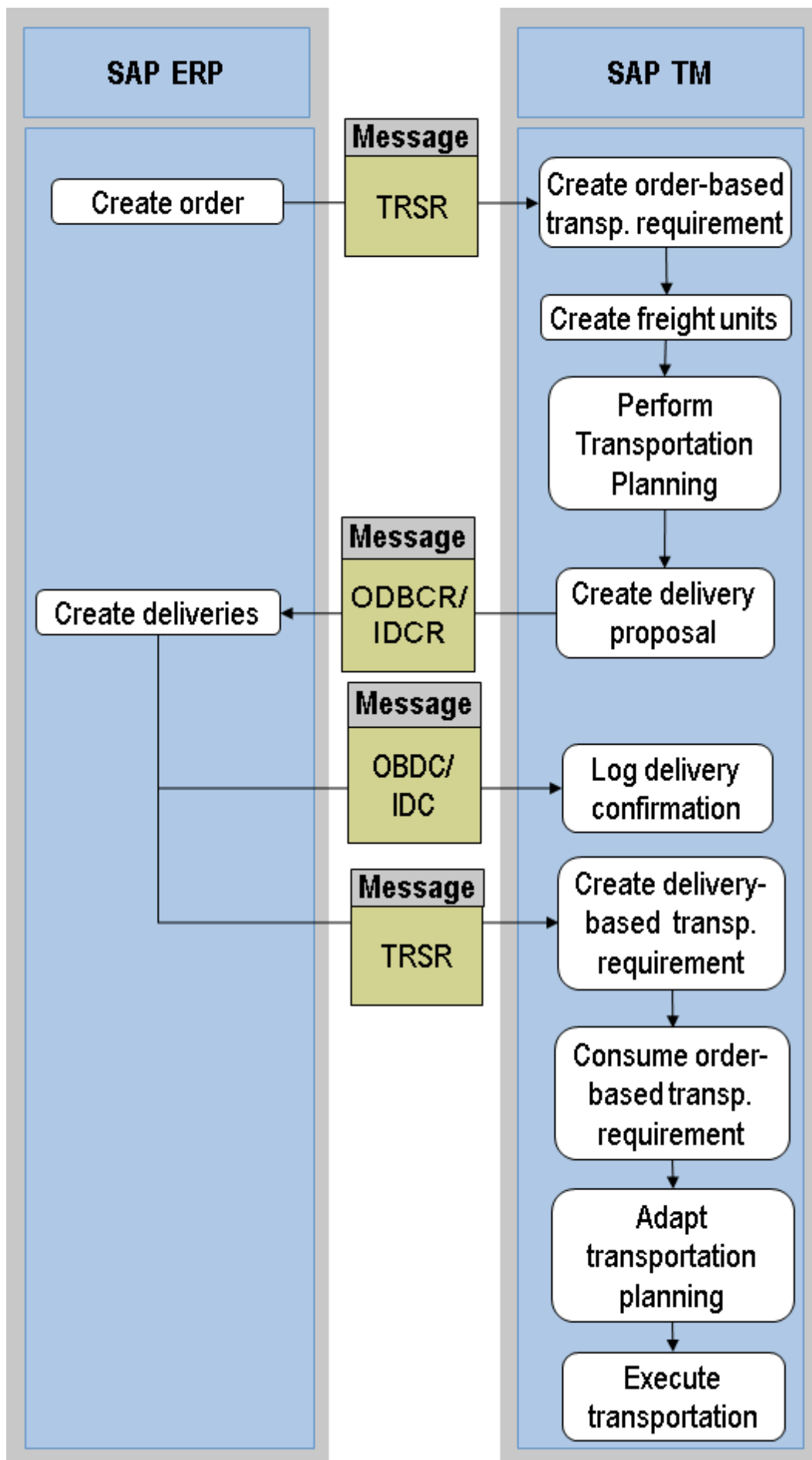
Integration of ERP Orders and Deliveries in Transp. Planning

Use

You can integrate orders and deliveries from SAP ERP with Transportation Management (TM).

Process

The following graphic shows the standard process for integrating orders and deliveries from SAP ERP with transportation planning in TM, and is followed by a description of the process. Variants of this process are possible, and are described in subsequent sections.



Integration of ERP Orders and Deliveries with SAP TM

The following abbreviations are used for message types in the above figure:

- TRSR: TransportationRequestSUITERequest

- ODBC: Outbound Delivery Bulk Create Request
- IDC: InboundDeliveryCreateRequest
- ODBC: Outbound Delivery Bulk Confirmation
- IDC: InboundDeliveryConfirmation_V1

The process runs as follows:

1. Create order (SAP ERP)

An order is created in SAP ERP containing items and schedule lines. When the order has been created, SAP ERP sends the order to TM. For more information, see [Integration of ERP Orders](#).

2. Create order-based transportation requirement (TM)

When TM receives an order from SAP ERP, TM creates an order-based transportation requirement. For more information about order-based transportation requirements, see [Order-Based Transportation Requirement](#).

3. Create freight units (TM)

Before transportation planning can take place, TM builds freight units (FUs) based on the order-based transportation requirement. TM considers items and schedule line items when creating freight units. For more information about creating freight units, see [Creation and Editing of Freight Units](#).

4. Perform transportation planning (TM)

TM performs transportation planning based on the freight units. The planning results include pickup dates and delivery dates, resource assignments, and routes. For more information about transportation planning, see [Planning](#).

5. Create delivery proposal (TM)

TM creates delivery proposals based on the transportation planning results. TM sends these proposals to SAP ERP to create a delivery. It is also possible to create delivery proposals based on freight units, if transportation planning has not been performed. For more information, see [Creation of Delivery Proposals](#).

6. Create deliveries (SAP ERP)

When SAP ERP receives the request to create a delivery, SAP ERP creates the delivery. This can be an inbound delivery or an outbound delivery. For more information about the process of creating deliveries in SAP ERP based on delivery proposals from TM, see [Creation of ERP Deliveries from TM](#). It is also possible to create deliveries in SAP ERP without having sent a proposal from TM. For more information, see [Integration of ERP Deliveries](#).

7. Log delivery confirmation (TM)

When TM receives a confirmation of the deliveries that have been created in SAP ERP, TM writes this information to the application log. If errors occurred during delivery creation an alert is generated in TM.

8. Create delivery-based transportation requirement (TM)

When TM receives a delivery from SAP ERP, TM creates a delivery-based transportation requirement. For more information about delivery-based transportation requirements, see [Delivery-Based Transportation Requirement](#).

9. Consume order-based transportation requirement

When TM creates the delivery-based transportation requirement, the delivery-based transportation requirement consumes the transportation demand of the corresponding order-based transportation requirement or requirements. That is, TM determines the corresponding order-based transportation requirement items and reassigns the freight units to the delivery-based transportation requirement.

For more information, see [Consumption of Order-Based Transportation Requirements](#).

10. Adapt transportation planning

If necessary, TM can re-plan the transportation demands of the delivery-based transportation requirement and its freight units.

11. Execute transportation

Transportation is executed based on the planning results.

i Note

The TM document flow for a specific ERP order or delivery can be displayed in the SAP ERP system (available as of SAP enhancement package 6 for SAP ERP 6.0 support package 4). TM provides SAP ERP with information about the documents created for an order or delivery (for example, freight units, freight orders, and freight settlement documents) and the status of the documents (for example, life cycle status and execution status). This gives the user an overview of the entire ERP-TM process for an individual ERP document and enables the user to track the status of document processing in TM from the SAP ERP system.

Integration of ERP Orders

Use

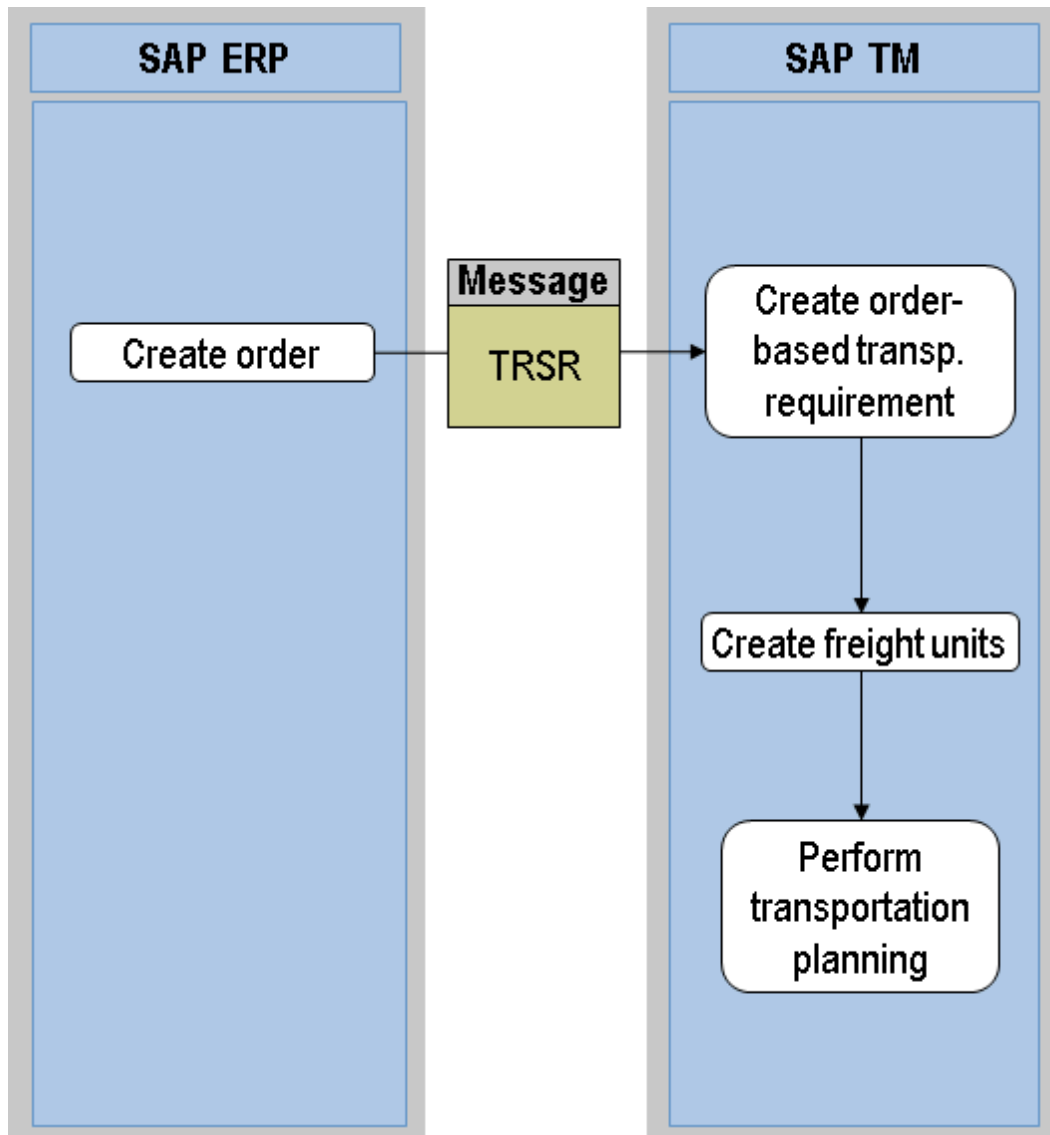
In this process, a customer bases their transportation processes on an order created in SAP ERP.

Prerequisites

For information about setting up SAP ERP and TM for order integration, see [Configuring Integration of Orders and Deliveries](#).

Process

The following figure shows the process for transportation planning based on an SAP ERP order, and is followed by a description of the process:



Transportation Planning Based on SAP ERP Order

In the above figure, the abbreviation TRSR means TransportationRequestSUITERequest.

The process runs as follows:

1. Create order (SAP ERP)

An order is created in SAP ERP containing items and schedule lines. The item contains information regarding the material and the requested quantity, for example. The quantity of an item can be subdivided into multiple schedule lines, which, for example, can have different delivery dates. When the order has been created, SAP ERP sends the order to TM.

2. Create order-based transportation requirement (TM)

The order number and order item numbers from SAP ERP are stored as references in the order-based transportation requirement (on header level or item level respectively). In TM, the schedule lines from the SAP ERP order are stored as items of the type **Schedule Line** in the order-based transportation requirement

3. Create freight units (TM)

TM creates freight units containing items that correspond to the schedule line items of order-based transportation requirements. Depending on your settings in Customizing for the order-based transportation requirement type, TM can build freight units automatically when the order-based transportation requirement is created, or the user can trigger the creation manually. TM can use either the requested quantities or the confirmed quantities in an order-based transportation requirement when creating freight units. Which quantity is used, depends on your settings in Customizing for the order-based transportation requirement type.

TM only creates freight units for transportation-relevant items. For example, if an order-based transportation requirement contains an item for a bill of material, TM creates a freight unit (or units) for either the header item only or for the component items only. This is dependent on the Customizing settings for the material in SAP ERP that determine which items are relevant for goods movement.

TM can consolidate the transportation demand of an order-based transportation requirement with the demand of another order-based transportation requirement into the same freight unit. However, particular attributes can prevent TM from consolidating order-based transportation requirements into the same freight unit, for example if order combination is not allowed for an order-based transportation requirement. If complete delivery is specified for an order, or items belong to the same delivery group, TM consolidates the demand into a freight unit. For more information about consolidation and incompatibilities that are considered when creating freight units, see [Freight Unit Building Rule](#) and [Incompatibilities](#).

4. Perform transportation planning (TM)

TM can consolidate FUs from multiple order-based transportation requirements, even if the base documents in SAP ERP have different order types. For example, the transportation demands of purchase orders and sales orders can be consolidated during transportation planning. Furthermore, planning can be carried out either using own resources or by tendering and subcontracting.

Result

After transportation planning, TM can either execute transportation or trigger the creation of deliveries in SAP ERP. For more information, see [Creation of ERP Deliveries from TM](#).

You can display individual overviews of order-based freight orders, order-based air freight bookings, and order-based ocean freight bookings in the [Transportation Cockpit](#) app, which you open from the launchpad.

Order-Based Transportation Requirement

Definition

Represents a transportation demand based on an order that was created in an SAP ERP system. The order-based transportation requirement can be used as a basis for transportation planning.

The order-based transportation requirement corresponds to a single order in SAP ERP and cannot be edited in Transportation Management (TM).

Structure

The transportation requirement has the following structure:

- Tabs for general data, terms and conditions, business partners, location and dates/times, notes, and administrative data.

Contains information that is relevant for all items of an order-based transportation requirement.

- Document flow

Shows a list of preceding and subsequent documents that are related to the order-based transportation requirement.

- Items

This table provides a hierarchical view of the product items, schedule lines, service items, and text items. In the order-based transportation requirement, the product items correspond to items in the ERP order. For each item, you can display more detailed information on a number of tab pages.

Statutes in the Order-Based Transportation Requirement

The following statuses are available in the order-based transportation requirement:

- **Lifecycle Status**

Status	Description
New	The order-based transportation requirement has been created. No freight units have been created for the order-based transportation requirement yet.
In Planning	The order-based transportation requirement is in the planning process and at least one freight unit has been created for the transportation requirement.
Planned	All the freight units that are assigned to the order-based transportation requirement have been planned.
In Execution	The order-based transportation requirement is in the execution process. At least one follow-on document (not necessarily a freight unit) is in execution. For more information about the execution of a transportation requirement, see the execution status of the transportation requirement or of the follow-on documents.
Completed	Processing of the order-based transportation requirement has been completed. Execution has finished. This status is also set when the freight units that were assigned to the order-based transportation requirement have been consumed by a delivery-based transportation requirement.
Canceled	The order-based transportation requirement has been canceled.

- **Execution Status**

Status	Description
Execution Not Started	The execution of the order-based transportation requirement has not yet started.
In Execution	The order-based transportation requirement is currently being executed. At least one of the planning documents assigned to the order-based transportation requirement is in the execution process.
Executed	The execution of the order-based transportation requirement has been completed. The execution of all planning documents assigned to the order-based transportation requirement has been completed.

- **Consumption Status**

Status	Description
Not Consumed	No freight units have been consumed by a delivery-based transportation requirement.
Consumed Partially	Some of the freight units that were assigned to the order-based transportation requirement have been consumed or partially consumed by a delivery-based transportation requirement.
Consumed Completely	All of the freight units that were assigned to the order-based transportation requirement have been consumed by a delivery-based transportation requirement.

Blocks

An order-based transportation requirement can contain the following blocks:

- Planning block
- Execution block
- Delivery block

You can configure the way in which TM responds to delivery block reasons from SAP ERP. You can specify whether the TM system should create a planning block or an execution block or both, according to the defined SAP ERP block reason.

For more information on this see Customizing for Transportation management under [Integration](#) > [Logistics Integration](#) > [External TM System Integration](#) > [Define Blocking of Transportation Requirements](#) > [Define Blocks Based on Delivery Block](#)

Credit Check

You can set up credit control for sales orders in ERP. A sales order has a credit status that reflects the outcome of the credit limit check on that document. If the credit status of the document is not “approved” TM receives a delivery block with blocking reason 01 (“Credit Limit”) for the document.

Integration

The order-based transportation requirement is integrated with the following objects in TM:

- Freight unit

You can create freight units based on an order-based transportation requirement.

- Delivery-based transportation requirement

One or more delivery-based transportation requirements can consume the transportation demand of one or more order-based transportation requirements.

Creation of ERP Deliveries from TM

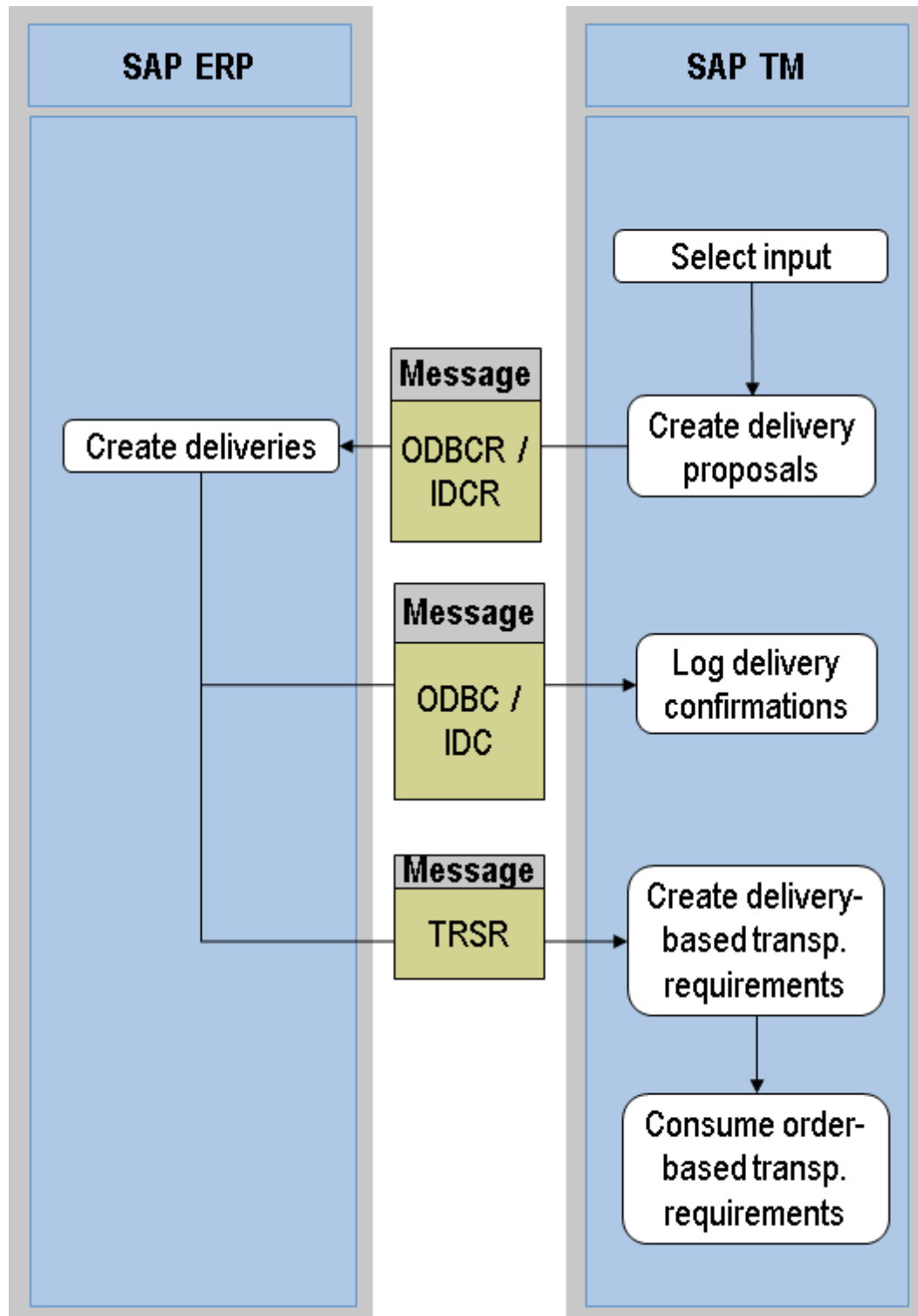
Use

You use this process if you want to create a delivery proposal in Transportation Management (TM) and send the proposal to SAP ERP where the delivery is created. The delivery proposal is based on the results of transportation planning such as consolidation

and dates, taking into account transportation-planning-relevant information such as resource availability and transportation durations. TM can also trigger the creation of deliveries prior to transportation planning.

Process

The following figure shows the process for creating delivery proposals in TM and the subsequent creation of deliveries in SAP ERP and is followed by a description of the process:



Creation of ERP Deliveries from SAP TM

The following abbreviations are used for message types in the above figure:

- ODBCR: Outbound Delivery Bulk Create Request
- IDCR: InboundDeliveryCreateRequest

- ODBC: Outbound Delivery Bulk Confirmation
- IDC: InboundDeliveryConfirmation_V1
- TRSR: TransportationRequestSUITERequest

The process runs as follows:

1. Select input (TM)

The user can start delivery creation by using a background report or interactively. For more information, see [Creation of Delivery Proposals](#).

TM considers data for delivery creation from the order-based transportation requirements and from the assigned freight units, freight orders, and freight bookings.

2. Generate delivery proposals (TM)

Based on the chosen input, TM creates delivery proposals. A proposal consists of groups of freight units that can be delivered together. Depending on the input method used, TM either sends the proposals automatically to SAP ERP to create deliveries, or the user can check the proposals and then send them to SAP ERP.

If the user specified a delivery profile that requires the planning results to be fixed for planned freight units, freight orders, or freight bookings, the system fixes the planning results during delivery proposal creation. Otherwise, the system does not fix the planning results.

After the delivery proposals have been created and sent to SAP ERP, the system creates an entry in the application log. The application log entry includes the delivery proposals that were sent to SAP ERP. Furthermore, the freight units for which a delivery proposal is sent to SAP ERP are marked with a send timestamp.

3. Create deliveries (SAP ERP)

When SAP ERP receives the request to create deliveries, SAP ERP creates the deliveries. These can be inbound or outbound deliveries. SAP ERP might split the proposed deliveries further, for example, based on additional split criteria. However, SAP ERP will not change the dates that were proposed by TM nor consolidate proposed deliveries. When SAP ERP creates the deliveries, SAP ERP sends a confirmation of delivery creation to TM, and a request to create a delivery-based transportation requirement in TM.

4. Log delivery confirmations (TM)

TM creates an entry in the application log. The application log entry includes the system messages that were created in SAP ERP during delivery creation as well as the number and document IDs of deliveries created in SAP ERP. If an error occurred during delivery creation in SAP ERP, then TM generates an alert.

When a confirmation is received, TM sets a confirmation timestamp in the relevant freight units. The timestamp is used to monitor whether SAP ERP has responded.

5. Create delivery-based transportation requirements (TM)

For more information, see [Integration of ERP Deliveries](#).

6. Consume order-based transportation requirements (TM)

For more information, see [Consumption of Order-Based Transportation Requirements](#).

Creation of Delivery Proposals

Use

Transportation Management (TM) can propose inbound deliveries and outbound deliveries and send these proposals to SAP ERP to trigger the creation of deliveries.

Prerequisites

- An order-based transportation requirement exists.
- Freight units exist for the order-based transportation requirement.
- The order-based transportation requirement is not blocked for delivery.
- A delivery is expected for the order in the ERP system. This can be determined, for example, by the order type or shipping conditions.
- If you want to use a selection profile to select the relevant input for the delivery proposal, you have defined a selection profile (see [Selection Profile](#)).
- If you want to use a delivery profile during delivery creation, you have defined a delivery profile (see [Delivery Profile](#)).
- If you want to send delivery proposals to SAP ERP from the freight order UI, you have assigned a delivery profile to the relevant freight order types. You do this in Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types**.
- If you want to handle actual amounts that differ from the planned amount, you have to activate this functionality for the relevant OTR type in Customizing for **Transportation Management** under **Integration** > **Logistics Integration** > **External TM System Integration** > **Order-Based Transportation Requirement** > **Define Order-Based Transportation Requirement Types**.

When using the background report, a delivery proposal must not have already been made for the freight units. This includes freight units for which a delivery failed to be created.

Features

Background Report

The user can create delivery proposals for order-based transportation requirements by using a background function (using transaction SE38 to run program /SCMTMS/DLV_BATCH). The user can enter selection criteria or use selection profiles. In both cases, the user can also enter a delivery profile. The system selects the documents, creates the delivery proposals, and sends them automatically to SAP ERP.

If a freight unit has previously been used for a delivery proposal, but no delivery-based transportation requirement was ultimately created, the user cannot use the function to create a new delivery proposal for that freight unit. Instead, the user has to create the delivery proposal interactively.

For more information about the background function, see [Creation of Deliveries in SAP ERP](#).

Interactive Delivery Creation

The user can create deliveries interactively as follows:

- From the launchpad, using the **Delivery Proposal Builder** app.

The user specifies one or more profiles according to which the system selects and displays a list of business documents. The user then selects the relevant documents in the list and triggers delivery proposal creation. The user can review the delivery proposals before sending them to SAP ERP.

- From the following overviews:

- Overview of freight orders or freight bookings in the **Freight Orders** app, which you open from the launchpad.
- Overview of order-based transportation requirements or order-based freight units in the **Order-Based Transportation Requirements** app, which you open from the launchpad.
- Overview of order-based freight orders or order-based freight bookings in the **Transportation Cockpit** app, which you open from the launchpad.

The user selects the relevant business documents in the overview and chooses the **Create Deliveries in ERP** pushbutton. The system displays a list of freight units related to the selected business documents. The user selects the relevant freight units and triggers delivery proposal creation. The user can review the delivery proposals before sending them to SAP ERP.

- From the **Transportation Cockpit**

The user triggers delivery proposal creation for specific business documents after carrying out planning in the **Transportation Cockpit**. The user selects the business documents in the **Transportation Cockpit**, specifies a delivery profile, and chooses the **Create Delivery Proposals** pushbutton. The system automatically creates and displays the delivery proposals. The user can review the delivery proposals before sending them to SAP ERP.

The user can also trigger the creation and sending of delivery proposals for a specific freight order by choosing **Follow Up > Send Delivery Proposals to ERP** in the freight order UI. The system automatically creates the delivery proposals and sends them to SAP ERP. In this case, the user cannot review the delivery proposals before they are sent to SAP ERP. Note that if the user generates the delivery proposal from a freight order UI, the system creates the delivery proposals according to the delivery profile assigned to the freight order type.

Source of Data

TM collects data from the relevant documents as follows:

- Dates

TM uses dates from the freight order or freight booking. If the order-based transportation requirement is not planned, that is no freight order or freight booking exists, then TM uses dates from the freight units.

- Quantities

TM uses quantities from the freight units.

- ERP document references

TM retrieves document references (such as order document number) from the order-based transportation requirement.

Consolidation

If order-based transportation requirements have been planned, and freight orders or freight bookings exist for the transportation requirements, TM uses only one freight order or one freight booking for each delivery proposal. That is, only freight units that belong to the same freight order or freight booking can be consolidated into a single delivery proposal. In this way, freight units from various order-based transportation requirements can be consolidated into a single delivery proposal if they belong to the same freight order or freight booking. If, however, freight units from a single order-based transportation requirement were distributed across multiple freight orders or freight bookings, then multiple deliveries are proposed for the freight units of this one order-based transportation requirement.

If order combination is not allowed for an order-based transportation requirement, TM does not group freight units from this transportation requirement together with freight units from other order-based transportation requirements. If freight units for such order-based transportation requirements have been consolidated into the same freight order or freight booking during planning, TM does not include them in the same delivery proposal, but instead creates multiple delivery proposals.

In addition, freight units can only be consolidated into the same delivery proposal if the following data is the same:

- SAP ERP system from which the order was received
- Source location
- Destination location
- Incoterms
- Shipping conditions
- SAP ERP document type, such as purchase order or sales order
- Start stop (if planning has already been done)
- Destination stop (if planning has already been done)

If an order-based transportation requirement has not been planned in TM, and no freight order or freight booking exists, TM can also create delivery proposals for the corresponding freight units. In this case, TM creates delivery proposals according to the settings in the delivery profile. For more information about the delivery profile, see [Delivery Profile](#).

Bill of Materials (BOMs)

Either a BOM header or BOM items can be goods-movement-relevant. If a BOM header is goods-movement relevant, TM can, if necessary, distribute the quantity across multiple delivery proposals. If BOM items are delivery relevant, TM expects a common delivery group for all items and does not distribute the items or their quantities across multiple delivery proposals.

Consumption of Schedule Lines in SAP ERP

The delivery proposal in TM takes into account that SAP ERP consumes order schedule lines in chronological order. Therefore, TM does not propose a delivery for a schedule line before another schedule line of the same order item that has an earlier delivery date has been used for a delivery proposal. If freight unit selection results in this situation, TM excludes the affected freight units from the selection.

If an error occurs in SAP ERP when creating a delivery, then SAP ERP does not create deliveries that correspond to later schedule lines of the same order item, or the delivery is created but excludes the later schedule lines of that item.

Execution and Planning Blocks in SAP ERP

You can use the delivery profile to determine how, during delivery creation, the TM system deals with objects that are blocked for planning or execution. You can choose either or both of the following options:

- Exclude freight units or freight orders that are blocked for planning
- Exclude freight units or freight orders that are blocked for execution

i Note

Previously the system excluded all objects with the status “Blocked for Execution”, from delivery creation. The option to exclude objects with the status “Blocked for Planning” was not previously available.

The system considers freight units and freight orders to check whether they are blocked for planning or blocked for execution. If a freight order or its assigned freight units are blocked, the system – depending on the settings you have chosen in the delivery profile and depending on how you start delivery creation – prevents the user from creating a delivery. The system runs the check based only on the object (freight orders, bookings, transportation units, freight units) that you have selected as the starting point for creation of delivery proposals.

❖ Example

A freight unit is assigned to a pre-carriage and a main-carriage freight order. The main-carriage freight order is blocked for execution, but the freight unit has no blocks. The delivery profile has been configured to exclude freight units or freight orders that are blocked for execution.

If you start creation of delivery proposals with the pre-carriage freight order, the system allows creation of deliveries.

If you start creation of delivery proposals with the main-carriage freight order, which is blocked for execution, the system prevents you from creating a delivery.

The system checks for blocks in freight orders only when you start delivery creation in a freight order context, for example, from the freight order POWL or the freight order UI or by using a freight order selection profile in the report "Create Deliveries in ERP". Otherwise only the blocks in freight units are considered.

Delivery Proposals with Freight Document Hierarchies

With transportation units, TM provides a new three-tier hierarchy of freight documents. This can comprise a freight order on the highest level, transportation units on the middle level, and freight units on the lowest level. For two-tier hierarchies, the highest level can include freight orders, freight bookings and transportation units. On the lowest level, you can have freight units and automatically generated transportation units. In addition, automatically generated transportation units and automatically generated freight orders can represent a one level hierarchy.

You can create delivery proposals for each of these hierarchy types. If not explicitly stated, the term **freight unit** can also mean **automatically generated freight order** or **automatically generated transportation unit**, in order to represent objects on the lowest hierarchy level. In the same way the term **freight order** can also mean **freight booking** or **transportation unit** in order to represent objects on the highest hierarchy level.

Delivery Proposals for Bulk Material

Bulk materials are fluids and materials like sand or coal, that are given in base units of measure of volume or weight. You can use TM to deal with the differences in actual and planned quantity that occur naturally during the loading of liquid or dry bulk material.

In the Freight Order UI you can enter the volume and/or the weight of the actual quantities that were measured during or after loading. If these quantities differ from the planned quantities that were received from the OTR, the system creates a delivery proposal with the actual quantities. Subsequently, the ERP system creates a delivery accordingly. Differing quantities do not lead to split Freight Units.

Integration of ERP Deliveries

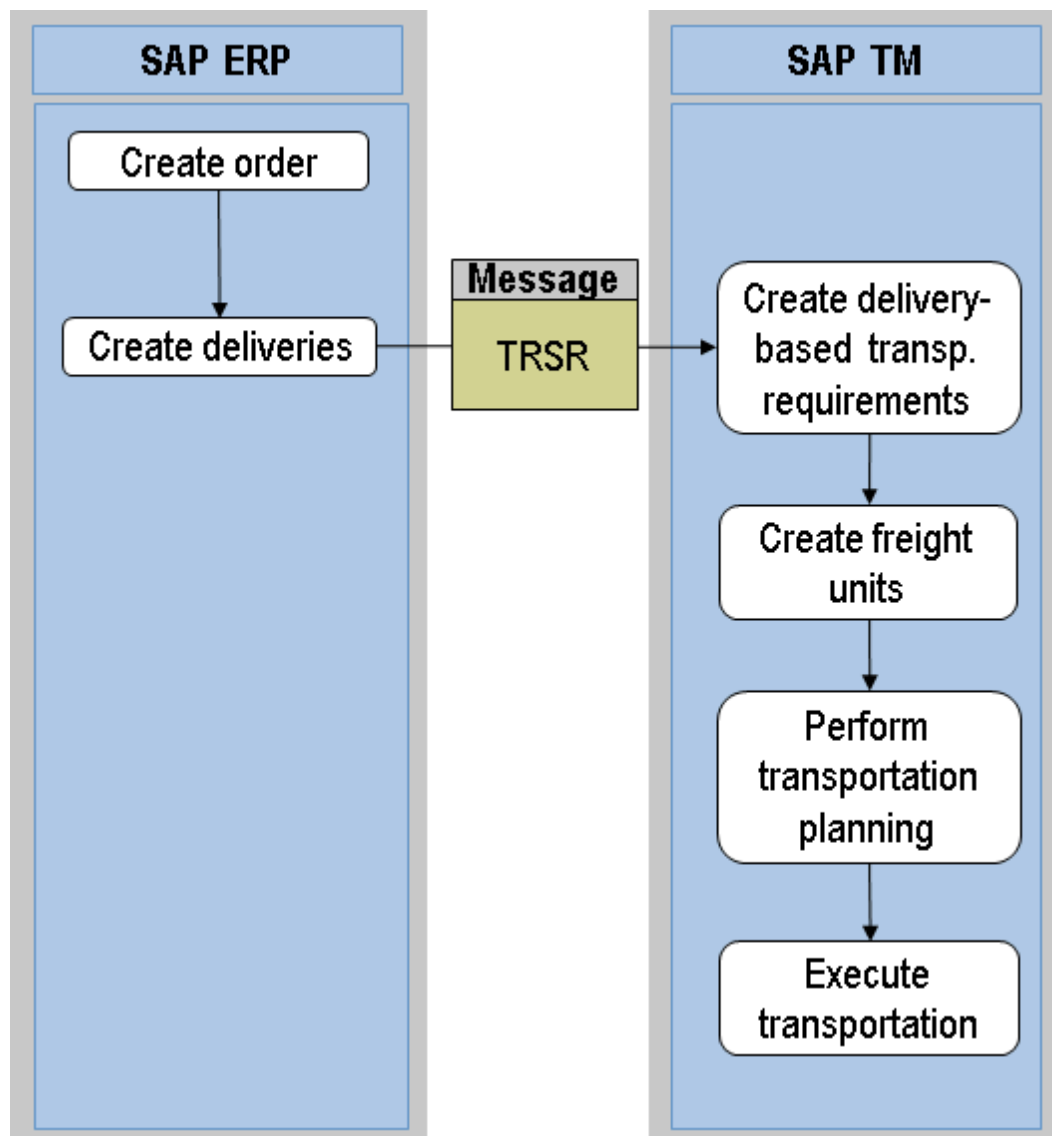
Use

You use this process if a customer creates a delivery in SAP ERP and sends it to Transportation Management (TM). In this process, a delivery-based transportation requirement is created in TM based on the delivery from SAP ERP. If the corresponding order was also sent to TM, the transportation demand of the order-based transportation requirements is consumed.

Process

Integration of ERP Deliveries Without Sending ERP Order to TM

The following figure shows the process flow for a delivery that is created in the SAP ERP system and sent to TM, where a corresponding order-based transportation requirement has **not** been created in TM:



Integration of ERP Deliveries Without Sending ERP Order to SAP TM

In the above figure, the abbreviation TRSR means `TransportationRequestSUITERequest`.

The process runs as follows:

1. Create order (SAP ERP)

An order is created in SAP ERP, but is not sent to TM.

2. Create deliveries (SAP ERP)

Deliveries for the order are created in SAP ERP. SAP ERP sends the deliveries to TM.

3. Create delivery-based transportation requirements (TM)

When TM receives the deliveries, TM creates delivery-based transportation requirements for these deliveries.

4. Create freight units (TM)

Since no corresponding order-based transportation requirements exist for the delivery-based transportation requirement, TM creates new freight units for the delivery-based transportation requirement, according to the freight unit building rule specified.

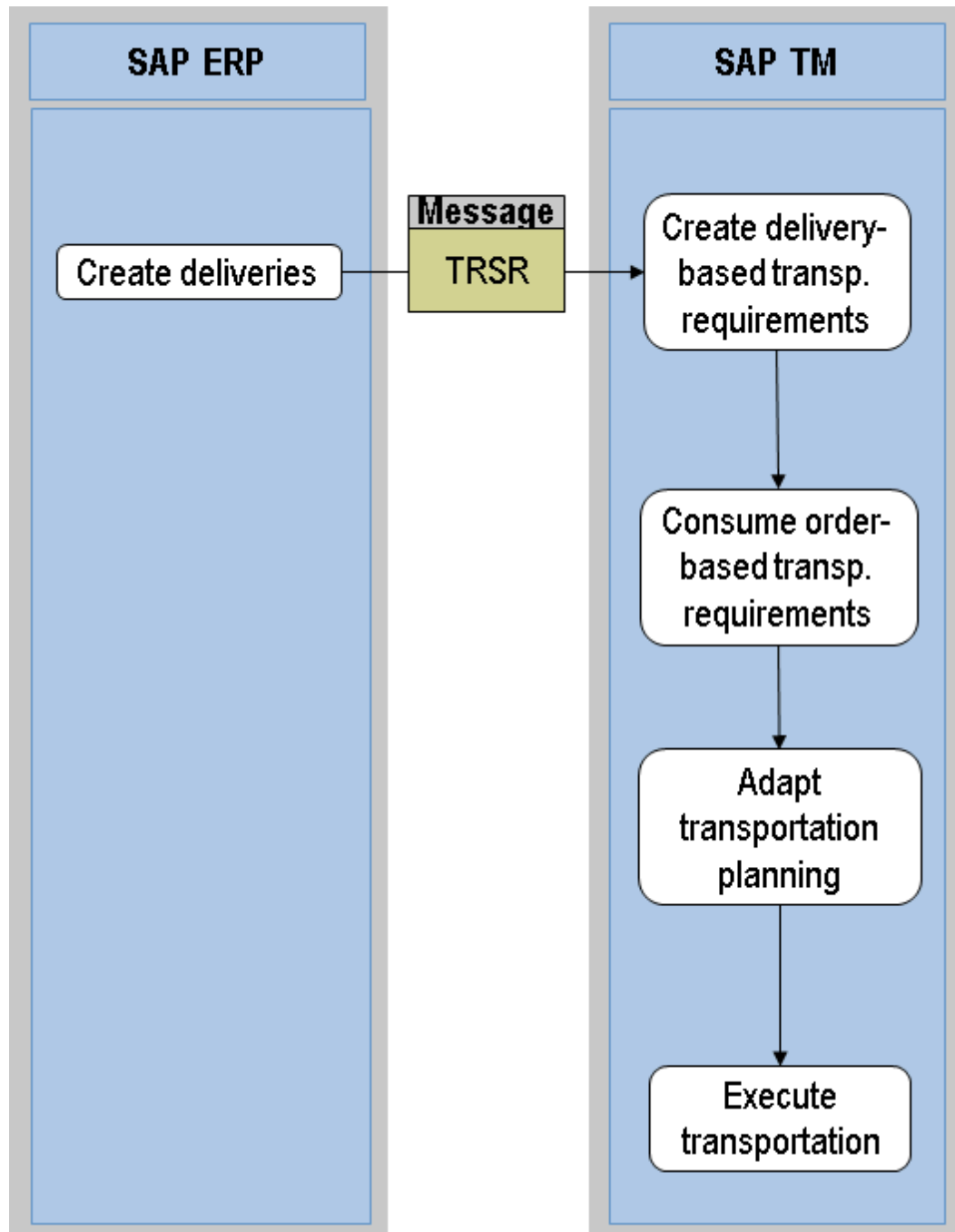
5. Perform transportation planning (TM)

6. Execute transportation (TM)

Integration of ERP Deliveries With Order Integration

The following figure shows the process flow for a delivery that is created in the SAP ERP system and sent to TM where a corresponding order-based transportation requirement exists in TM. There are two alternative processes that can **precede** this process:

- An order has been sent to TM and TM has sent a delivery proposal to SAP ERP (see [Creation of ERP Deliveries from TM](#)).
- An order has been sent to TM, but TM has **not** sent a delivery proposal to SAP ERP.



Integration of ERP Deliveries With Order Integration

In the above figure, the abbreviation TRSR means `TransportationRequestSUITERequest`.

The process runs as follows:

1. Create deliveries (SAP ERP)

Deliveries are created in SAP ERP. SAP ERP sends the deliveries to TM.

2. Create delivery-based transportation requirements (TM)

When TM receives the deliveries, TM creates delivery-based transportation requirements for these deliveries.

3. Consume order-based transportation requirements (TM)

Since one or more corresponding order-based transportation requirements exist in TM, the delivery-based transportation requirements consume the transportation demand of the order-based transportation requirement.

For more information, see [Consumption of Order-Based Transportation Requirements](#).

4. Adapt transportation planning (TM)

Depending on change controller setting, TM can adjust the transportation planning that had previously been performed for the order-based transportation requirement. For more information about the change controller, see [Change Controller](#).

5. Execute transportation planning (TM)

i Note

You can display individual overviews of delivery-based freight orders, delivery-based air freight bookings, and delivery-based ocean freight bookings in the [Transportation Cockpit](#) app, which you open from the launchpad.

Delivery-Based Transportation Requirement

Definition

Represents a transportation demand based on a delivery that was created in SAP ERP. The delivery-based transportation requirement (DTR) in Transportation Management (TM) corresponds to a single delivery in SAP ERP.

Structure

The delivery-based transportation requirement has the following structure:

- Tabs for general data, terms and conditions, business partners, location and dates/times, notes, and administrative data.

Contains information that is relevant for all items of a delivery-based transportation requirement.

- Document flow

Shows a list of preceding and subsequent documents that are related to the delivery-based transportation requirement.

- Items

This table shows the product items, service items, text items, and handling units. In the delivery-based transportation requirement, handling units can contain product items and other handling units. For each item, you can display more detailed information on a number of tab pages.

Statuses in the Delivery-Based Transportation Requirement

The following statuses are available in the delivery-based transportation requirement:

- **Lifecycle Status**

Status	Description
New	The delivery-based transportation requirement has been created. No freight units have been assigned to the delivery-based transportation requirement yet.
In Planning	The delivery-based transportation requirement is in the planning process and at least one freight unit has been created for the transportation requirement.
Planned	All the freight units that are assigned to the delivery-based transportation requirement have been planned.
In Execution	The delivery-based transportation requirement is in the execution process. At least one follow-on document (not necessarily a freight unit) is in execution. For more information about the execution of a transportation requirement, see the execution status of the transportation requirement or of the follow-on documents.
Completed	Processing of the delivery-based transportation requirement has been completed. Execution has finished.
Canceled	The delivery-based transportation requirement has been canceled.

- **Execution Status**

Status	Description
Execution Not Started	The execution of the delivery-based transportation requirement has not yet started.
In Execution	The delivery-based transportation requirement is currently being executed. At least one of the planning documents assigned to the order-based transportation requirement is in the execution process.
Executed	The execution of the delivery-based transportation requirement has been completed. The execution of all planning documents assigned to the delivery-based transportation requirement has been completed.

Blocks

A delivery-based transportation requirement can contain the following blocks:

- Planning block
- Execution block

You can configure the way in which TM responds to delivery block or shipment planning block reasons from SAP ERP. You can specify whether the TM system should create a planning block or an execution block or both, according to the defined SAP ERP block reason.

For more information on this see Customizing for Transportation management under:

- [Integration](#) > [Logistics Integration](#) > [External TM System Integration](#) > [Define Blocking of Transportation Requirements](#) > [Define Blocks Based on Delivery Block](#)
- [Integration](#) > [Logistics Integration](#) > [External TM System Integration](#) > [Define Blocking of Transportation Requirements](#) > [Define Blocks Based on ERP Shipment Planning Block](#)

Credit Check

You can set up credit control for deliveries in SAP ERP. An order or delivery has a credit status that reflects the outcome of the credit limit check on that document. If the credit status of the document is not “approved” TM receives a delivery block with blocking reason 01 (“Credit Limit”) for the document.

Packaging Information

During the packing process in SAP ERP, handling units can be created to reflect items such as containers, rail cars, or pallets. In TM you can now configure the system to use the appropriate item category to create the packaging information in TM.

For more information on this see Customizing for Transportation management under [Integration](#) > [Logistics Integration](#) > [External TM System Integration](#) > [Delivery-Based Transportation Requirement](#) > [Define Usage of ERP Packaging Information](#)

Integration

The delivery-based transportation requirement is integrated with the following objects in TM:

- Freight units

TM can create freight units based on a delivery-based transportation requirement, or consume existing freight units.

- Order-based transportation requirement

One or more delivery-based transportation requirements consume the transportation demand of one or more order-based transportation requirements.

Consumption of Order-Based Transportation Requirements

Use

When a delivery-based transportation requirement (DTR) is created in Transportation Management (TM) based on a delivery that was created in SAP ERP, the DTR consumes the freight units assigned to the relevant order-based transportation requirements (OTR) if the corresponding orders were sent to TM.

Features

Determination of Freight Units

TM uses the ERP order (base document) item number that is referenced in the DTR item to find the corresponding OTR items (the base document number and item number is also referenced in the OTR items). In this way, the system can determine the relevant freight units, which are assigned to these OTR items.

Reassigning Freight Units to the Delivery-Based Transportation Requirement

TM reassigns the freight units from the OTRs to the DTRs. If multiple freight unit items are determined, TM assigns the freight unit items to the items of the DTR in chronological order (according the dates of the schedule lines that the freight unit items

reference). Once a freight unit is assigned to a DTR, the freight unit cannot be assigned to another DTR.

If TM does not determine any freight units, TM can build new freight units for the DTR, if automatic freight unit building is allowed for the DTR type. This situation might occur for example, if deliveries are created in the ERP system without a preceding delivery proposal being sent from TM.

Splitting Freight Units

If a freight unit is to be reassigned, but not all freight unit items or quantities are relevant for the DTR, TM splits the freight unit. One of the resulting freight units contains only the relevant freight unit items and is assigned to the DTR. The other resulting freight unit contains the non-relevant freight unit items and is assigned to the OTR.

Furthermore, if the quantity in the DTR item is less than the quantity in the freight unit, TM also splits the relevant freight unit.

If there are other differences between the freight unit or item and the DTR, TM adapts the freight units accordingly so that the information in the freight unit corresponds to the transportation requirement.

Updating the Open Quantity of the Order-Based Transportation Requirement

The open quantity in an OTR represents the quantity that has not been assigned to a DTR. When freight units are reassigned to a DTR, the open quantity of the OTR is reduced accordingly.

Updating the Status of the Order-Based Transportation Requirement

Initially the OTR has the consumption status **Not Consumed**. If at least one OTR item or a part of an item has been consumed by a DTR item, the consumption status of the OTR will change to **Consumed Partially**. If all items in an OTR have been consumed, the system changes the consumption status to **Consumed Completely**. Once the consumption status is **Consumed Completely**, the system changes the lifecycle status to **Completed**.

Handling Units

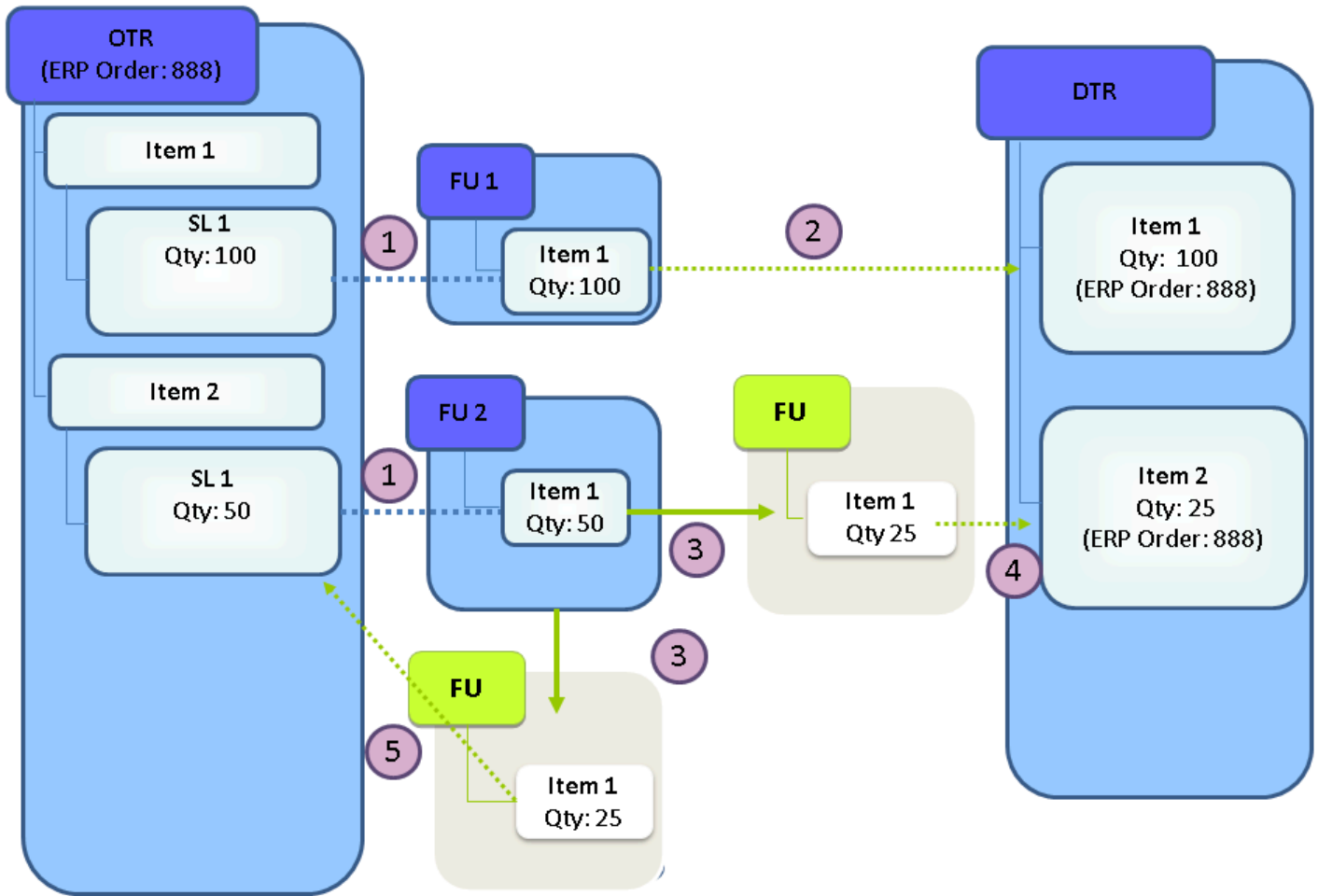
The item structure of the DTR may differ from the structure in the OTR due to the handling unit information in the delivery received from SAP ERP. If a handling unit has been created, a handling unit is added to the DTR. This handling unit item is transportation relevant and includes the complete quantity of both the packaging and goods that are to be transported. The handling unit item can contain one or more product items or packaging items. The freight units (or parts thereof) that are determined for the items of this handling unit item are consolidated into one single freight unit item, which is assigned to the handling unit item.

i Note

ERP-TM integration for deliveries does not support cross-delivery handling units.

Example

The following figure shows a simple example of how an OTR might be consumed by a DTR and is followed by an explanation:



OTR Consumed by DTR

1. Determine freight units

The relevant freight units are determined based on the order item references that are contained in the DTR items and the OTR items.

2. Reassign freight unit (FU 1)

The first DTR item fully consumes the freight unit (FU 1) assigned to OTR item 1.

The open quantity of OTR item 1 is reduced to 0

3. Split freight unit (FU 2)

The quantity in FU 2 is greater than the quantity in item 2 of the DTR. The DTR item consumes the necessary quantity of the FU. Therefore, the freight unit (FU 2) is split.

4. Assign freight unit to DTR

The freight unit containing only one item with the correct quantity, is assigned to the DTR item

5. Assign freight unit to OTR

The freight unit containing the remaining quantity is assigned to OTR item 2. The open quantity of OTR item 2 is reduced to 25.

Posting Goods Issue from TM

Use

With this feature you can post goods issue for SAP ERP outbound deliveries from TM.

The TM system can post a goods issue to the ERP system if all of the following conditions are met:

- a transport is started
- all delivery items in ERP that are relevant for picking have been picked
- a plant and storage location have been assigned
- the goods issue has not already been posted

i Note

You cannot cancel the goods issue from TM. If it is necessary to cancel the goods issue this has to be done manually in ERP.

Integration

For more information about the integration of this function into overall background processing, see [Background Processing](#)

Prerequisites

An outbound delivery has been sent from SAP ERP to TM and a delivery-based transportation requirement (DTR) has been created in TM.

Features

In SAP ERP, the ERP inbound service `OutboundDeliveryERPChangeRequestConfirmation_In_V1` is used to post a goods issue for an outbound delivery in ERP. In TM this service is replicated as an outbound service:

`OutboundDeliveryERPChangeRequestConfirmation_Out_V1`. This TM outbound service does not support the complete functionality of the ERP inbound service but only the subset needed for goods issue posting.

Activities

You can use transaction SE38 to run the program `/SCMTMS/POST_GI_BATCH` to post goods issue to SAP ERP.

The system checks the goods movement status of the delivery in the ERP system and selects only deliveries for which no goods issue has been posted. For all such deliveries the system checks the execution status for the related DTR in the TM system. The system triggers goods issue for all DTRs for which execution has started or completed.

Synchronization of Transportation Planning Status

Use

This feature synchronizes the transportation planning status of a delivery in SAP ERP with the planning status of a delivery-based transportation requirement (DTR) in TM. For every change to the planning status in TM an update is sent to SAP ERP.

Prerequisites

You have enabled the sending of delivery updates from TM to SAP ERP delivery for the relevant delivery split/update type of the DTR in Customizing for [Transportation Management](#) under [Integration](#) [Logistics Integration](#) [External TM System Integration](#) [Delivery-Based Transportation Requirement](#) [Define Delivery Split/Updates Types](#).

An outbound delivery has been sent from SAP ERP to TM and a DTR has been created in TM.

Features

In SAP ERP, a delivery contains a transportation planning status at header level. There are four possible transportation planning statuses:

- Not Relevant

The delivery is not relevant to shipment.

- Not Processed

The delivery is not currently assigned to a shipment.

- Partially Processed

The delivery has been assigned to at least one stage of a shipment.

- Completely Processed

The delivery has been assigned to all required stages. All stages required for the delivery have been planned.

i Note

Prior to the release of enhancement package 1 (SP01) of TM9 this status could only be set automatically from SAP ERP shipment processing.

In TM a corresponding planning status exists on DTR header level. The four possible planning statuses are:

- Planning Not Started

No freight units (FU) have been created.

- In Planning

All FUs have been created; no freight order assignment has been made.

- Partially Planned

At least one FU has been assigned to a freight order.

- Planned

All stages of all FUs have been planned.

The correspondence of statuses is as follows:

TM Planning Status	SAP ERP Transportation Planning Status
Planning Not Started	Not Processed
In Planning	Not Processed
Partially Planned	Partially Processed

TM Planning Status	SAP ERP Transportation Planning Status
Planned	Completely Processed

Integration of Changed and Deleted ERP Orders and Deliveries

Use

Logistics integration with SAP ERP supports changes to and deletion of orders in SAP ERP, and changes to, deletion, and splitting of deliveries in SAP ERP. If Transportation Management (TM) receives an updated order or delivery, the change controller propagates the changes to the freight units of the corresponding transportation requirement. Once freight units have been adapted, the change controller can process subsequent documents further depending on strategies. For more information, see [Change Controller](#).

i Note

A user cannot change the data in an order-based transportation requirement or delivery-based transportation requirement nor cancel or delete these documents in Transportation Management (TM). Changes can only be made in the SAP ERP system and sent to TM.

Features

For more information about how the change controller handles freight units assigned to changed or deleted transportation requirements, see the following sections:

- [Processing Changed or Deleted Orders from ERP](#)
- [Processing Changed, Deleted, or Split Deliveries from ERP](#)

Deleting and Canceling Freight Units

If, based on a changed or deleted transportation requirement, a freight unit becomes empty, the system either deletes or cancels the freight unit. Whether the freight unit is deleted or canceled, depends on your settings for the freight unit type.

Processing Changed or Deleted Orders from ERP

Use

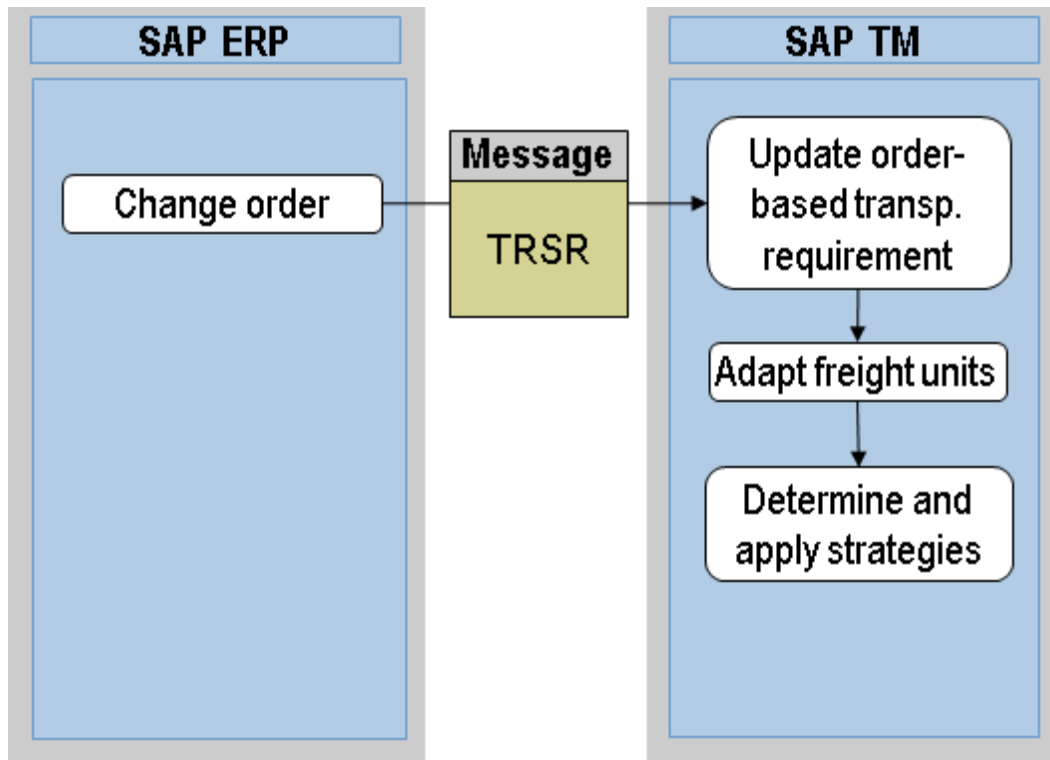
If a user changes the data in an order in SAP ERP (for example, quantities, dates and locations can be changed), SAP ERP sends these changes to Transportation Management (TM) and the corresponding order-based transportation requirement is updated.

If an order is deleted in SAP ERP, the corresponding order-based transportation requirement is canceled in TM.

Process

Update of an Order

The following figure shows the process flow when an order is changed in SAP ERP, and is followed by a description:



Update of Order in SAP ERP

In the above figure, the abbreviation TRSR means TransportationRequestSUITERequest.

The process runs as follows:

1. Change order (SAP ERP)

If changes are made to an order in SAP ERP, an update is sent to TM

2. Update order-based transportation requirement (TM)

In TM, the corresponding order-based transportation requirement is updated. If the quantity has changed, the open quantity (and, if affected by this, the consumption status) is adjusted accordingly.

3. Adapt freight units (TM)

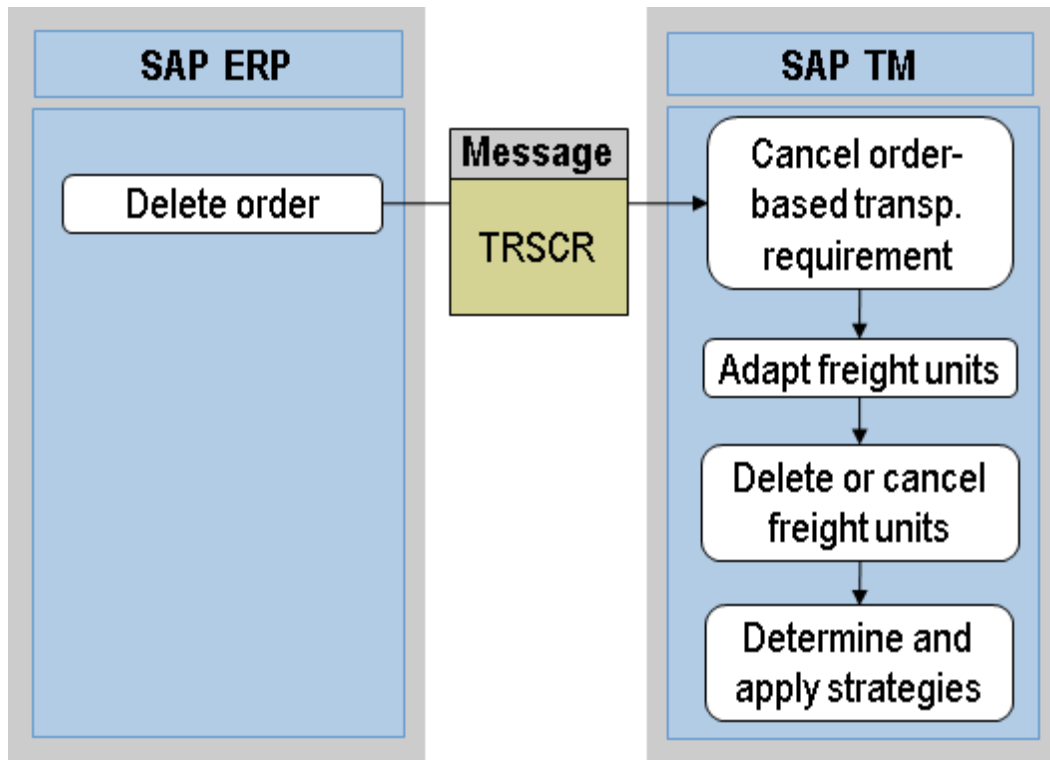
TM updates the freight units according to the assigned freight unit building rule. If a delivery has already been created (and thus, the freight units have already been reassigned to the delivery-based transportation requirement), TM creates new freight units in case of a quantity increase to cover the resulting open quantity of the order-based transportation requirement. Note that the delivery-based transportation requirement and its freight units are not affected by the updated order.

4. Determine and apply strategies (TM)

The change controller uses strategies to adapt subsequent documents.

Deletion of an Order

The following figure shows the process flow when an order is deleted in SAP ERP, and is followed by a description:



Deletion of Order in SAP ERP

In the above figure, the abbreviation TRSCR means TransportationRequestSUITECancellationRequest.

The process runs as follows:

1. Delete order (SAP ERP)

SAP ERP sends a request to cancel the order-based transportation requirement in TM. Note that if a delivery has already been created in SAP ERP, the order cannot be deleted in SAP ERP

2. Cancel order-based transportation requirement (TM)

In TM, the corresponding order-based transportation requirement is canceled.

3. Adapt freight units

If the freight unit was only relevant for the canceled order-based transportation requirement, the freight unit becomes empty. If the freight unit was assigned to multiple order-based transportation requirements, the quantity in the freight unit is reduced according to the canceled order-based transportation requirement.

4. Delete or cancel freight units (TM)

If the freight unit is empty, TM either deletes or cancels the freight unit according to your settings for the freight unit type.

5. Determine and apply strategies (TM)

Processing Changed, Deleted, or Split Deliveries from ERP

Use

If a delivery is changed, split, or deleted in SAP ERP, this information is sent to Transportation Management (TM). TM adjusts the corresponding delivery-based transportation requirement based on this information.

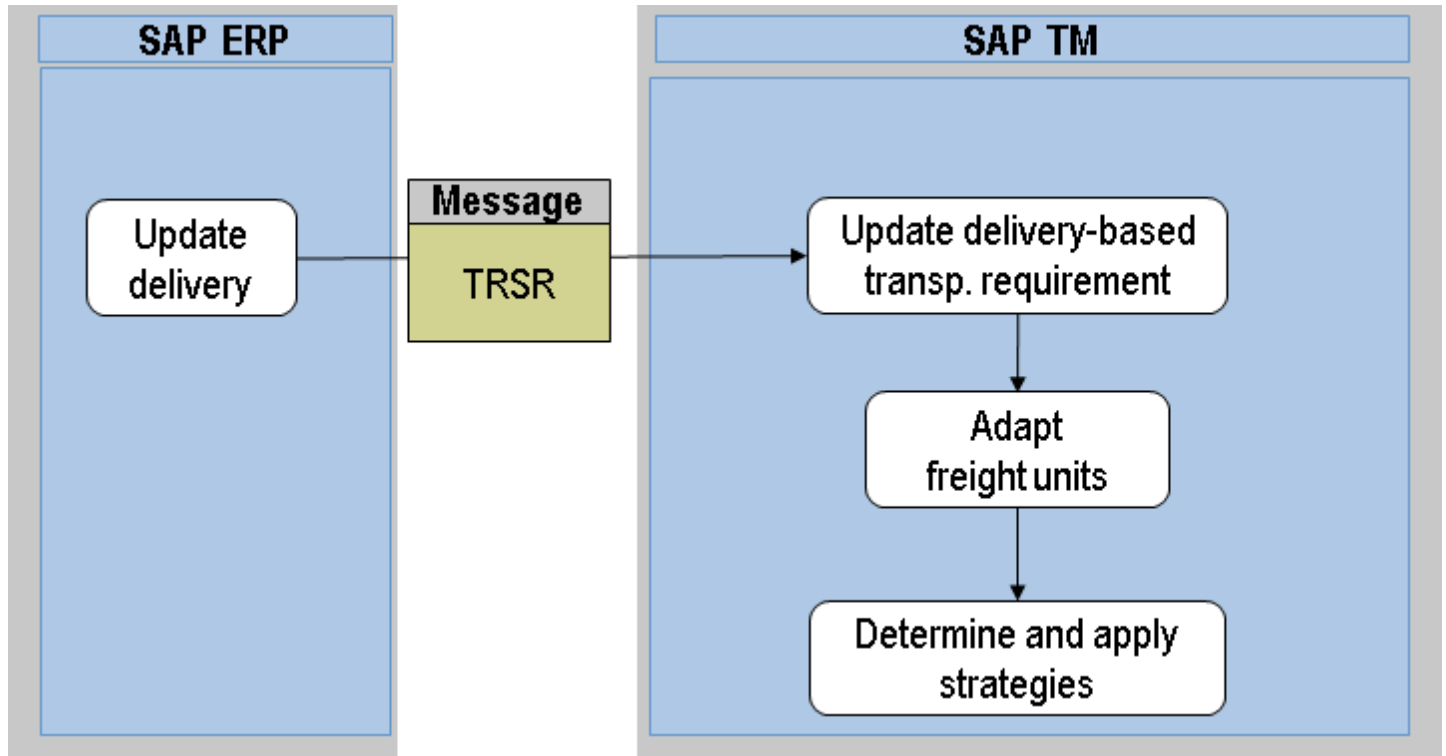
Note that changes or splits can originate in SAP ERP or in the SAP Extended Warehouse Management (SAP EWM) application. Changes or splits in SAP EWM are always transferred to TM via SAP ERP.

The following sections describe how TM handles these changes.

Process

Update of a Delivery

The following figure shows the process for updating a delivery in SAP ERP after delivery creation and is followed by a description.



Update of Delivery in SAP ERP

In the above figure, the abbreviation TRSR means TransportationRequestSUITERequest.

The process runs as follows:

1. Update delivery (SAP ERP)

ERP sends a changed delivery to TM.

2. Update delivery-based transportation requirement (TM)

In TM, the corresponding delivery-based transportation requirement is updated.

3. Adapt freight units (TM)

The change controller adjusts the freight units according to the changes in the delivery-based transportation requirement.

If the quantity of the delivery-based transportation requirement has changed **and** the delivery-based transportation requirement is assigned to freight units that were consumed from an order-based transportation requirement, the change controller processes the freight units as follows:

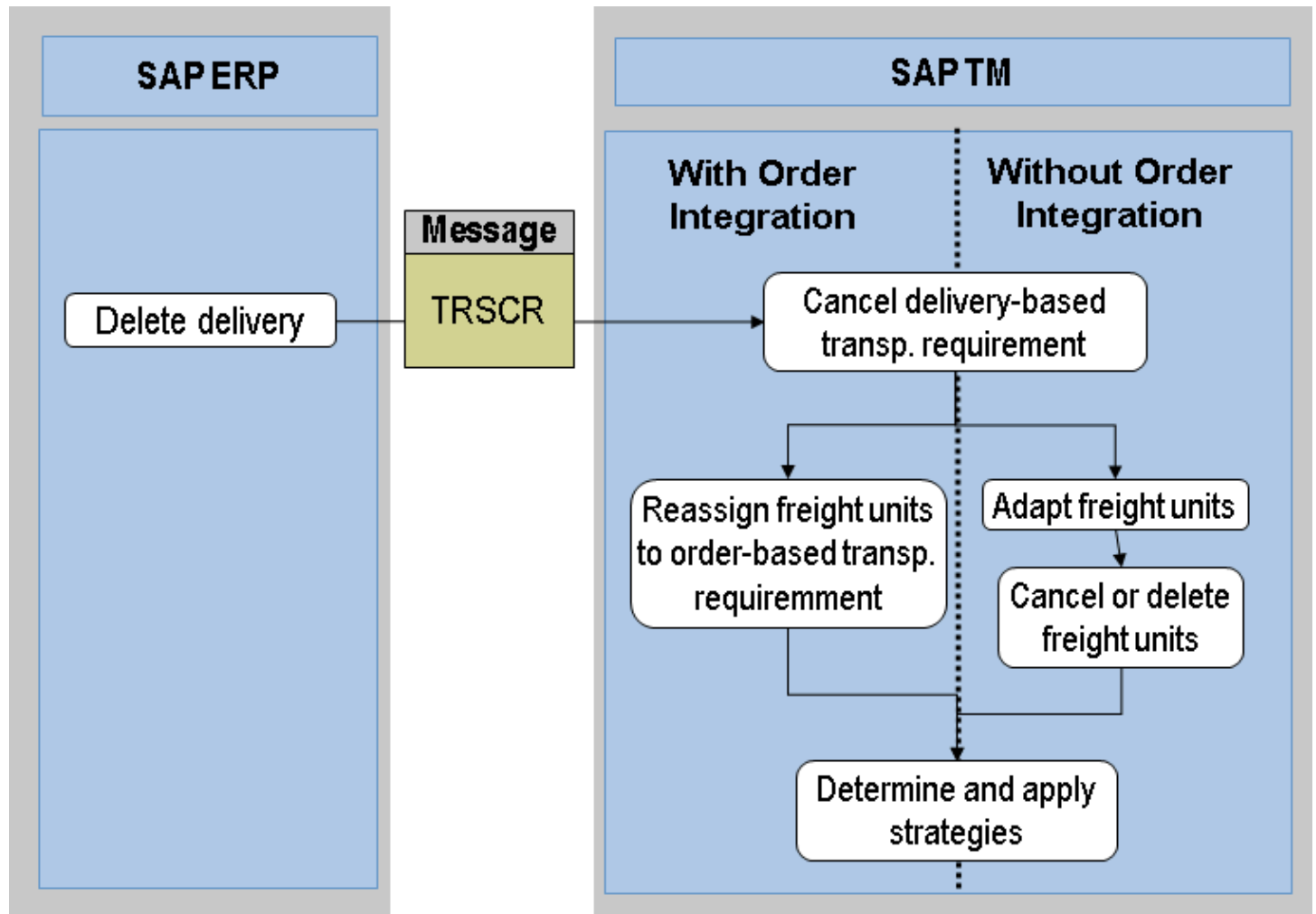
- o If the quantity of the delivery-based transportation requirement has **decreased**, the open quantity of the corresponding order-based transportation requirements is increased accordingly; and the freight units of the delivery-based transportation requirement are split. Freight units are retained by the delivery-based transportation requirement to cover the reduced quantity of the delivery-based transportation requirement, and the remaining freight units are assigned to the corresponding order-based transportation requirement.
- o If the quantity of the delivery-based transportation requirement has **increased**, and there are still open quantities in the corresponding order-based transportation requirement, the system reassigns the freight units from the order-

based transportation requirement to cover this increased delivery quantity. The system then reduces the open quantity of the order-based transportation requirements.

4. Determine and apply strategies (TM)

Deletion of a Delivery

The following figure shows the process for deleting a delivery in SAP ERP after delivery creation and is followed by a description:



Deletion of Delivery in SAP ERP

In the above figure, the abbreviation TRSCR means TransportationRequestSUITECancellationRequest.

The process runs as follows:

1. Delete delivery (SAP ERP)

ERP sends a request to cancel the delivery-based transportation requirement.

2. Cancel delivery-based transportation requirement (TM)

In TM, the corresponding delivery-based transportation requirement is canceled.

3. Reassign freight units to order-based transportation requirement (TM)

If a corresponding order-based transportation requirement exists, the freight units are reassigned to this order-based transportation requirement.

i Note

If the process runs without order integration (that is, no order-based transportation requirement exists), the freight units are adapted and, if necessary, are canceled or deleted depending on your settings for the freight unit type.

4. Determine and apply strategies (TM)

Split of a Delivery

For more information, see the following:

- [Delivery Splits Triggered by SAP ERP](#)
- [Delivery Splits and Updates Triggered by TM](#)

Delivery Splits Triggered by SAP ERP

Use

You can split an outbound delivery or inbound delivery in SAP ERP or in the SAP Extended Warehouse Management (SAP EWM) application and trigger the required changes in TM. If SAP EWM triggers the split, the delivery split is processed in SAP ERP first and then transferred to TM.

Prerequisites

- You have created a delivery-based transportation requirement (DTR) in TM based on an SAP ERP delivery.
- If required, you have defined a delivery split/update type in Customizing for **Transportation Management** under **Integration > Logistics Integration > External TM System Integration > Delivery-Based Transportation Requirement > Define Delivery Split/Update Types**.

In the delivery/split update type, you define how the system updates freight unit planning after the delivery split has been performed in TM. Note, however, that the system checks this setting only if the planning changes cannot be determined based on other criteria (for example, the application that triggered the split). For more information about the criteria and the sequence in which they are checked, see the field help for the **Handle Planning Results** field in the Customizing activity **Define Delivery Split/Update Types**.

- You have assigned the delivery split/update type to the delivery-based transportation requirement type in Customizing for **Transportation Management** under **Integration > Logistics Integration > External TM System Integration > Delivery-Based Transportation Requirement > Define Delivery-Based Transportation Requirement Types**.

Process

1. Split delivery (SAP ERP)

SAP ERP splits a delivery that has already been sent to TM. SAP ERP sends the following information in separate messages:

- The new delivery that has been created for the items split from the original delivery. The data sent includes a reference to the original delivery and the triggering application SAP ERP or SAP EWM.
- The changed original delivery. The data sent includes a reference to the new delivery.

2. Create new DTR and update original DTR (TM)

TM creates the new DTR first by reassigning the freight units from the original DTR to the new DTR. It then updates the original DTR.

TM receives a new or updated delivery using the [Maintain Transportation Request](#) (IntracompanyTransportationRequestRequest_In) enterprise service in the [Transportation Request Processing](#) process component.

3. Update freight unit planning (TM)

The system determines the required changes to freight unit planning for the freight units assigned to the new DTR and the original DTR. For example, the system discards the planning of the freight units assigned to the new DTR and keeps the planning of the freight units assigned to the original DTR.

Delivery Splits and Updates Triggered by TM

Use

You can split an outbound delivery in SAP ERP based on planning changes in TM. You can also update the goods issue date, loading date, transportation start date, and delivery date in the outbound delivery based on changed dates in TM.

When SAP ERP receives a request for a split or update, it makes the necessary changes in outbound delivery processing and sends the delivery data back to TM. TM can then make the required changes to the delivery-based transportation requirements (DTRs).

Note that TM can trigger the split or update of outbound deliveries only. It cannot trigger the split or update of inbound deliveries.

Prerequisites

- You have created a delivery-based transportation requirement (DTR) in TM based on an SAP ERP outbound delivery.
- The outbound delivery in SAP ERP does not have one of the following statuses:
 - Distribution status: [Distributed](#)
 - Goods movement status: [Partially Completed](#) or [Completed](#)
 - If the delivery has one of the above statuses, TM does not request the split or update.
- You have assigned a split profile to the DTR type in Customizing for [Transportation Management](#) under [Integration > Logistics Integration > External TM System Integration > Delivery-Based Transportation Requirement > Define Delivery-Based Transportation Requirement Types](#).

Process

Outbound Delivery Split

1. Perform transportation planning (TM)

You perform planning or change the existing planning of freight units that belong to a DTR that is based on an outbound delivery. When you save the planning results, the system determines that a delivery split is required. For example, freight units that belong to the DTR have been assigned to different freight orders, or only some of the FUs that belong to a DTR have been assigned to freight orders and the other FUs have not been assigned. Note that the split is required only if the first stage or the last stage of the transportation chain is affected.

TM determines whether the SAP ERP delivery can still be changed by checking the delivery status. If the delivery can be changed, TM sends the required delivery split to SAP ERP. TM sends the details of the items that are to be split from the original delivery (that is, IDs of the items in the original delivery, split quantity, unit of measure, and the required dates for the new delivery).

TM sends the split request using the [Request Outbound Delivery Split V1](#) (OutboundDeliveryERPSplitRequest_Out_V1) enterprise service in the [Transportation Request Processing](#) process component. Note that if a delivery is to be split into several deliveries, each delivery is sent to SAP ERP in a separate message.

2. Split outbound delivery (SAP ERP)

Based on a split request received from TM, SAP ERP creates a new outbound delivery and changes the original outbound delivery (that is, it removes the relevant items from the original delivery and creates a new delivery for these items). If you specified a split reason in the delivery split/update type, SAP ERP splits the delivery according to the settings in the corresponding split profile.

SAP ERP informs TM about whether the delivery split was successful. The confirmation is received in TM by the [Change Transportation Request Based on Outbound Delivery Split Confirmation V1](#) (OutboundDeliveryERPSplitConfirmation_In_V1) enterprise service in the [Transportation Request Processing](#) process component.

SAP ERP sends the following information in separate messages:

- The new delivery that has been created for the items split from the original delivery. The data sent includes a reference to the original delivery and the triggering application TM.
- The changed original delivery. The data sent includes a reference to the new delivery.

3. Create new DTR and update original DTR (TM)

TM creates the new DTR first by reassigning the freight units from the original DTR to the new DTR. It then updates the original DTR.

TM receives a new or updated delivery using the [Maintain Transportation Request](#) (IntracompanyTransportationRequestRequest_In) enterprise service in the [Transportation Request Processing](#) process component.

Outbound Delivery Update

1. Perform transportation planning (TM)

You perform planning or change the existing planning of freight units that belong to a DTR that is based on an outbound delivery. Dates are changed during planning. Alternatively, you change the dates of the relevant freight order. When you save the planning results or the changed freight order, TM determines that the dates in the freight order no longer match the dates in the DTR and the original SAP ERP delivery.

TM determines whether the SAP ERP delivery can still be changed by checking the delivery status. If the delivery can be changed, TM sends the changed dates to SAP ERP and requests the update of the outbound delivery.

TM sends the update request using the [Request Outbound Delivery Update](#) (OutboundDeliveryUpdateRequest_Out) enterprise service in the [Transportation Request Processing](#) process component.

2. Update outbound delivery (SAP ERP)

Based on the request received from TM, SAP ERP updates the relevant dates.

SAP ERP informs TM about whether the delivery update was successful. The confirmation is received in TM by the [Change Transportation Request Based on Outbound Delivery Update Confirmation](#) (OutboundDeliveryConfirmation_In_V1) enterprise service in the [Transportation Request Processing](#) process component.

SAP ERP sends the updated outbound delivery to TM. In TM, the delivery is received by the [Maintain Transportation Request](#) (IntracompanyTransportationRequestRequest_In) enterprise service in the [Transportation Request](#)

[Processing](#) process component.

3. Update DTR (TM)

Result

The planning data in TM and the SAP ERP delivery data is aligned.

Note the following:

- The triggering application is included in the message sent by SAP ERP. This enables TM to distinguish between a split that is triggered by TM (planning is correct) and a split triggered by SAP ERP or SAP EWM (planning is no longer correct and must be updated).
- If TM triggers a delivery split and a delivery update for the same delivery at the same time, the split and update is sent in the same message using the [Request Outbound Delivery Split V1](#) (OutboundDeliveryERPSplitRequest_Out_V1) enterprise service.

You can use a worklist to monitor DTRs for which a delivery split or update has not been completed yet using the [Display Delivery-Based Transportation Requirement](#) app, which you open from the launchpad.

Delivery Creation and Split for Package Units with Multiple Assignments

You can configure the system to allow a freight unit to be assigned to several package units when you create package units from a freight order.

This means that the items of a freight unit may be spread across multiple package units and multiple freight orders. If load planning is not finalized, this can lead to:

- unnecessary delivery splits
- unnecessary freight unit splits
- creation of unnecessary deliveries

To avoid unnecessary splits, or creation of unnecessary deliveries, the system does not consider a freight order for delivery creation until load planning for the freight units of that freight order is finalized. The system treats such freight units as unplanned freight units. When load planning is finalized for all freight units of a freight order, the system considers that freight order and can create, split, or update deliveries.

Prerequisites

You have specified that a freight unit can be assigned to multiple packages. For more information see Customizing for [Transportation Management](#) under [Freight Order Management](#) > [Transportation Unit](#) > [Define Transportation Unit Types](#) > [\(Assgnmt of Pred.Docs](#) (Assignment of Predecessor Documents) field)

Features

When load planning for a freight order is finalized, the system identifies all required delivery splits, and then splits and updates deliveries accordingly.

Related Information

[Package Unit](#)[Creating a Package Unit](#)[Load Planning](#)

Integration of SAP ERP Scheduling Agreements with TM

Use

A scheduling agreement is an outline agreement containing delivery quantities and dates within a specified time period. These are then entered as schedule lines in a delivery schedule.

You can integrate purchase scheduling agreements and Sales scheduling agreements from SAP ERP with TM. The integration is based on delivery schedules. You can control the communication between the two systems by using specific time horizons and by specifying which delivery schedule types (Forecast or JIT) should be transferred.

Communication is based on eSOA services, supporting both process integration (PI) based communication and web services reliable messaging (WS-RM).

The integration can be triggered by the creation or a change of delivery schedules. You can use a batch process to manage the future transfer of delivery schedules

Prerequisites

Integration of Purchase Scheduling Agreements

You have completed Customizing for integration of purchase scheduling agreements. For more information see Customizing for SAP ERP under ► [Integration with other SAP Components](#) ► [Transportation Management](#) ► [Logistics Integration](#) ► [Define Settings for Integration of MM Scheduling Agreements](#) ►.

Integration of Sales Scheduling Agreements

You have completed Customizing for integration of sales scheduling agreements. For more information see Customizing for SAP ERP under ► [Integration with other SAP Components](#) ► [Transportation Management](#) ► [Logistics Integration](#) ► [Define Settings for Integration of Sales Scheduling Agreements](#) ►.

Transfer of Purchasing Documents

You have completed Customizing for transfer of purchasing documents. For more information see Customizing for SAP ERP under ► [Integration with other SAP Components](#) ► [Transportation Management](#) ► [Logistics Integration](#) ► [Activate Transfer of Purchasing Documents](#) ►.

Controlling the Sales Document Transfer

You have customized the processing of sales scheduling agreements to determine the relevant TM control key, specifying which documents in a document chain are to be transferred to TM. For more information see Customizing for SAP ERP under ► [Integration with other SAP Components](#) ► [Transportation Management](#) ► [Order Integration](#) ► [Activate Transfer of Sales Documents](#) ►.

Features

You can work with the following types of scheduling agreements:

- Sales scheduling agreements
 - Create sales scheduling agreements (and corresponding delivery schedules) in SAP ERP and transfer them initially to TM. You can then create transportation requirements in TM.
 - Change sales scheduling agreements (and corresponding delivery schedules) in SAP ERP and transfer the changes to TM. You can then change the corresponding transportation requirements in TM.
 - Create delivery proposals in TM for freight units (FU), on the basis of sales delivery schedules. You can then process the delivery proposal and create outbound deliveries in SAP ERP.
- Materials Management (MM) scheduling agreements
 - Create MM delivery schedules in SAP ERP and transfer them to TM. You can then create transportation requirements in TM.
 - Change MM delivery schedules in SAP ERP and transfer the changes to TM. You can then change the corresponding transportation requirements in TM.
 - Create delivery proposals in TM for FUs which are based on MM delivery schedules. Process the delivery proposal and create inbound deliveries in SAP ERP.

Restrictions

Sales scheduling agreements:

- It is out of scope to send sales scheduling agreements via batch report RSNAST00. The messages for sales scheduling agreements need to be created during the save process of the document.
- Sales order scheduling (SOS) is not supported for sales scheduling agreements.
- Packing instructions are not reflected in the interface for scheduling agreements.
- Scheduling agreements with delivery orders (Standard document type LZM, VBAK-VBKLT = 'C') are out of scope.
- Replenishment deliveries based on inbound sequenced JIT calls (Standard document type LZJ) are out of scope.
- Scheduling agreements with external service agents are out of scope.
- The use of bills of material is out of scope.
- In the integration scenario, SAP ERP transfers either forecast (FRC) delivery schedules or just-in-time (JIT) delivery schedules to TM. The simultaneous transfer of both FRC and JIT delivery schedules is not supported. Material requirements planning (MRP) for delivery schedule types that require both FRC delivery schedules and JIT delivery schedules to be delivery-relevant is not supported. This means that for delivery schedule types A and B, in which the schedule lines of the forecast delivery schedule become relevant for delivery after the JIT delivery schedule horizon, integration is not possible.

Purchase scheduling agreements:

- Usage of pickup sheets: the customer can send a message to the vendor which represents a pickup sheet. A pickup sheet is sent to the vendor and is the basis for transportation planning as the pickup is arranged according to this document. At the vendor a delivery order is created out of the pickup sheet information. The pickup is planned in the TM system of the customer. With respect to TM-integration this process is out of scope.
- Acknowledgements of internet SA releases have no influence on which data is transferred to the TM system.
- Replenishment with summarized JIT calls is out of scope.
- Replenishment based on outbound sequenced JIT calls is out of scope.

- If delivery creation is triggered by the ECC system the delivery creation must be consistent to the schedule line information which has been transferred to TM. This means that a delivery creation process in ECC must not include schedule lines which have not yet been transferred to the TM system or will never be transferred. Especially the ECC system must prevent delivery of FRC schedule lines if these are not configured to be transferred to TM. The same applies to JIT schedule lines.

Supplier Returns Processing Based on QM Notifications

This **example** describes a process for handling supplier returns.

Prerequisites

If you use this process within Advanced Shipping and Receiving, see [Prerequisites for Advanced Shipping and Receiving](#)

Customer Side: Inbound Process

1. Creating a purchase order

Create a purchase order in Materials Management (MM). The system automatically creates freight units (FU).

2. Creating a freight order

Create a freight order in Transportation Management (TM) and insert the FU.

3. Creating an inbound delivery

Create an inbound delivery in TM. You can create the delivery in the [Create Freight Order](#) app by selecting [Follow Up](#) [Create Delivery](#).

When goods are delivered, you post goods receipt.

4. Executing quality management

After putaway, you subject the goods to quality inspection. When any given number of goods fail inspection, you create a QM notification and a returns handling process is initiated.

Customer Side: Returns Handling

1. Creating a quality management (QM) notification

Start with a QM notification.

You can run transaction QM01 to call up the [Create Notification](#) window. To create a complaint, use notification type Q2.

Enter the purchasing document ID and item number.

Under tab [Subject](#), you can enter the quantity of goods complained.

2. Creating a returns delivery

Next, go to [Change Notification](#).

To create a returns delivery, first you must transfer the goods to a blocked stock. You can do that under [Enter Transfer Posting](#). Enter the complaint quantity and storage location. Save the changes.

You can then create a returns delivery. Select the reason for the movement, for example, [damaged](#).

You can find the returns delivery under [Change Outbound Delivery](#).

3. Posting goods issue

To proceed, post goods issue under [Outbound Delivery](#).

A new FU with the indicated quantity of goods is automatically assigned to the original purchase document.

Supplier Side: Outbound Process

1. Delivering goods based on open quantity

The supplier can now execute a new delivery based on the quantity indicated in the returns delivery.

Related Information

[Advanced Shipping and Receiving](#)

[Integration with Advanced Shipping and Receiving](#)